

CompTIA Server+

Server Identity and Access Management

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- Lab Topology
- Exercise 1 - Password and Account Lockout Policies
- Exercise 2 - User Accounts and Groups
- Exercise 3 - Permissions and Access Controls
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- Exercise 5 - Multifactor Authentication & Single Sign-on
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Introduction

Server+

Password Policies

Account Lockout Policies

Auditing

Multifactor authentication (MFA)

Single Sign-on (SSO)

Welcome to the **Server Identity and Access Management** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

A business's infrastructure, personnel, and data security rely heavily on effective access control. With the use of software like Microsoft Active Directory and Group Policy Management, administrators will be able to make new user accounts and easily manage the privileges these accounts require. When properly set up, these technologies can permit a review of past network activity. This section will investigate, create, and set up policies that may be implemented to better protect the network.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Password and Account Lockout Policies

- Exercise 2 - User Accounts and Groups
- Exercise 3 - Permissions and Access Controls
- Exercise 4 - Auditing
- Exercise 5 - Multifactor Authentication & Single Sign-on

After completing this module, you should be able to:

- Explore, Define and Configure Password Policies
- Explore, Define and Configure Account Lockout Policy
- Apply Password Policy
- Update Group Policy
- Explore, Define and Configure Active Directory User and Computers Accounts
- Explore, Define and Configure User Groups and Role Based Access Control
- Explore, Define, and Configure Delegation
- Audit User Activity
- Audit Logins
- Audit Group Membership

After completing this module, you should have further knowledge of:

- Segregation of Duties
- Multifactor Authentication
- Single Sign-On

Exam Objectives

The following exam objectives are covered in this module:

3.3 Explain important concepts pertaining to identity and access management for server administration

- User accounts
- User groups
- Password policies
- Permissions and access controls
- Auditing
- Multifactor authentication (MFA)
- Single sign-on (SSO)

Lab Duration

It will take approximately **1 hour and 30 minutes** to complete this lab.

The completion time of this module exceeds our usual 1-hour timeframe. Please ensure you have taken into consideration how long this module will take to complete.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)
- **PLABDM01** - (Windows Server 2022 - Domain Member Server)
- **PLABWIN10** - (Windows 10 - Domain Member Workstation)
- **PLABALMA** - (Alma Linux 8.7 - Stand-alone Linux Workstation)
- **PLABUBUNTU** - (Linux Ubuntu - Stand-alone Linux Server)

Click **Next** to proceed to the first exercise.

Exercise 1 - Password and Account Lockout Policies

Password and account lockout policies and their configuration are crucial to the safety of the network. Too weak, and the network can be at risk. Too strong, and users will find ways to circumvent it.

In this exercise, you will explore different options for password and account lockout policies.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore, Define and Configure Password Policies
- Explore, Define and Configure Account Lockout Policy
- Apply Password Policy
- Update Group Policy

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Explore, Define and Configure Password Policies

A strong password policy will help ensure users use passwords with adequate strength to keep their computers and the network safe. Changing the password often, not reusing passwords, and having complex passwords with combinations of letters, numbers and symbols are recommended. This will help to stop attacks like dictionary attacks which try combinations of common words to crack the password.

In this task, you will explore, define, and configure these password policies.

Step 1

Connect to **PLABDC01**.

The **Server Manager** window is displayed.

Click the **Tools** menu and select **Group Policy Management**.

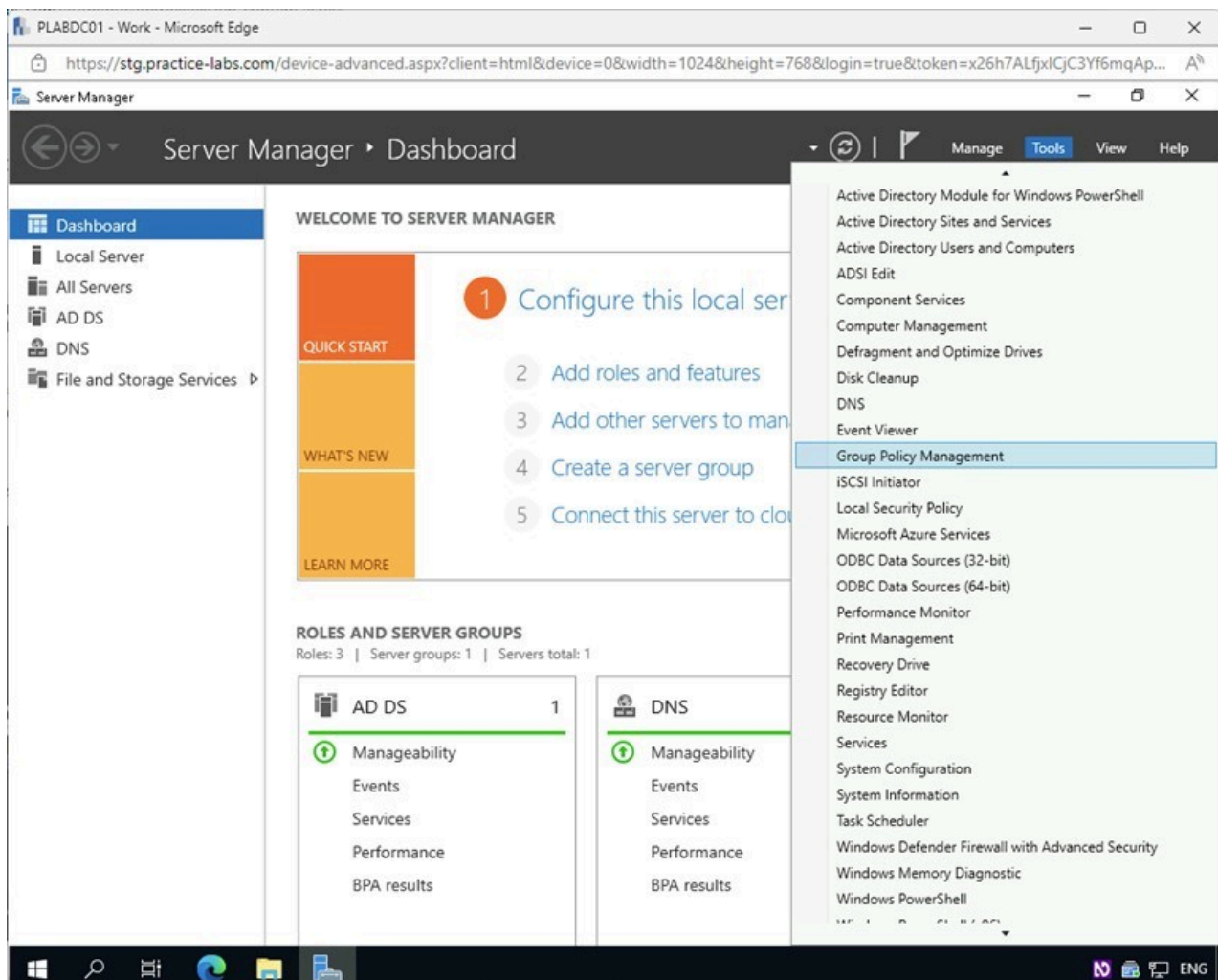


Figure 1.1 Screenshot of PLABDC01: Displaying the Server Manager and selecting Tools > Group Policy Management.

Step 2

In the **Group Policy Management** window, expand **Forest: practicelabs.com > Domains**.

Right-click on **practicelabs.com** and select **Create a GPO in this domain, and Link it here....**

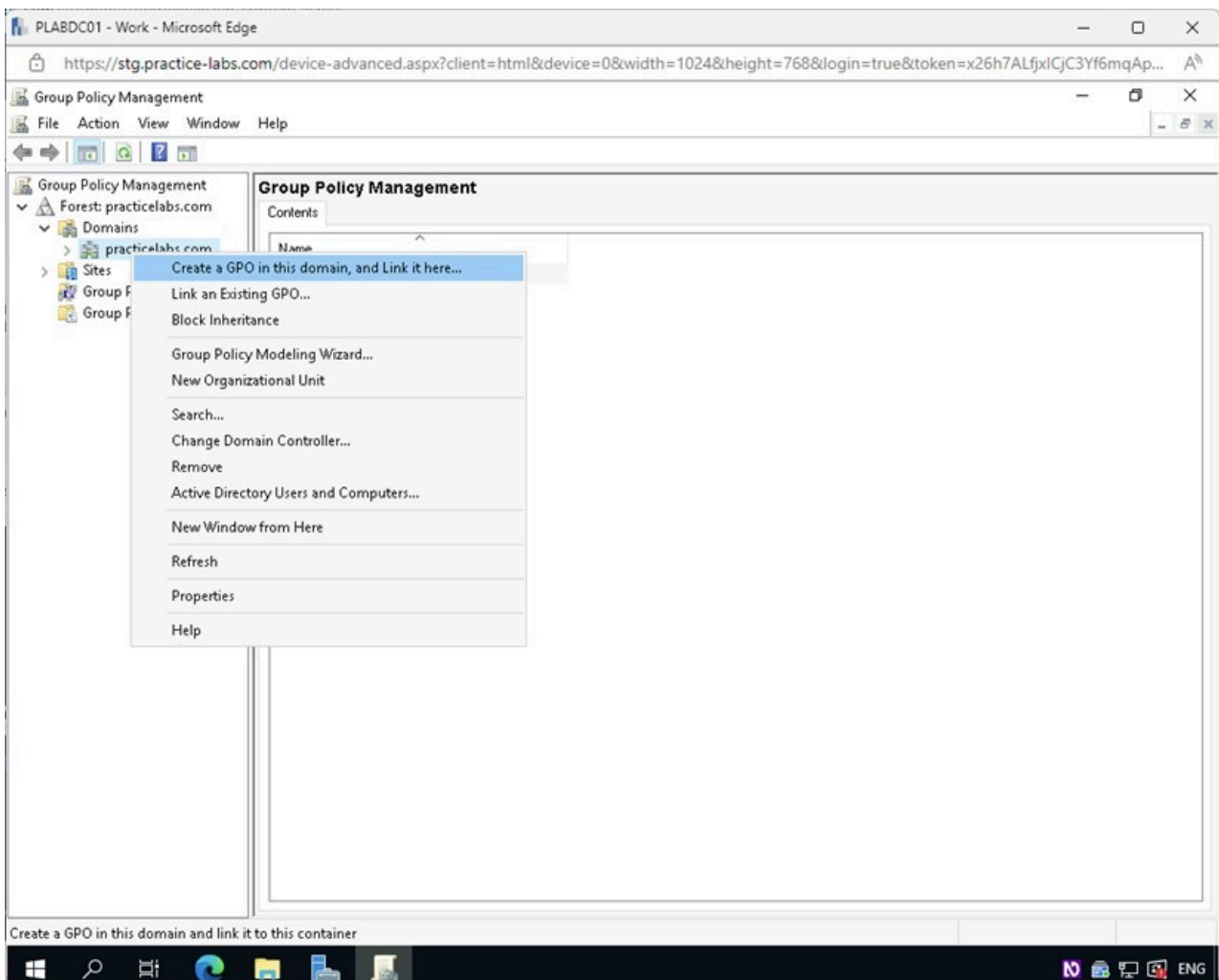


Figure 1.2 Screenshot of PLABDC01: Displaying the Group Policy Management window. practicelabs.com is right-clicked, and Create a GPO in this domain, and Link it here... is selected.

Step 3

The **New GPO** pop-up window appears.

Type the following into the **Name** text box and click **OK**:

Password Policy

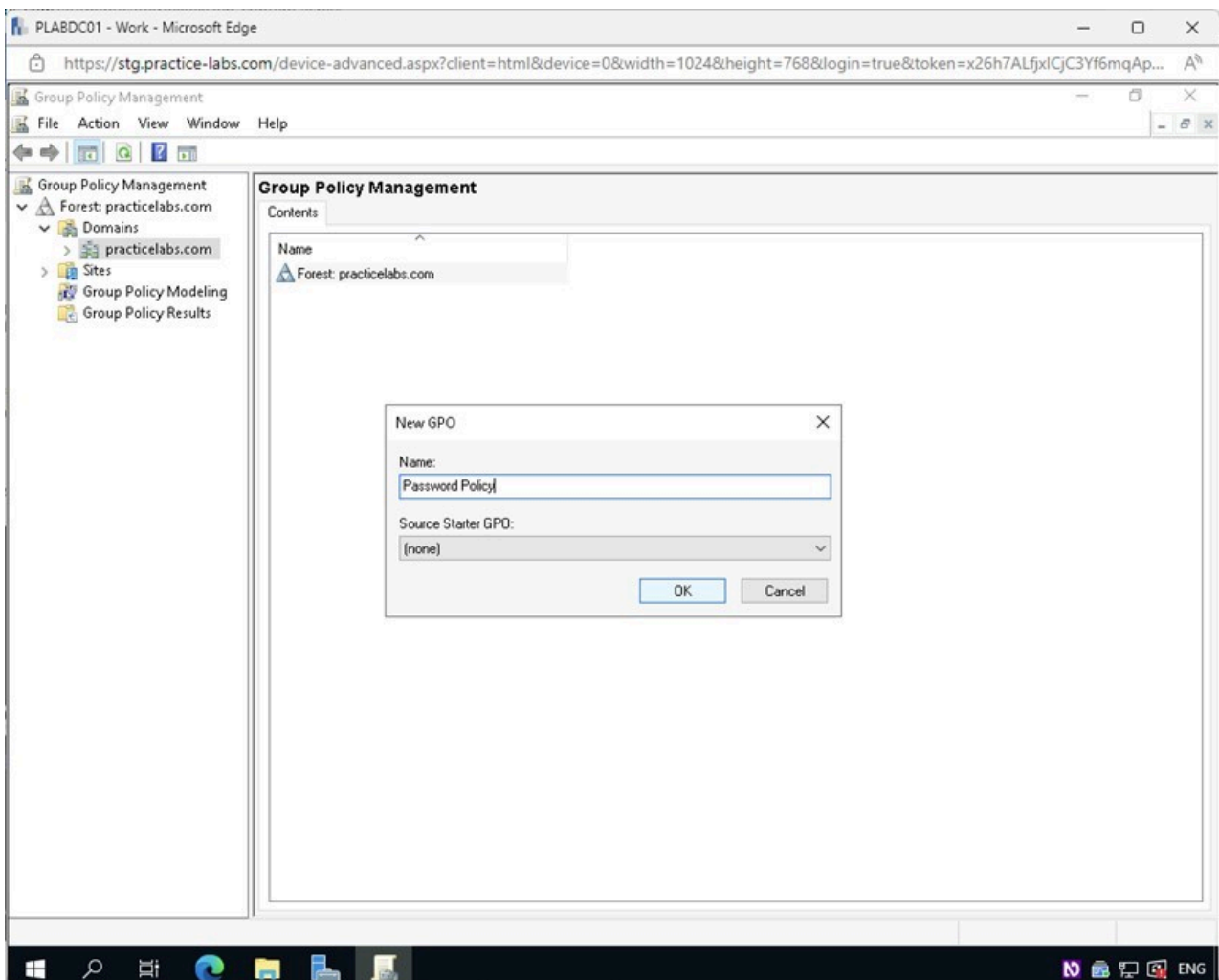


Figure 1.3 Screenshot of PLABDC01: Displaying the New GPO pop-up, showing the required Name entered and the OK button selected.

Step 4

Back on the **Group Policy Management** window, expand the **practicelabs.com** node.

Notice the new GPO **Password Policy** has been created.

Right-click on **Password Policy** and select **Edit**.

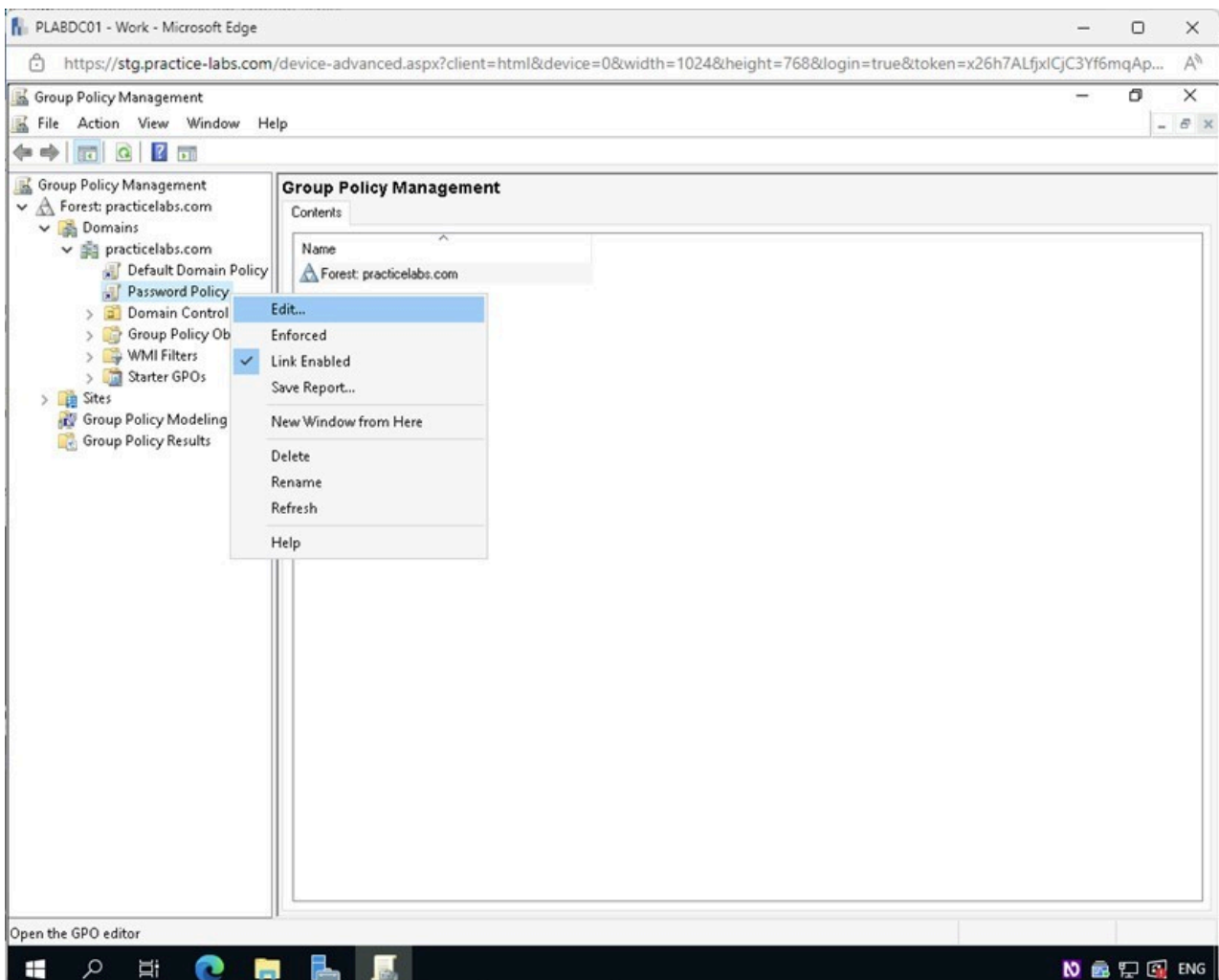


Figure 1.4 Screenshot of PLABDC01: Displaying the Group Policy Management window. Password Policy is right-clicked, and Edit selected.

Step 5

In the **Group Policy Management Editor** window, expand the following nodes on the left pane:

Computer Configuration > Policies > Windows Settings > Security Settings > Account Policies.

Select **Password Policy**.

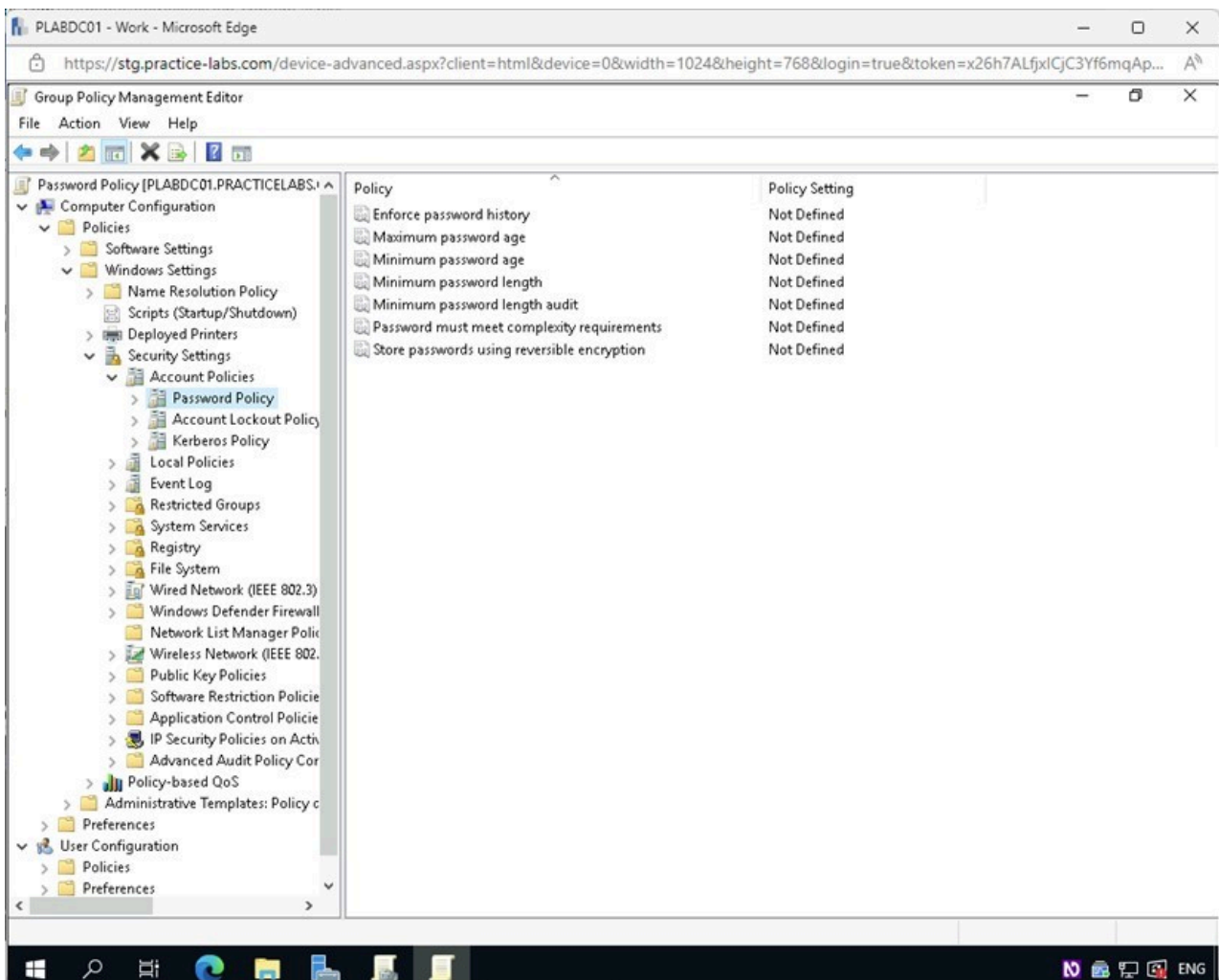


Figure 1.5 Screenshot of PLABDC01: Displaying the Group Policy Management Editor pane, showing Password Policy selected on the left pane.

Step 6

In the right details pane, double-click on the **Enforce password history** Policy.

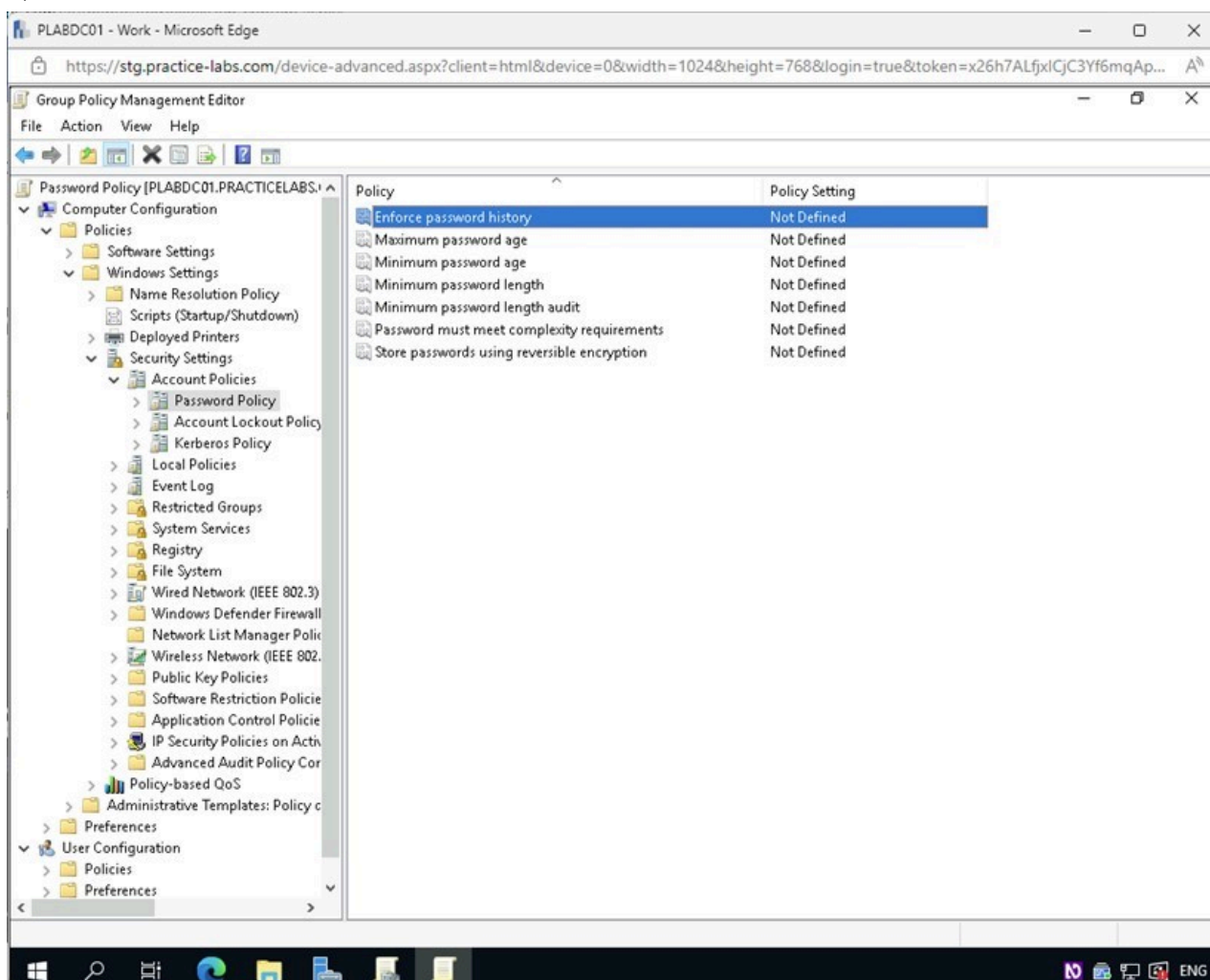


Figure 1.6 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Enforce password history Policy selected.

Step 7

On the **Enforce password history Properties** dialog box, tick the **Define this policy setting** checkbox.

Change the **Keep password history for** field to:

5

Select the **Explain** tab.

Note: The **Keep password history for** field will initially appear as **Do not keep password history**. Once you've changed the value, it changes to **Keep password history for**.

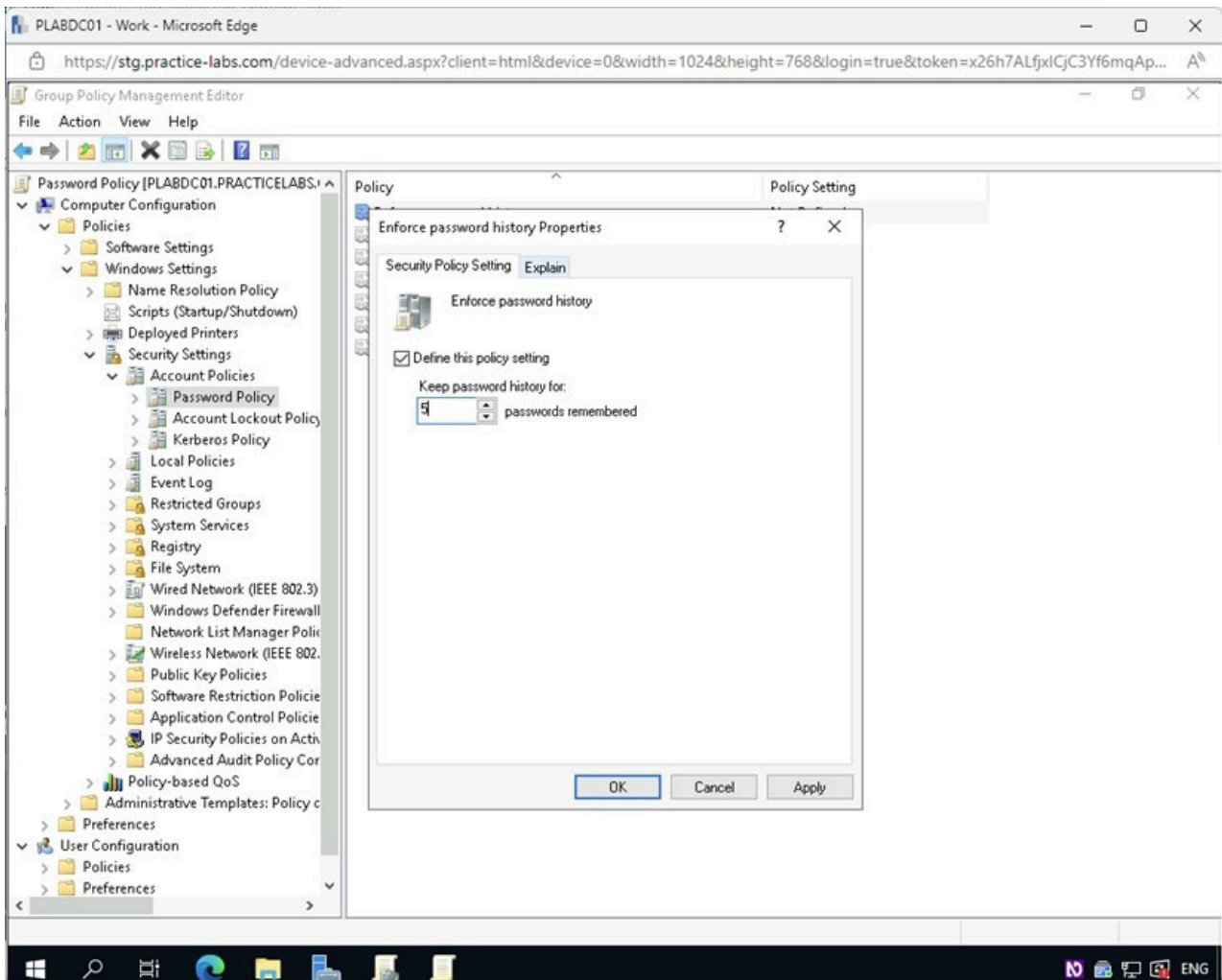


Figure 1.7 Screenshot of PLABDC01: Displaying the Enforce password history Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 8

In the **Explain** tab, you can view detailed information on the **Enforce password history** policy.

Once you've read the information available, click **OK**.

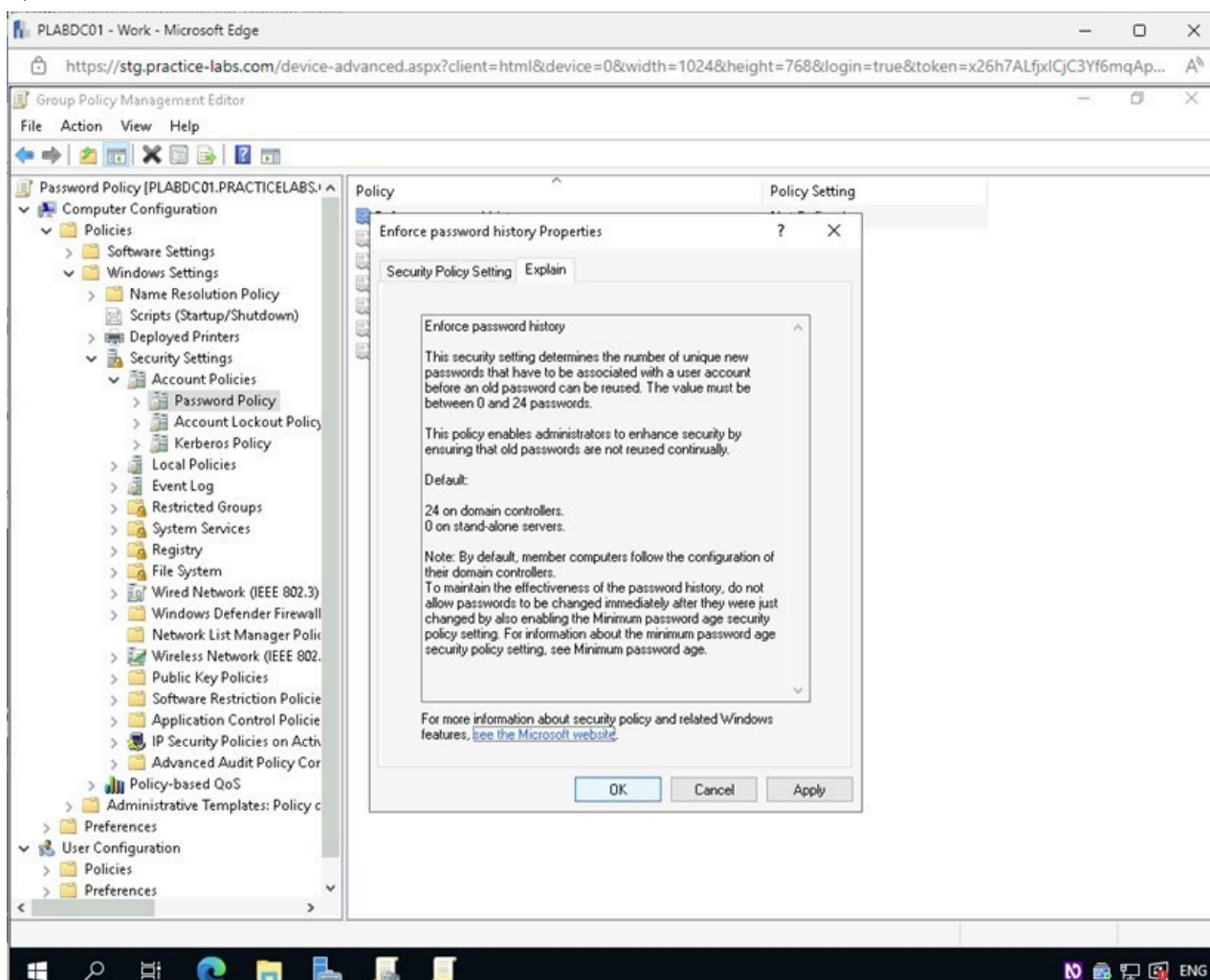


Figure 1.8 Screenshot of PLABDC01: Displaying the Enforce password history Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 9

Now that you have seen the **Enforce Password History** Policy let's look at the **Maximum password age**.

Back on the **Group Policy Management Editor** window, double-click on the **Maximum password age** Policy.

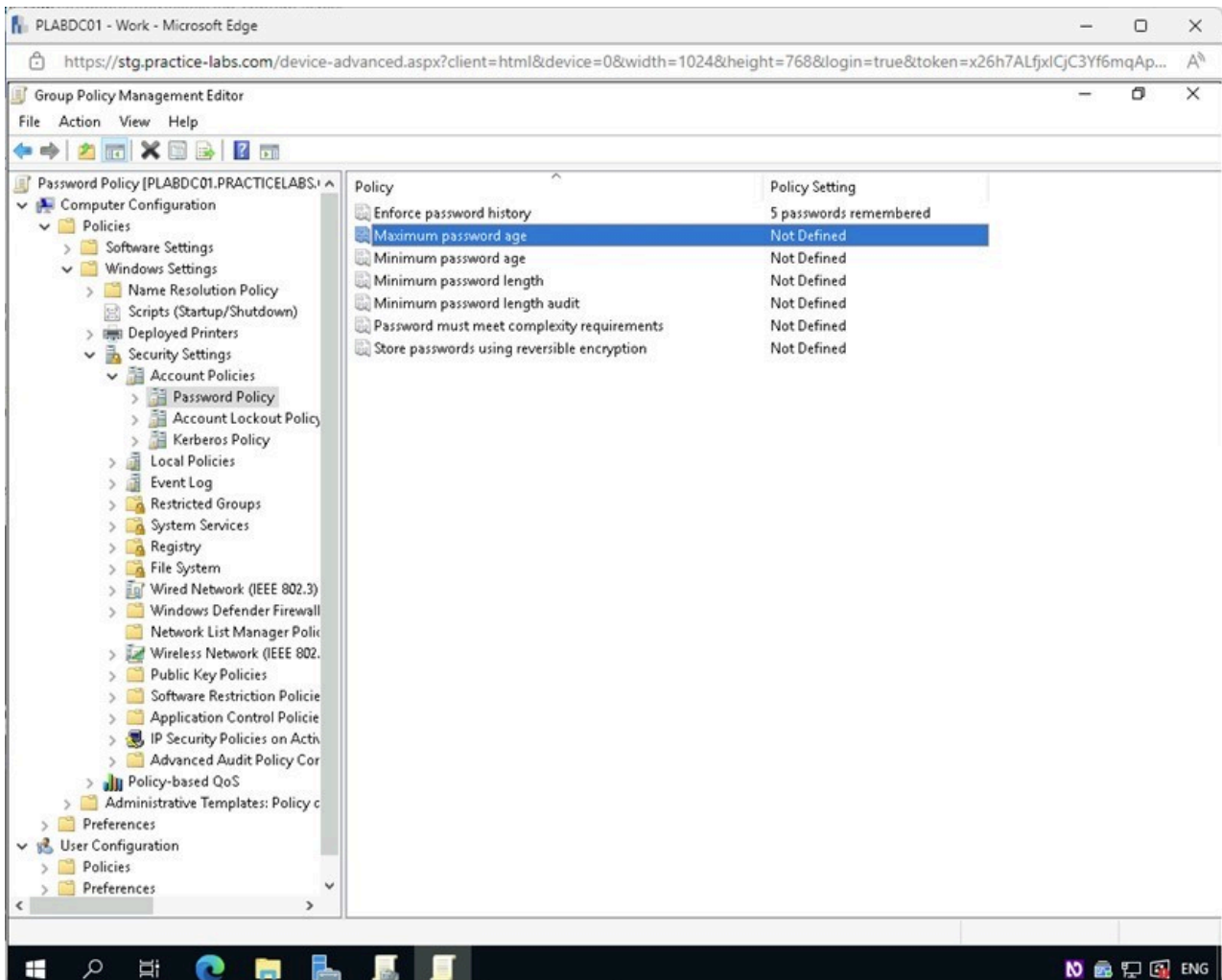


Figure 1.9 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Maximum password age Policy selected.

Step 10

On the **Maximum password age Properties** dialog box, tick the **Define this policy setting** checkbox.

Change the **Password will expire in** field to:

60

Select the **Explain** tab.

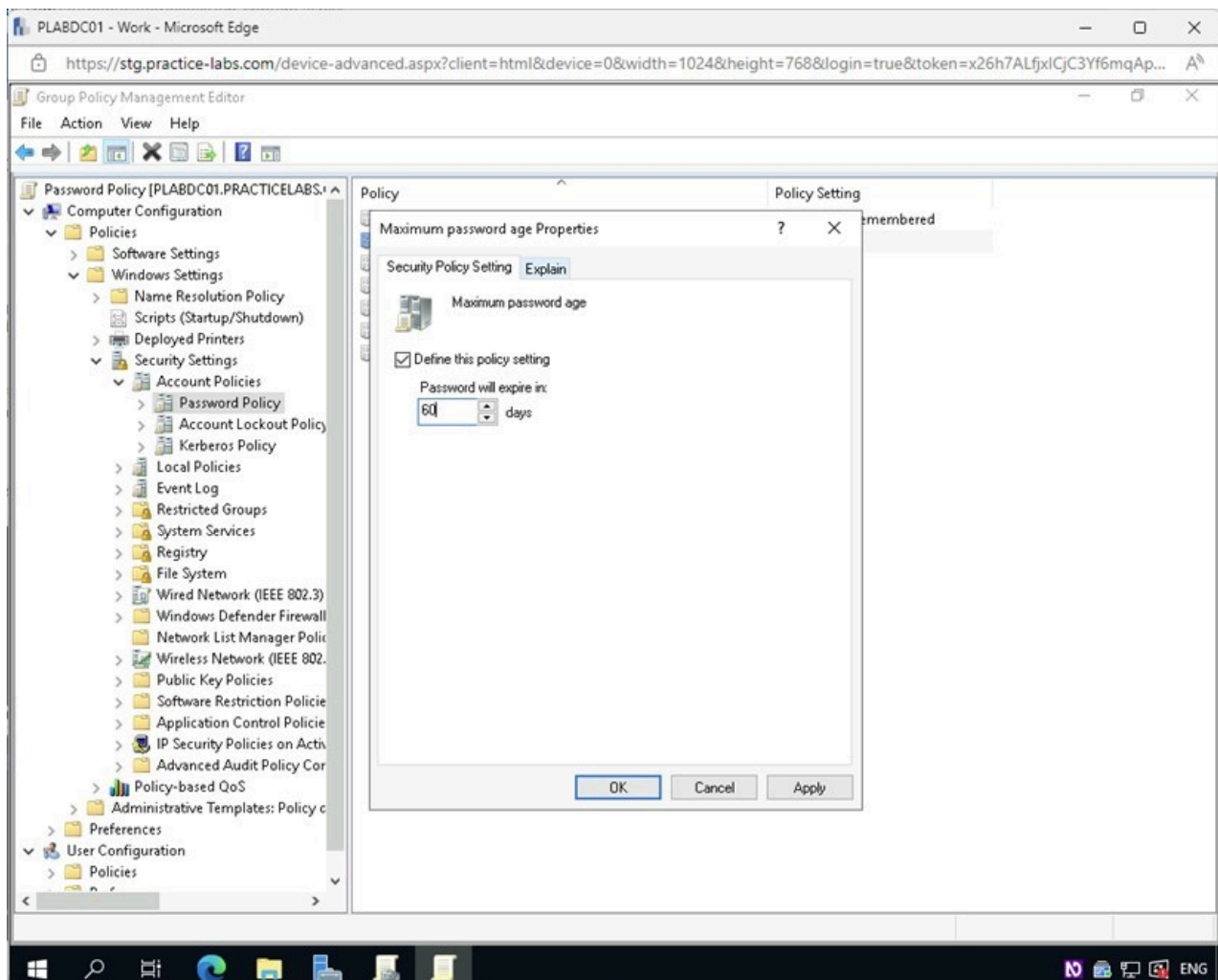


Figure 1.10 Screenshot of PLABDC01: Displaying the Maximum password age Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 11

In the **Explain** tab, you can view detailed information on the **Maximum password age** policy.

Once you've read the information available, click **OK**.

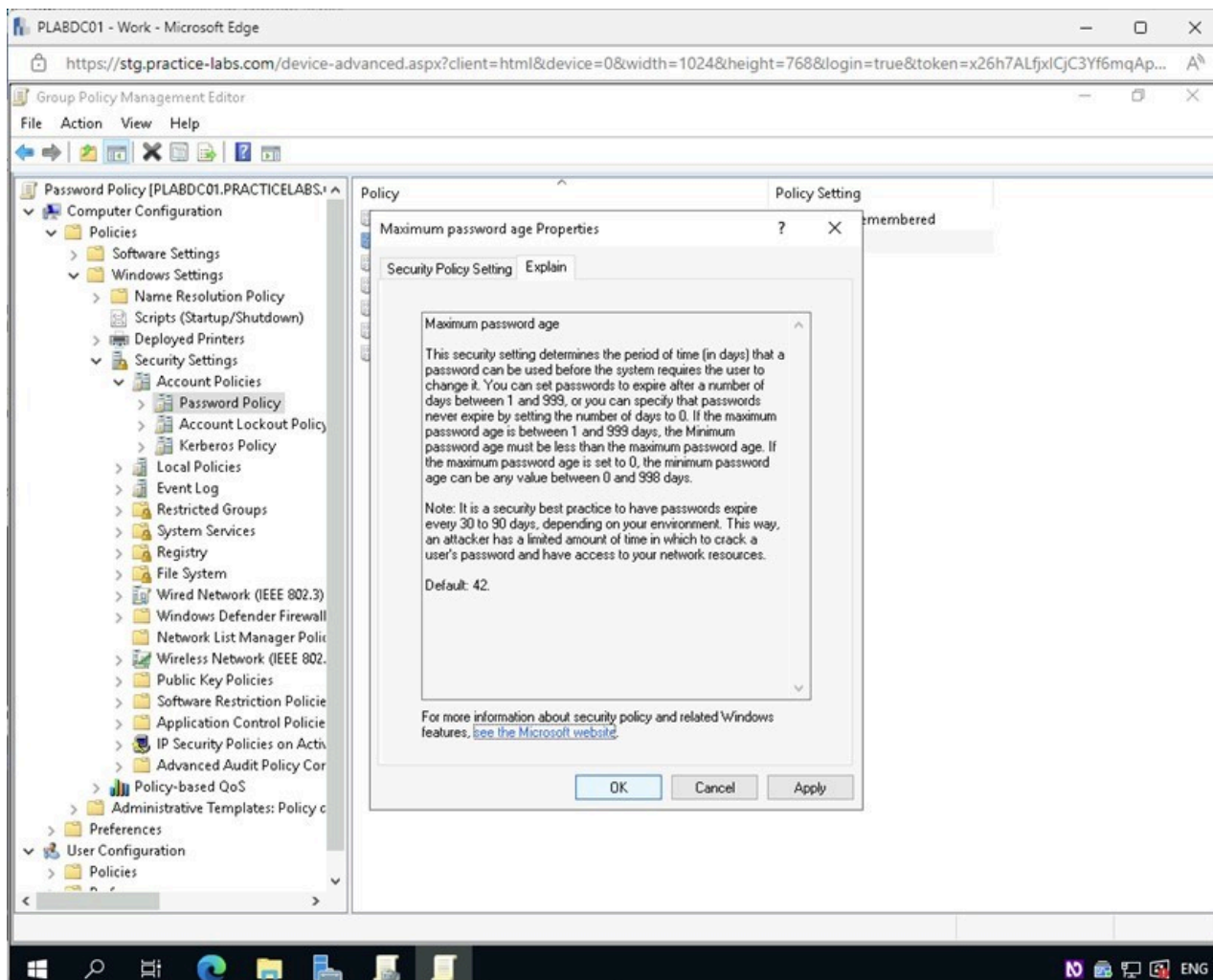


Figure 1.11 Screenshot of PLABDC01: Displaying the Maximum password age policy Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 12

Setting the **Maximum Age** to **60** automatically sets the **Minimum password age** to **30 days**.

Click **OK** on the **Suggested Value Changes** pop-up window.

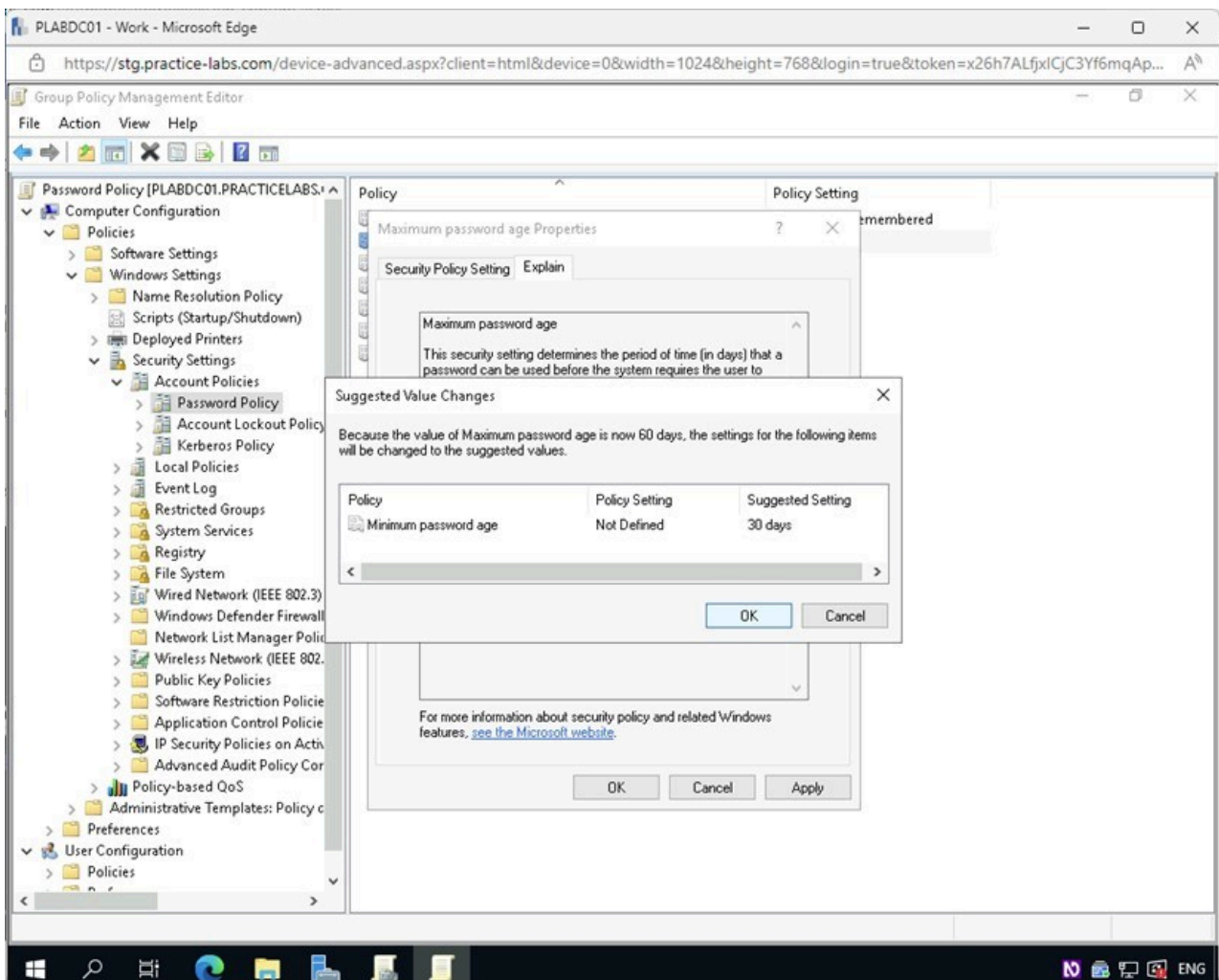


Figure 1.12 Screenshot of PLABDC01: Displaying clicking OK on the Suggested Value Changes pop-up window.

Step 13

Back on the **Group Policy Management Editor** window, double-click on the **Minimum password age** Policy.

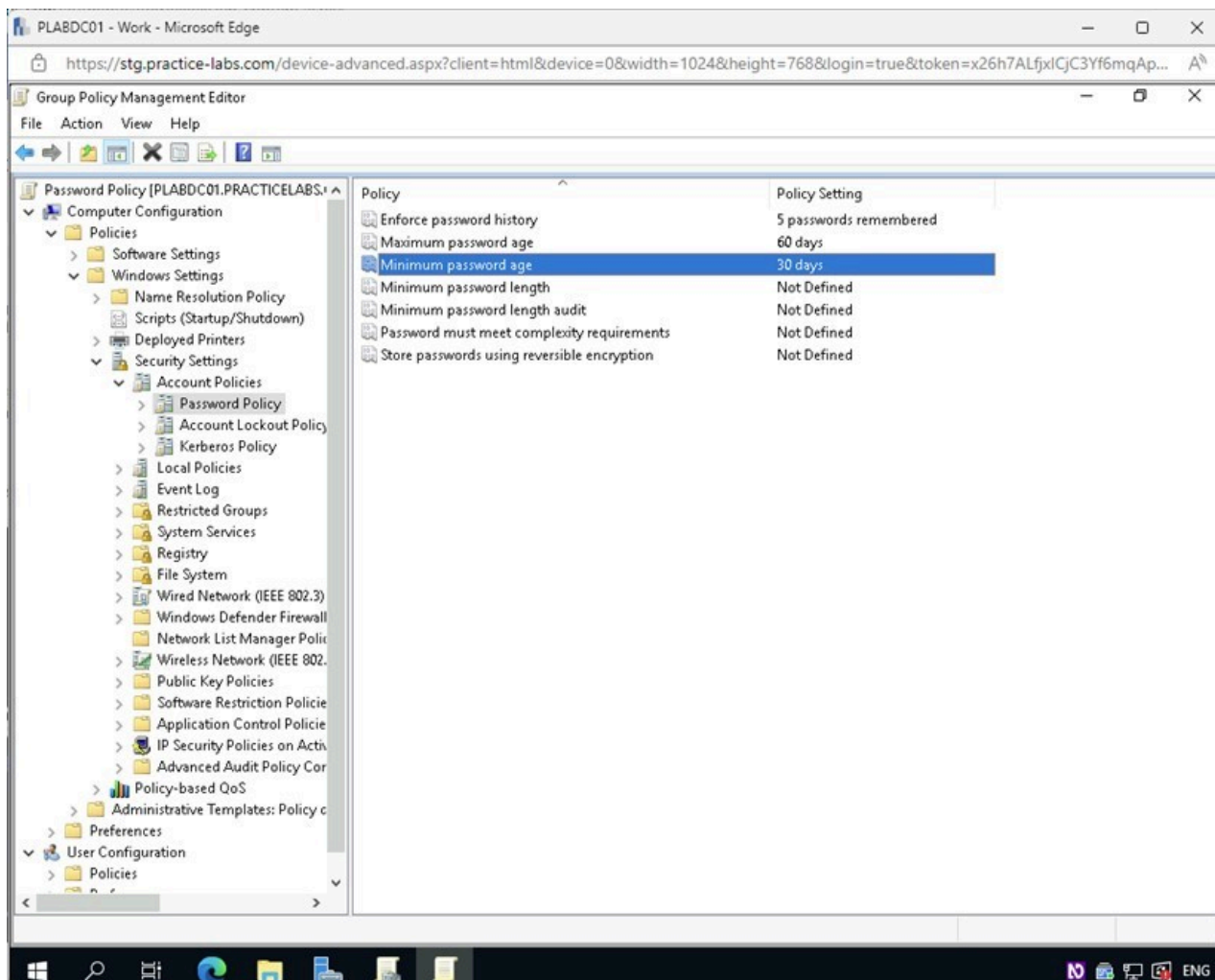


Figure 1.13 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Minimum password age Policy selected.

Step 14

On the **Minimum password age Properties** dialog box, select the **Explain** tab.

Here you can view detailed information on the **Minimum password age** policy.

Once you've read the information available, click **OK**.

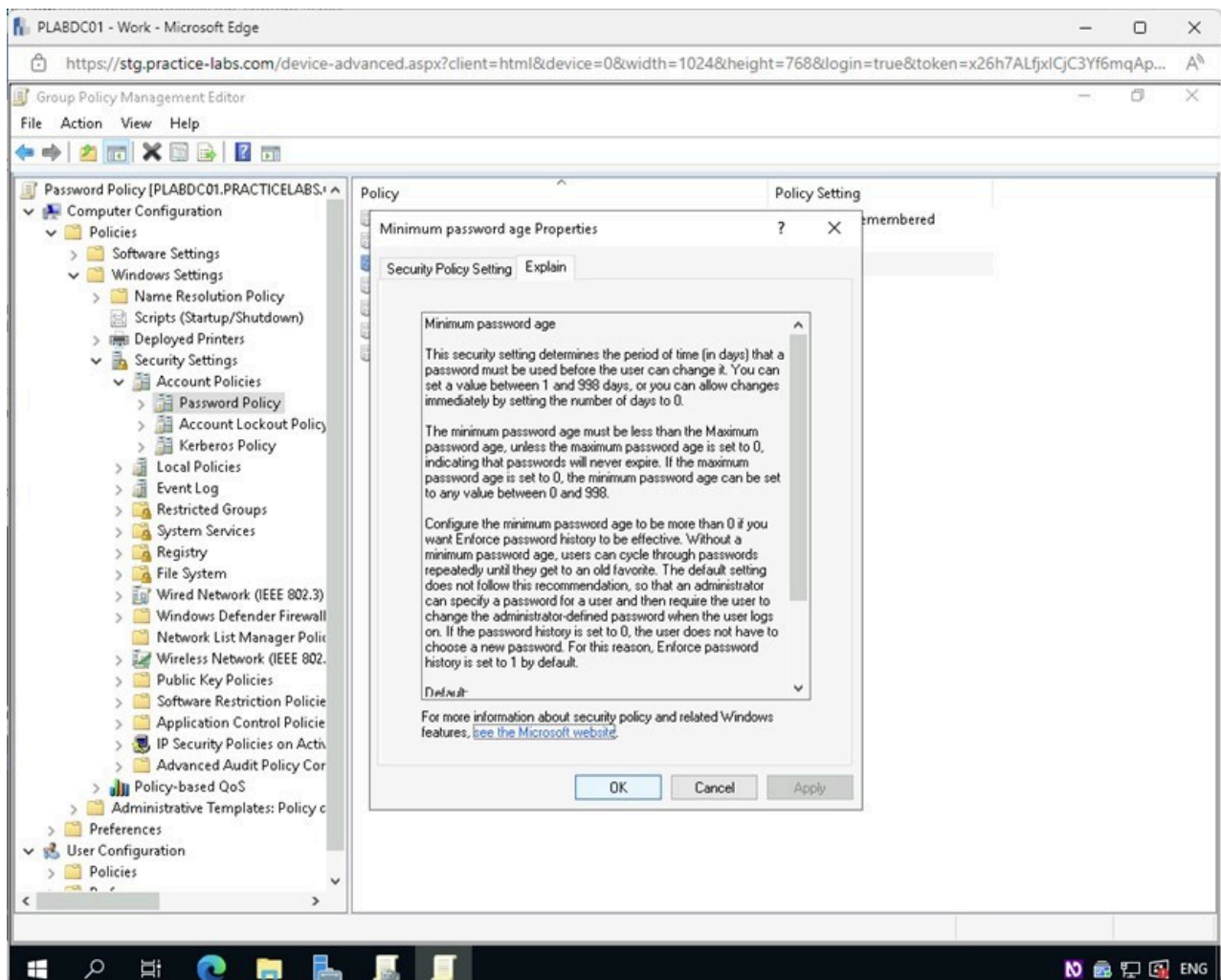


Figure 1.14 Screenshot of PLABDC01: Displaying the Minimum password age Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 15

Back on the **Group Policy Management Editor** window, double-click on the **Minimum password length** Policy.

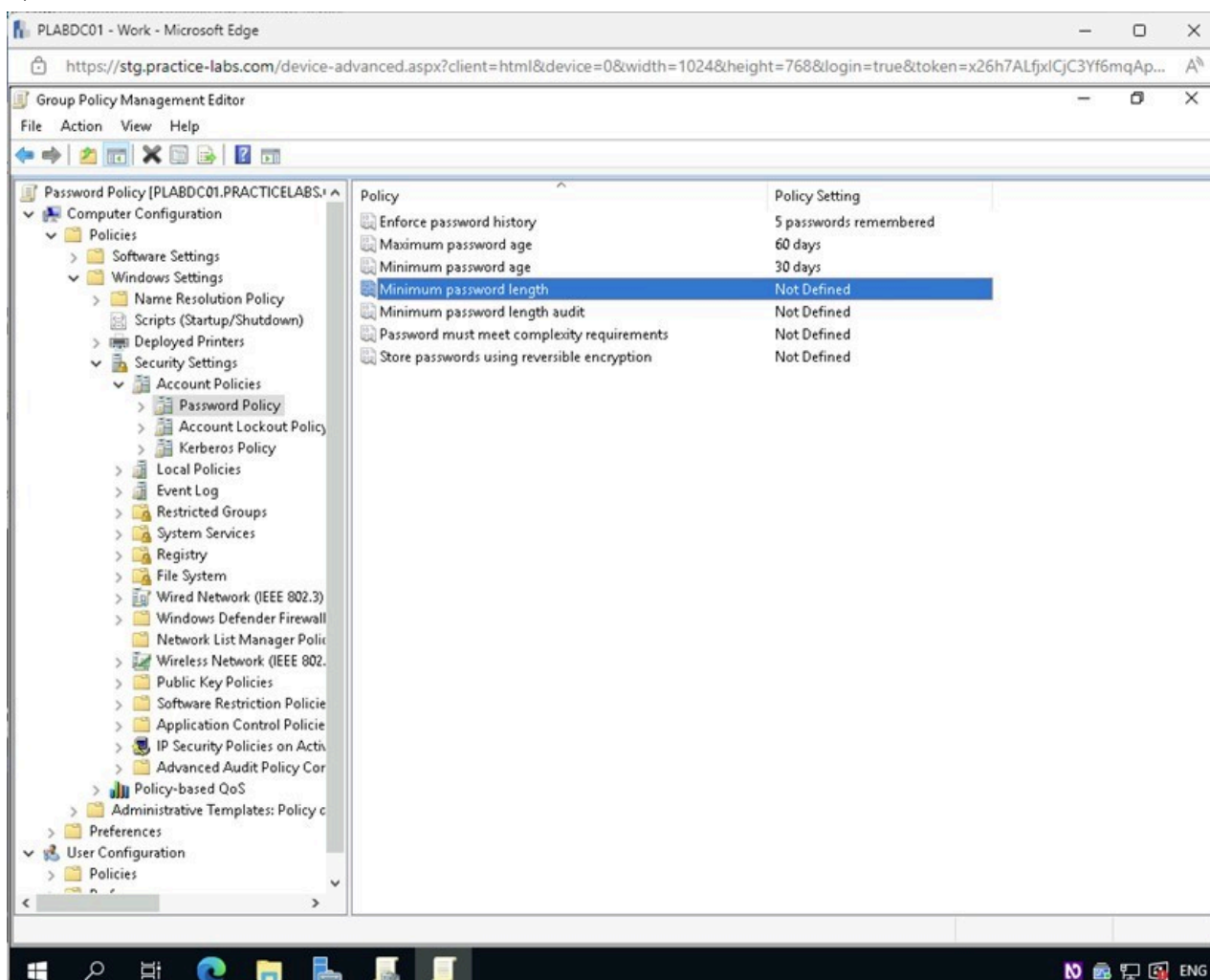


Figure 1.15 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Minimum password length Policy selected.

Step 16

On the **Minimum password length Properties** pane, tick the **Define this policy setting** checkbox.

Change the **Password must be at least** field to:

10

Select the **Explain** tab.

Note: The **Password must be at least** field will initially appear as **No password required**. Once you've changed the value, it changes to **Password must be at least**.

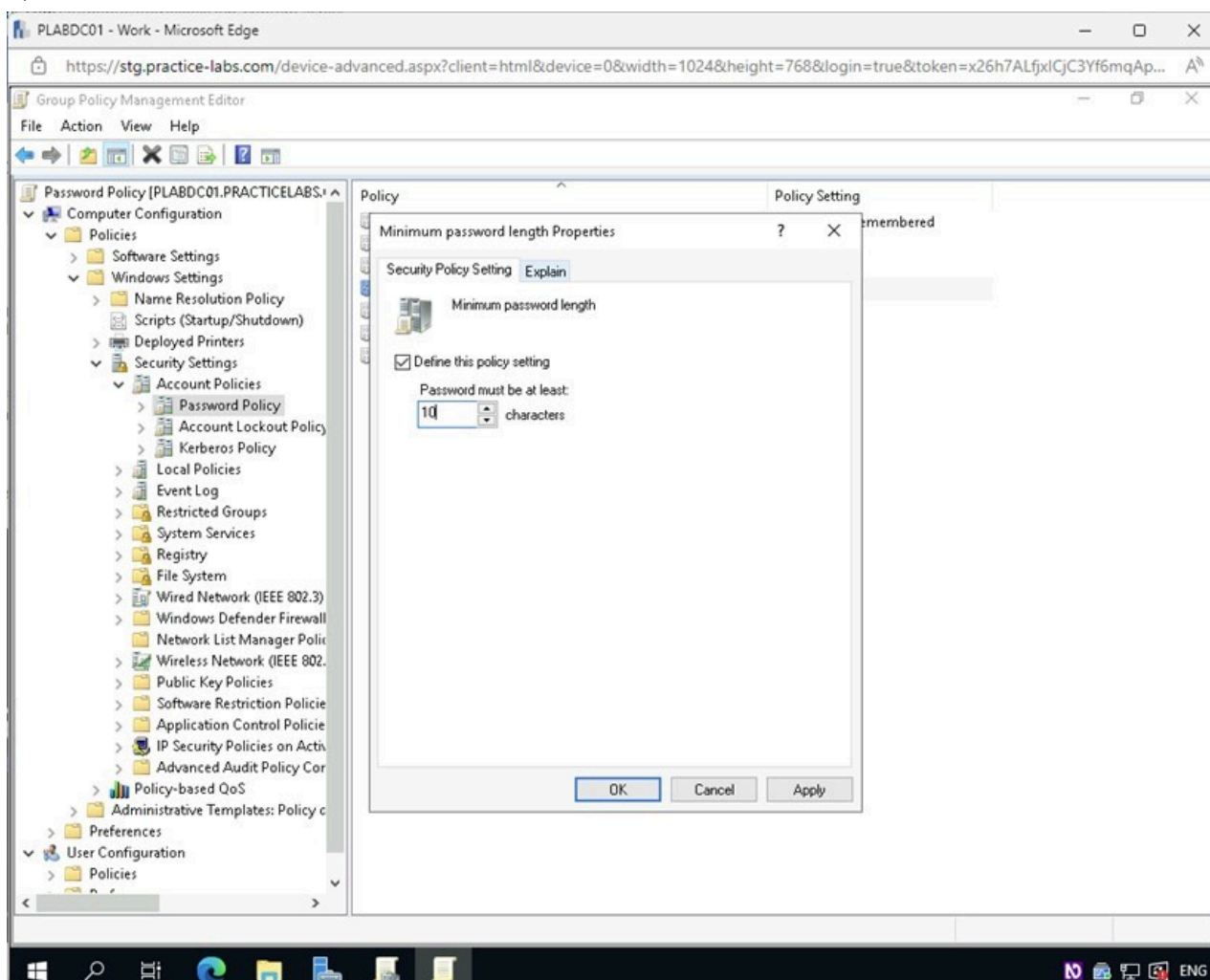


Figure 1.16 Screenshot of PLABDC01: Displaying the Minimum password length Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 17

In the **Explain** tab, you can view detailed information on the **Minimum password length** policy.

Once you've read the information available, click **OK**.

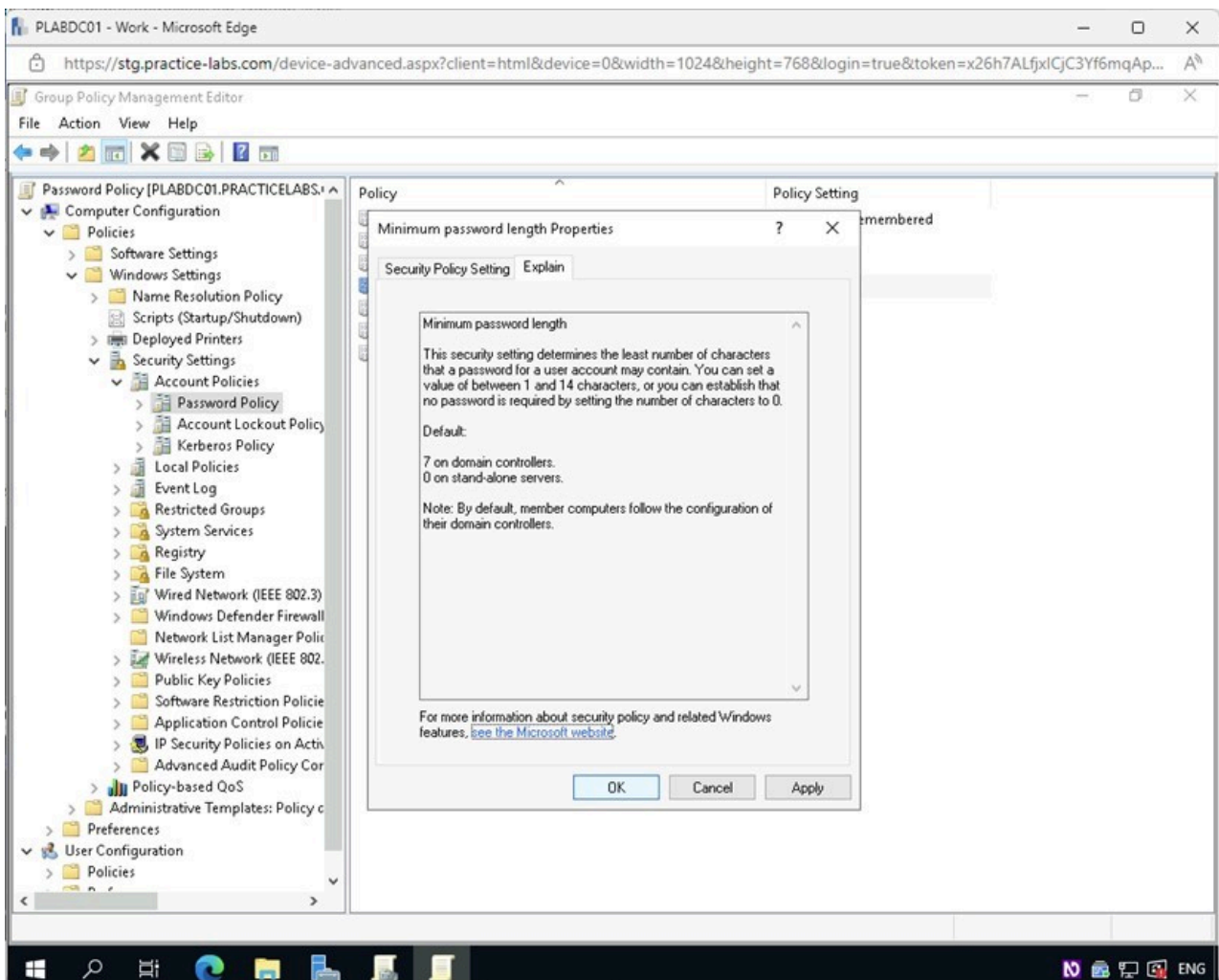


Figure 1.17 Screenshot of PLABDC01: Displaying the Minimum password length Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 18

Back on the **Group Policy Management Editor** window, double-click on the **Password must meet complexity requirements** Policy.

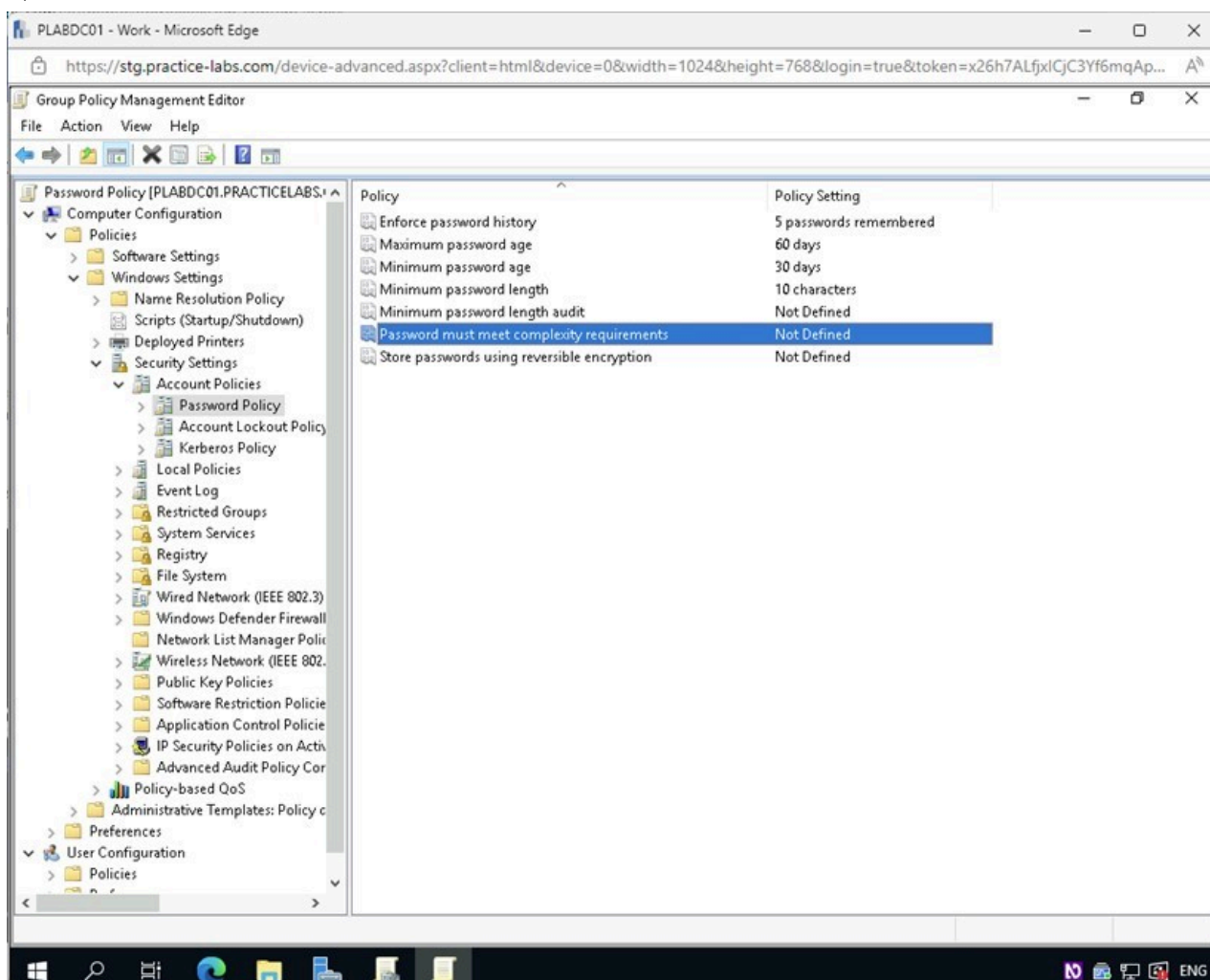


Figure 1.18 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Password must meet complexity requirements Policy selected.

Step 19

On the **Password must meet complexity requirements Properties** dialog box, tick the **Define this policy setting** checkbox.

Select the **Enabled** radio button option.

Select the **Explain** tab.

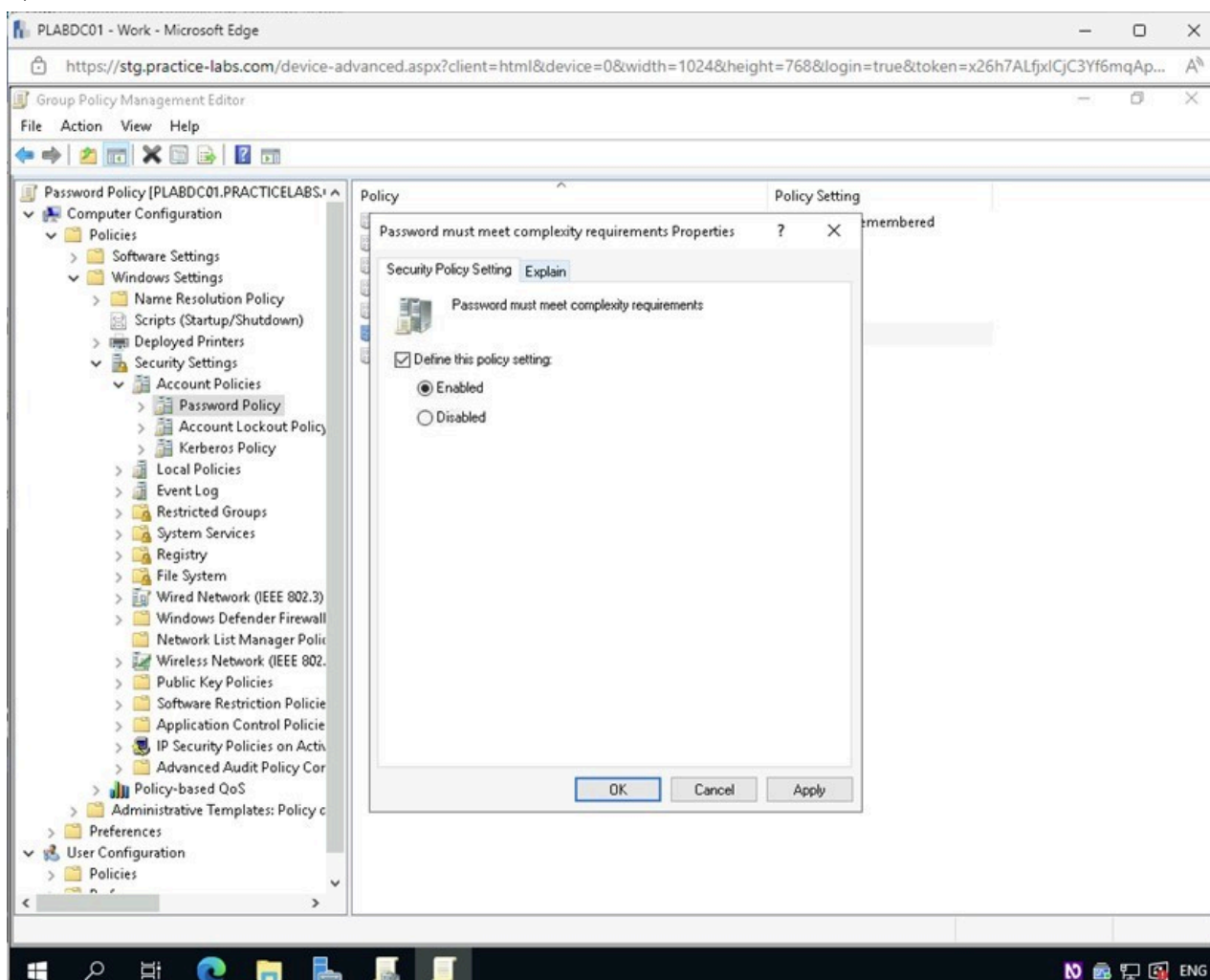


Figure 1.19 Screenshot of PLABDC01: Displaying the Password must meet complexity requirements Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 20

In the **Explain** tab, you can view detailed information on the **Password must meet complexity requirements** policy.

Once you've read the information available, click **OK**.

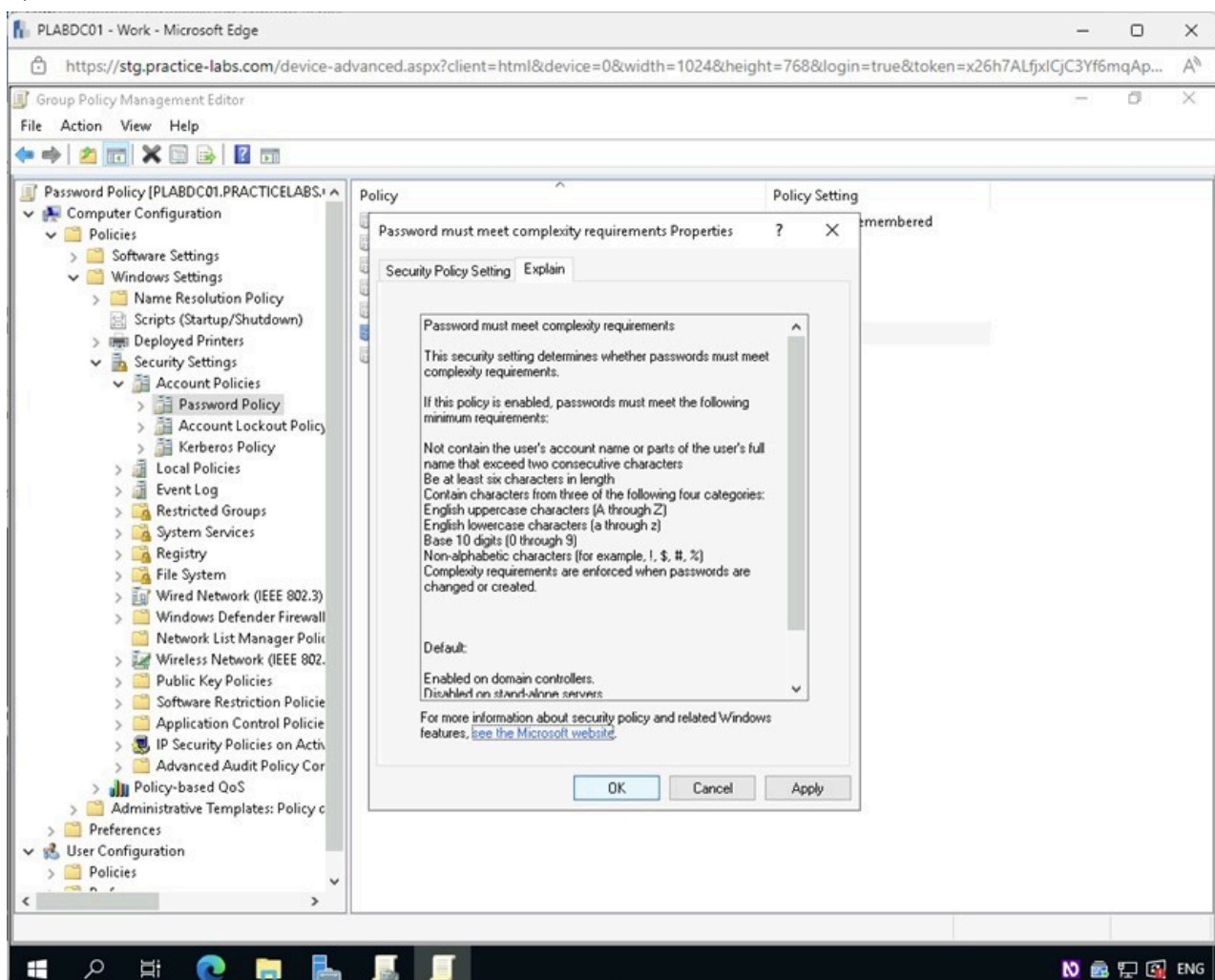


Figure 1.20 Screenshot of PLABDC01: Displaying the Password must meet complexity requirements Properties dialog box, showing the Explain tab and the OK button highlighted.

Keep the **Group Policy Management Editor** window open.

Task 2 - Explore, Define and Configure Account Lockout Policy

Account lockout is another security measure that should be set up alongside a robust password policy. When set up properly, attacks like Brute Force password cracking can be thwarted. The maximum number of failed login attempts before a resource lock is set by the system. You may either manually unlock it after a specific amount of time has passed or set a timer to allow additional tries.

In this task, you will learn about account lockout policies.

Step 1

Ensure you are connected to **PLABDC01** and the **Group Policy Management Editor** window is open.

Select **Account Lockout Policy** in the left pane.

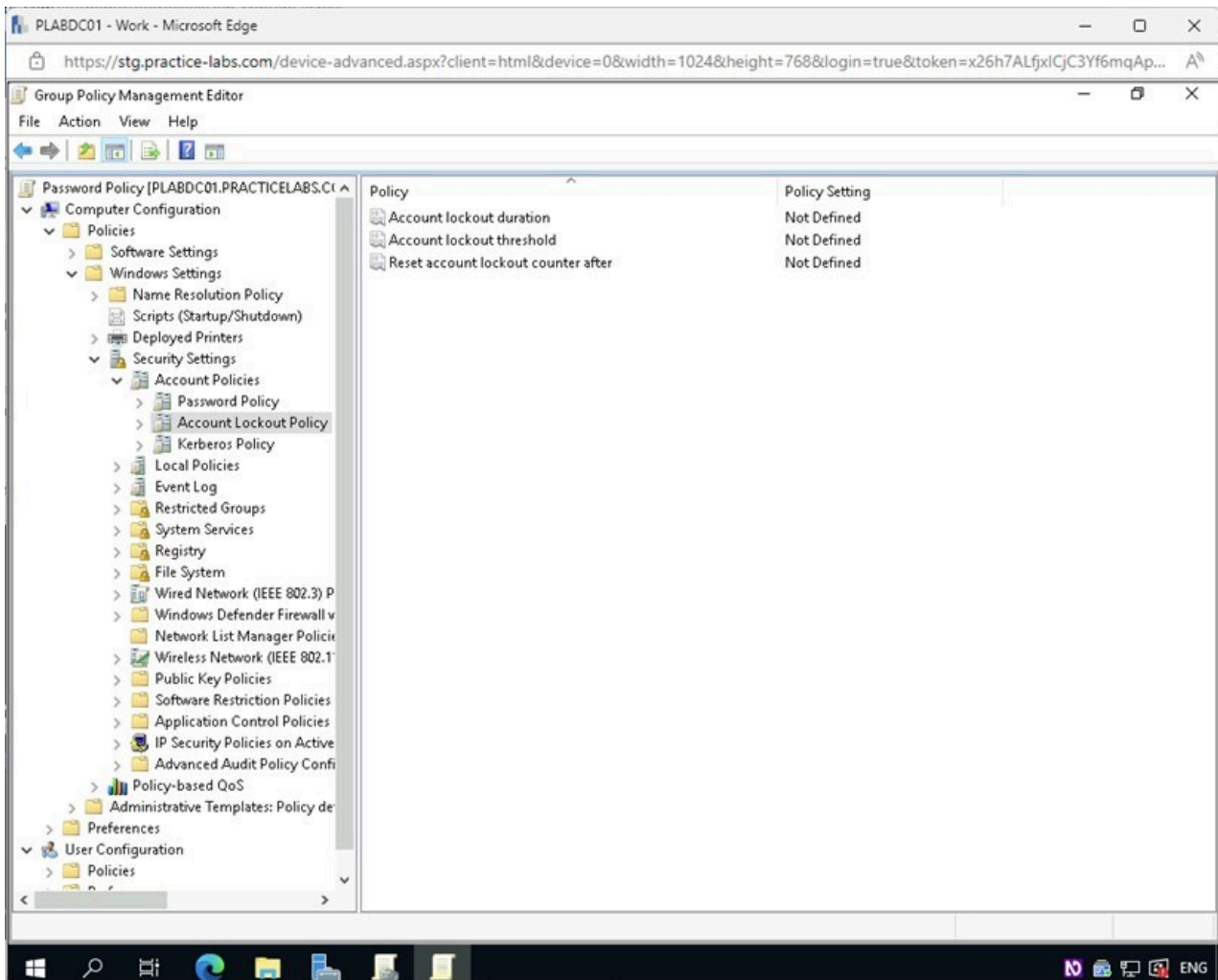


Figure 1.21 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window and selecting Account Lockout Policy on the left pane.

Step 2

In the right details pane, double-click on the **Account lockout duration** Policy.

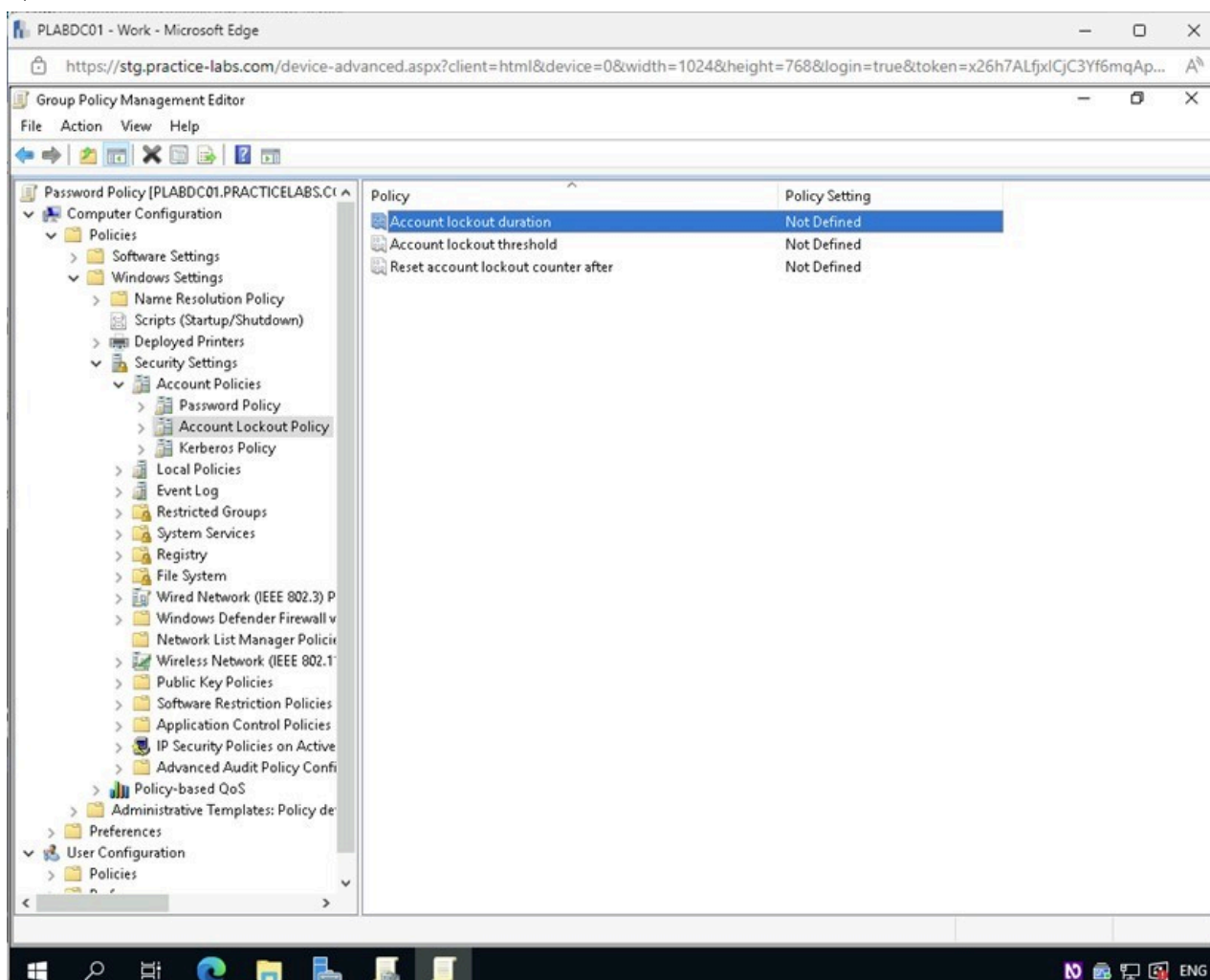


Figure 1.22 Screenshot of PLABDC01: Displaying the Group Policy Manager Editor window, showing the Account lockout duration policy selected.

Step 3

On the **Account lockout duration Properties** dialog box, tick the **Define this policy setting** checkbox.

Change the **Account is locked out for** field to:

30

Select the **Explain** tab.

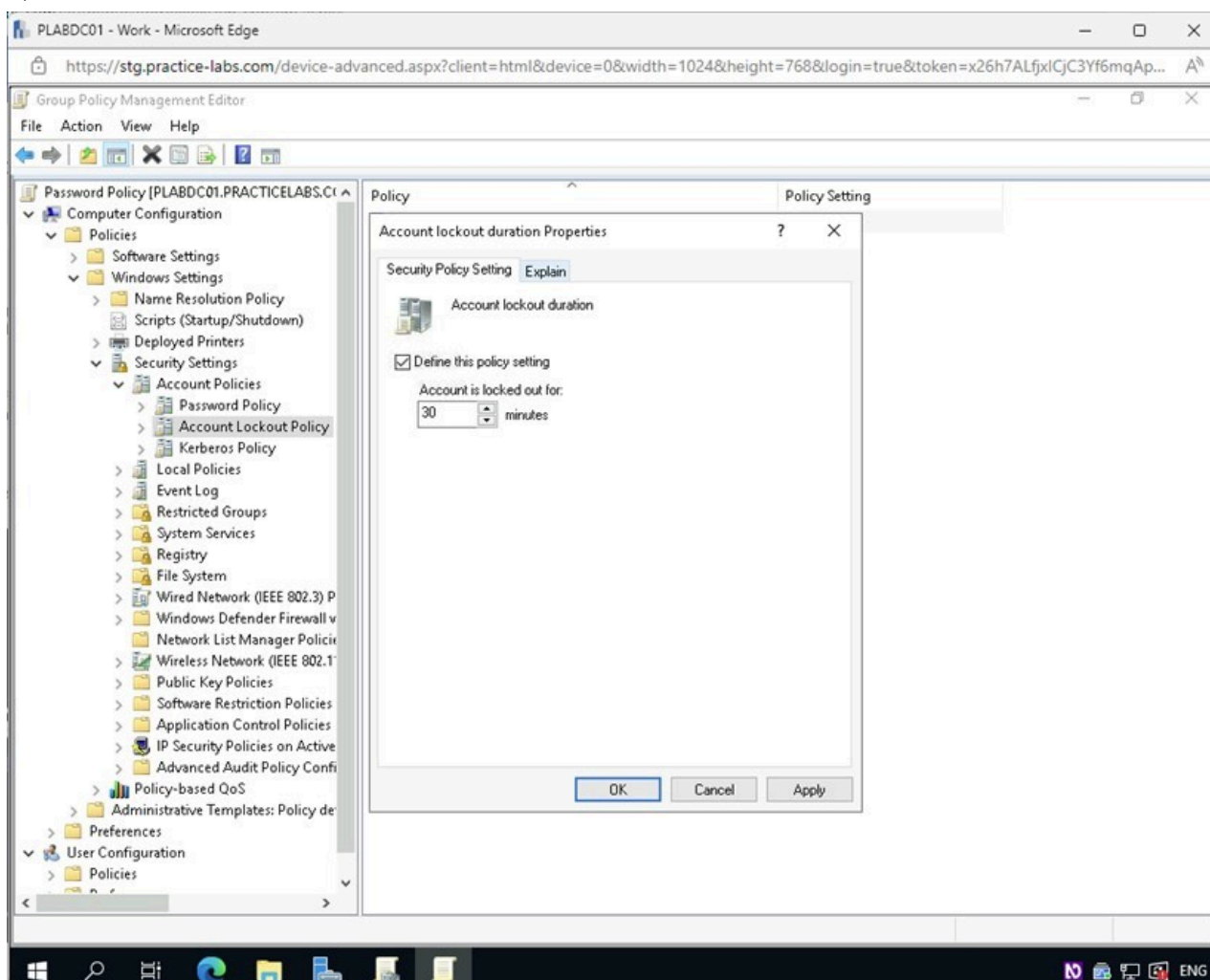


Figure 1.23 Screenshot of PLABDC01: Displaying the Account lockout duration Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 4

In the **Explain** tab, you can view detailed information on the **Account lockout duration** policy.

Once you've read the information available, click **OK**.

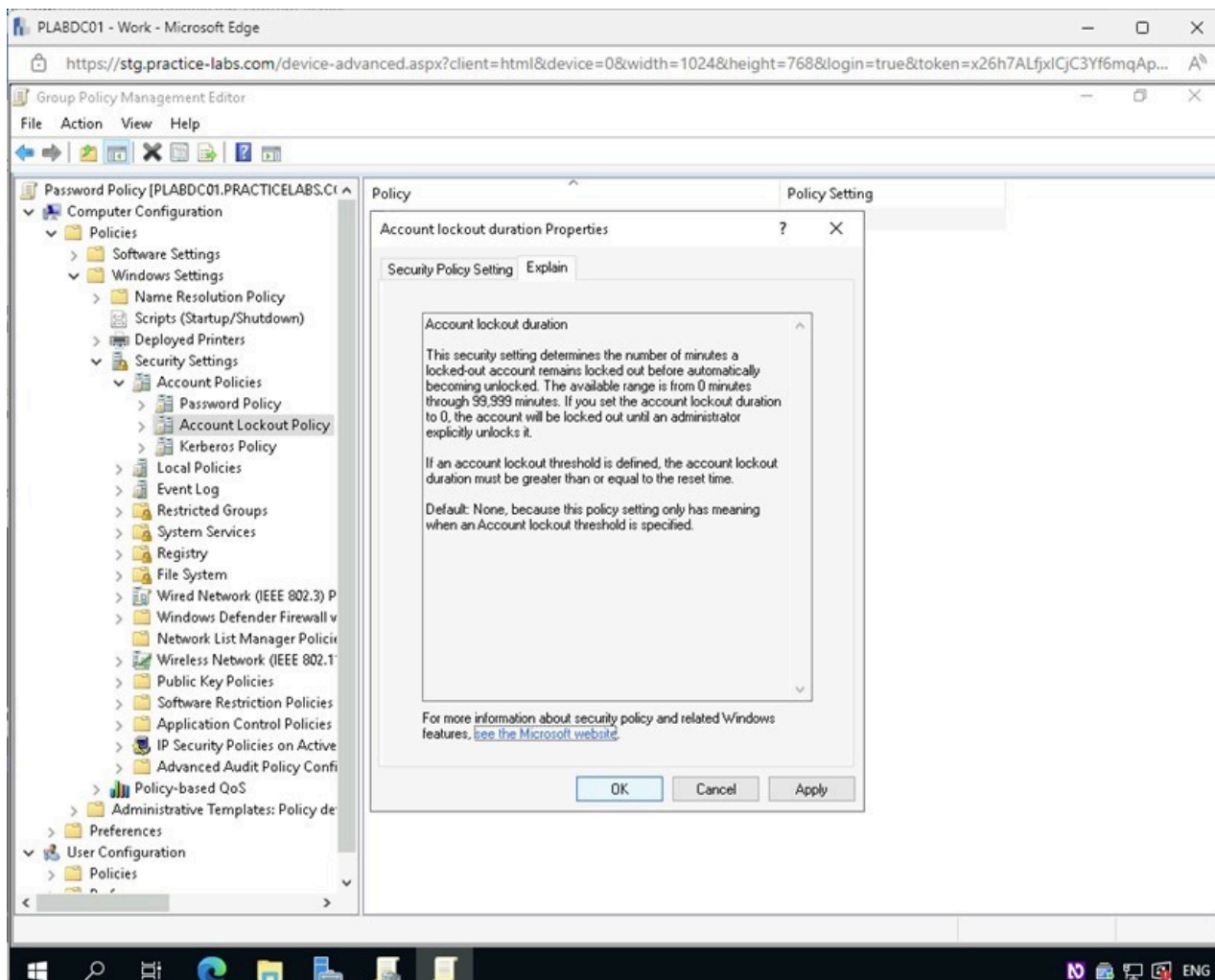


Figure 1.24 Screenshot of PLABDC01: Displaying the Account lockout duration policy Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 5

Setting the **Account lockout duration** automatically sets the **Account lockout threshold** to **5 invalid logon attempts** and the **Reset account lockout counter after** to **30 minutes**.

Click **OK** on the **Suggested Value Changes** pop-up window.

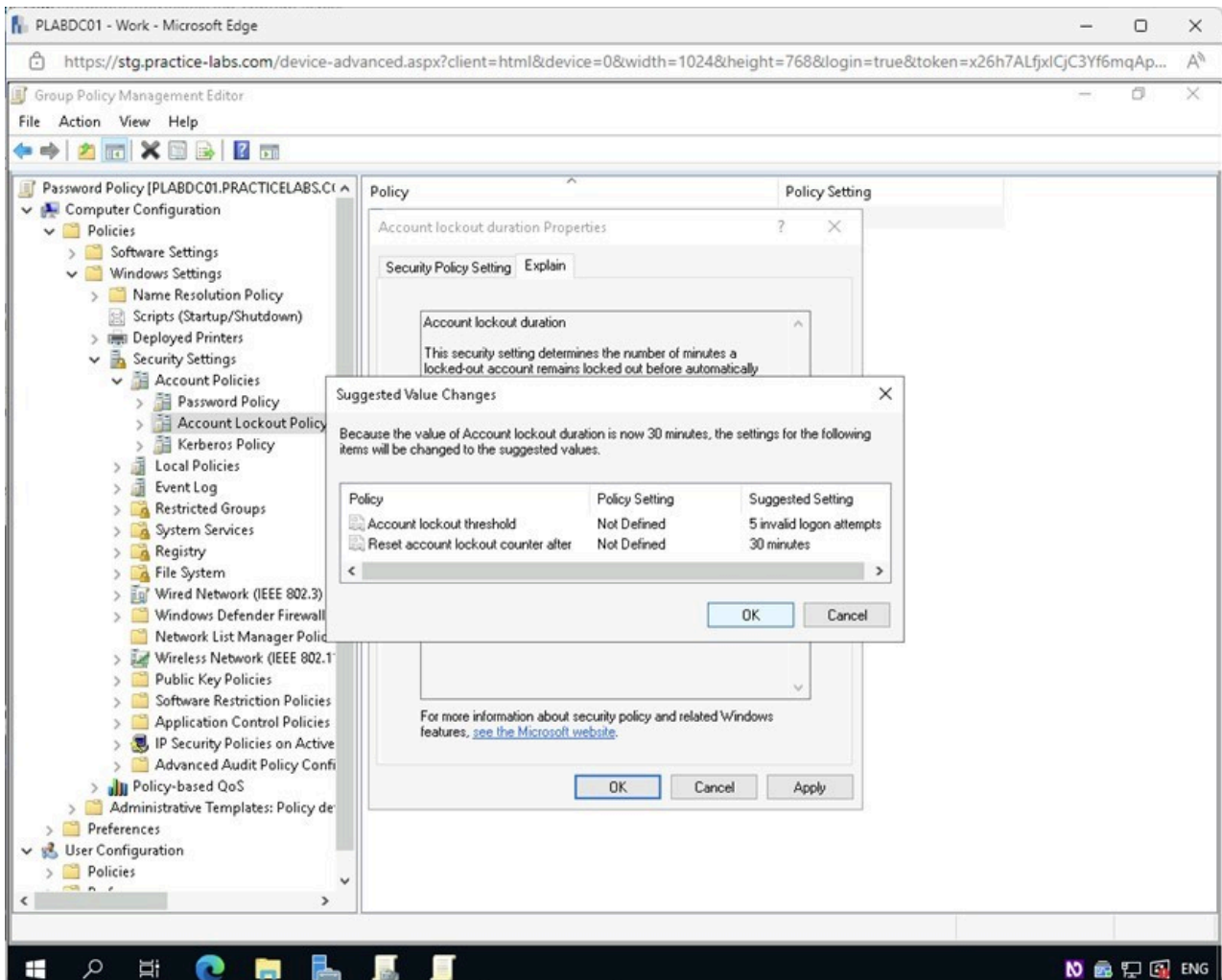


Figure 1.25 Screenshot of PLABDC01: Displaying selecting OK on the Suggested Value Changes pop-up window.

Step 6

Back on the **Group Policy Management Editor** window, double-click on the **Account lockout threshold** Policy.

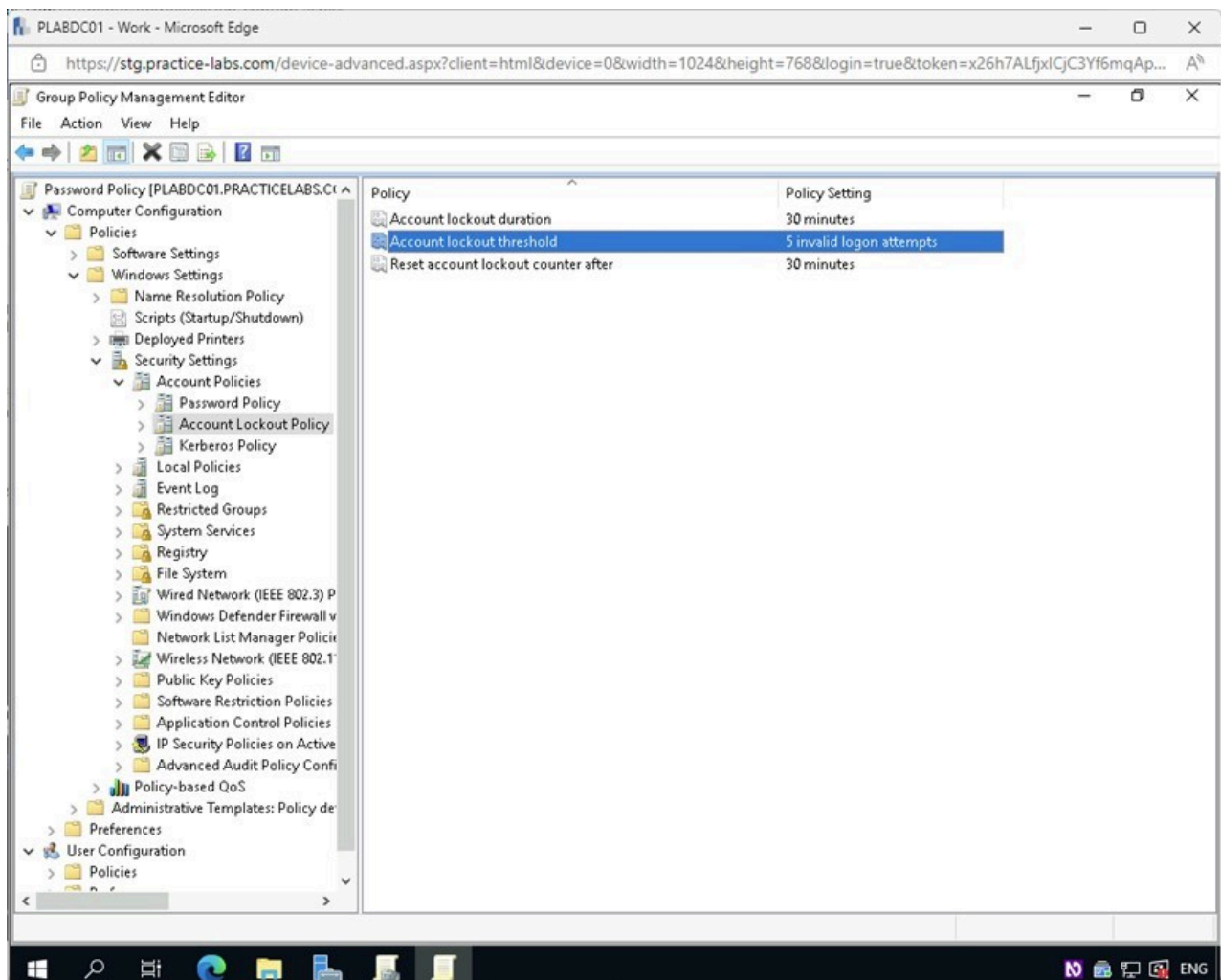


Figure 1.26 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Account lockout threshold policy selected.

Step 7

On the **Account lockout threshold Properties** dialog box, select the **Explain** tab.

Here you can view detailed information on the **Account lockout threshold** policy.

Once you've read the information available, click **OK**.

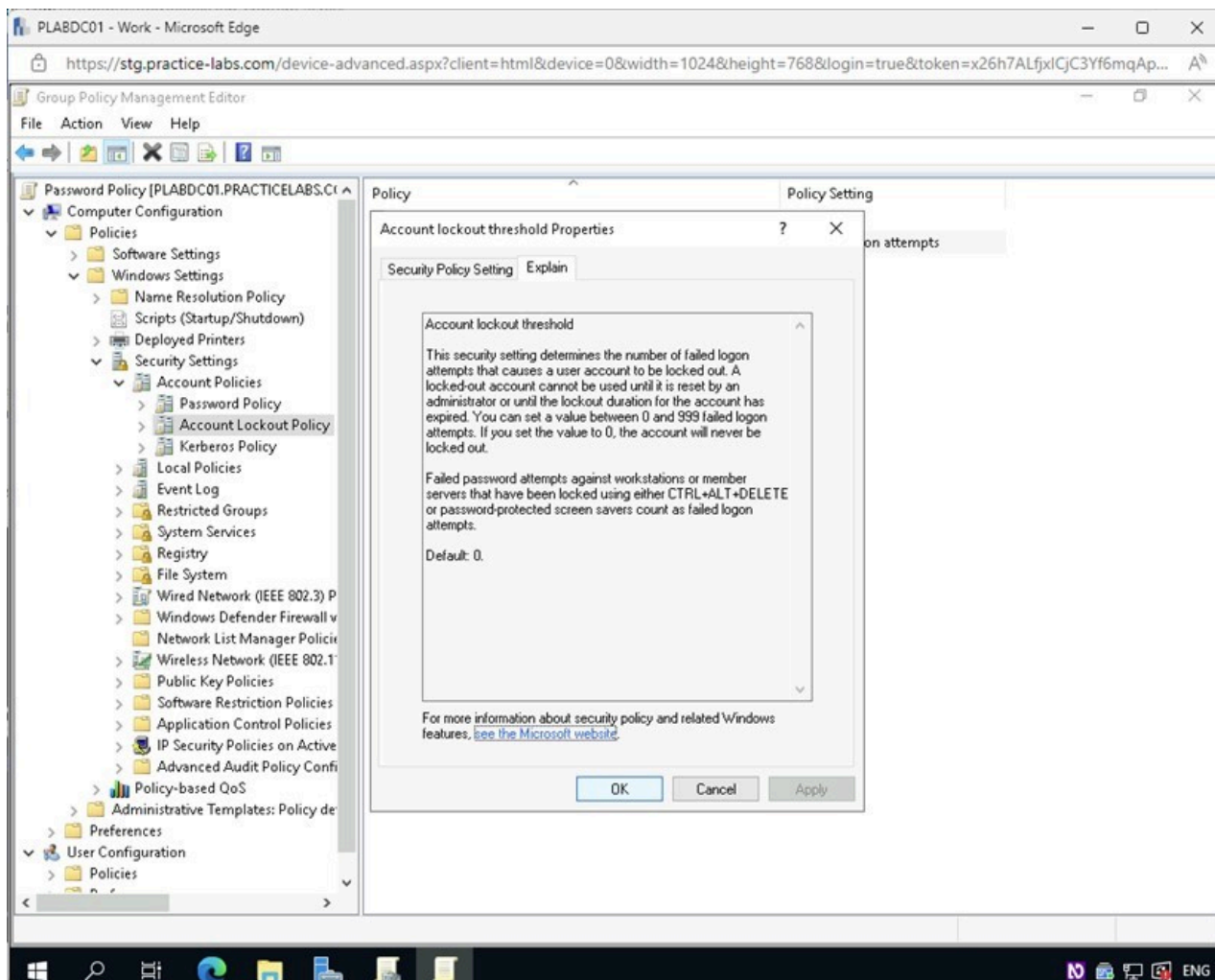


Figure 1.27 Screenshot of PLABDC01: Displaying the Account lockout threshold Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 8

Back on the **Group Policy Management Editor** window, double-click on the **Reset account lockout counter after Policy**.

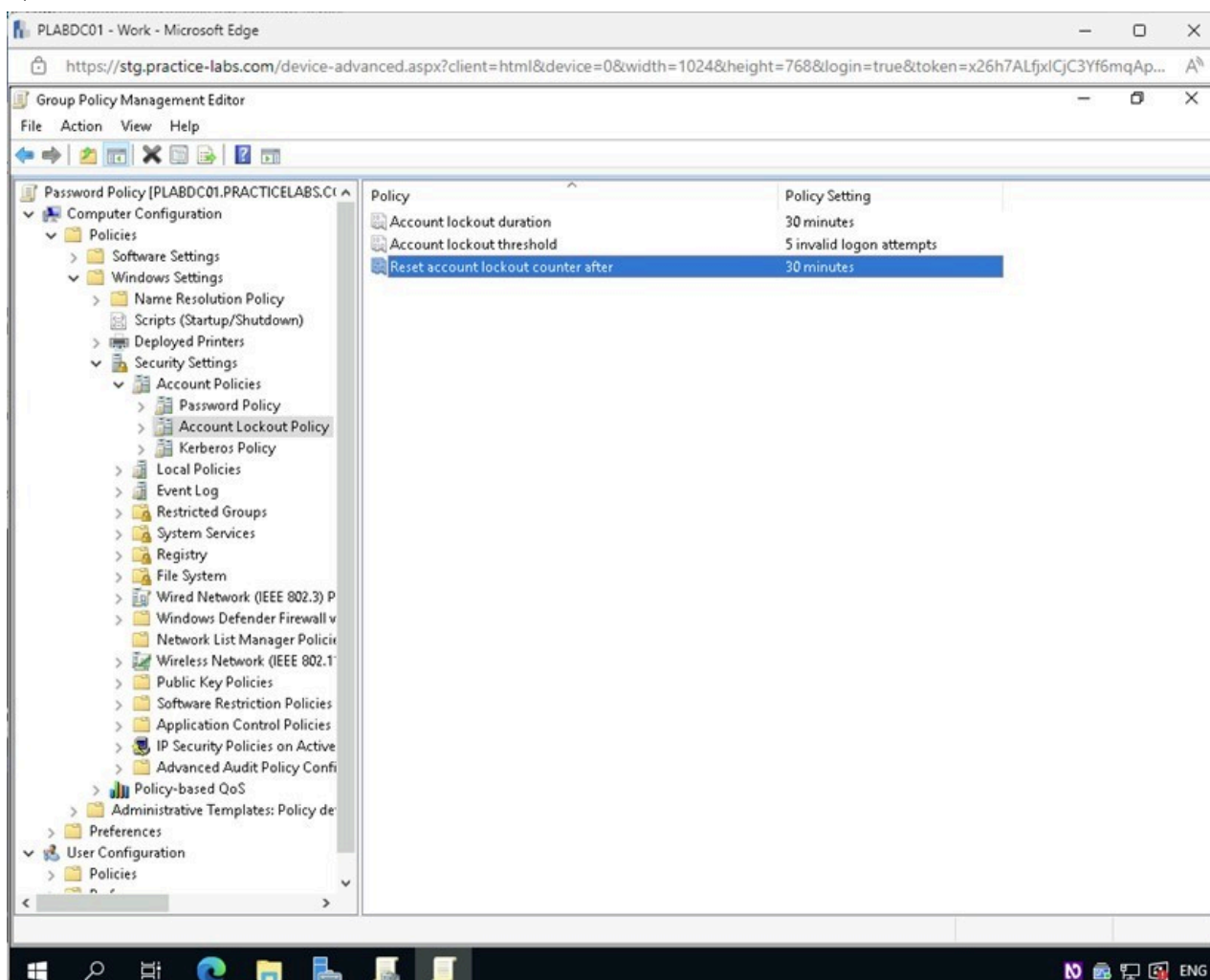


Figure 1.28 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing the Reset account lockout counter after Policy selected.

Step 9

On the **Reset account lockout counter after Properties** dialog box, select the **Explain** tab.

Here you can view detailed information on the **Reset account lockout counter after** policy.

Once you've read the information available, click **OK**.

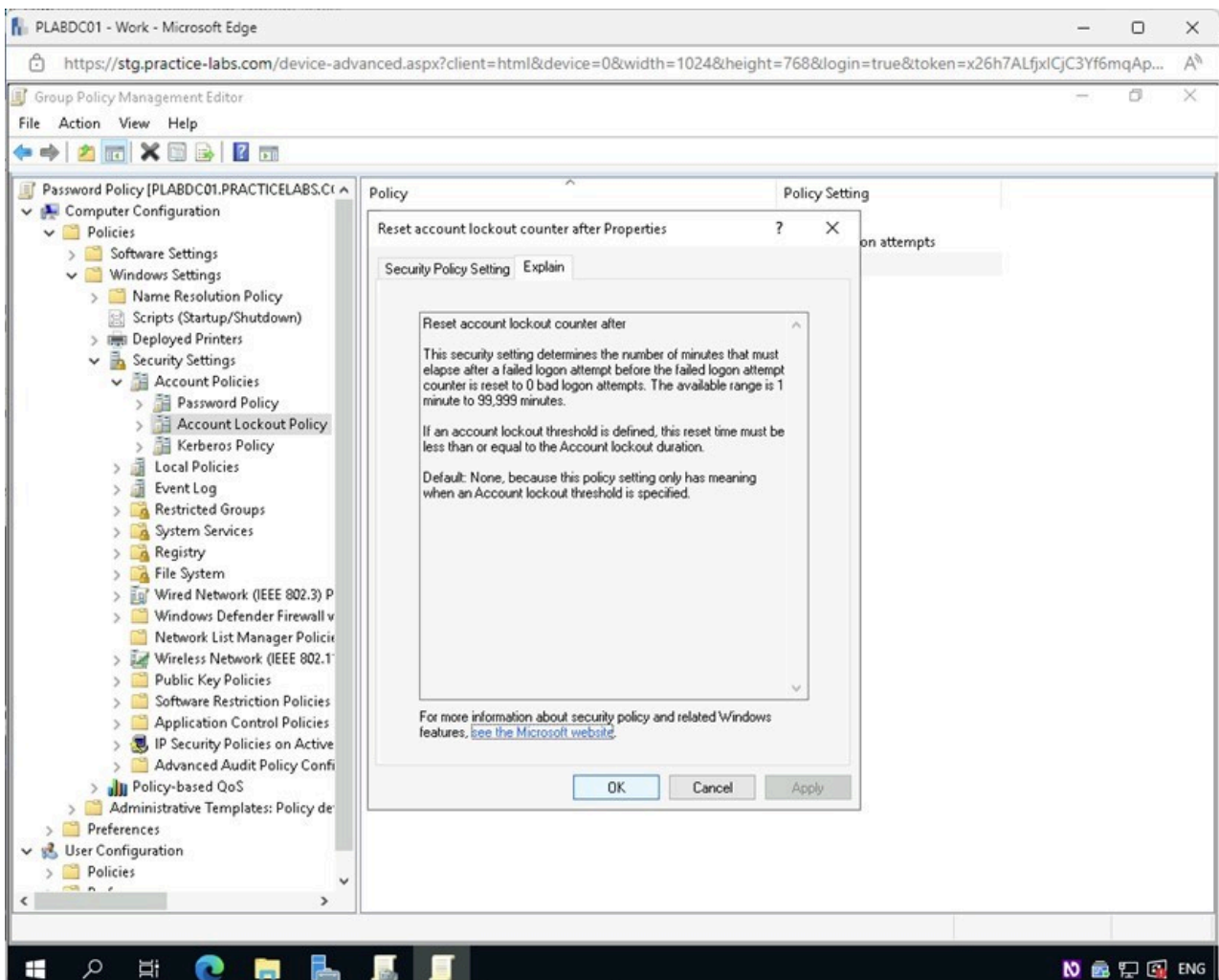


Figure 1.29 Screenshot of PLABDC01: Displaying the Reset account lockout counter after Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 10

Close the **Group Policy Management Editor** window.

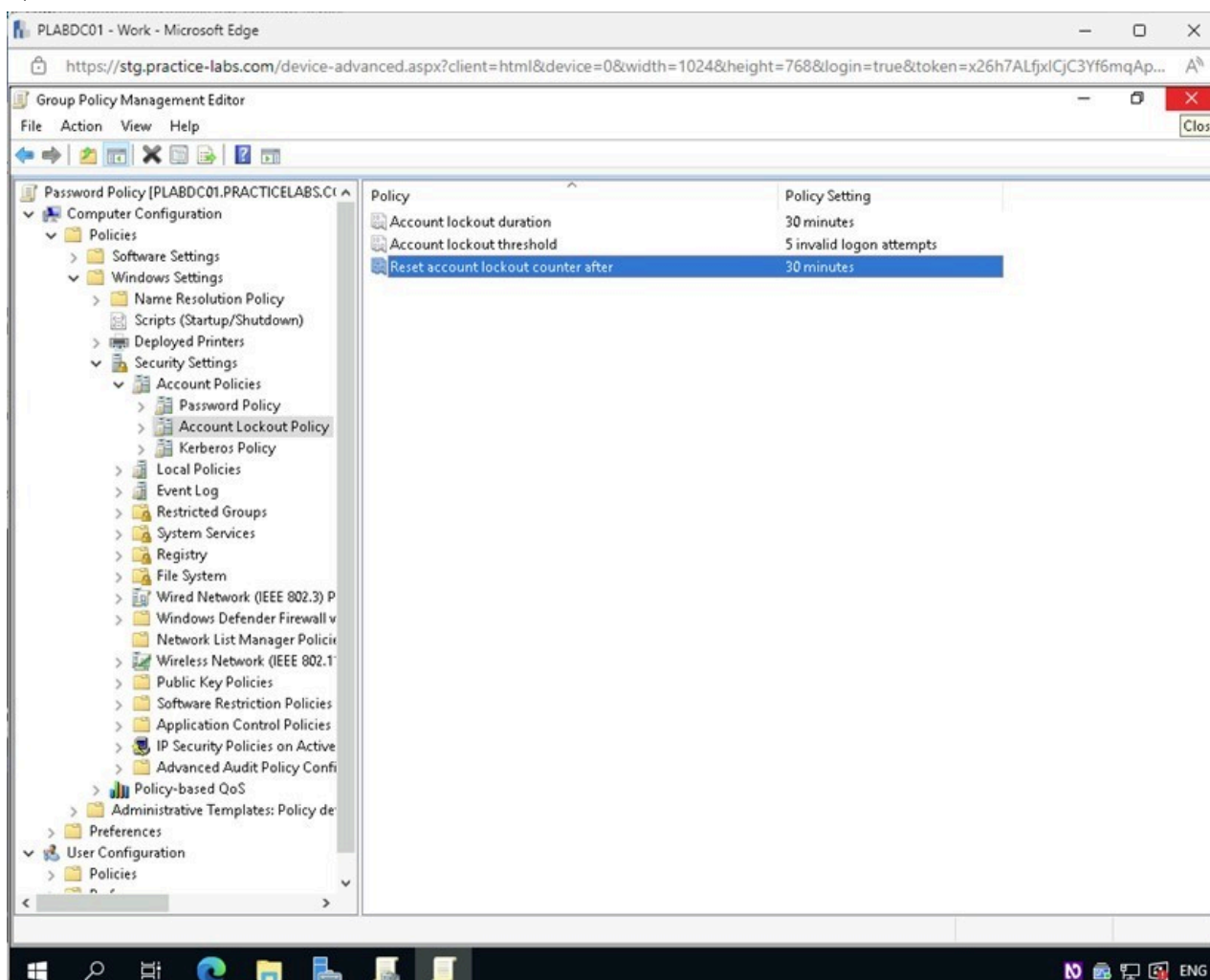


Figure 1.30 Screenshot of PLABDC01: Displaying closing the Group Policy Management Editor window.

Keep the **Group Policy Management** window open.

Task 3 - Apply Password Policy

You have now created the **Password Policy GPO**. This type of GPO is known as **Rule-Based Access Control**. Rule-based access control regulates who has access to locations, devices, or databases based on rules or permissions, regardless of job or position in an organization. Rule-based access control lets individuals enter organization premises depending on pre-set rules. For example, an administrator may restrict building access on specific days.

Rule-based access control requires a lot of configurations. Because rules must be constantly verified and altered, these systems might be difficult to maintain or require more effort.

In this task, you will apply the password policy to a group.

Step 1

In **PLABDC01**, the **Group Policy Management** window is open.

Select **Password Policy** on the left pane.

If required, click **OK** on the **Group Policy Management Console** pop-up window.

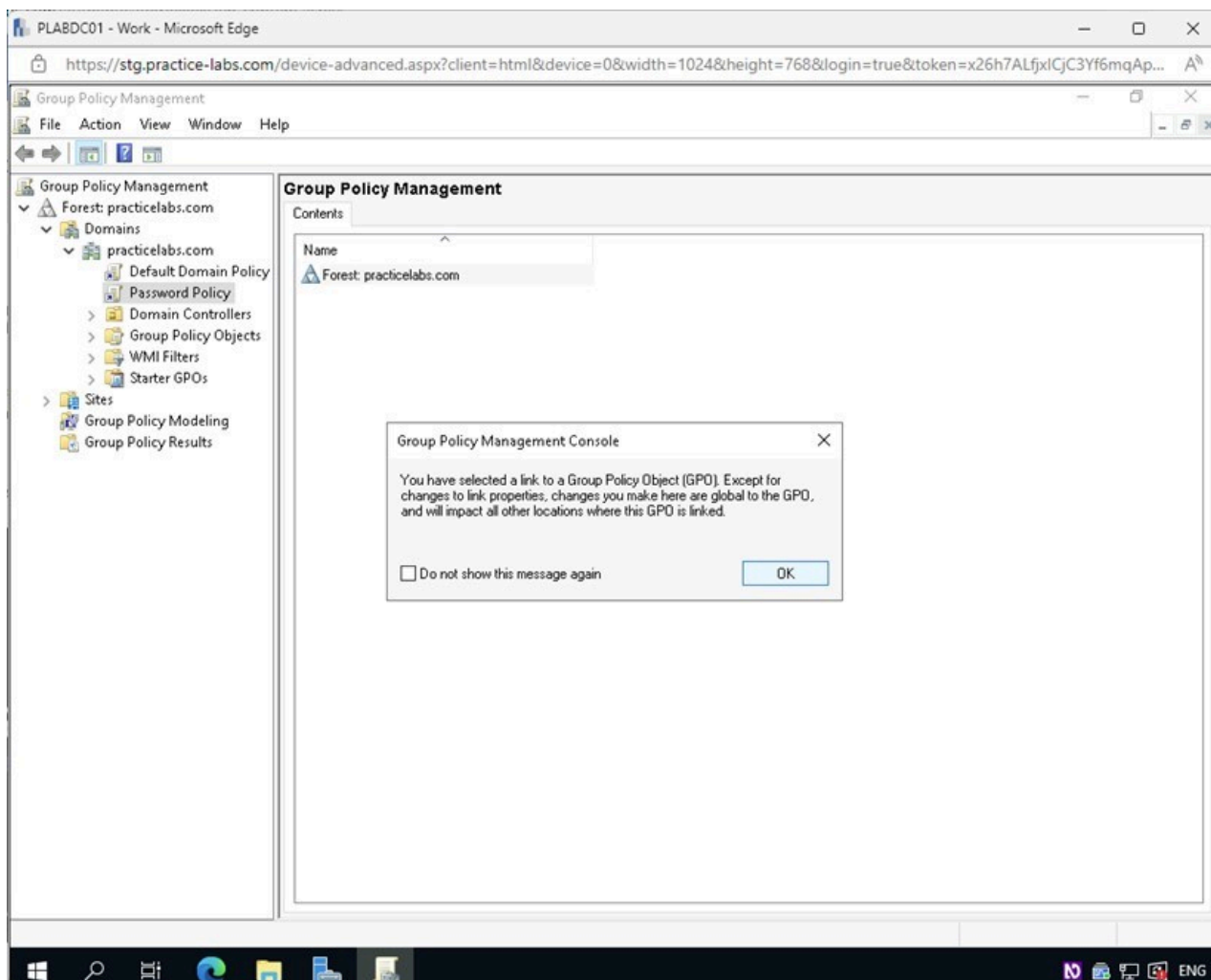


Figure 1.31 Screenshot of PLABDC01: Displaying the Group Policy Management window, showing OK selected on the Group Policy Management Console pop-up window.

Step 2

In the right pane, under the **Security Filtering** section, click **Add**.

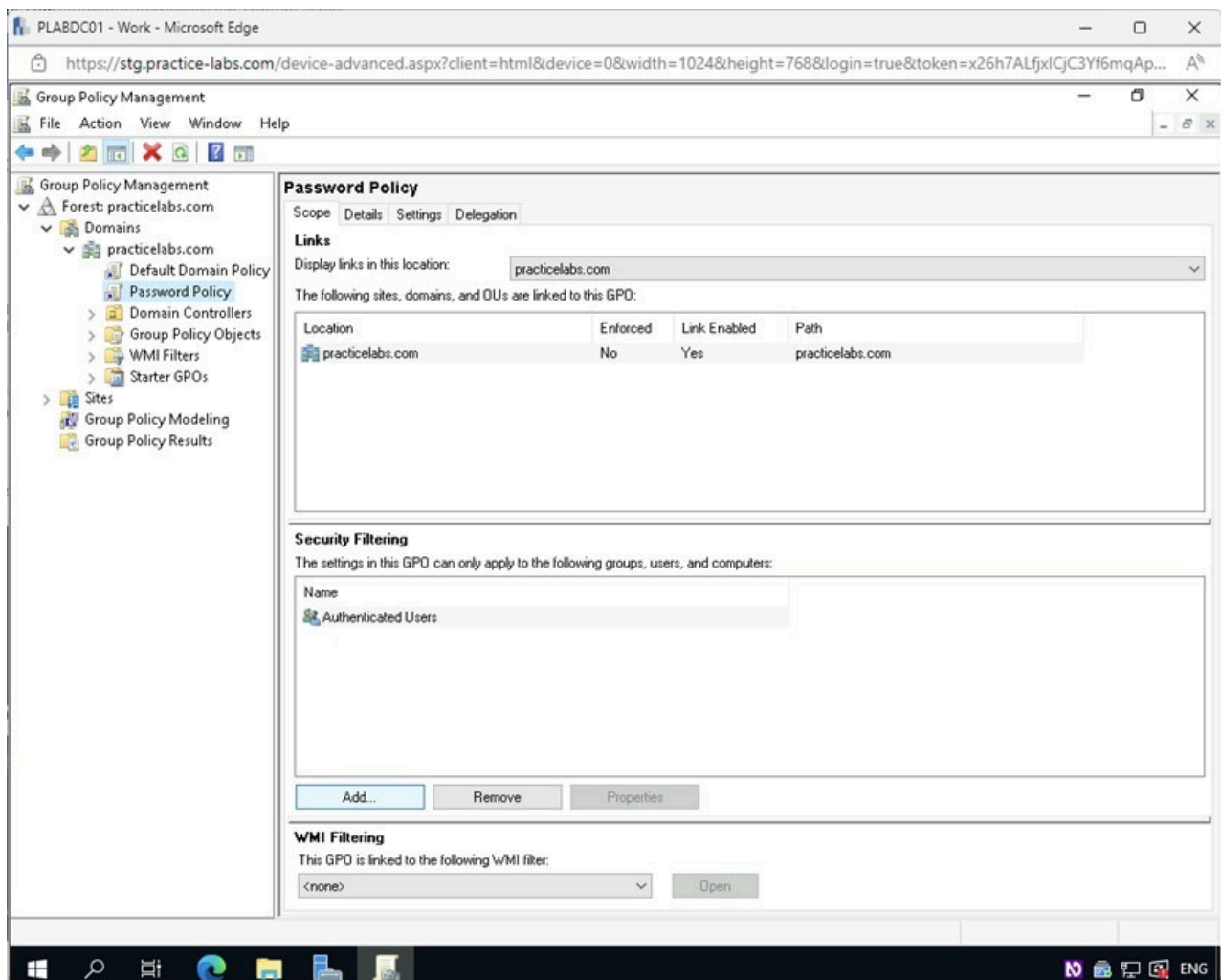


Figure 1.32 Screenshot of PLABDC01: Displaying the Group Policy Management window. Add button under the Security Filtering section is selected.

Step 3

On the **Select User, Computer, or Group** pop-up window, click the **Advanced** button.

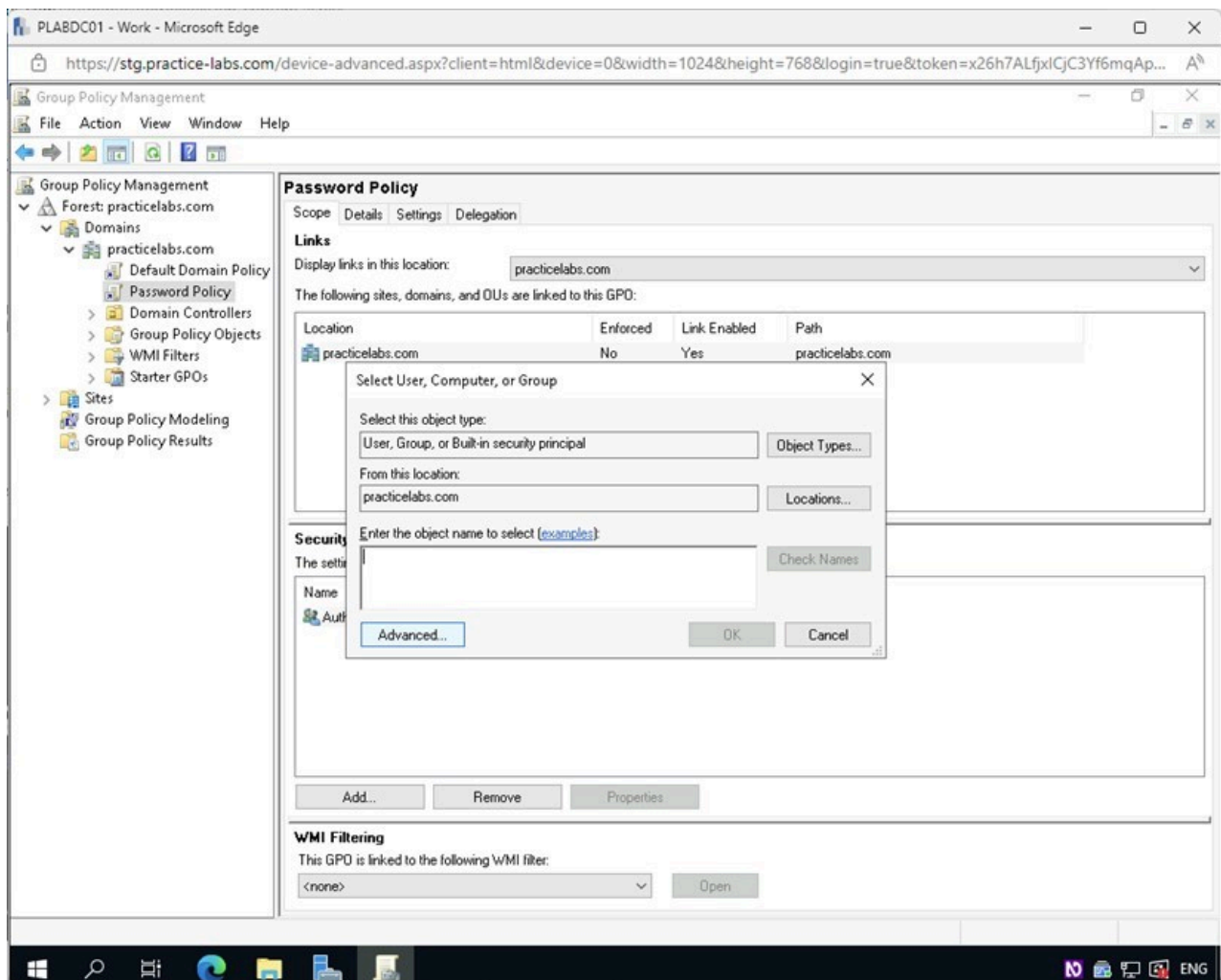


Figure 1.33 Screenshot of PLABDC01: Displaying clicking the Advanced button in the Select User, Computer, or Group pop-up window.

Step 4

On the **Select User, Computer, or Group** pop-up window, click the **Find Now** button.

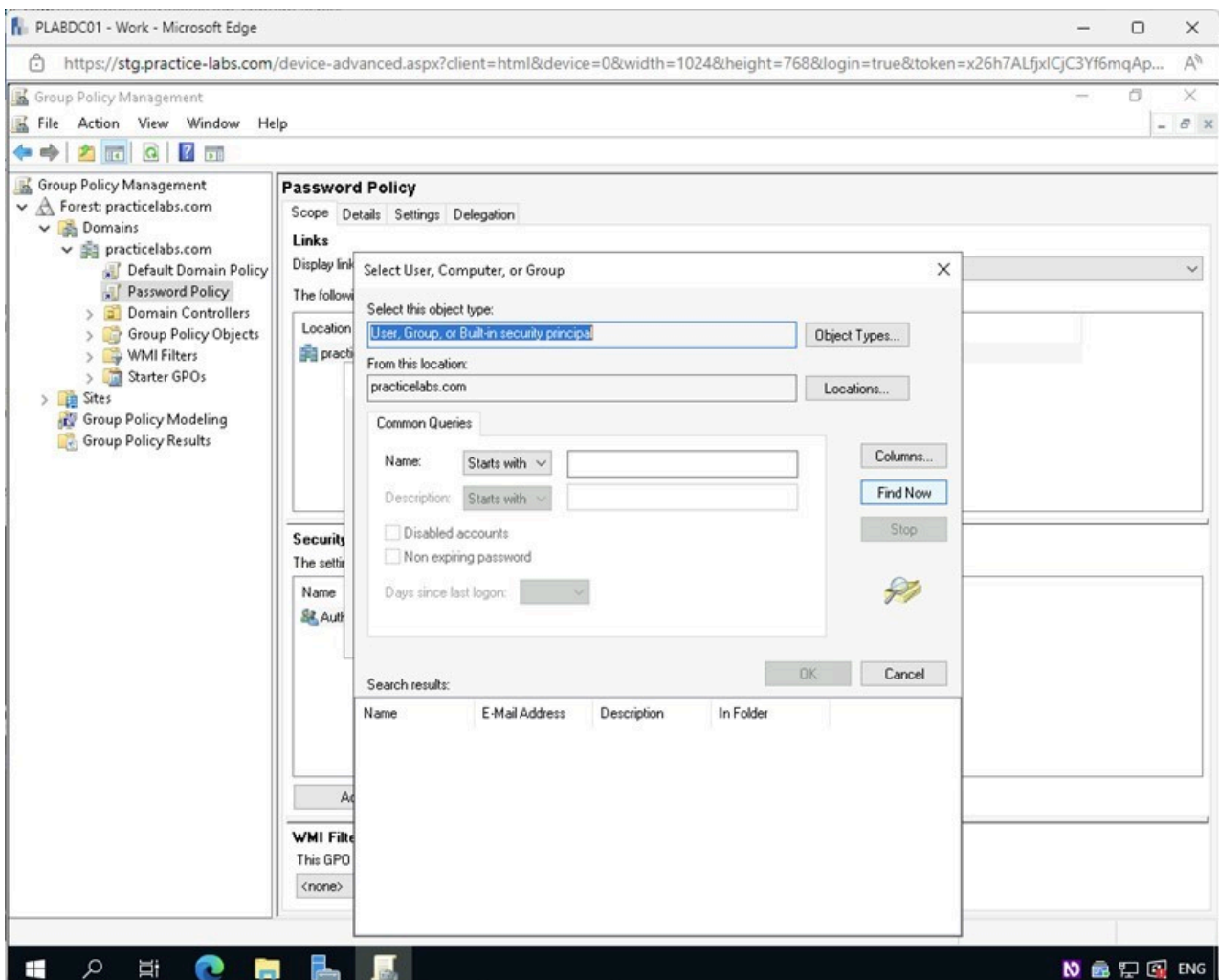


Figure 1.34 Screenshot of PLABDC01: Displaying clicking the Find Now button in the Select User, Computer, or Group pop-up window.

Step 5

Names of groups will appear in the **Search results** section.

Scroll down and select **Domain Users - All domain users**.

Click **OK**.

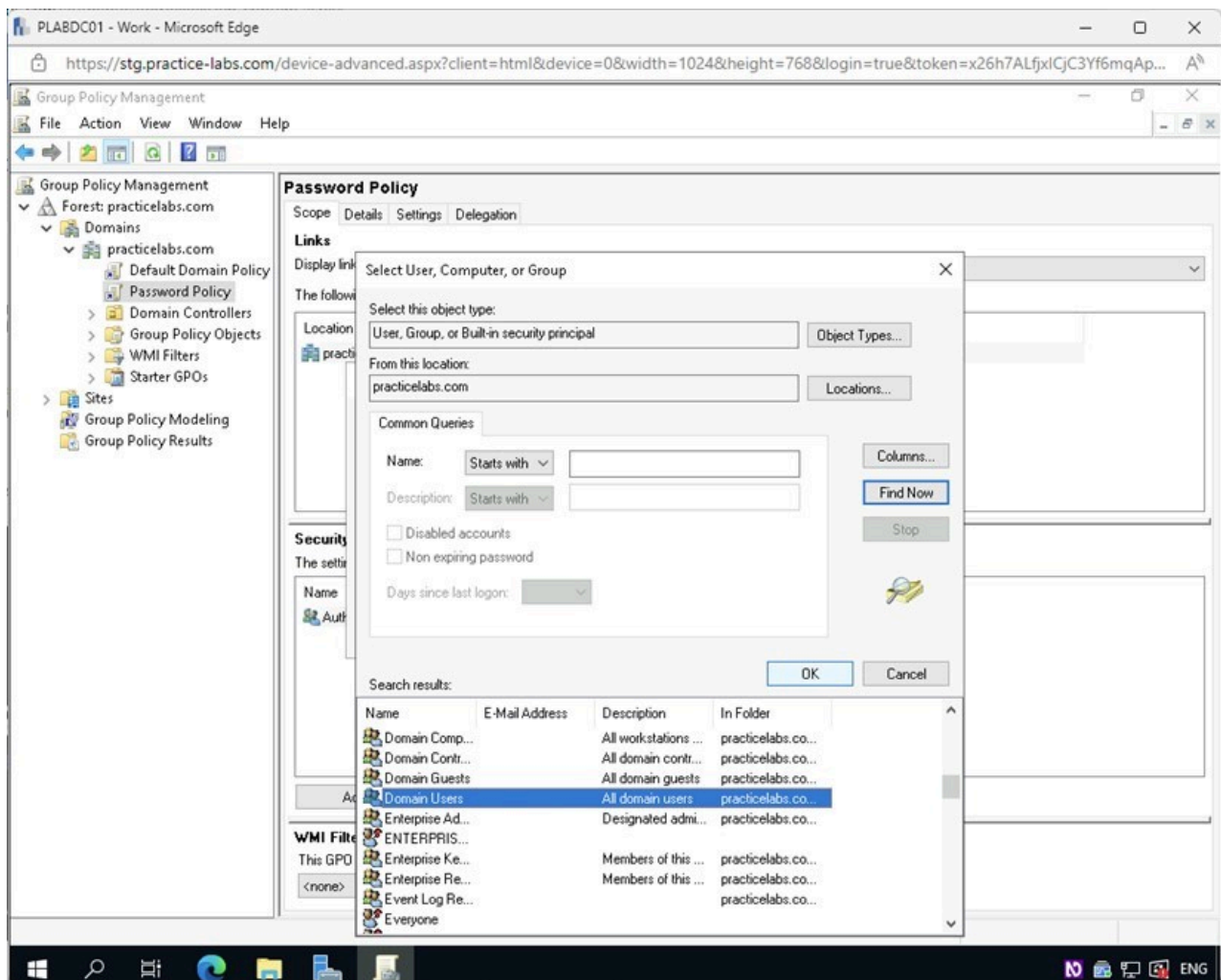


Figure 1.35 Screenshot of PLABDC01: Displaying the Select User, Computer, or Group pop-up window, showing Domain Users selected.

Step 6

Back on the **Select User, Computer, or Group** pop-up window, click **OK**.

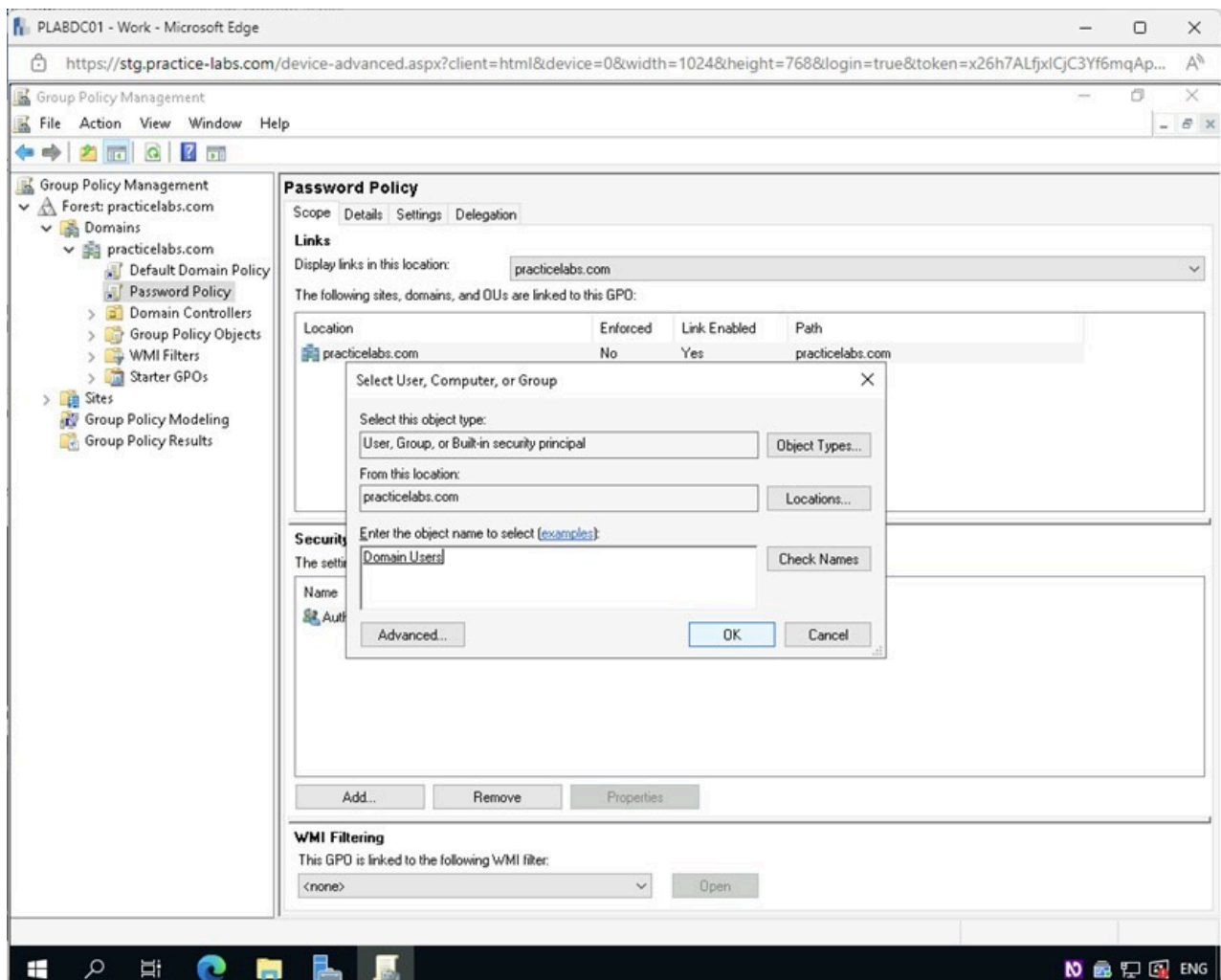


Figure 1.36 Screenshot of PLABDC01: Displaying clicking OK on the Select User, Computer, or Group pop-up window.

Step 7

Notice that the **Domain Users** have now been added under the **Security Filtering** section.

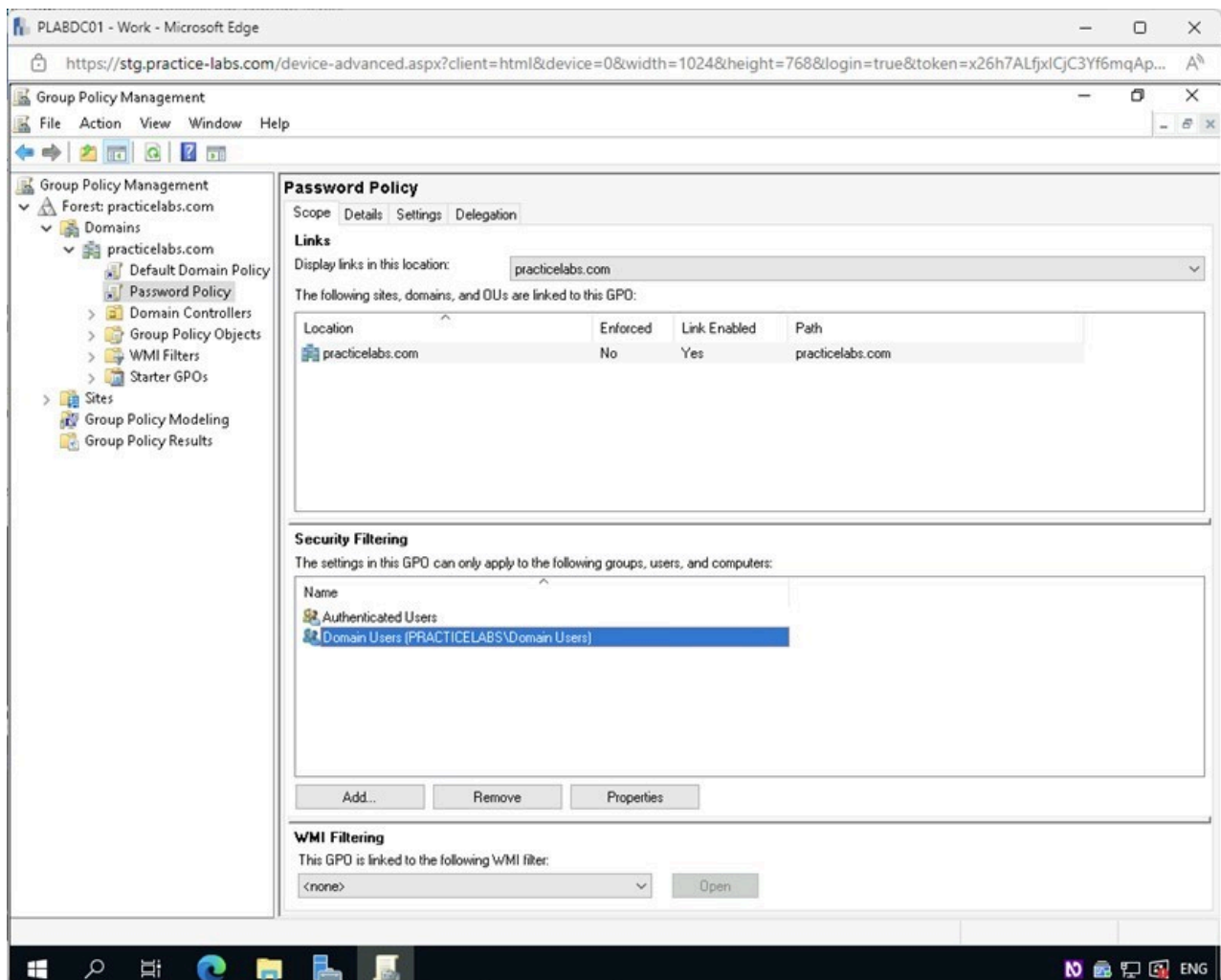


Figure 1.37 Screenshot of PLABDC01: Displaying the Group Policy Management window, showing Domain Users added under the Security Filtering section.

Step 8

Close the **Group Policy Management** window.

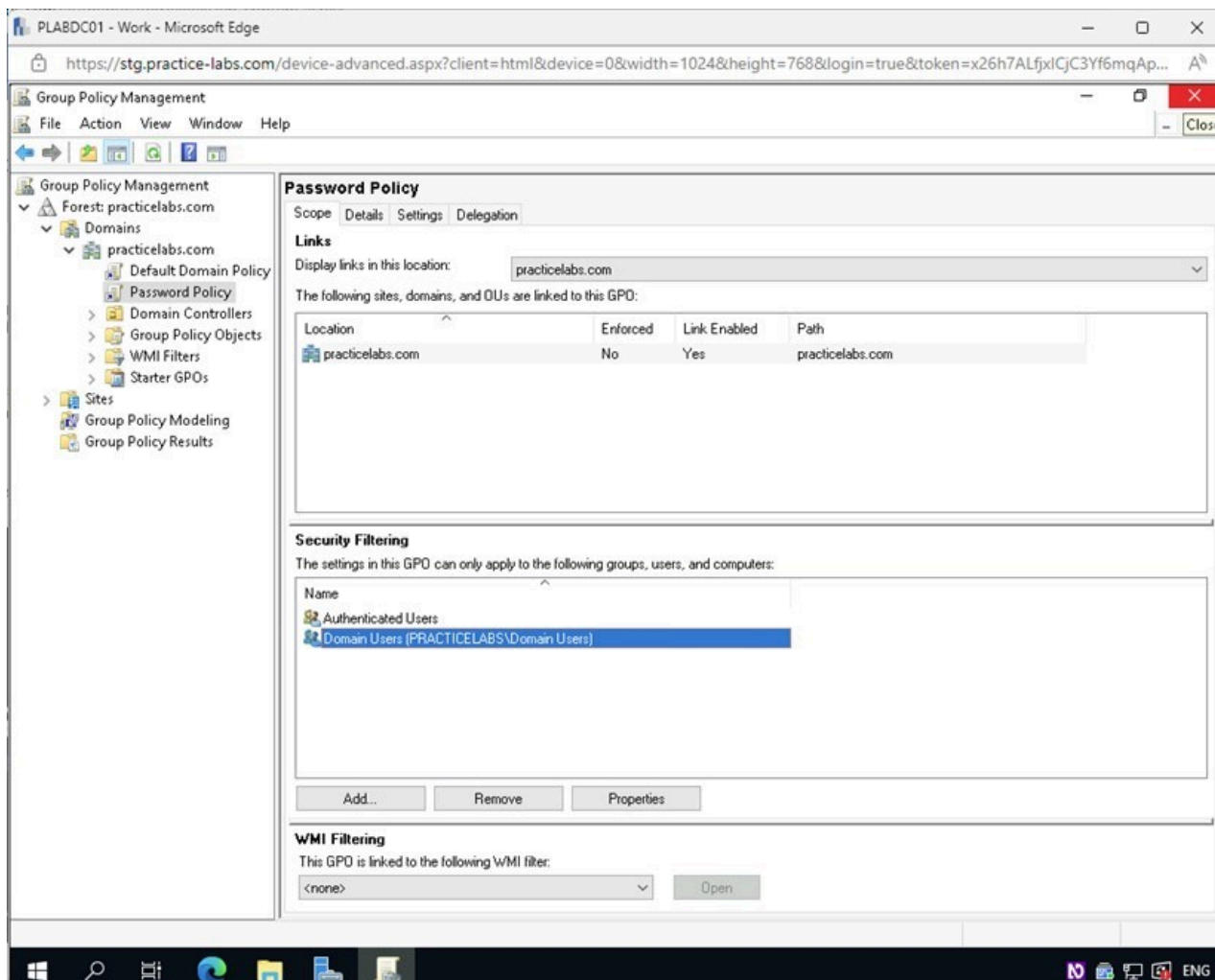


Figure 1.38 Screenshot of PLABDC01: Displaying closing the Group Policy Management window.

Task 4 - Update Group Policy

Group policy will update when the device restarts but can be manually done within a command line interface.

In this task, you will update the group policy.

Step 1

Ensure you are connected to **PLABDC01**.

Right click on the **Windows Start** charm and select **Windows PowerShell (Admin)**.

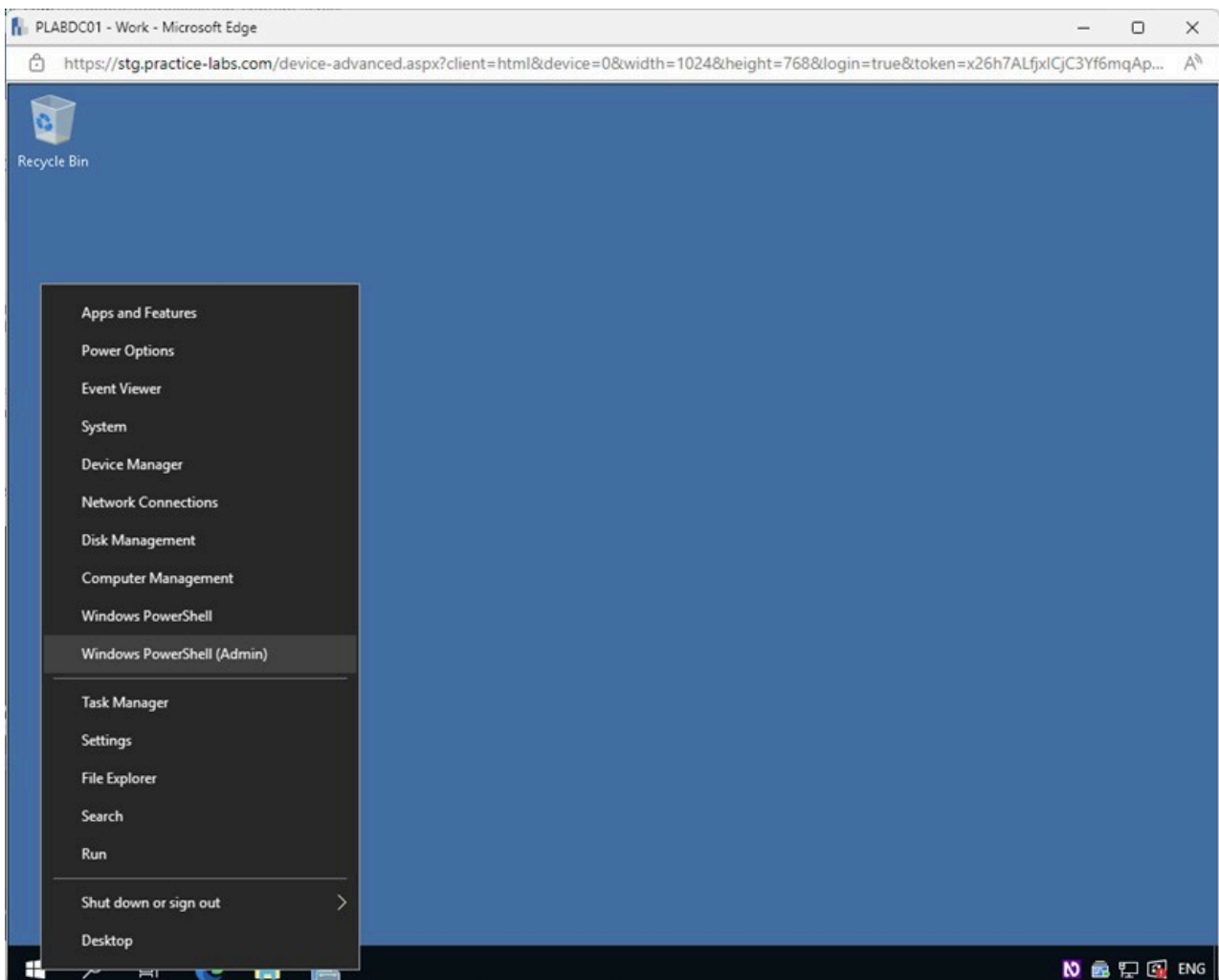


Figure 1.39 Screenshot of PLABDC01: Displaying right-clicking on the Windows Start charm and selecting Windows PowerShell (Admin).

Step 2

In the **Windows PowerShell** window, type the following:

```
gpupdate /force
```

Press **Enter**.

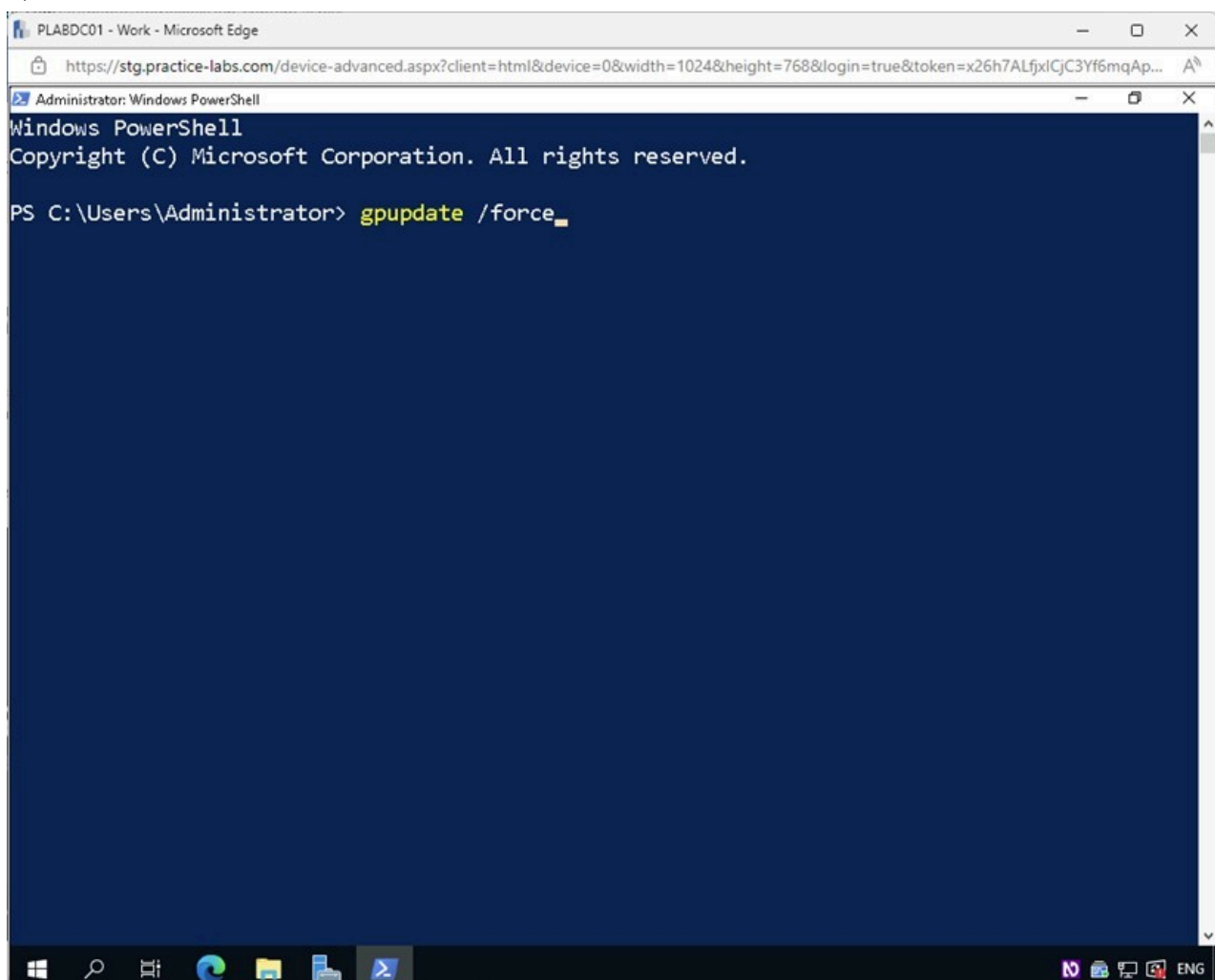


Figure 1.40 Screenshot of PLABDC01: Displaying the Windows PowerShell window, showing the required command typed in.

Step 3

Notice the Group Policy is now updated, and the Password Policy is applied.

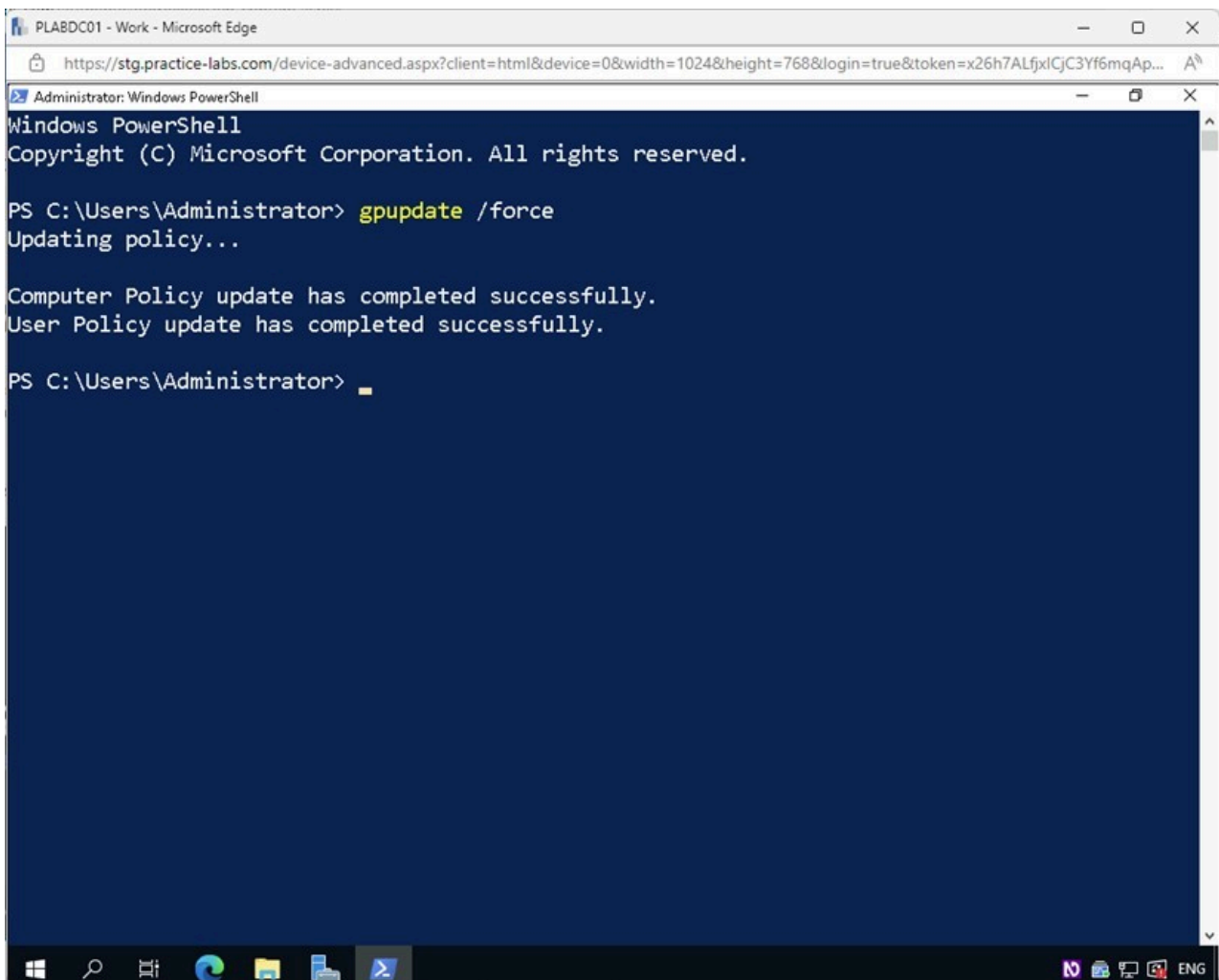


Figure 1.41 Screenshot of PLABDC01: Displaying the Windows PowerShell window, showing the Group policy has been updated.

Step 4

Close the **Windows PowerShell** window.

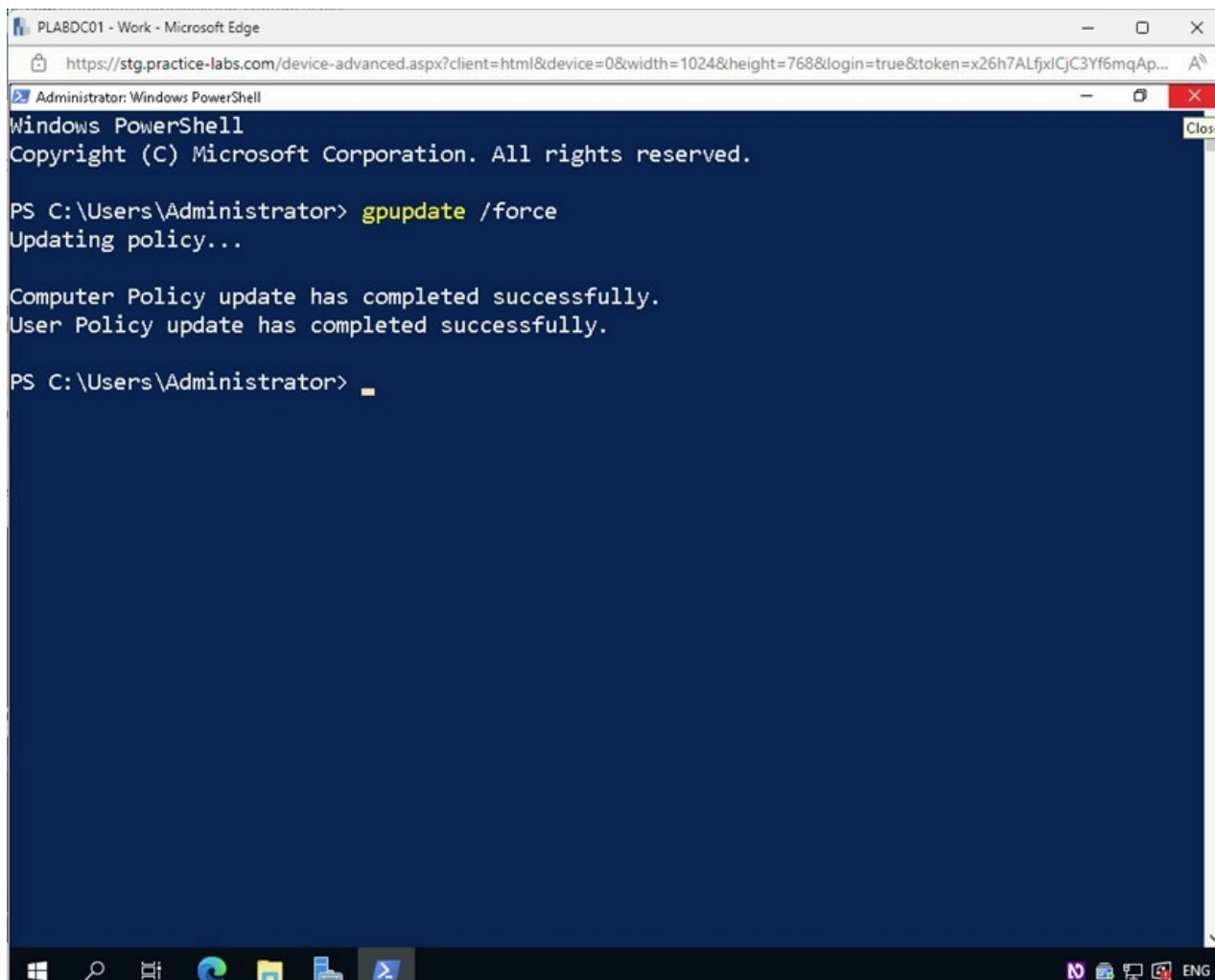


Figure 1.42 Screenshot of PLABDC01: Displaying closing the Windows PowerShell window.

Keep the **Server Management** window open.

Exercise 2 - User Accounts and Groups

Active Directory facilitates control over user accounts, group memberships, and even whole organizational hierarchies. This allows users to be added to groups and their privileges to be enforced. Rule-based access control is the term for this type of system. Although granular permission settings are configurable, the administrative burden of keeping them in check may rapidly increase.

In this exercise, you will learn about user accounts and groups.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore, Define and Configure Active Directory User and Computers Accounts
- Explore, Define and Configure User Groups and Role Based Access Control

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Explore, Define and Configure Active Directory User and Computers Accounts

In this task, you will learn about Active Directory users and computer accounts.

Step 1

Ensure you are connected to **PLABDC01** and the **Server Manager** window is open.

Click **Tools** and then select **Active Directory Users and Computers**.

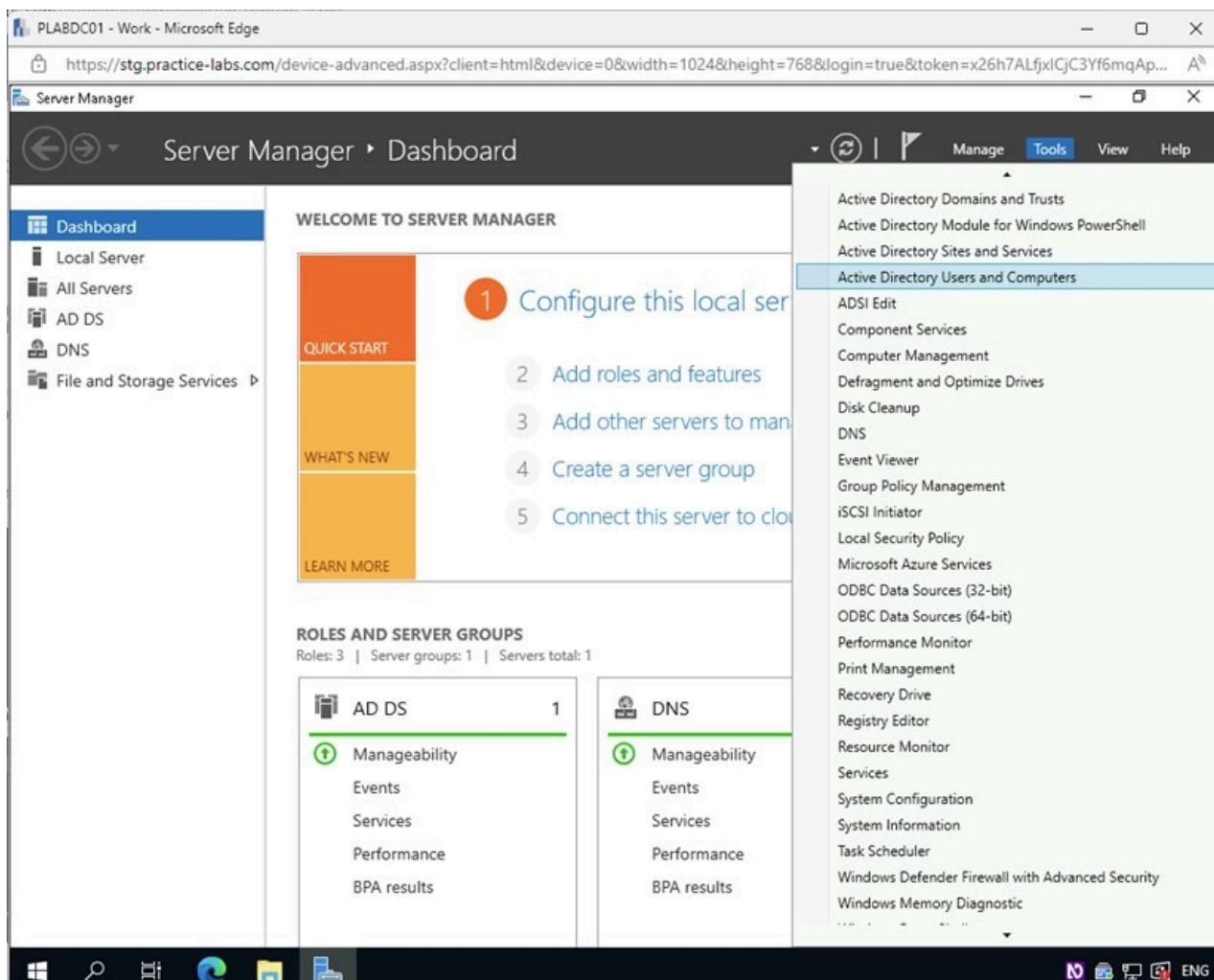


Figure 2.1 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Active Directory Users and Computers is selected.

Step 2

In the **Active Directory Users and Computers** window, expand the **practicelabs.com** node and select **Users**.

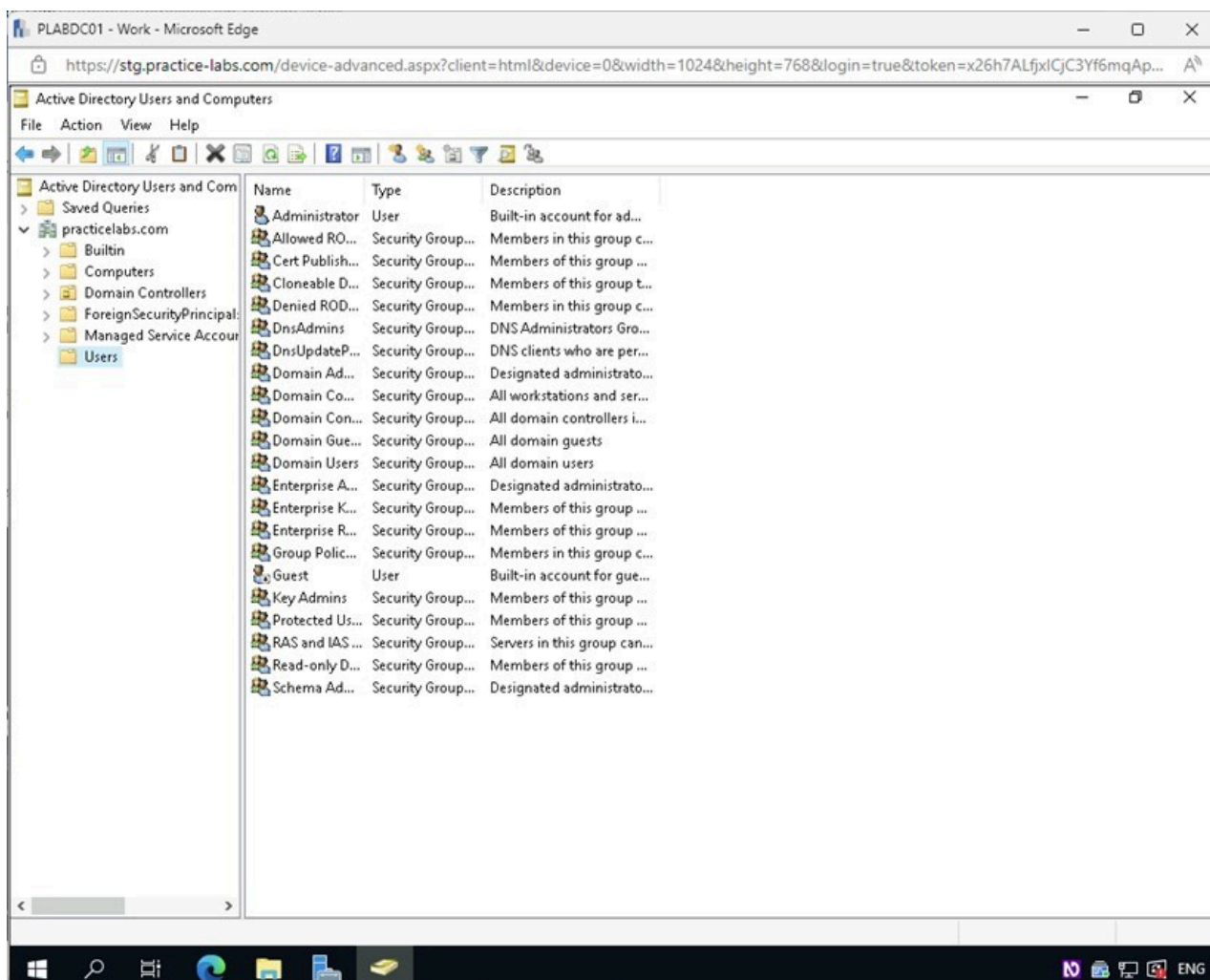


Figure 2.2 Screenshot of PLABDC01: Displaying the Active Directory Users and Computer Pane, showing the Users folder selected on the left pane.

Step 3

The **Users** for **practicelabs.com** are displayed in the right details pane.

Right-click anywhere on an empty space and select **New > User**.

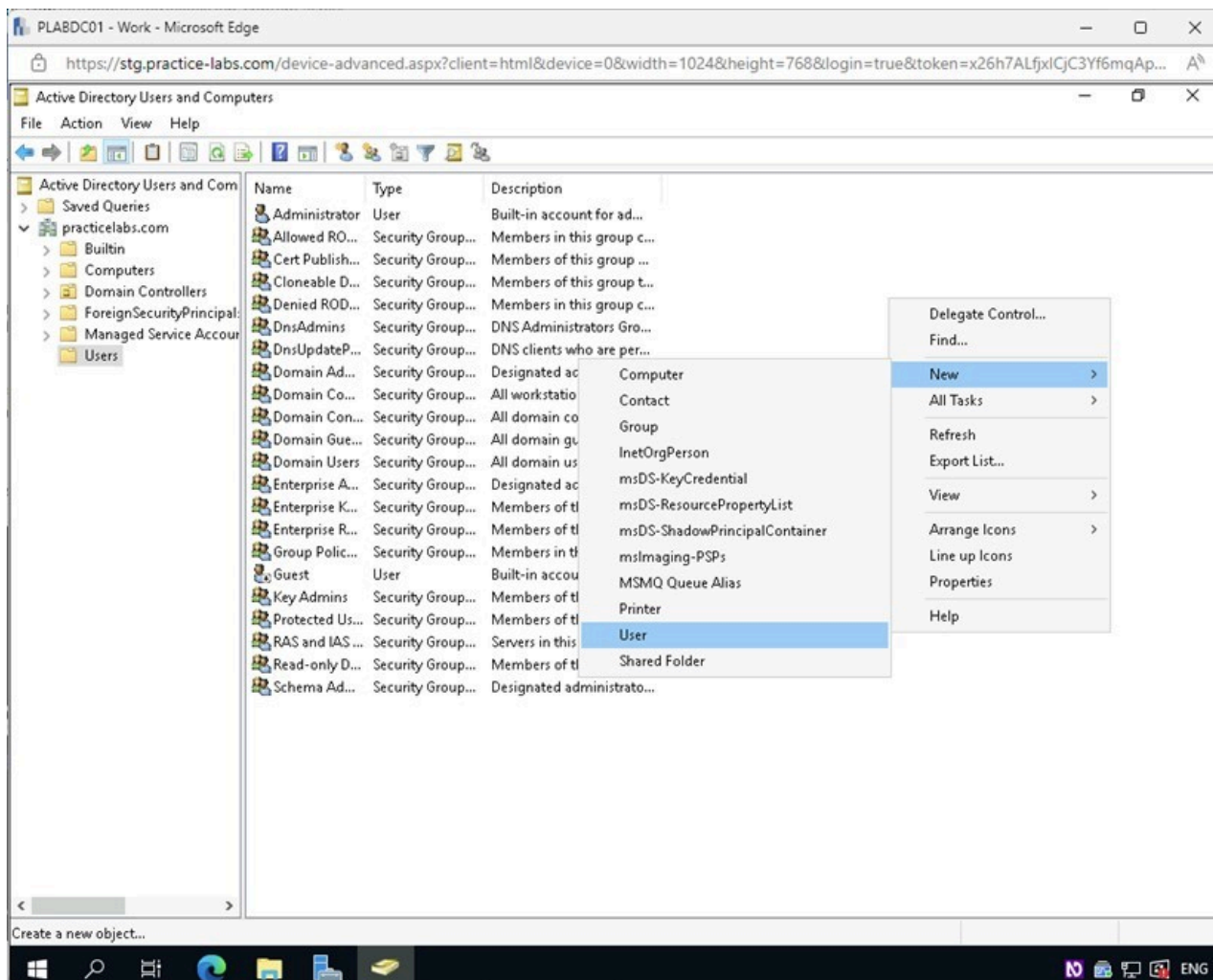


Figure 2.3 Screenshot of PLABDC01: Displaying the Active Directory Users and Computer Pane. New > User menu options are selected from the right-click context menu.

Step 4

From the **New Object - User** window, fill in the following fields:

First Name: Test
Last Name: User 1
User logon name: testuser1

Click **Next**.

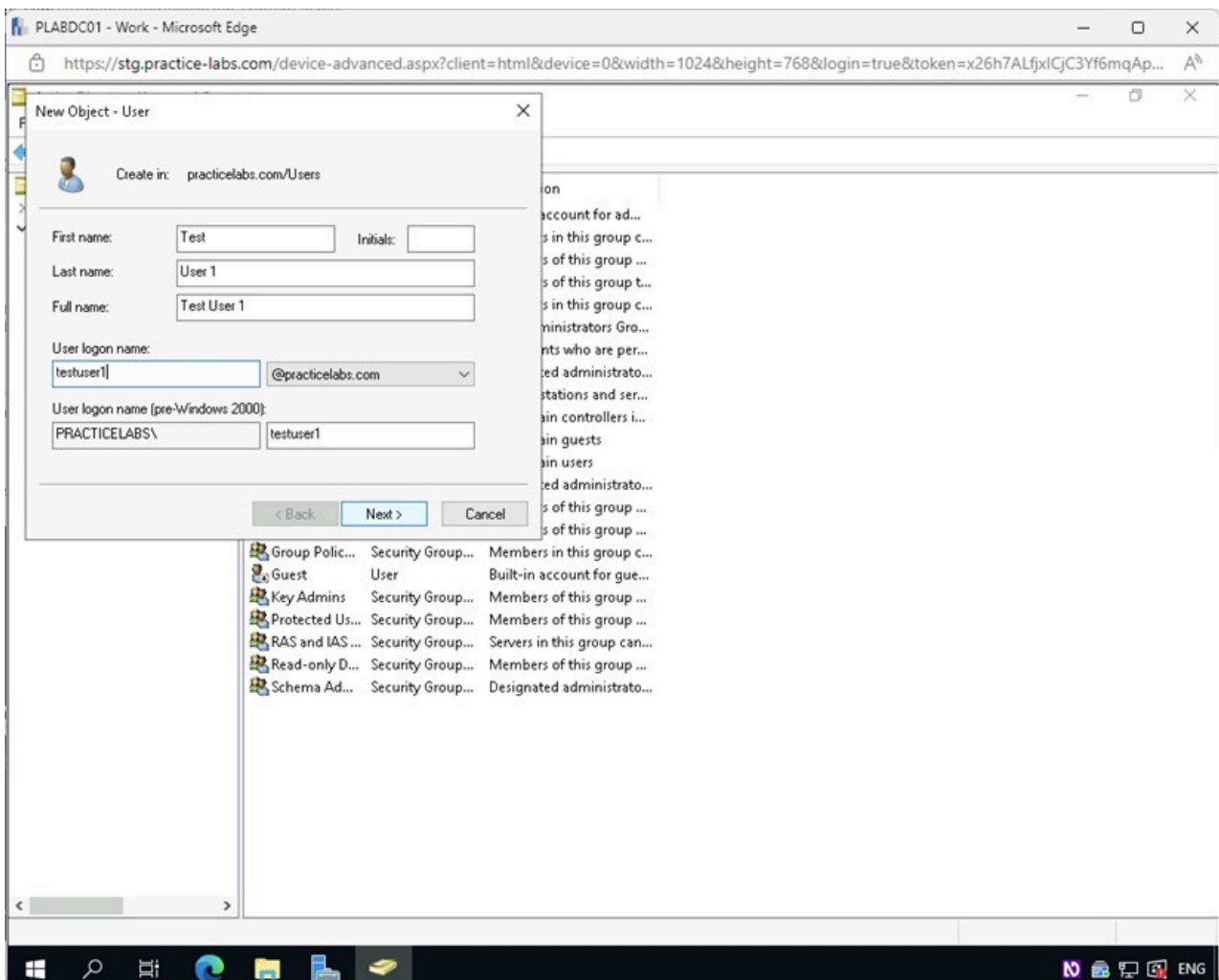


Figure 2.4 Screenshot of PLABDC01: Displaying the New Object - User window, showing the required fields typed in and the Next button highlighted.

Step 5

Type the following for the **Password** and **Confirm Password** fields, and click **Next**:

Passw0rd! 23

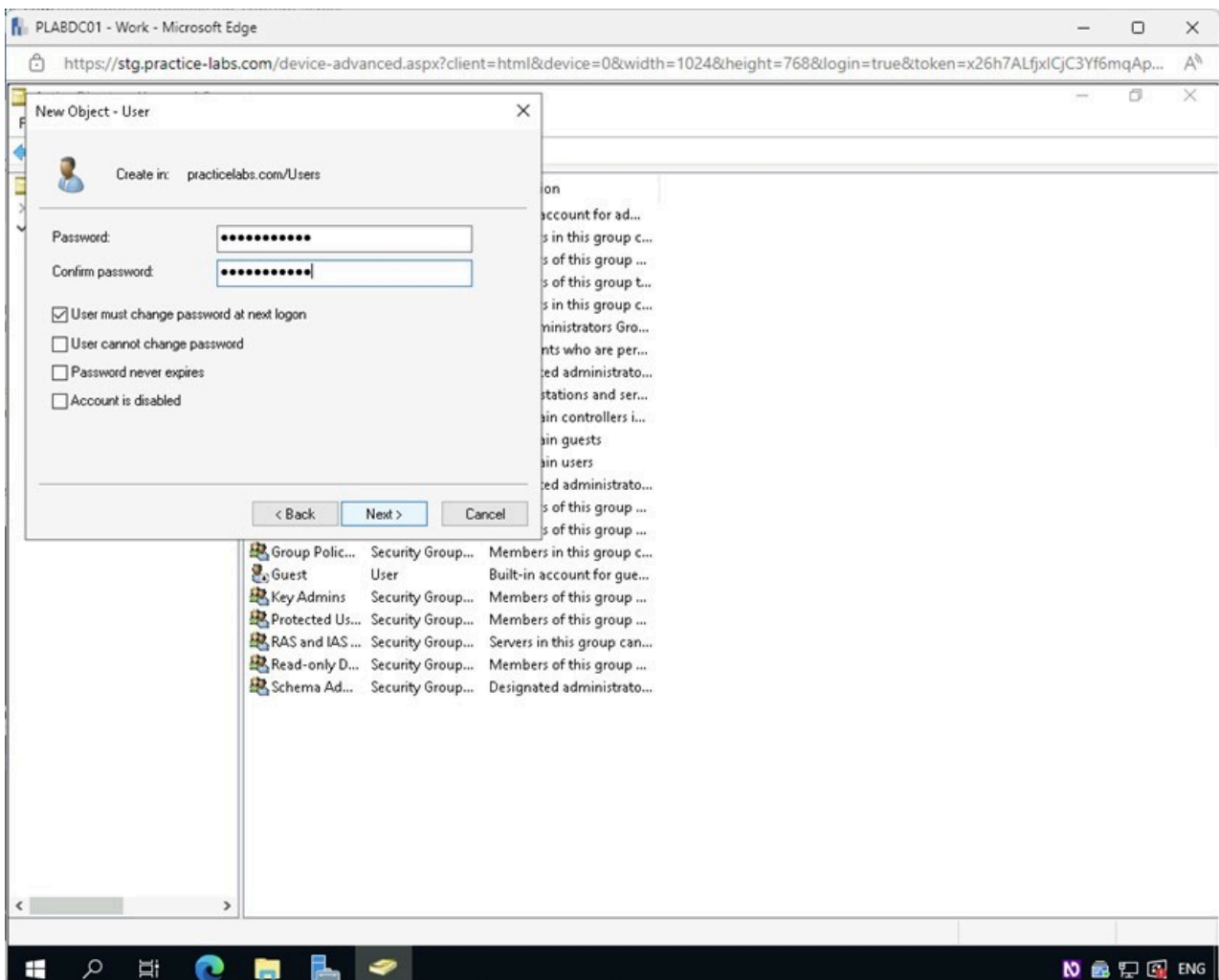


Figure 2.5 Screenshot of PLABDC01: Displaying the New Object - User window, showing the required password typed in and the Next button highlighted.

Note: The password created meets the requirements for length (10 characters) and complexity.

Step 6

In the **New Object - User** window, click **Finish**.

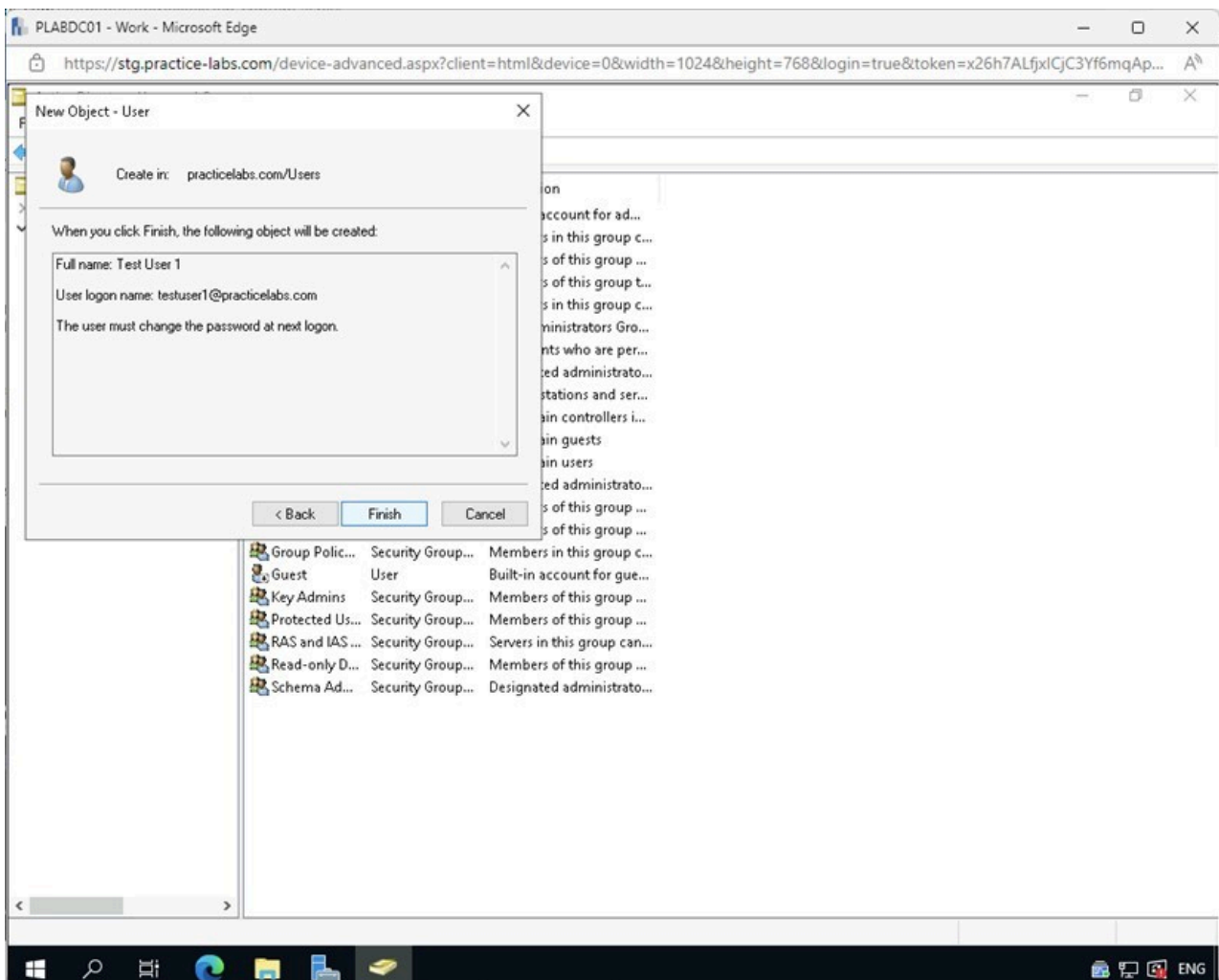


Figure 2.6 Screenshot of PLABDC01: Displaying select Finish in the New Object - User window.

Step 7

Back on the **Active Directory Users and Computers** window, notice that **Test User 1** has been added.

Right click **Test User 1** and select **Properties**.

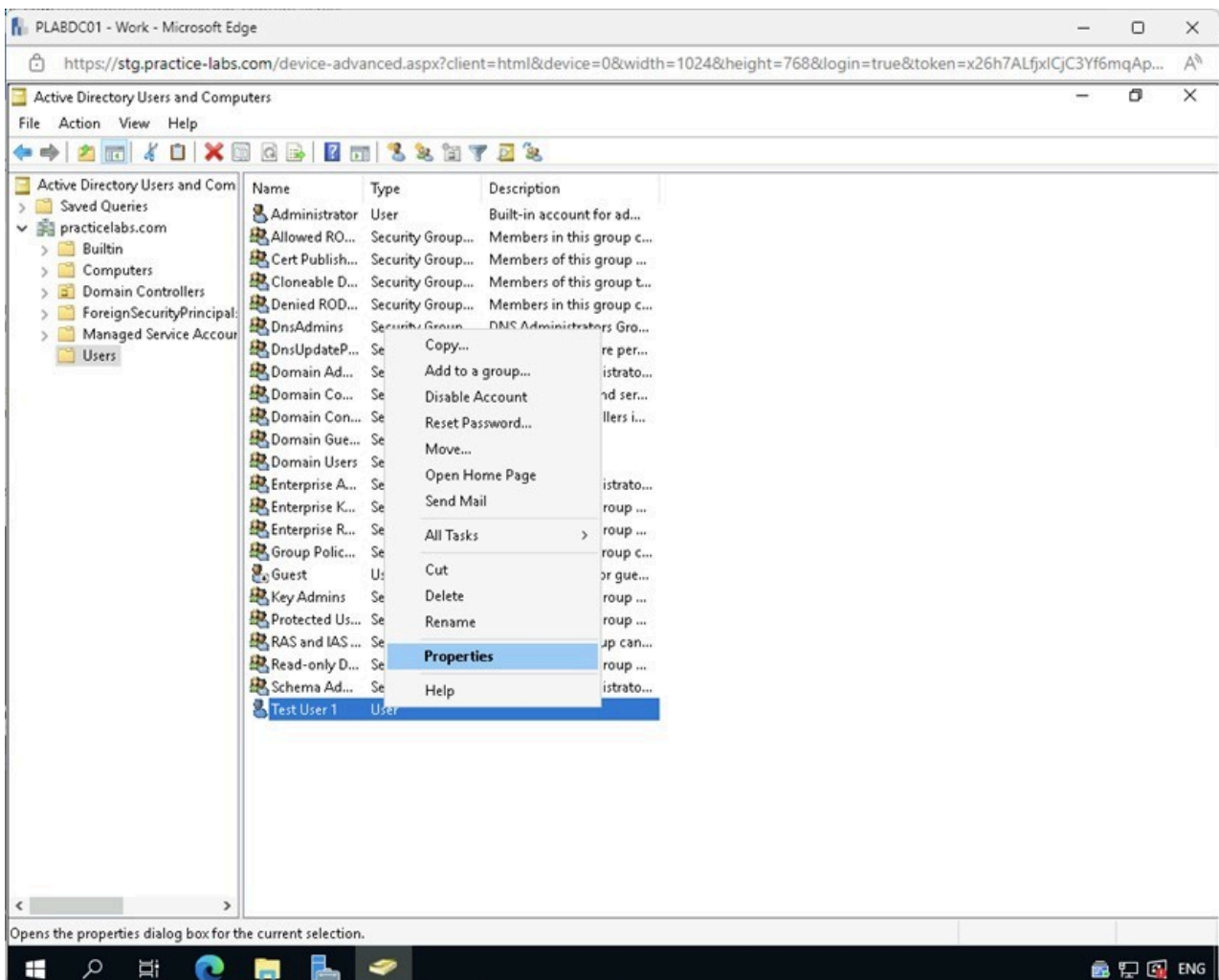


Figure 2.7 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window. Test User 1 is right-clicked and Properties selected.

Step 8

The **General** tab of the **Test User 1 Properties** window is displayed. The user's name, description, office, email, telephone number and web page can be configured here.

Select the **Account** tab.

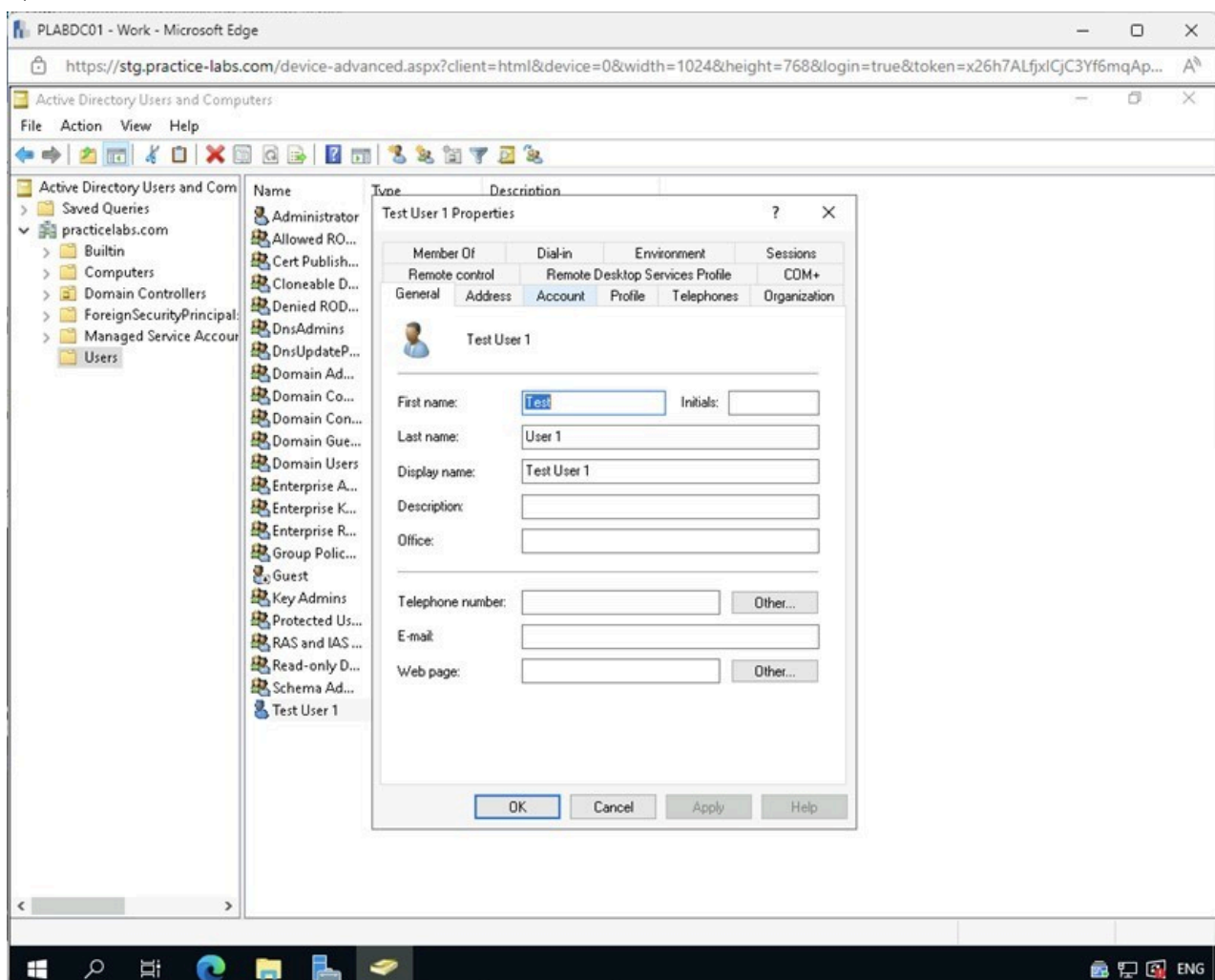


Figure 2.8 Screenshot of PLABDC01: Displaying the General tab of the Test User 1 Properties window. The Account tab is highlighted

Step 9

In the **Account** tab, Logon Hours, unlocking the account, and setting account expirations can be configured.

Click **OK**.

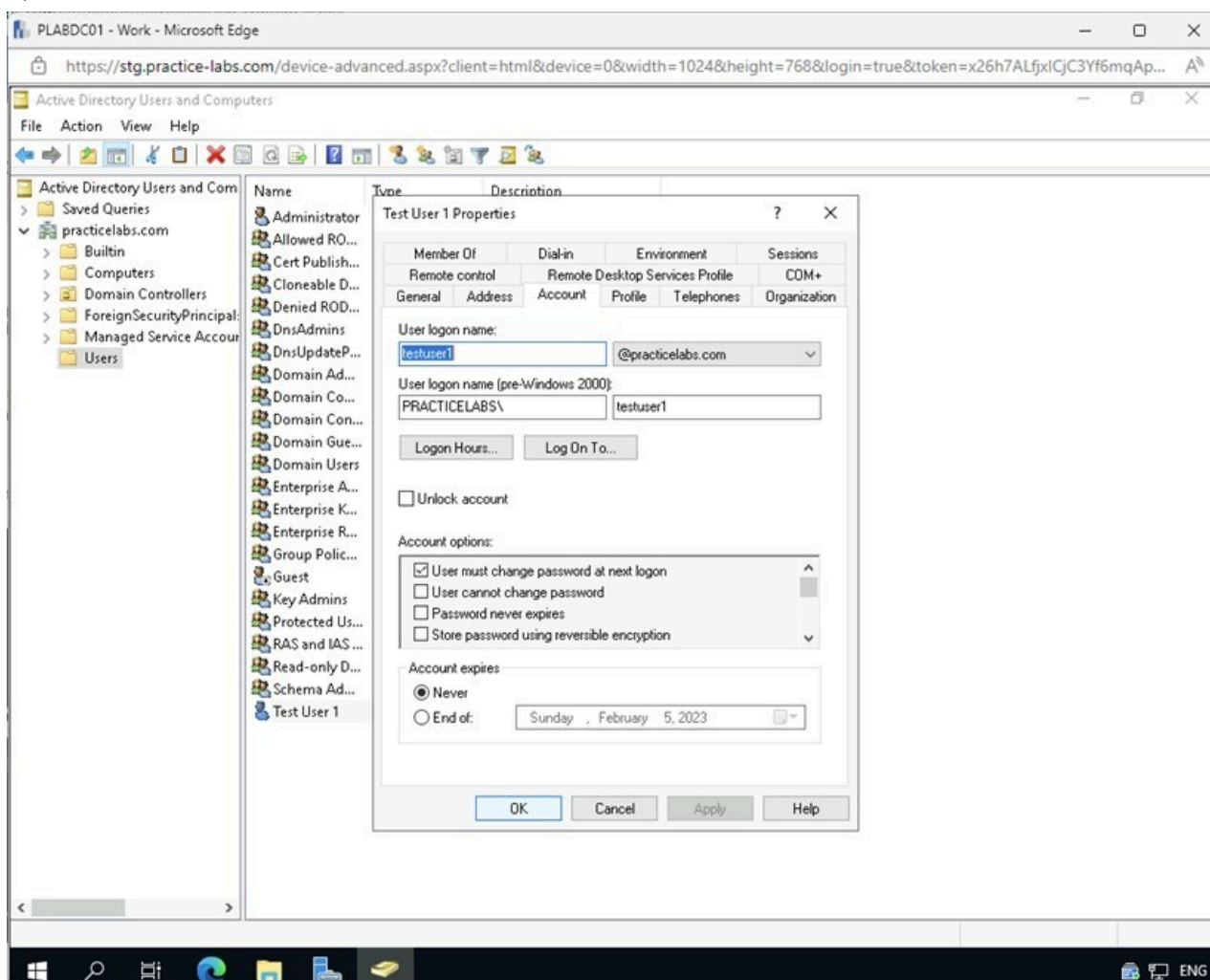


Figure 2.9 Screenshot of PLABDC01: Displaying the Account tab, showing the OK button selected.

Keep the **Active Directory Users and Computers** window open.

Task 2 - Explore, Define and Configure User Groups and Role Based Access Control

Role Based Access Control gives users restricted network access based on their organizational role. Only vital information necessary to do their jobs should be available to employees, known as the principle of least privilege. Access may be limited by authority, responsibility, and skill level. Folder access, file generation, and deletion are instances of computer-restricted activities.

Scope Based Access Control may allow for additional file and folder access. Scope-based access control grants users or directory groups permissions. The scope can create, modify, remove, and utilize a category's resources. Permissions can be assigned to each user or directory group.

In this task, you will configure user groups and role based access control.

Step 1

In **PLABDC01**, the **Active Directory Users and Computers** window is open.

Right-click **Test User 1** and select **Add to a group...**

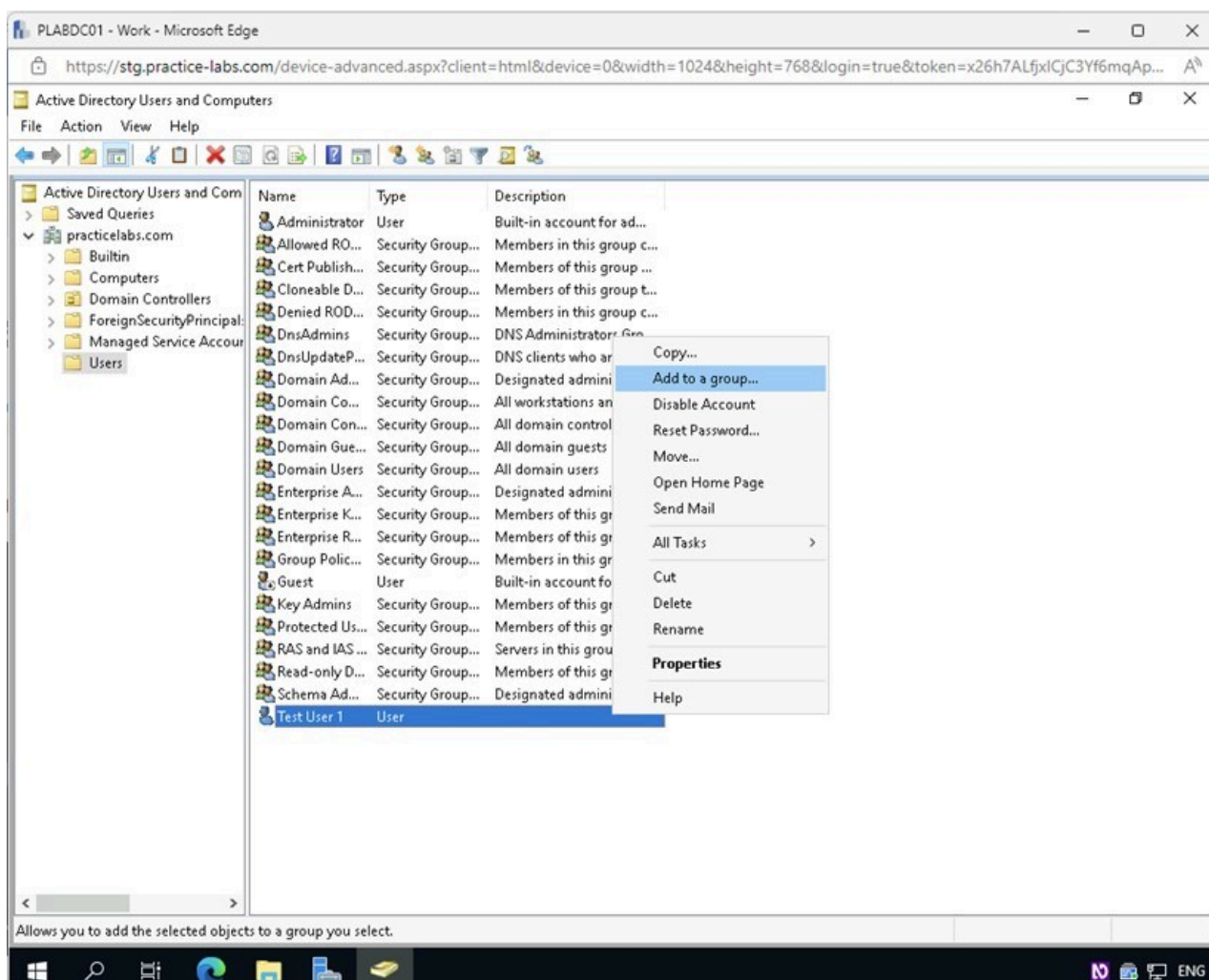


Figure 2.10 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window. Test User 1 is right-clicked, and Add to a group is selected.

Step 2

In the **Select Groups** window, select **Advanced**.

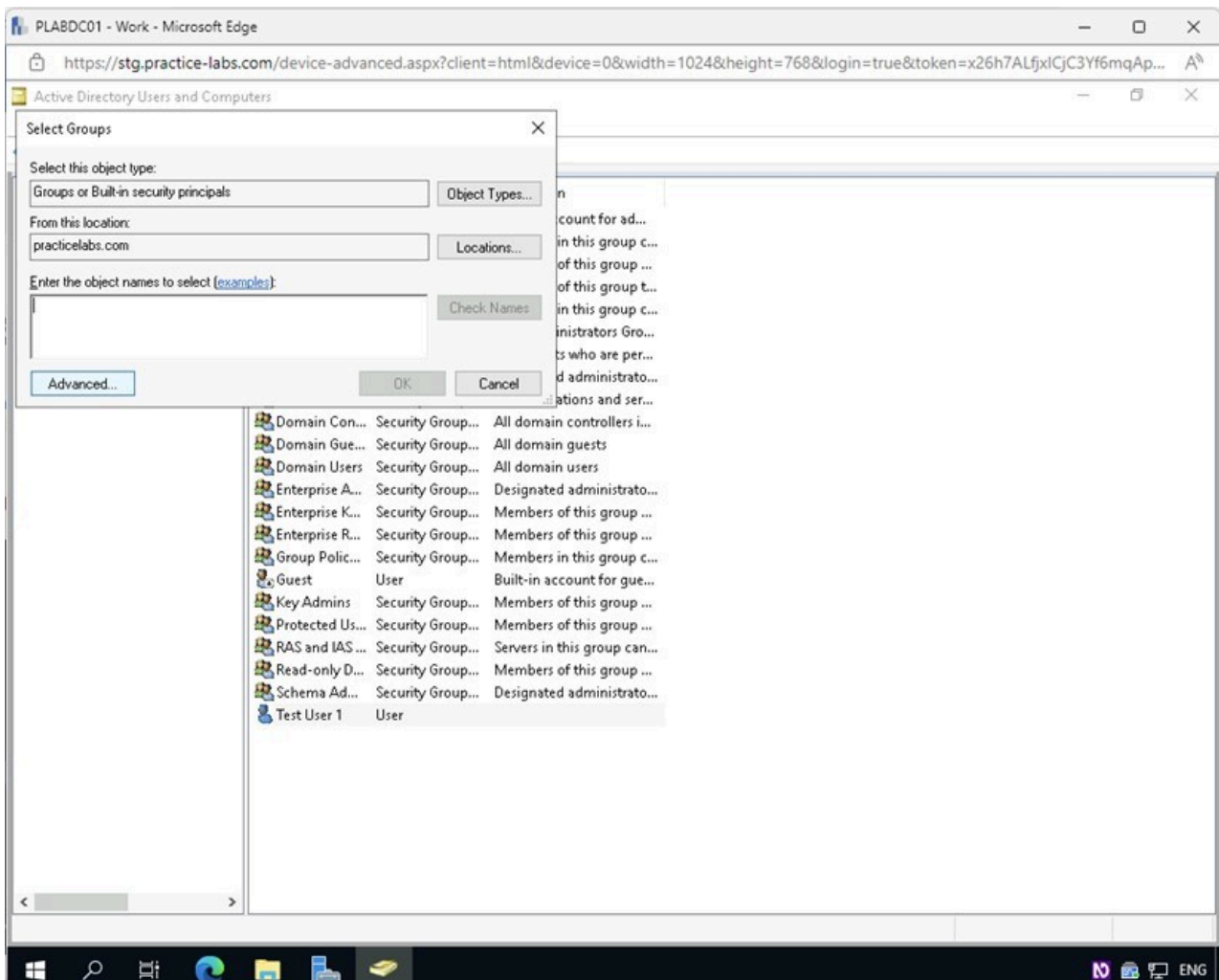


Figure 2.11 Screenshot of PLABDC01: Displaying the Select Groups window, showing Advanced selected.

Step 3

In the **Select Groups** window, click **Find Now** to scan for groups to add to the user.

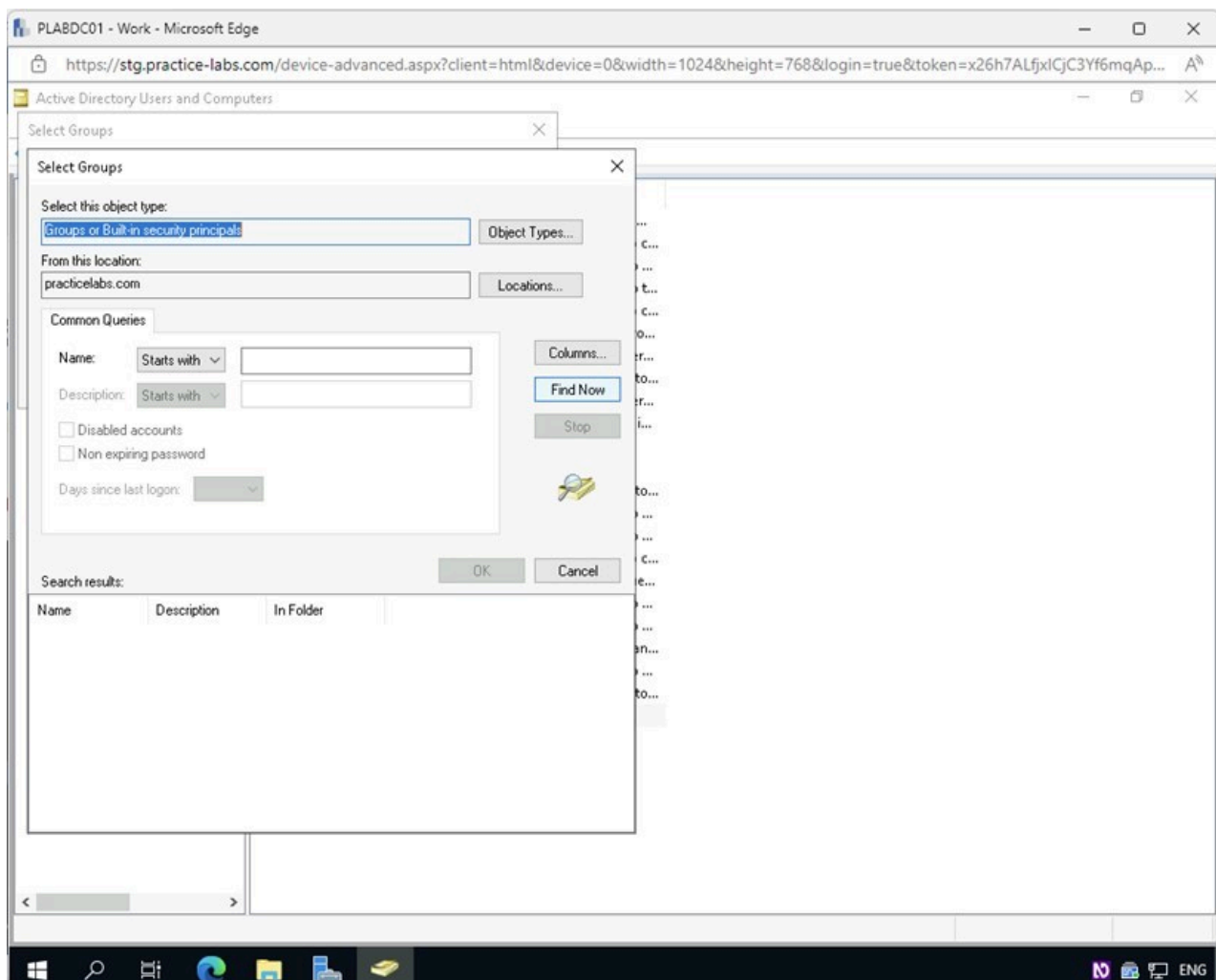


Figure 2.12 Screenshot of PLABDC01: Displaying the Select Groups window, showing Find Now selected.

Step 4

In the **Search results** section, scroll down and select **Administrators**.

Click **OK**.

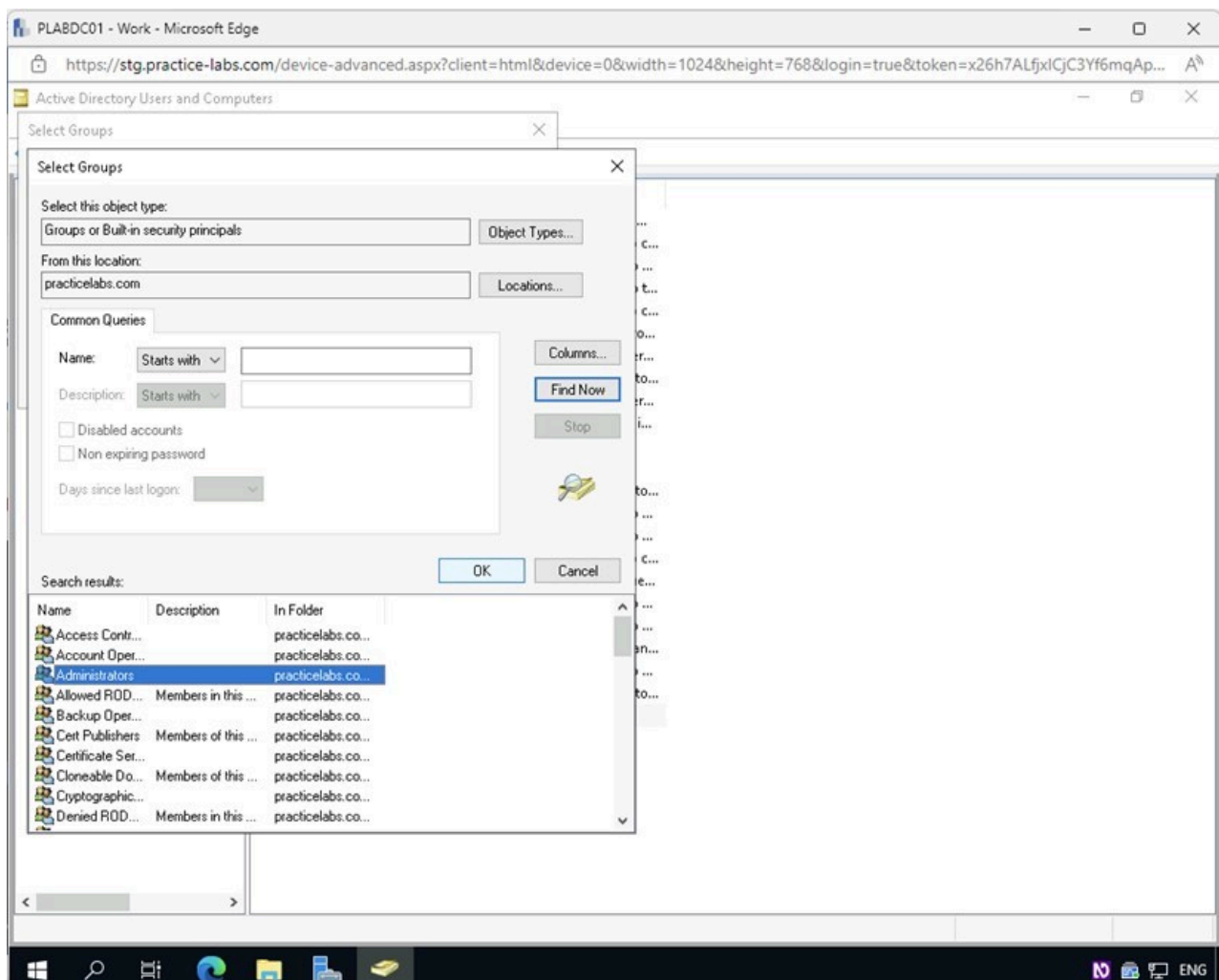


Figure 2.13 Screenshot of PLABDC01: Displaying the Select Groups dialog box. Administrators is selected, and OK is highlighted.

Step 5

Administrators is now displayed under the **Enter the object names to select** section.

Click **OK**.

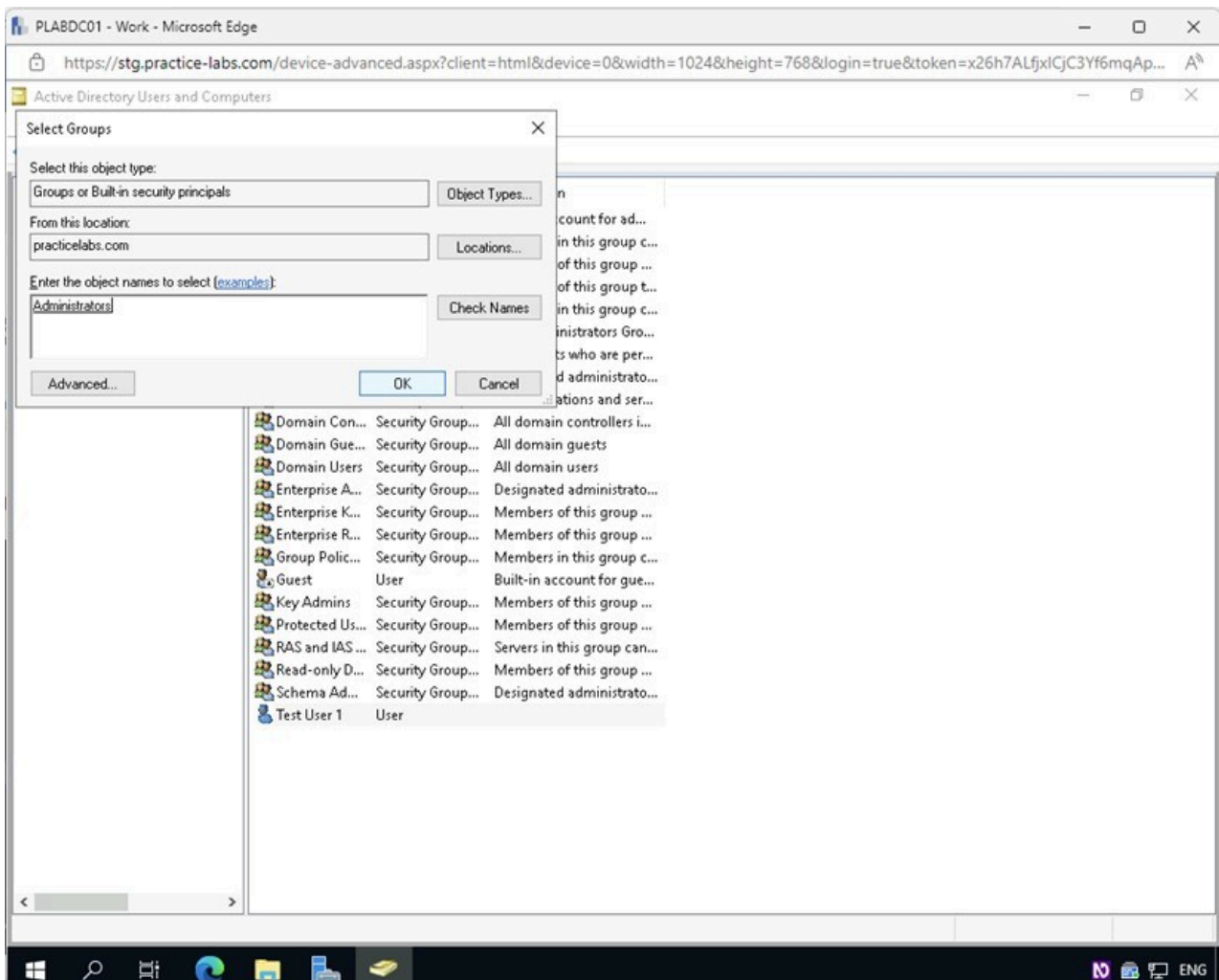


Figure 2.14 Screenshot of PLABDC01: Displaying selecting OK in the Select Groups window.

Step 6

Click **OK** on the **Active Directory Domain Services** pop-up window.

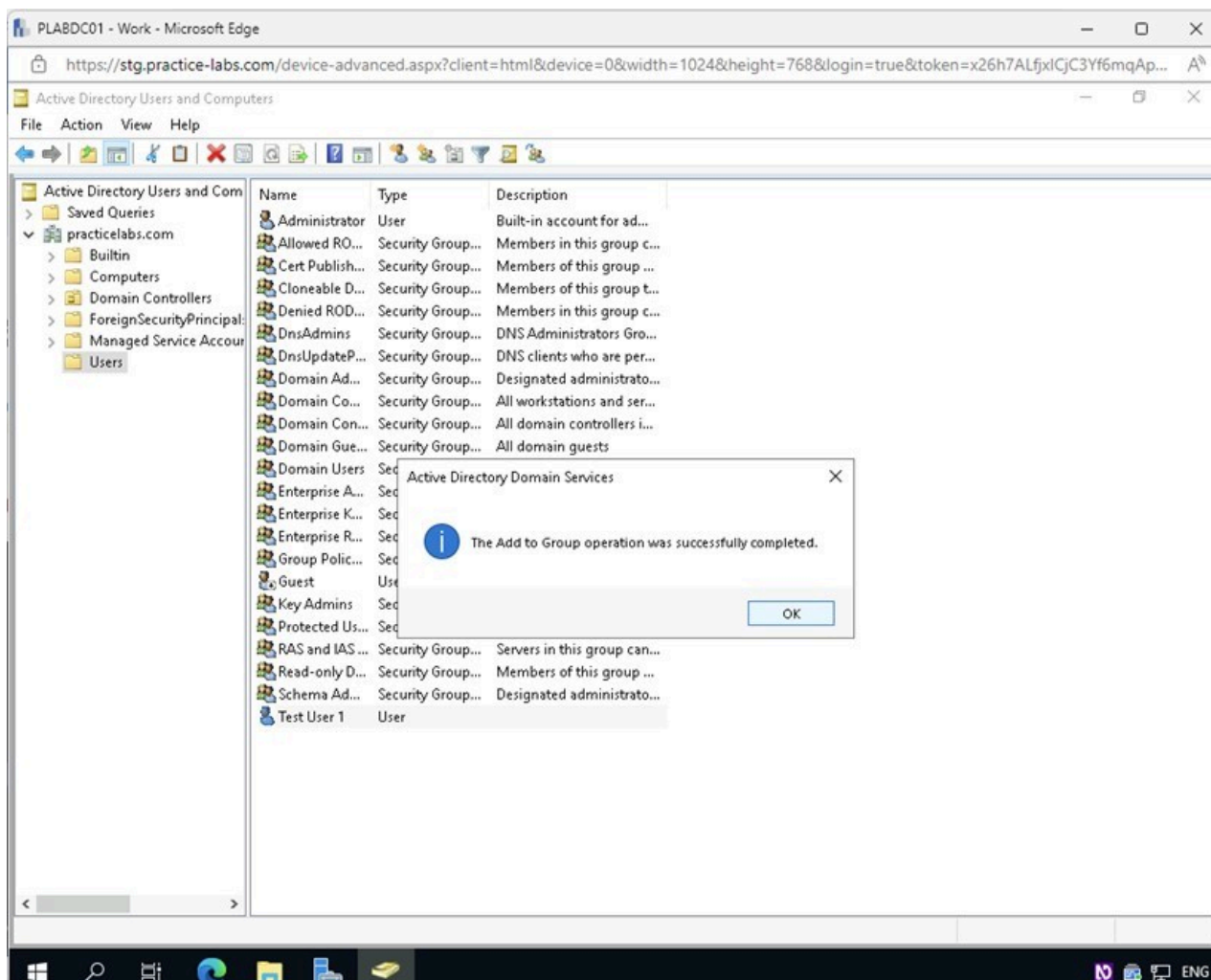


Figure 2.15 Screenshot of PLABDC01: Displaying selecting OK on the Active Directory Domain Services pop-up window.

Step 7

You will now view the group membership for the new **Test User 1**.

Right click **Test User 1** and select **Properties**.

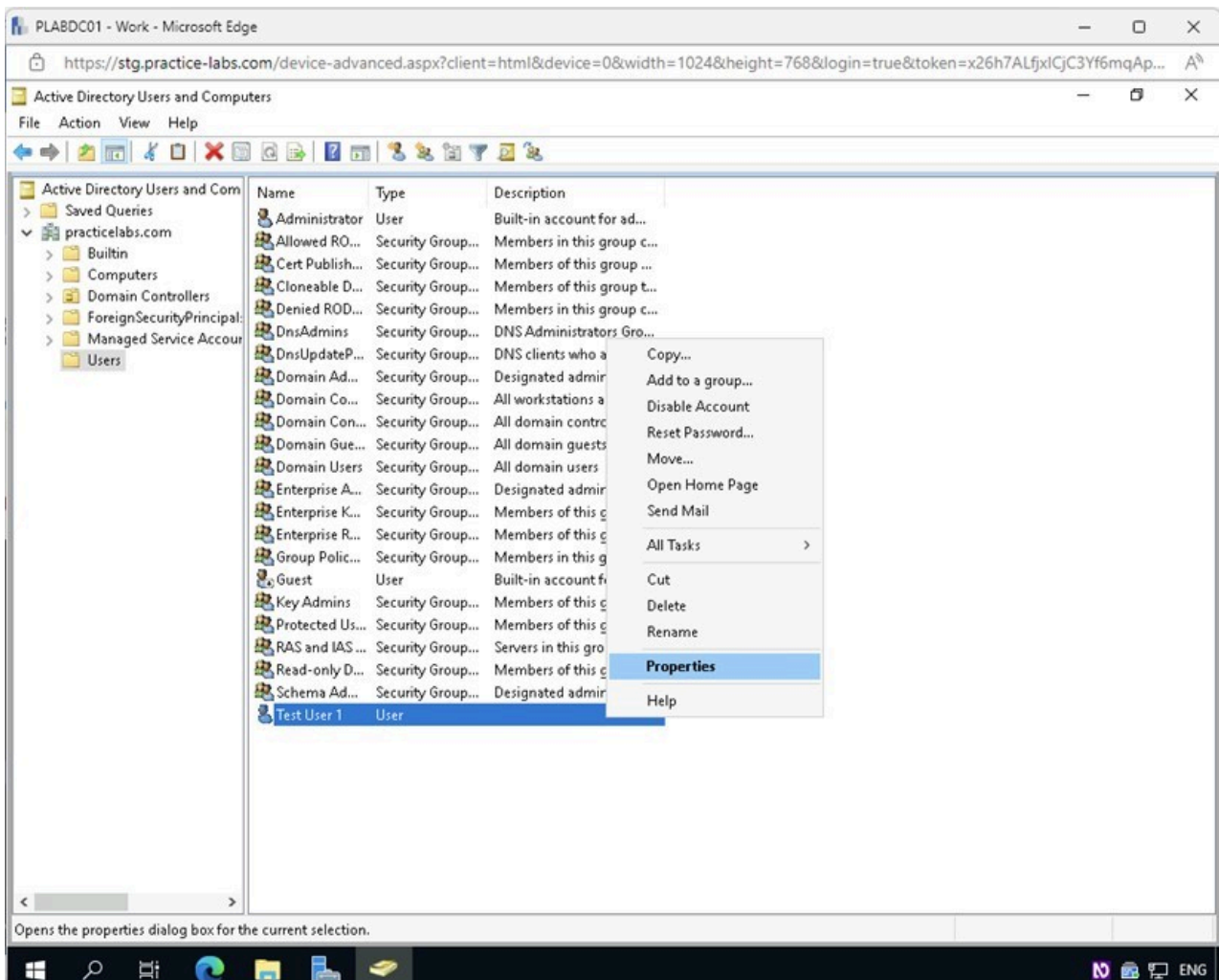


Figure 2.16 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window. Test User 1 is right-clicked and Properties selected.

Step 8

The **General** tab of the **Test User 1 Properties** window is displayed.

Select the **Member Of** tab.

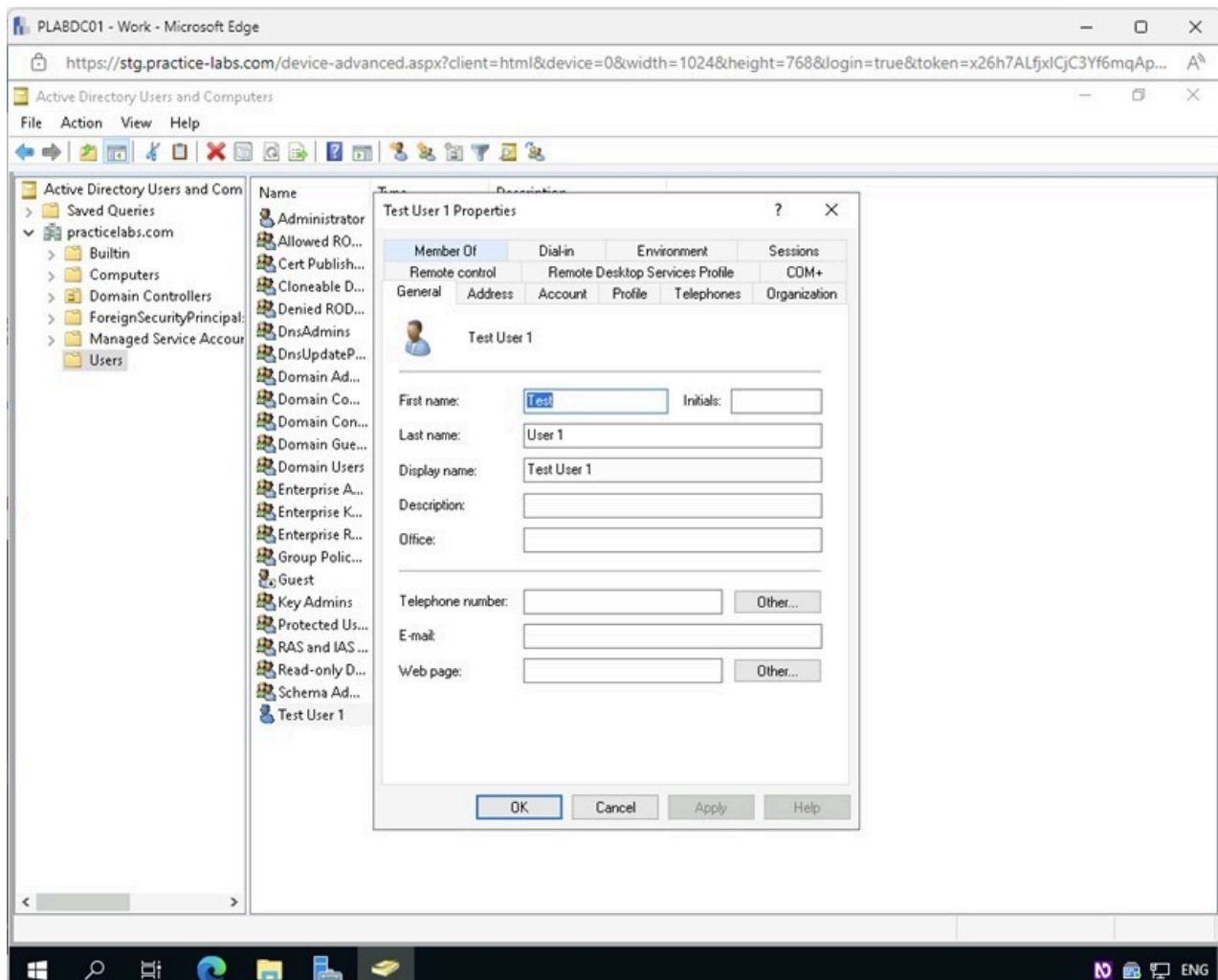


Figure 2.17 Screenshot of PLABDC01: Displaying the General tab of the Test User 1 Properties window, showing the Member Of tab selected.

Step 9

Here you can view the group membership for **Test User 1**.

Double-click on **Administrators** to learn more about the group.

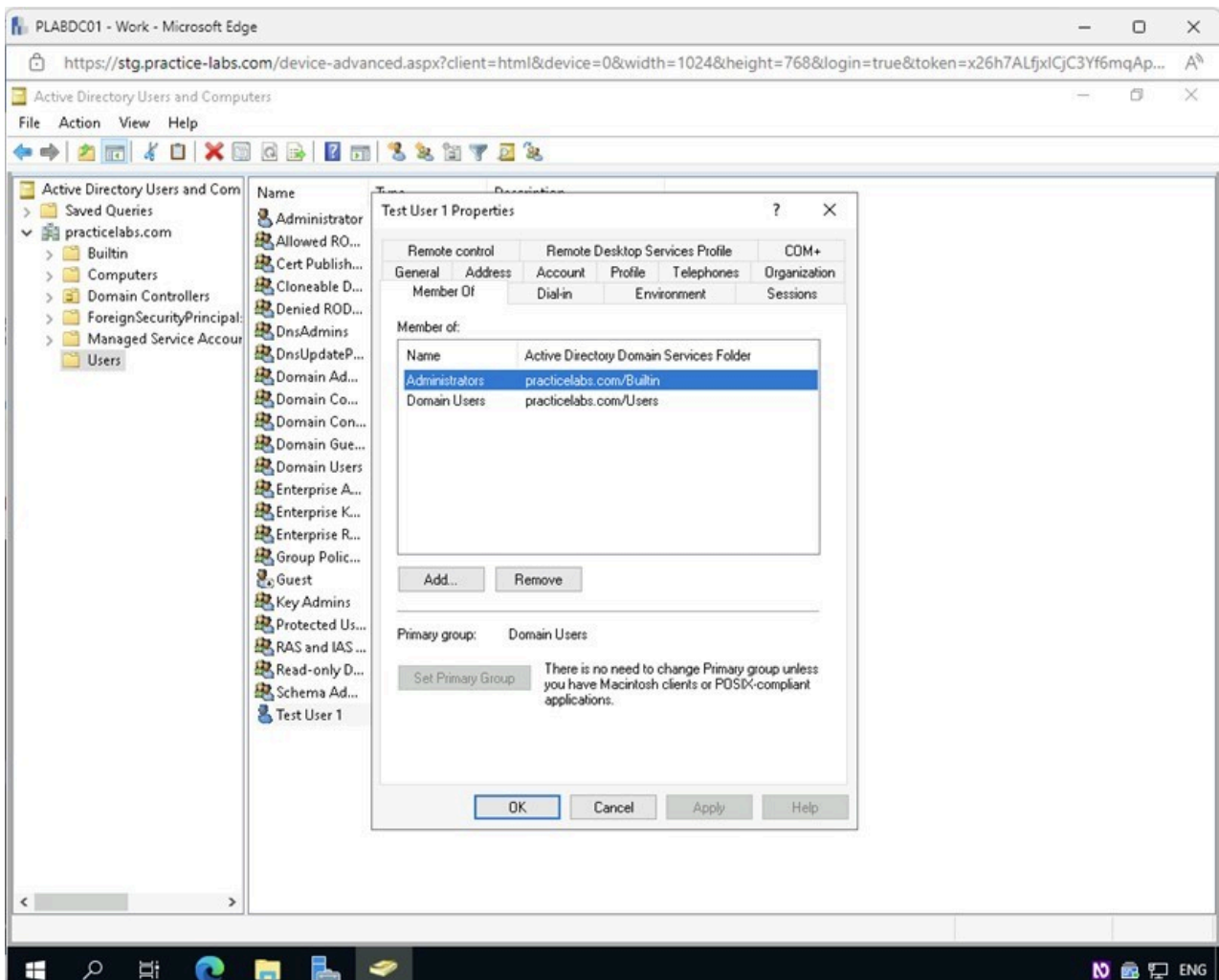


Figure 2.18 Screenshot of PLABDC01: Displaying the Test User 1 Properties-Member Of Tab, showing Administrators selected.

Step 10

In the **Administrator Properties** window, notice that Administrators have complete and unrestricted access to the computer/domain in the **Description** section.

Click **OK**.

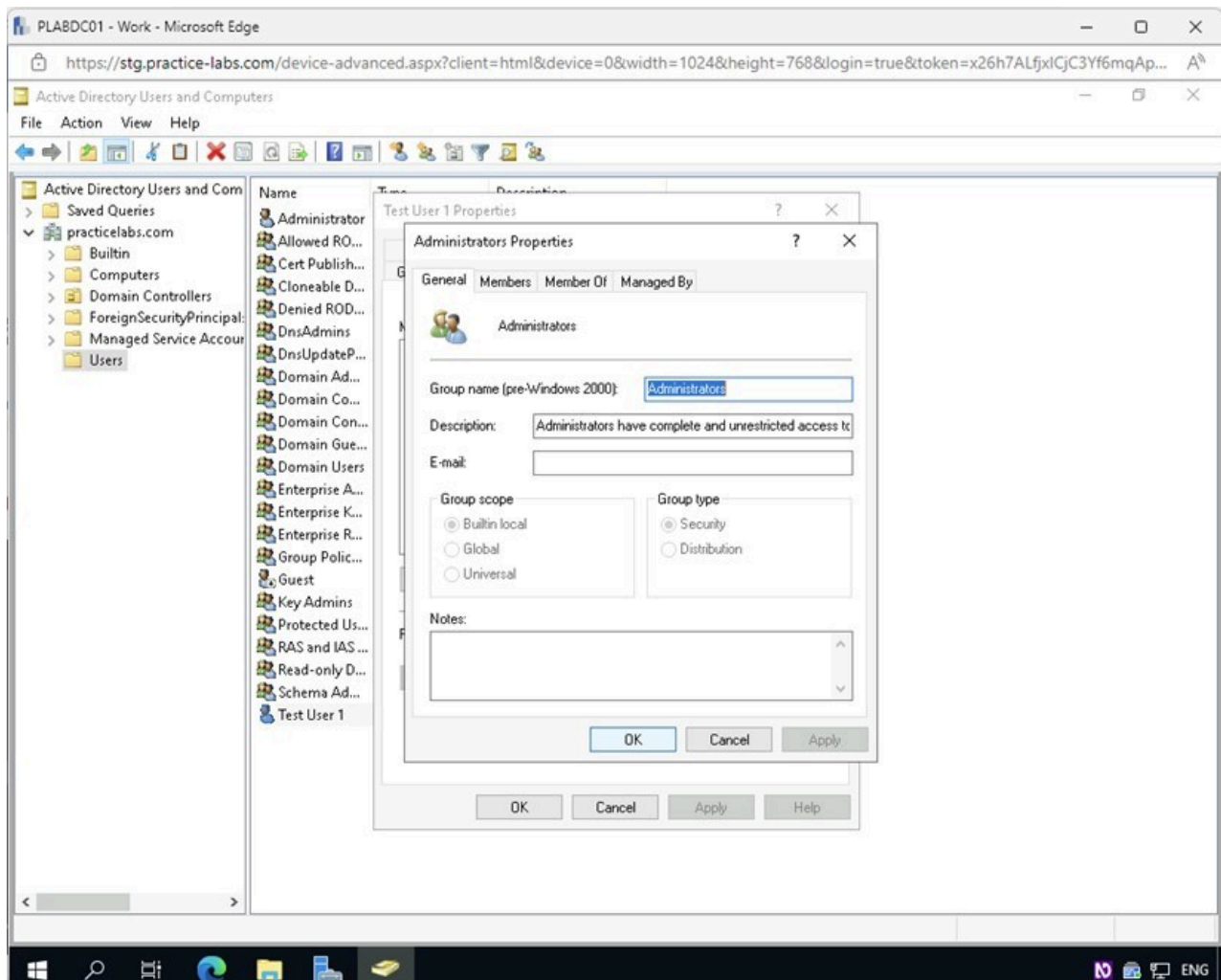


Figure 2.19 Screenshot of PLABDC01: Displaying the Administrators Properties window, showing the OK button selected.

Step 11

Next, you will add **Test User 1** to the **Remote Management Group** to allow remote network access.

Click **Add**.

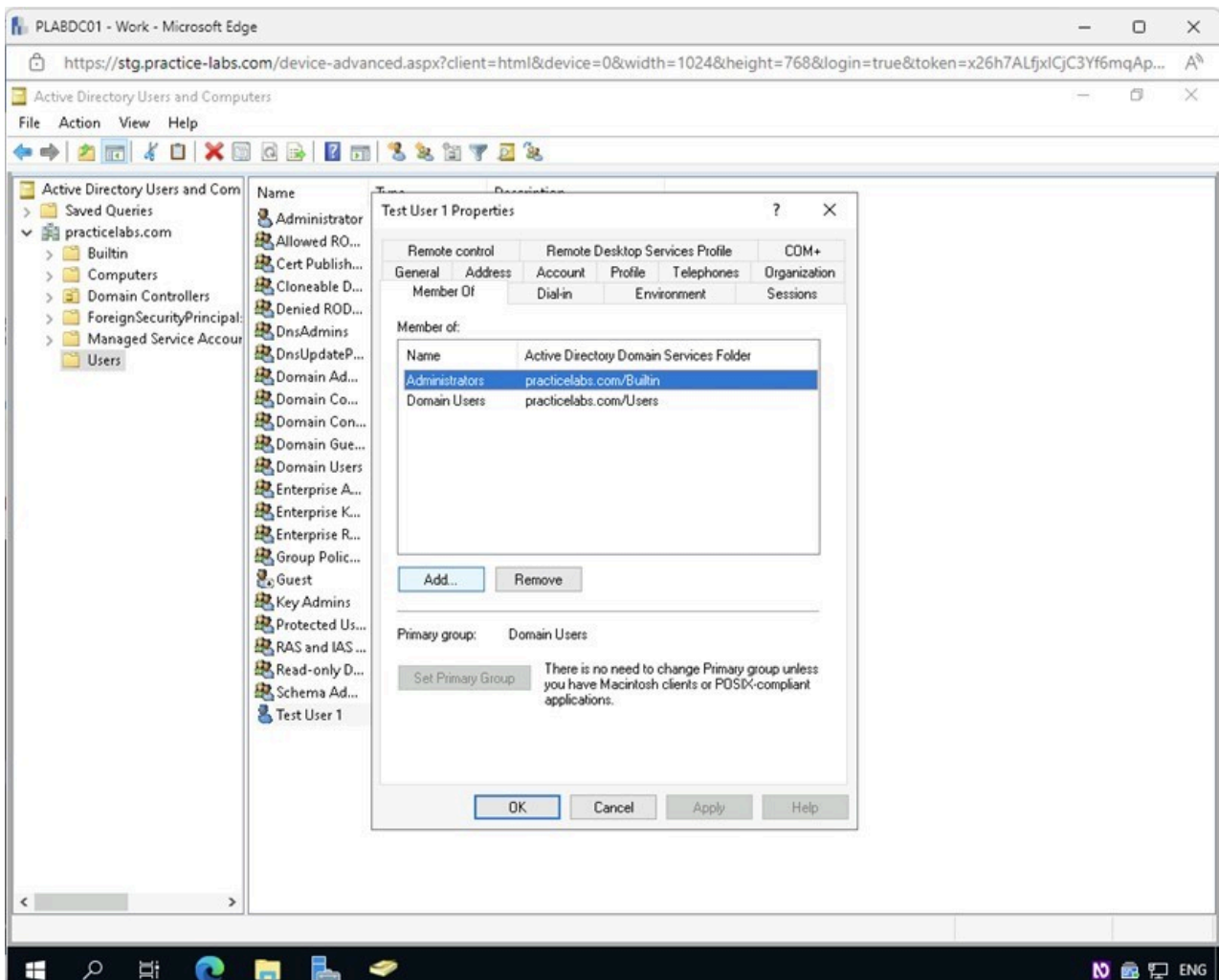


Figure 2.20 Screenshot of PLABDC01: Displaying the Test User 1 Properties window, showing Add selected.

Step 12

In the **Select Groups** window, click **Advanced**.

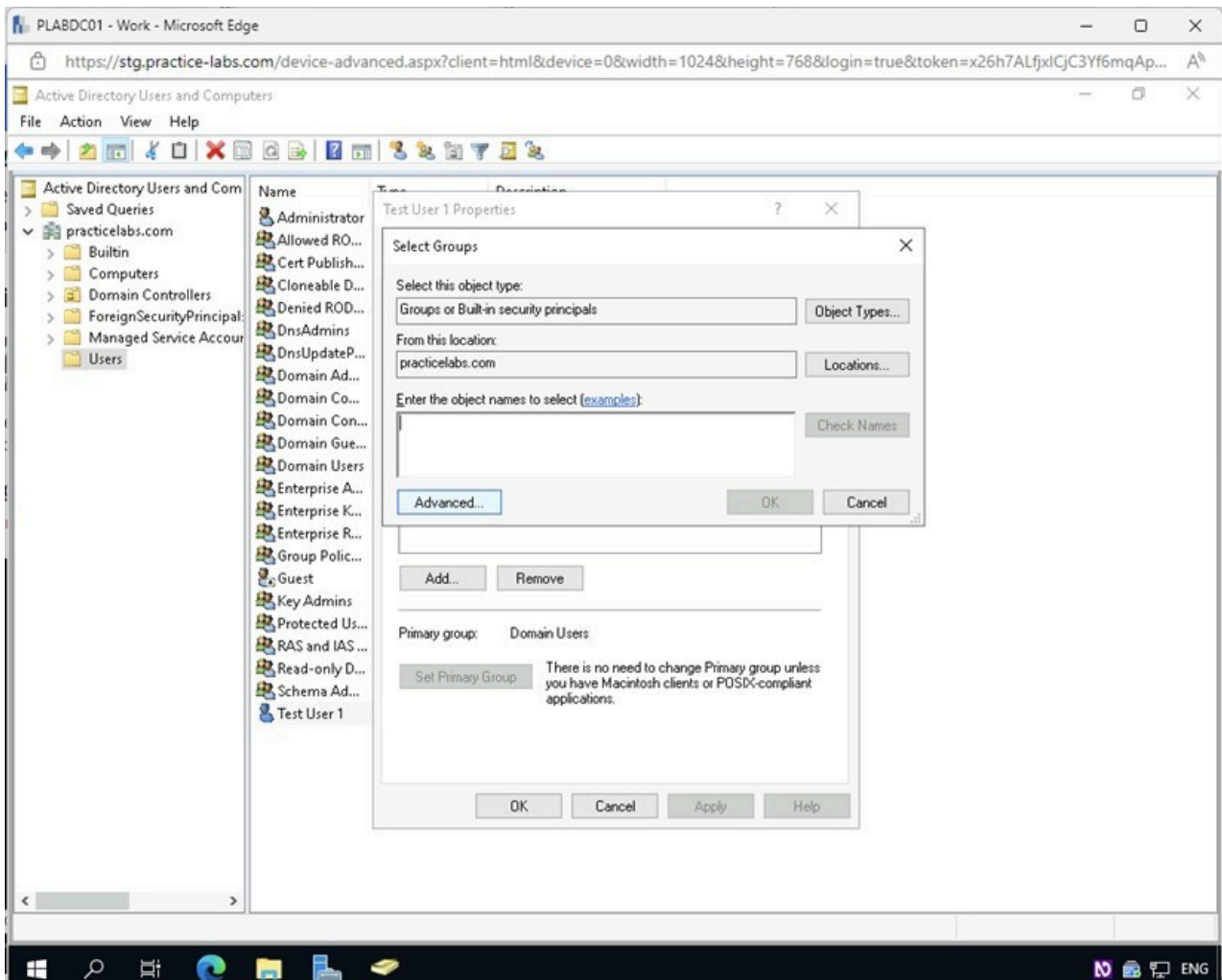


Figure 2.21 Screenshot of PLABDC01: Displaying selecting Advanced in the Select Groups window.

Step 13

From the **Select Groups** pop-up window, click **Find Now**.

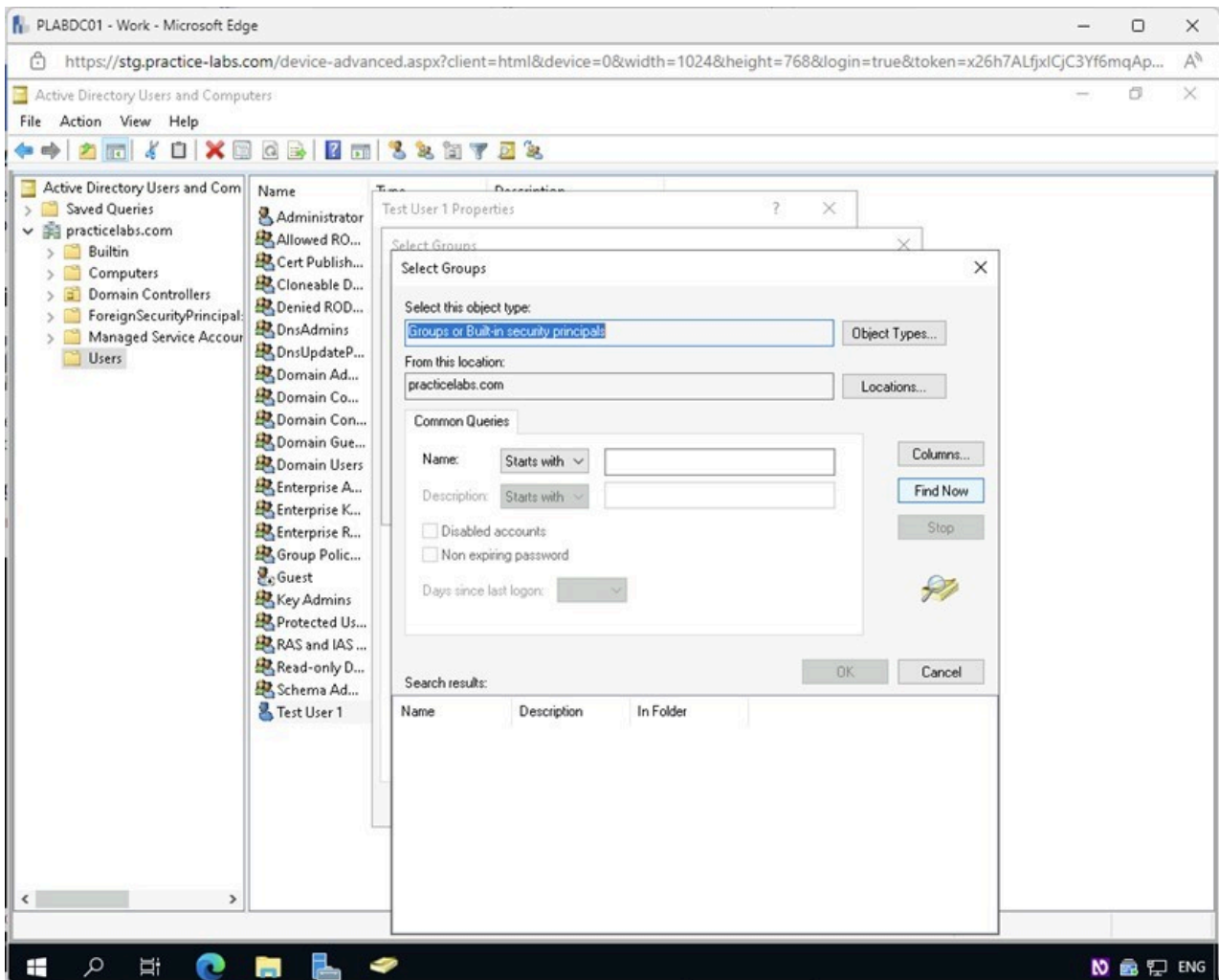


Figure 2.22 Screenshot of PLABDC01: Displaying selecting Find Now in the Select Groups pop-up window.

Step 14

In the **Select Groups** pop-up window, scroll down the list and select **Remote Management Users**.

Click **OK**.

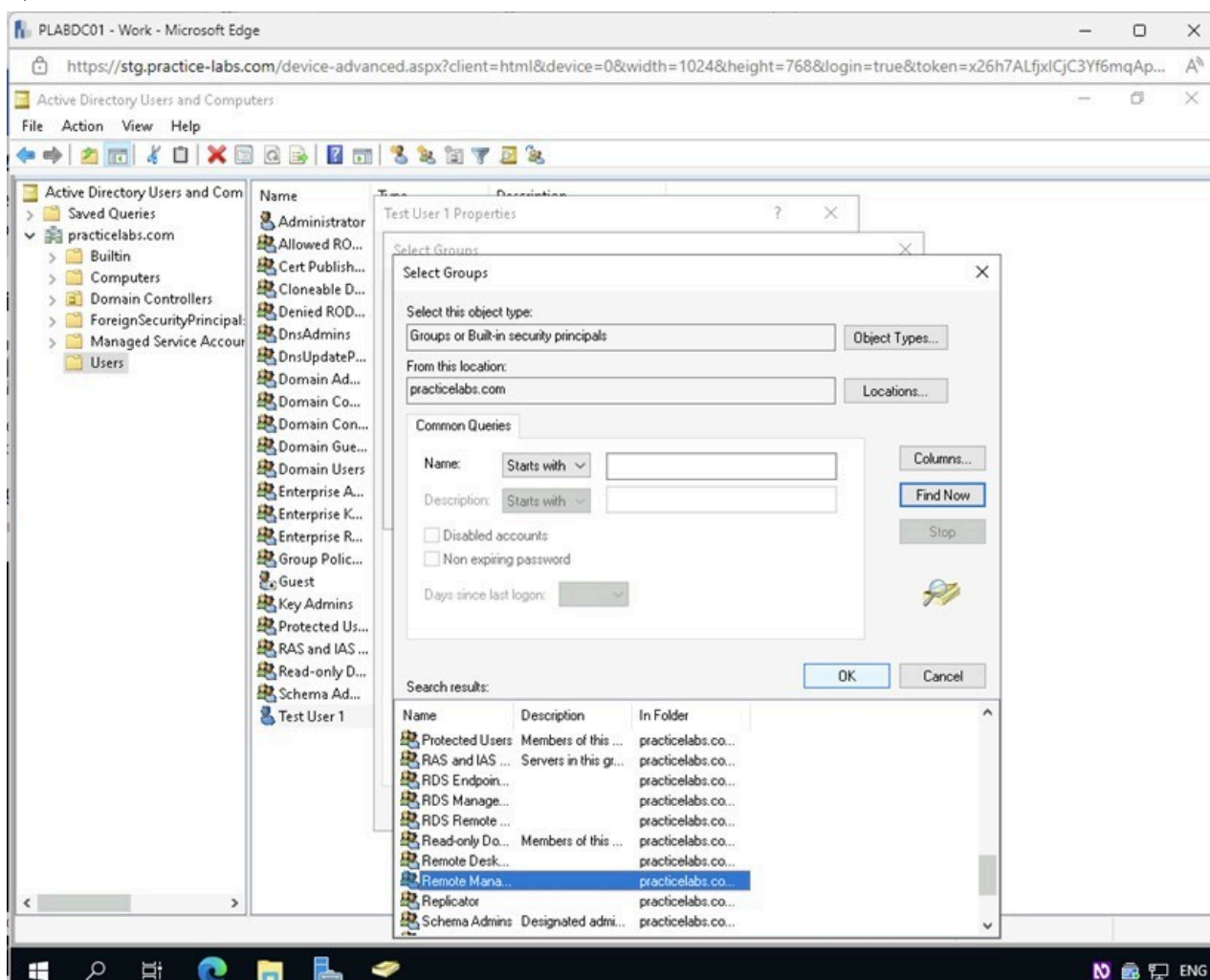


Figure 2.23 Screenshot of PLABDC01: Displaying the Select Groups pop-up window. Remote Management Users is selected, and OK highlighted.

Step 15

Remote Management Users group is now added under the **Enter the object names to select** section.

Click **OK**.

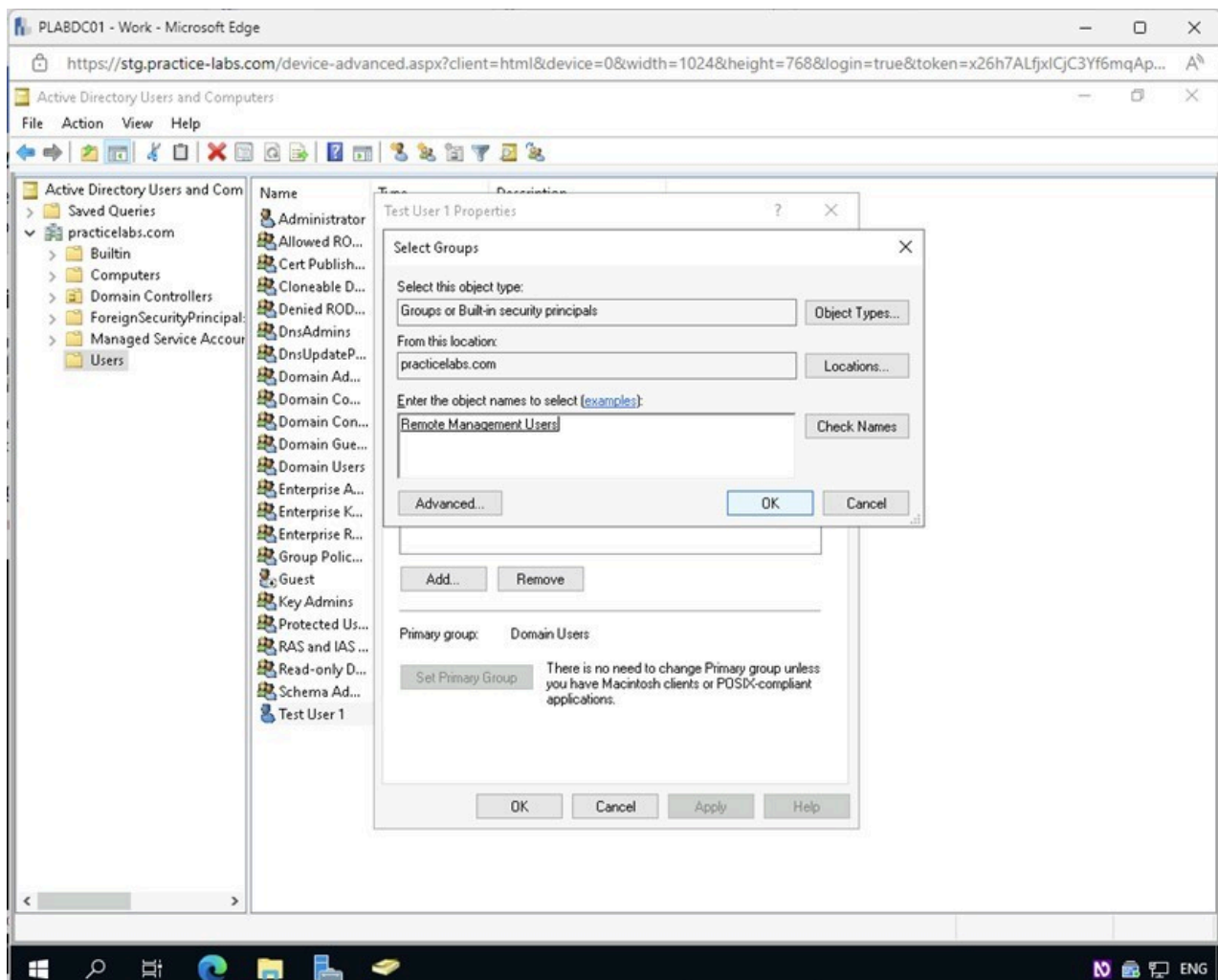


Figure 2.24 Screenshot of PLABDC01: Displaying selecting OK in the Select Groups window.

Step 16

Click **OK** in the **Test Users 1 Properties** window.

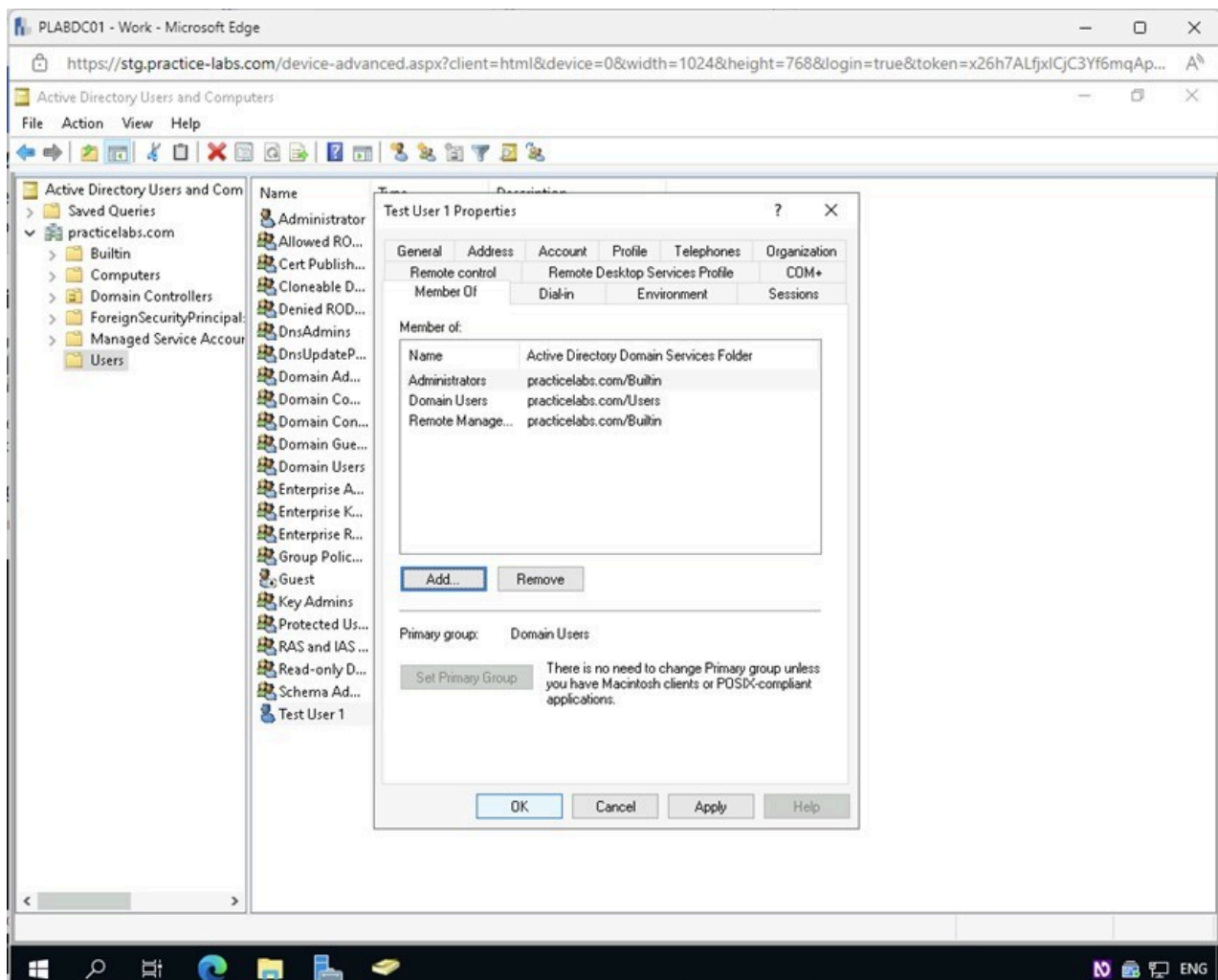


Figure 2.25 Screenshot of PLABDC01: Displaying selecting OK in the Test User 1 Properties window.

Step 17

Keep the **Active Directory Users and Computers** window open.

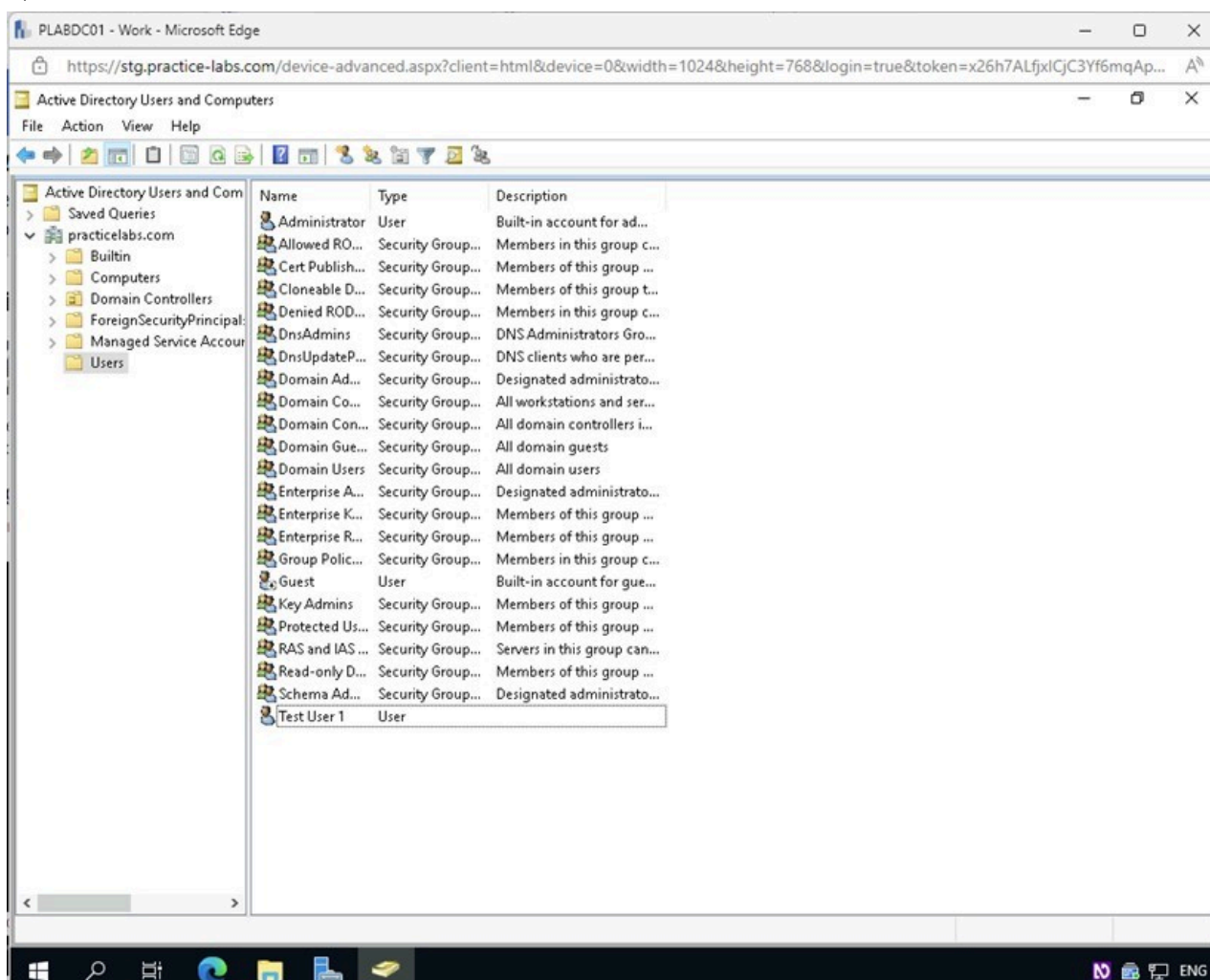


Figure 2.26 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window.

Exercise 3 - Permissions and Access Controls

Administrators can set up a wide variety of supplemental permissions and access control features to aid in the efficient and secure running of the network.

In this exercise, you will learn about delegation and job specialization.

Learning Outcomes

After completing this exercise, you should be able to:

- Explore, Define, and Configure Delegation

After completing this exercise, you should have further knowledge of:

- Segregation of Duties

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Explore, Define, and Configure Delegation

Administrators are responsible for a wide variety of tasks. When hiring new staff or promoting from within, a firm should seek employees who can learn new skills and advance in the company. Administrators will relieve part of their workload by delegating responsibilities to individuals that exhibit this potential more responsibility within the network. Commonly assigned duties may include making backups, regaining access to locked accounts, and creating new ones. This can be done either in the active directory or other 3rd party applications.

In this task, you will use the Delegation of Control Wizard and assign a responsibility to Test User 1.

Step 1

Ensure you are connected to **PLABDC01** and the **Active Directory Users and Computers** window is open.

In the left pane, right-click on **Users** and select **Delegate Control**.

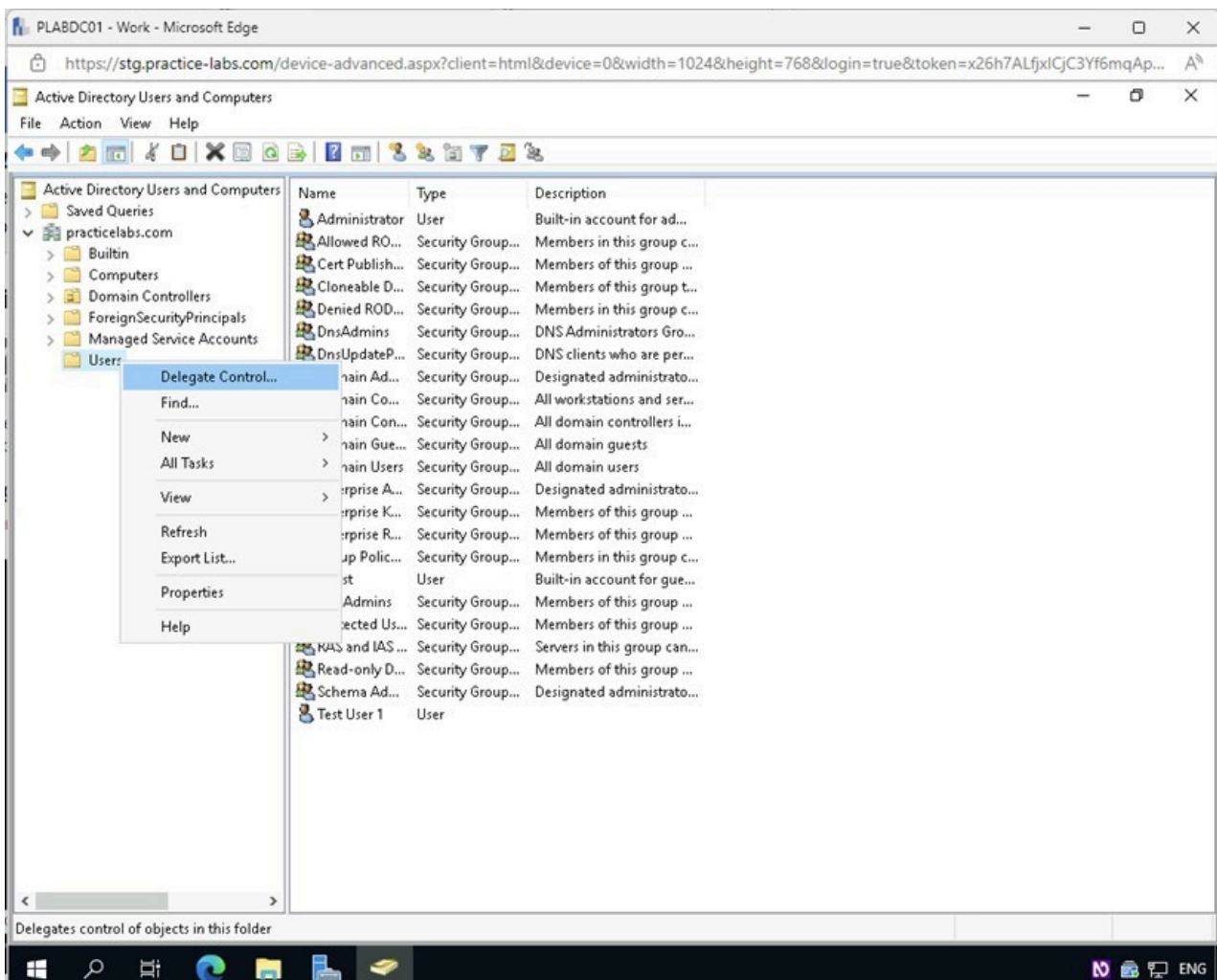


Figure 3.1 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window. Users folder is right-clicked, and Delegate Control is selected.

Step 2

In the **Delegation of Control Wizard - Welcome to the Delegation of Control Wizard** page, click **Next**.

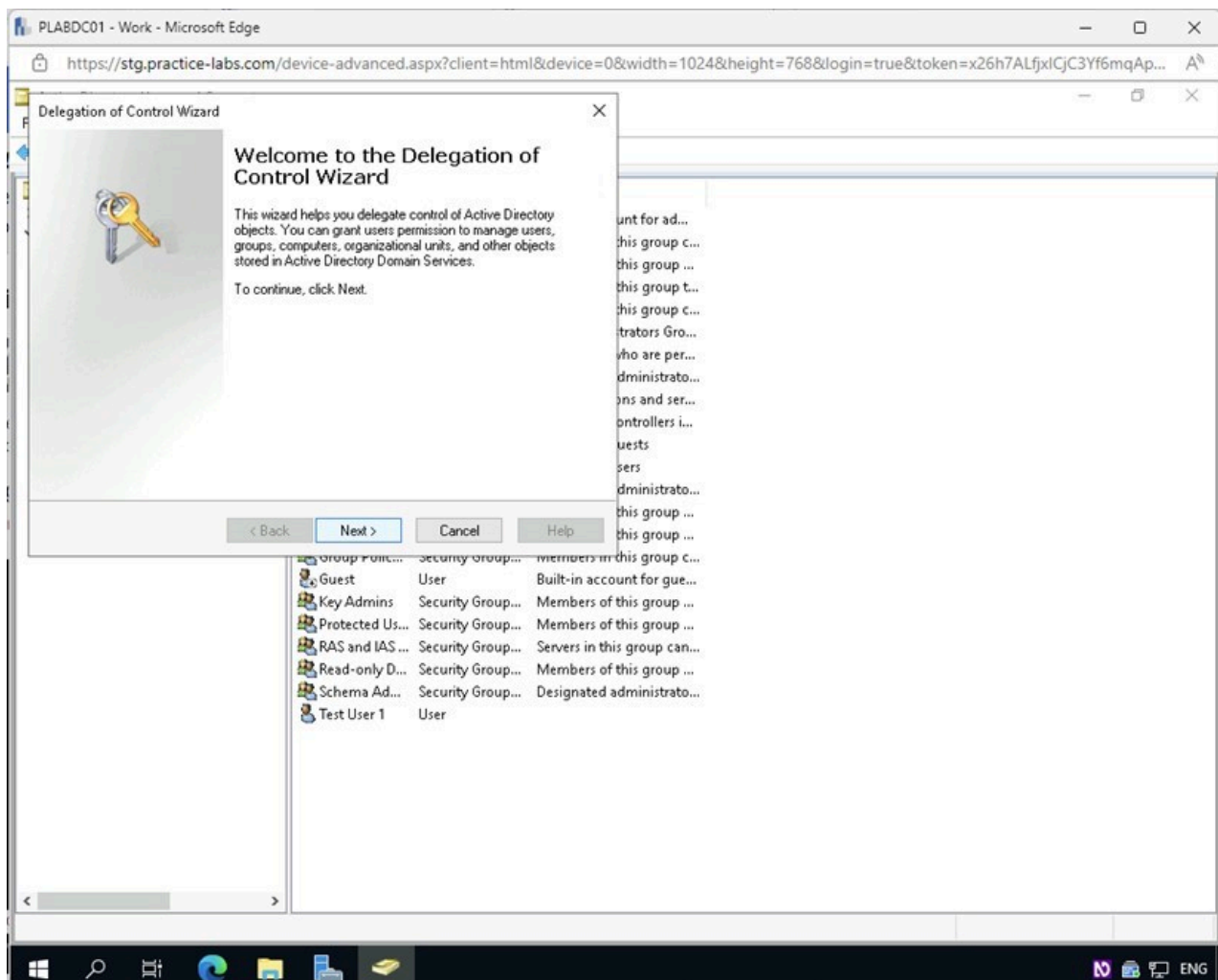


Figure 3.2 Screenshot of PLABDC01: Displaying selecting Next in the Delegation of Control Wizard - Welcome to the Delegation of Control Wizard page.

Step 3

On the **Users or Groups** page, click **Add**.

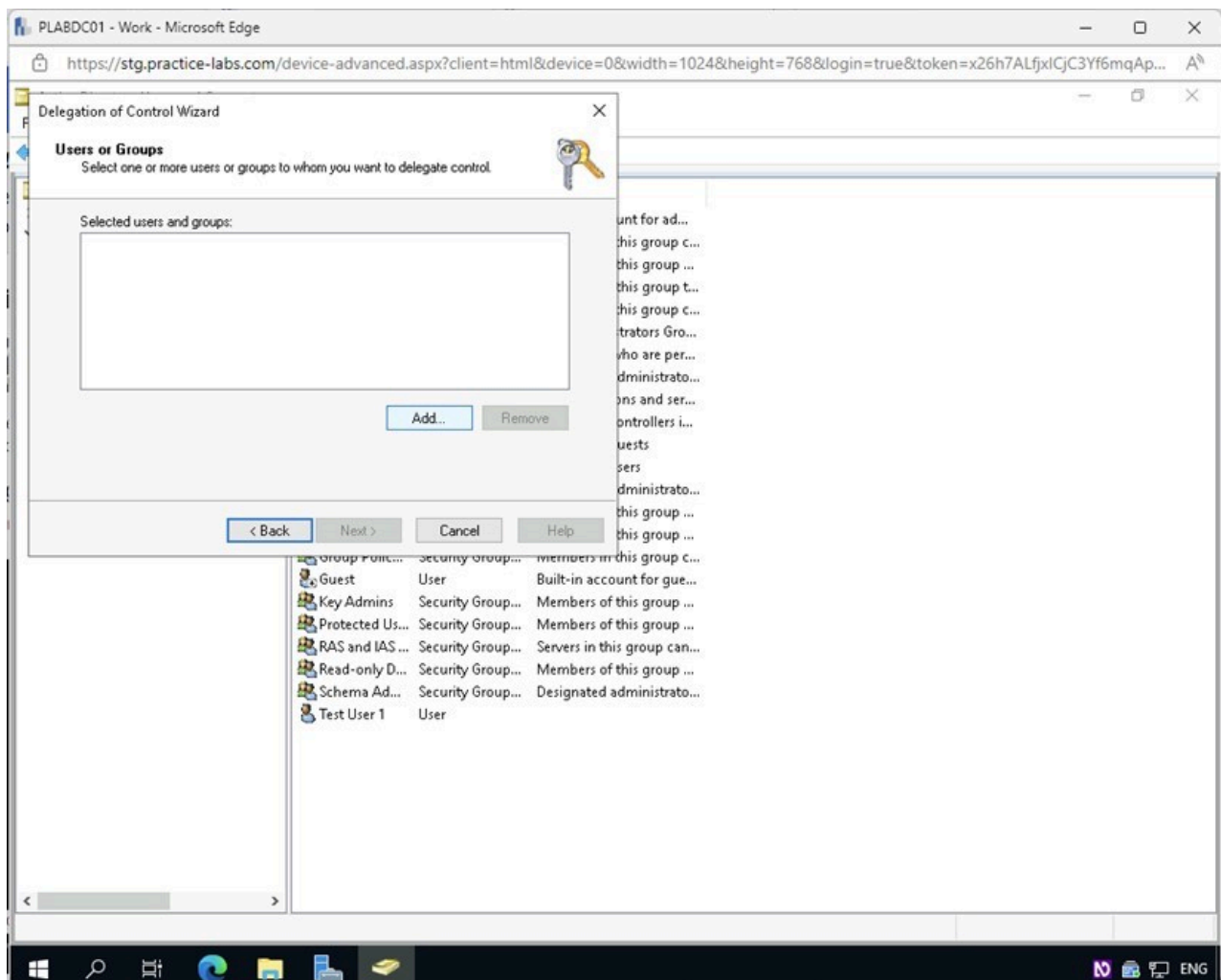


Figure 3.3 Screenshot of PLABDC01: Displaying selecting Add in the Users or Groups page.

Step 4

In the **Select Users, Computers, or Groups** window, type the following in the **Enter the object names to select** textbox:

test

Click **Check Names**.

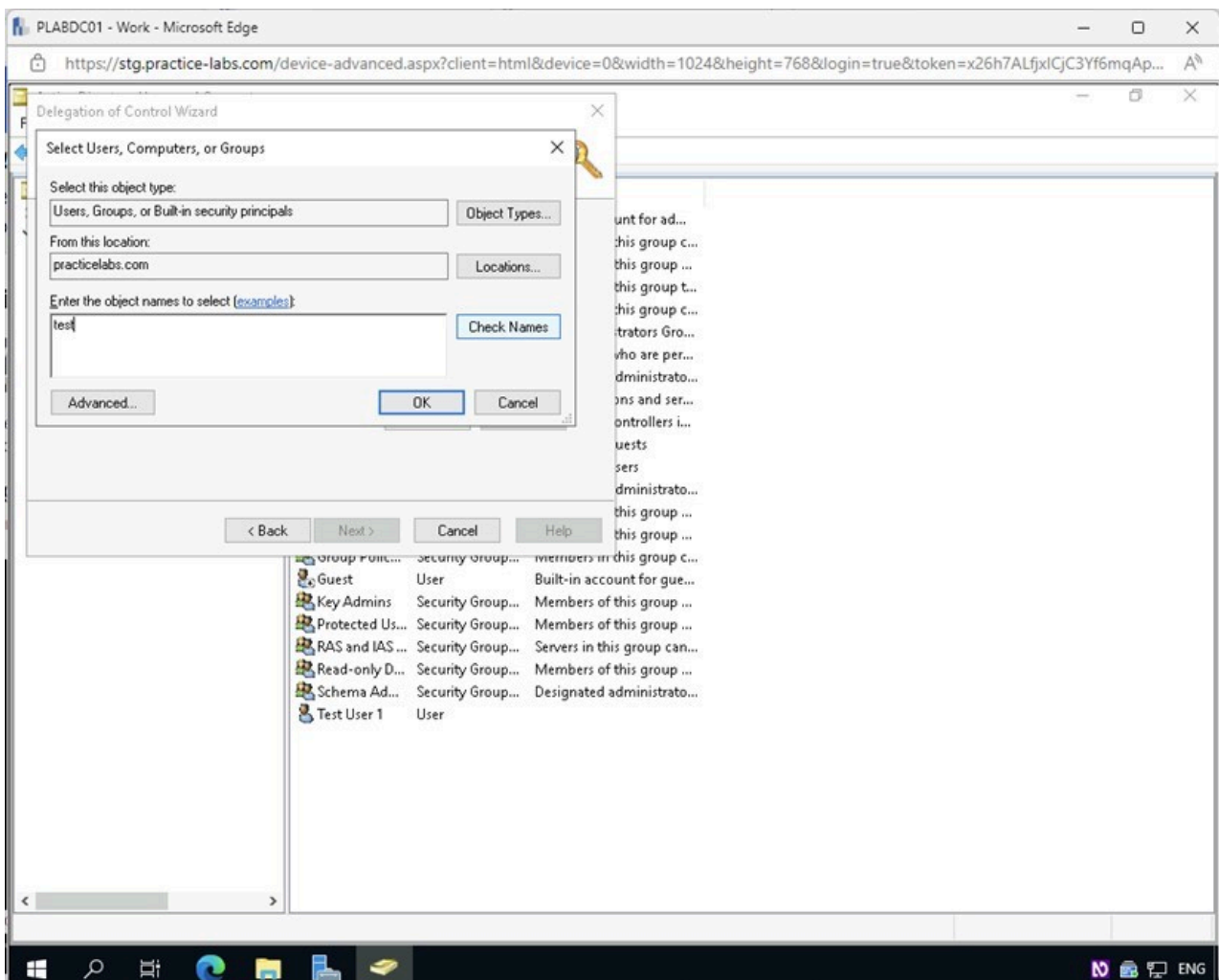


Figure 3.4 Screenshot of PLABDC01: Displaying the Select Users, Computers, or Groups window, showing the required object name entered and Check Names selected.

Step 5

Test User 1 is located and appears in the **Enter the object names to select** textbox.

Click **OK**.

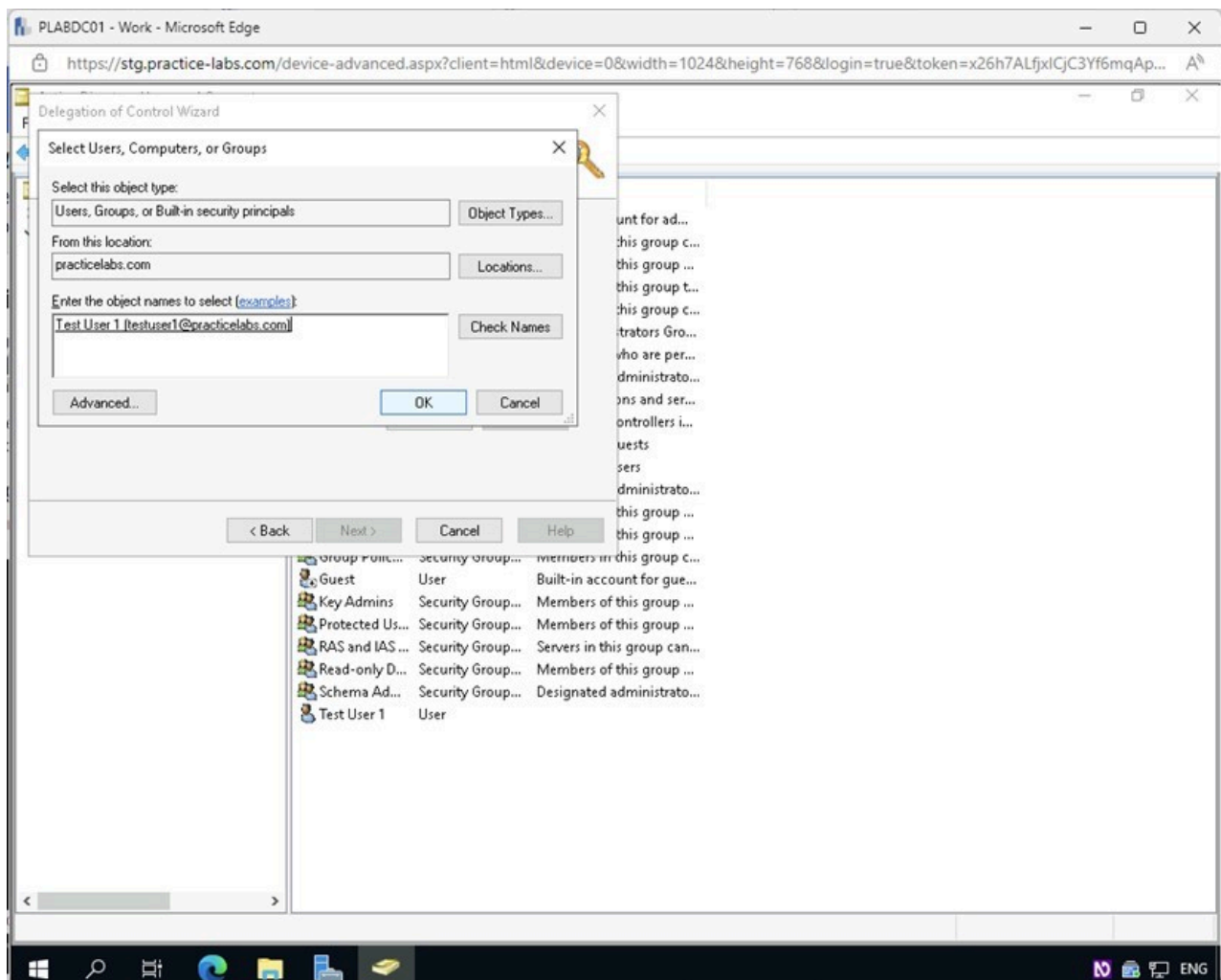


Figure 3.5 Screenshot of PLABDC01: Displaying selecting OK in the Select Users, Computers, or Groups window.

Step 6

Test User 1 is now added to the **Users or Groups** page.

Click **Next**.

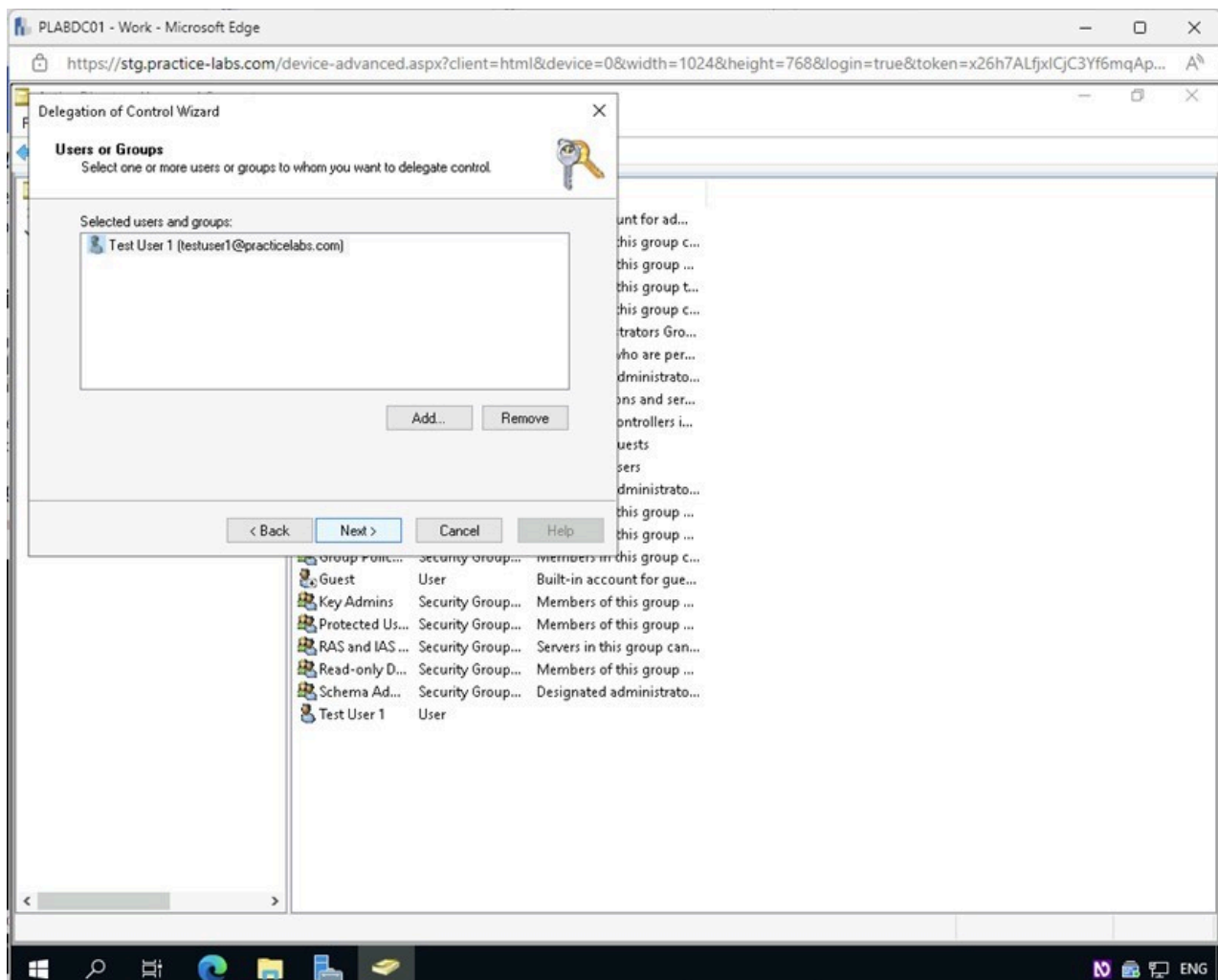


Figure 3.6 Screenshot of PLABDC01: Displaying selecting Next in the Users or Groups page.

Step 7

In the **Tasks to Delegate** page, tick the following checkboxes:

- **Create, delete, and manage user accounts**
- **Reset user passwords and force password change at next logon**

Click **Next**.

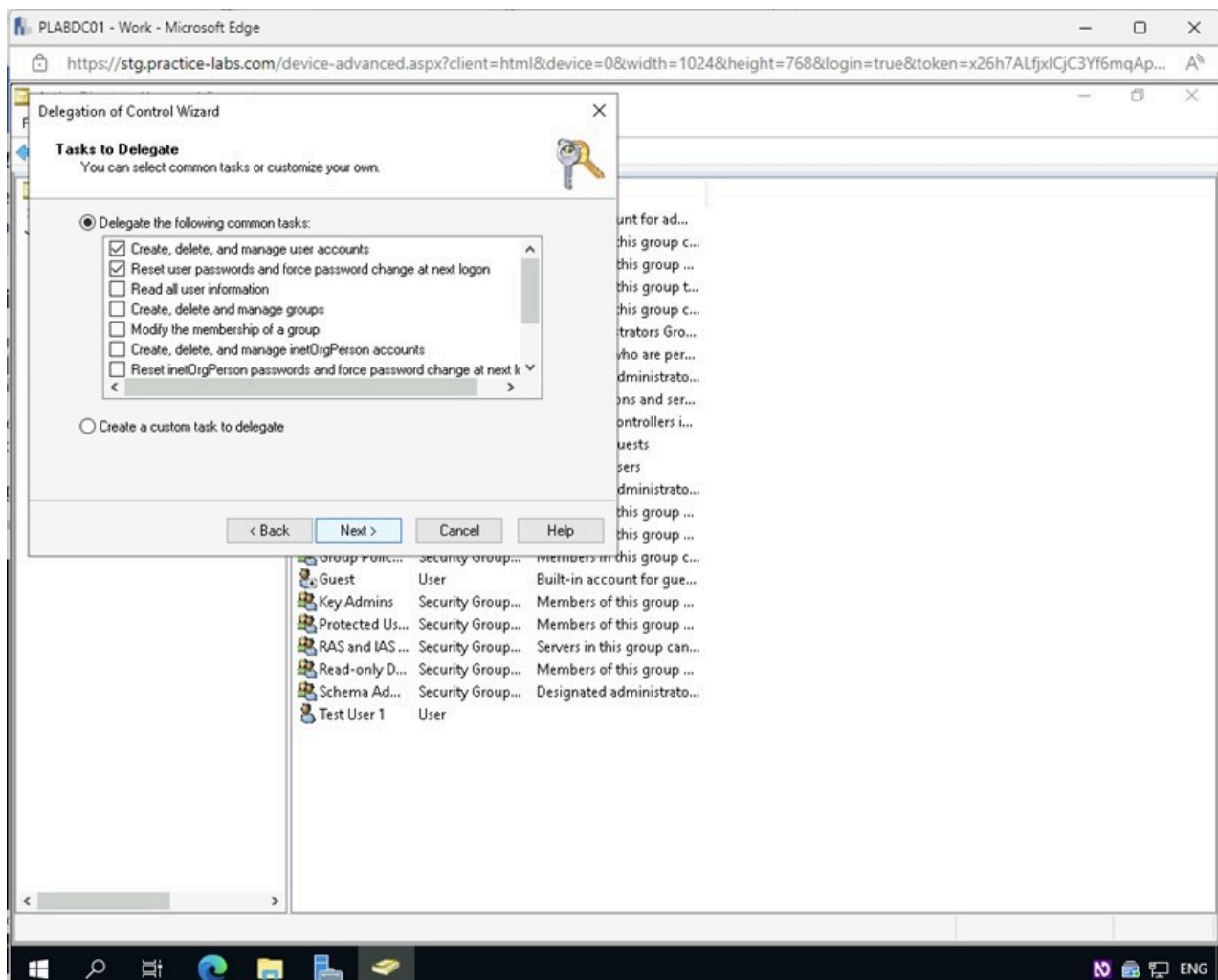


Figure 3.7 Screenshot of PLABDC01: Displaying the Tasks to Delegate page, showing the required settings performed and the Next button selected.

Step 8

In the **Completing the Delegation of Control Wizard** page, review the settings and click **Finish**.

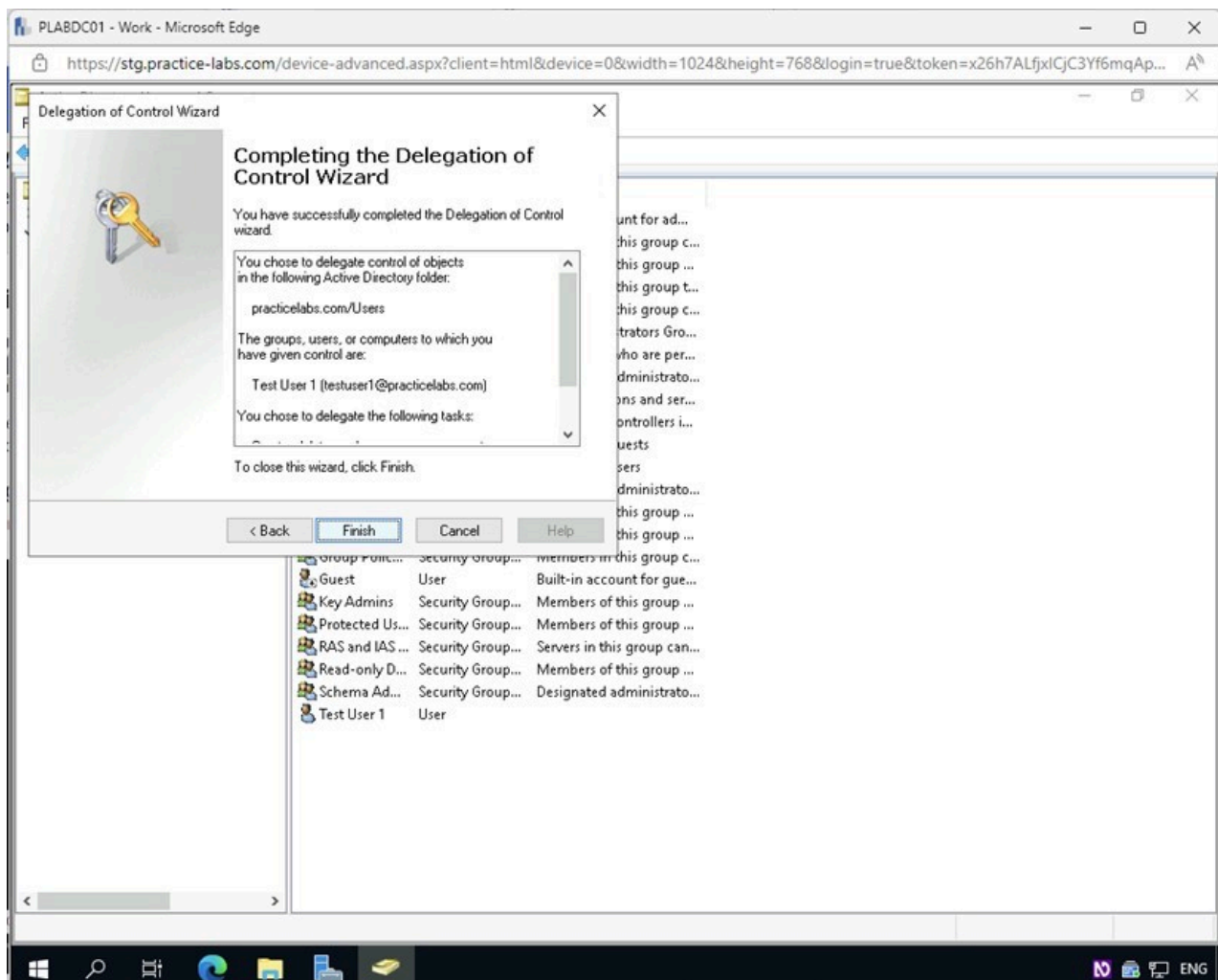


Figure 3.8 Screenshot of PLABDC01: Displaying selecting Finish in the Completing the Delegation of Control Wizard page.

Step 9

You have now delegated **Create, delete, and manage user accounts** and **Reset user passwords and force password change at next logon** to **Test User 1**.

Keep the **Active Directory Users and Computers** window open.

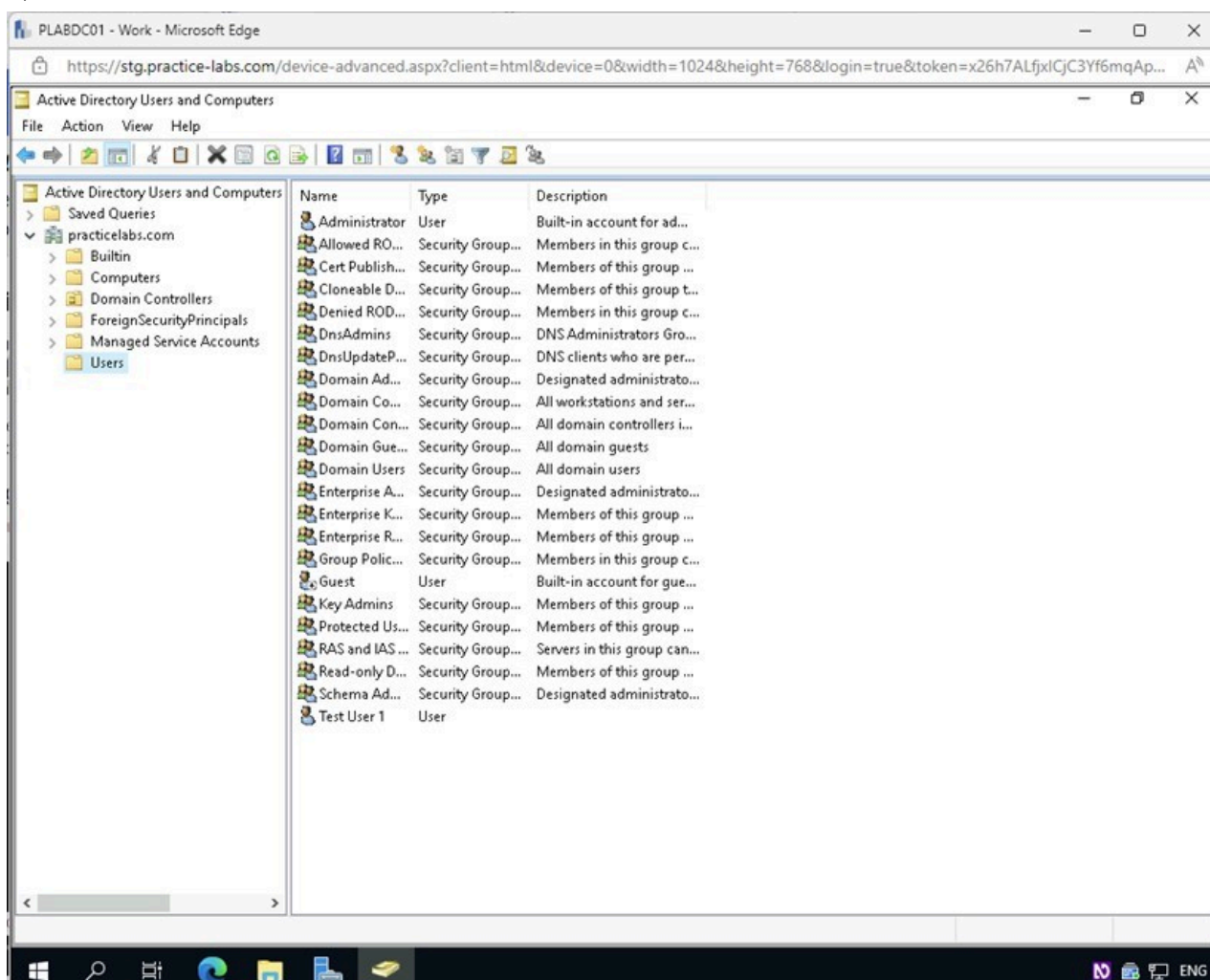


Figure 3.9 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers pane. Proceed to the next exercise.

Segregation of Duties

The term segregation of duties, also referred to as separation of responsibilities, is a crucial part of any reliable business management system. The goal is to divide responsibilities so that no individual has complete access to all sensitive and confidential information. It would be far more difficult to control the risks of fraud and mistakes without this separation of essential procedures and job responsibilities.

Exercise 4 - Auditing

Auditing, also known as Accounting in the AAA architecture, is the process of checking the network activity to ensure no data integrity has been compromised and no user's access has been granted beyond what is required under the concept of least privilege. Active Directory is a robust platform that enables the auditing of numerous computer-related activities on a network. By using auditing correctly, a non-refutable audit trail of a user's actions on the network may be

reconstructed. You should plan for extra storage capacity if your auditing needs increase. Due to the expense of storage, it is important to strike a balance in terms of what will be tracked.

In this exercise, you will learn to audit user activities, logins and group membership.

Learning Outcomes

After completing this exercise, you should be able to:

- Audit User Activity
- Audit Logins
- Audit Group Membership

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABDC01** - (Windows Server 2019 - Domain Controller)

Task 1 - Audit User Activity

The security posture of a company may be bolstered by keeping an eye on user behavior. There is a higher risk of data loss, theft, and compromise if users have unrestricted access to too much information. To further maintain data security, keeping a record of what is accessed and deleted by users is recommended.

In this task, you will audit user activities.

Step 1

In **PLABDC01**, the **Active Directory Users and Computers** window is open.

Click the **View** menu and select **Advanced Features**.

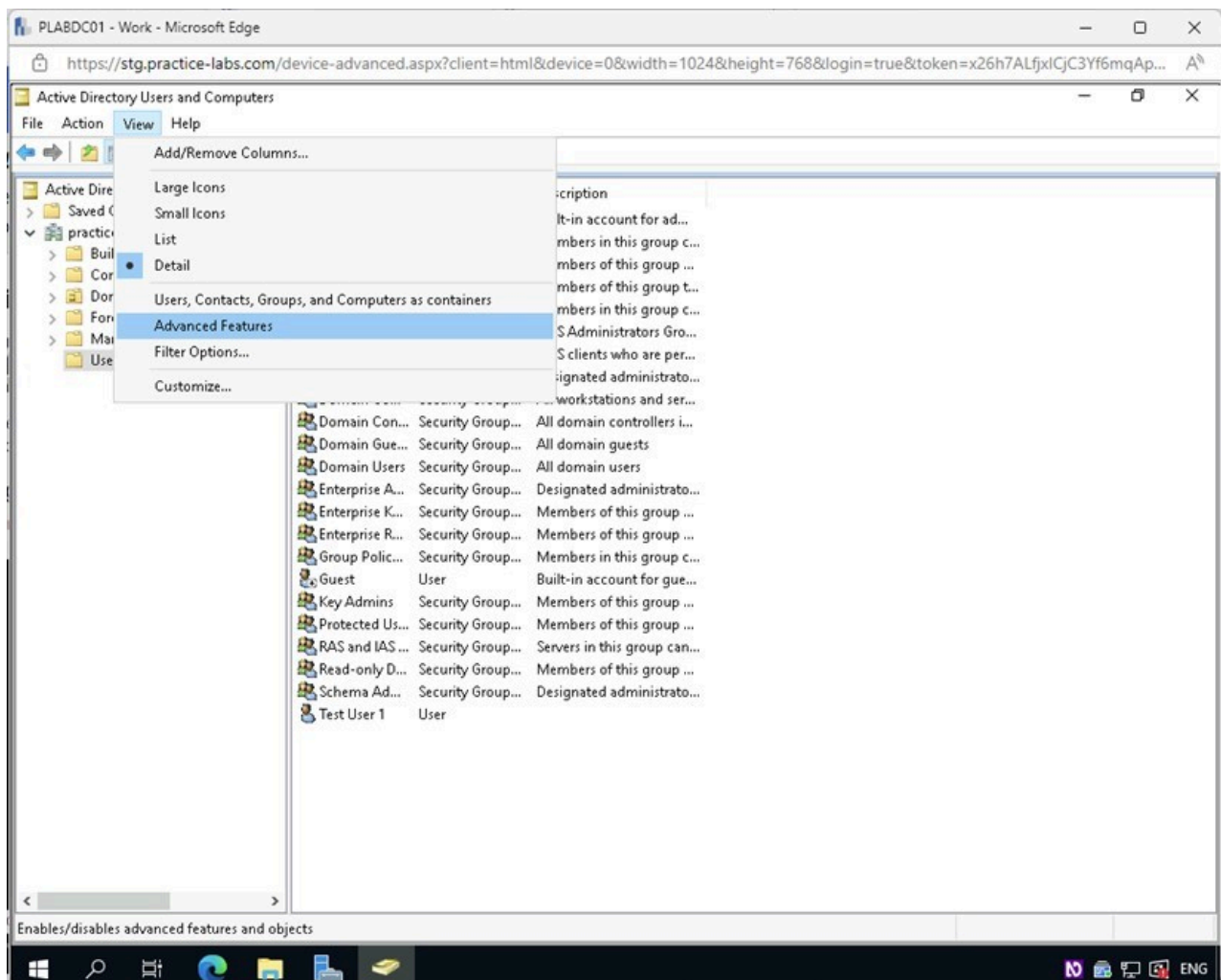


Figure 4.1 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window. The View menu is accessed, and Advanced Features is selected.

Step 2

Once the window refreshes, showing the different advanced features options available, right-click **Users** on the left pane and select **Properties**.

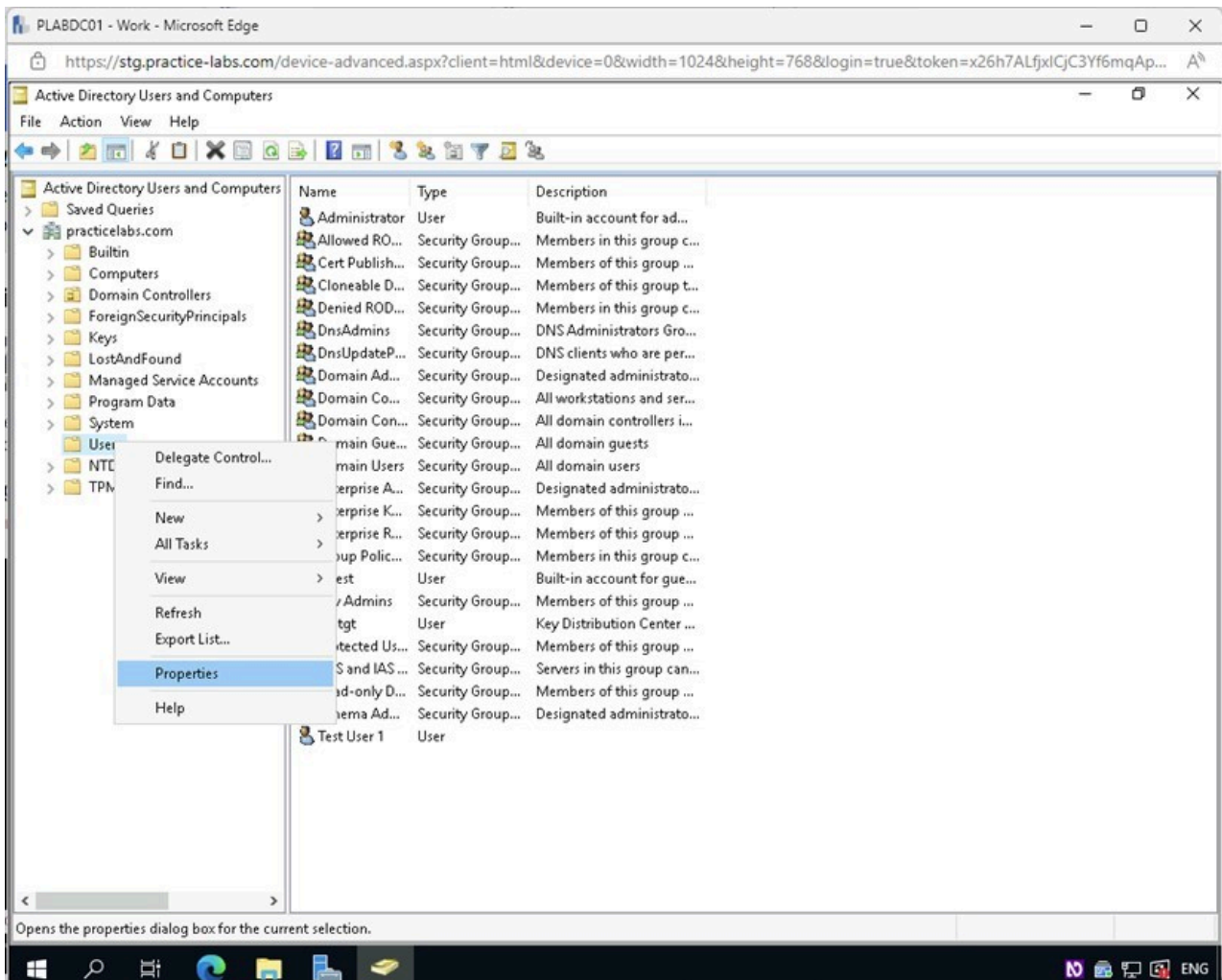


Figure 4.2 Screenshot of PLABDC01: Displaying the Active Directory Users and Computers window. Users is right-clicked, and Properties selected.

Step 3

In the **User Properties** window, select the **Security** tab.

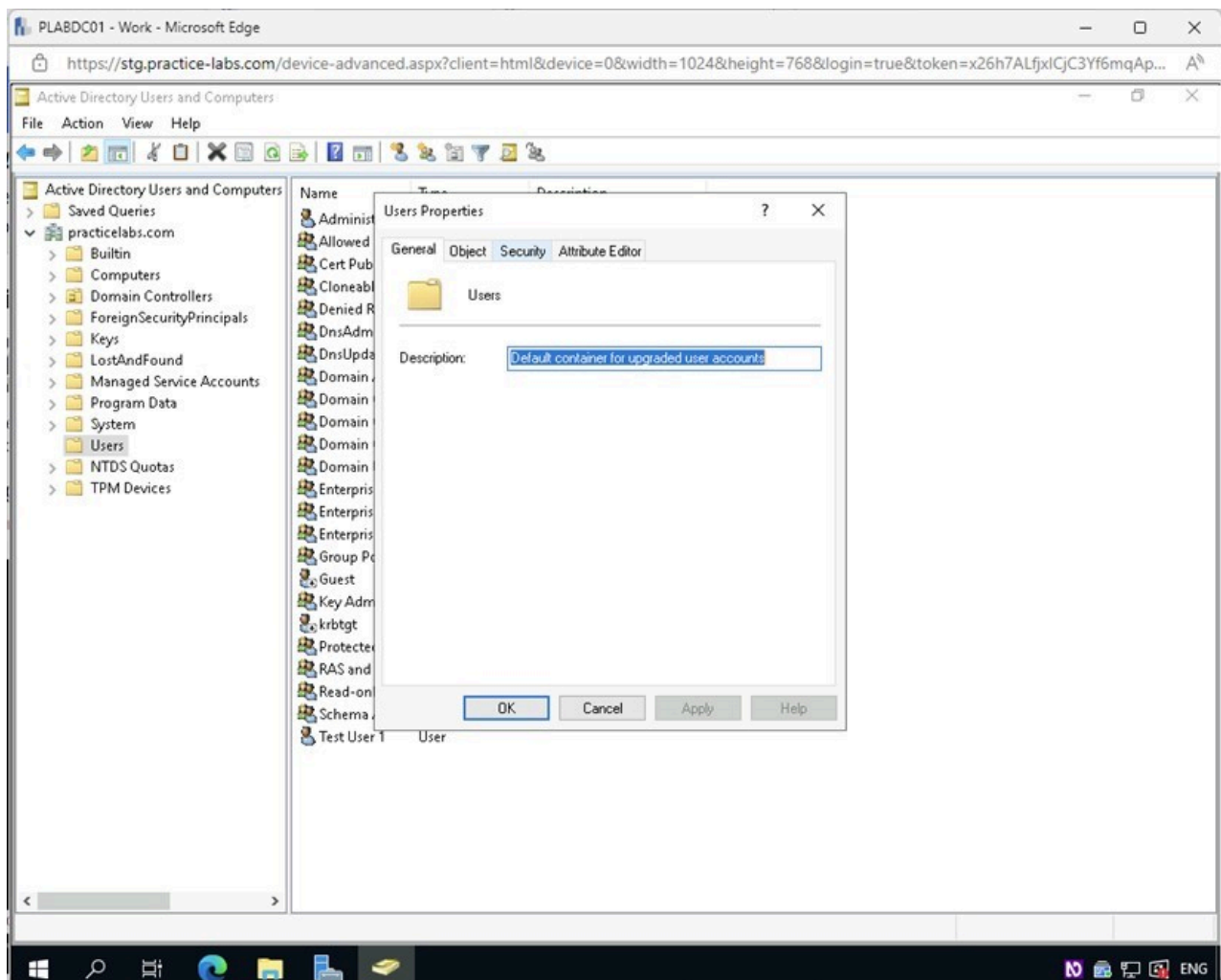


Figure 4.3 Screenshot of PLABDC01: Displaying selecting the Security tab in the User Properties window.

Step 4

In the **Users Properties - Security** tab, click **Advanced**.

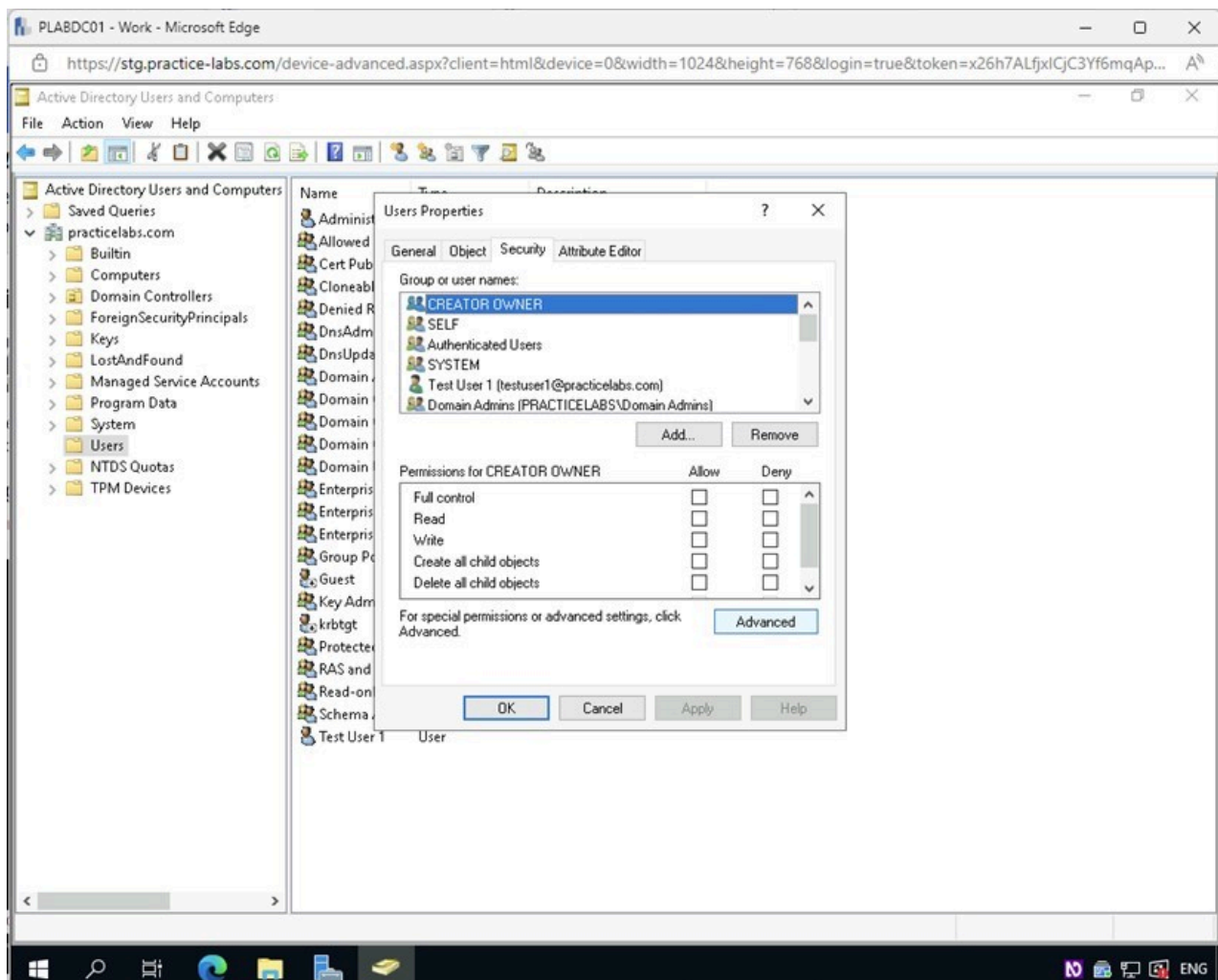


Figure 4.4 Screenshot of PLABDC01: Displaying selecting Advanced in the Users Properties - Security tab.

Step 5

In the **Advanced Security Settings for Users** window, select the **Auditing** tab.

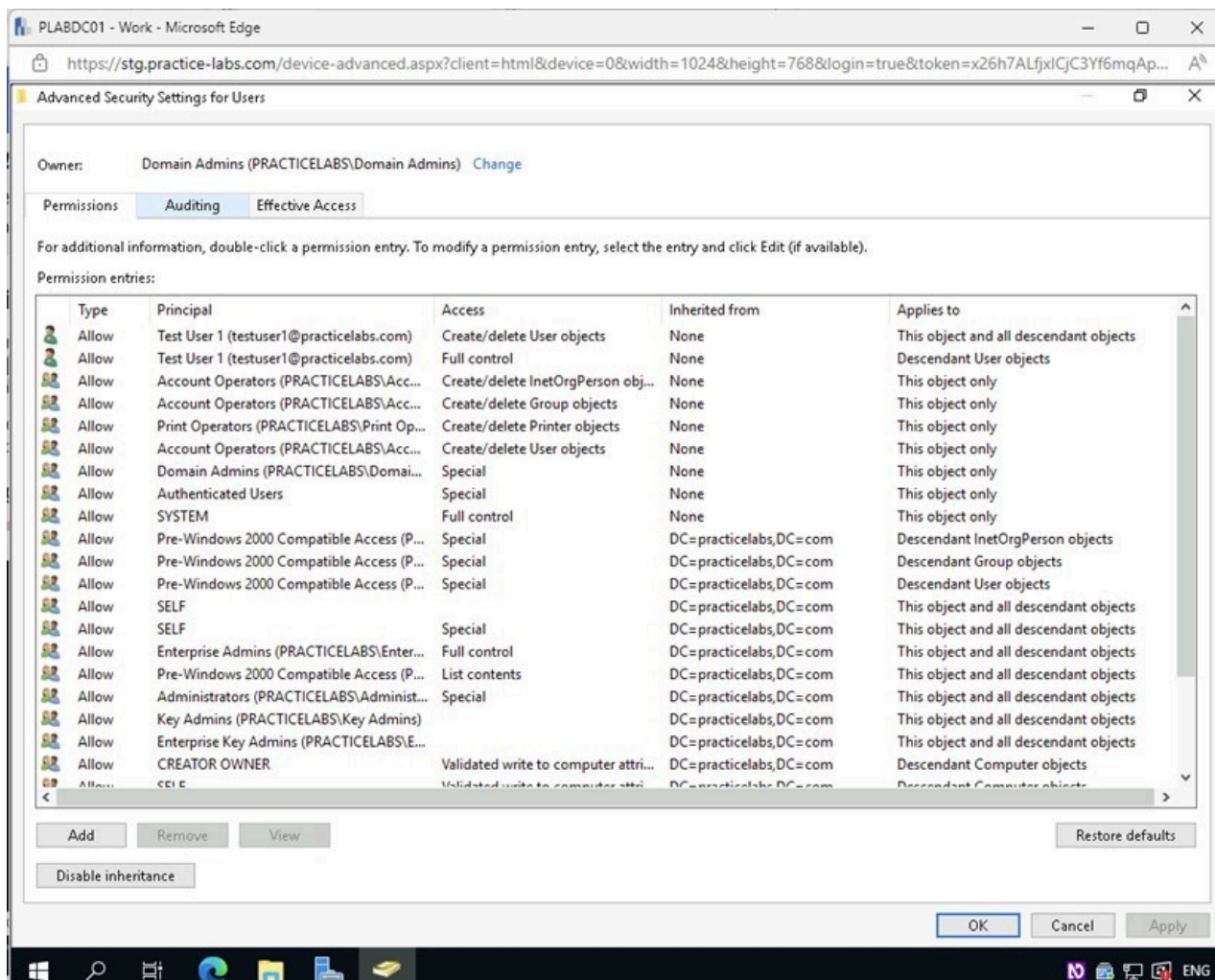


Figure 4.5 Screenshot of PLABDC01: Displaying select the Auditing tab in the Advanced Security Settings for Users window.

Step 6

In the **Advanced Security Settings for Users - Auditing** tab, click **Add**.

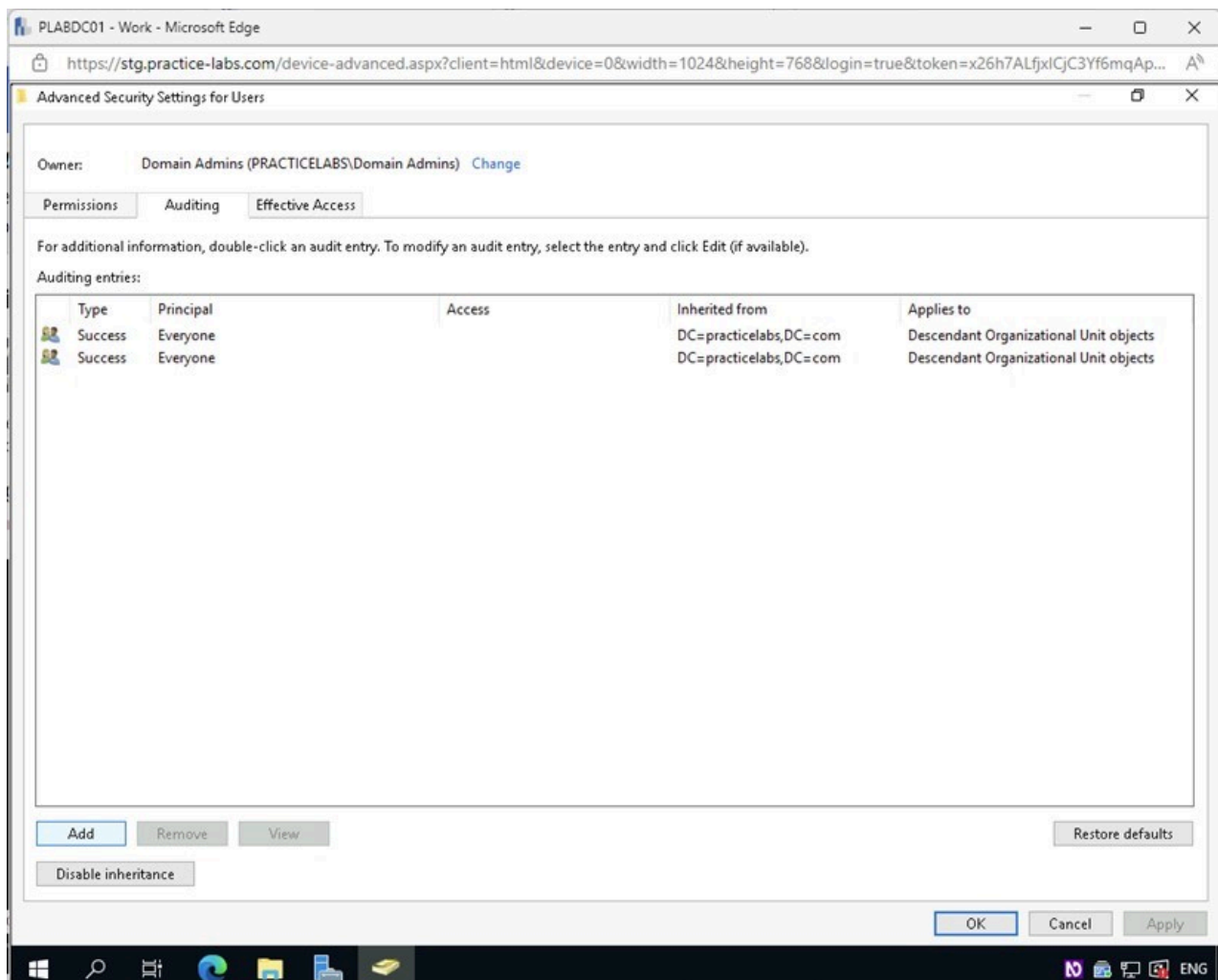


Figure 4.6 Screenshot of PLABDC01: Displaying selecting Add in the Advanced Security Settings for Users - Auditing tab.

Step 7

From the **Auditing Entry for Users** window, click the **Select a principal** link next to the **Principal** field.

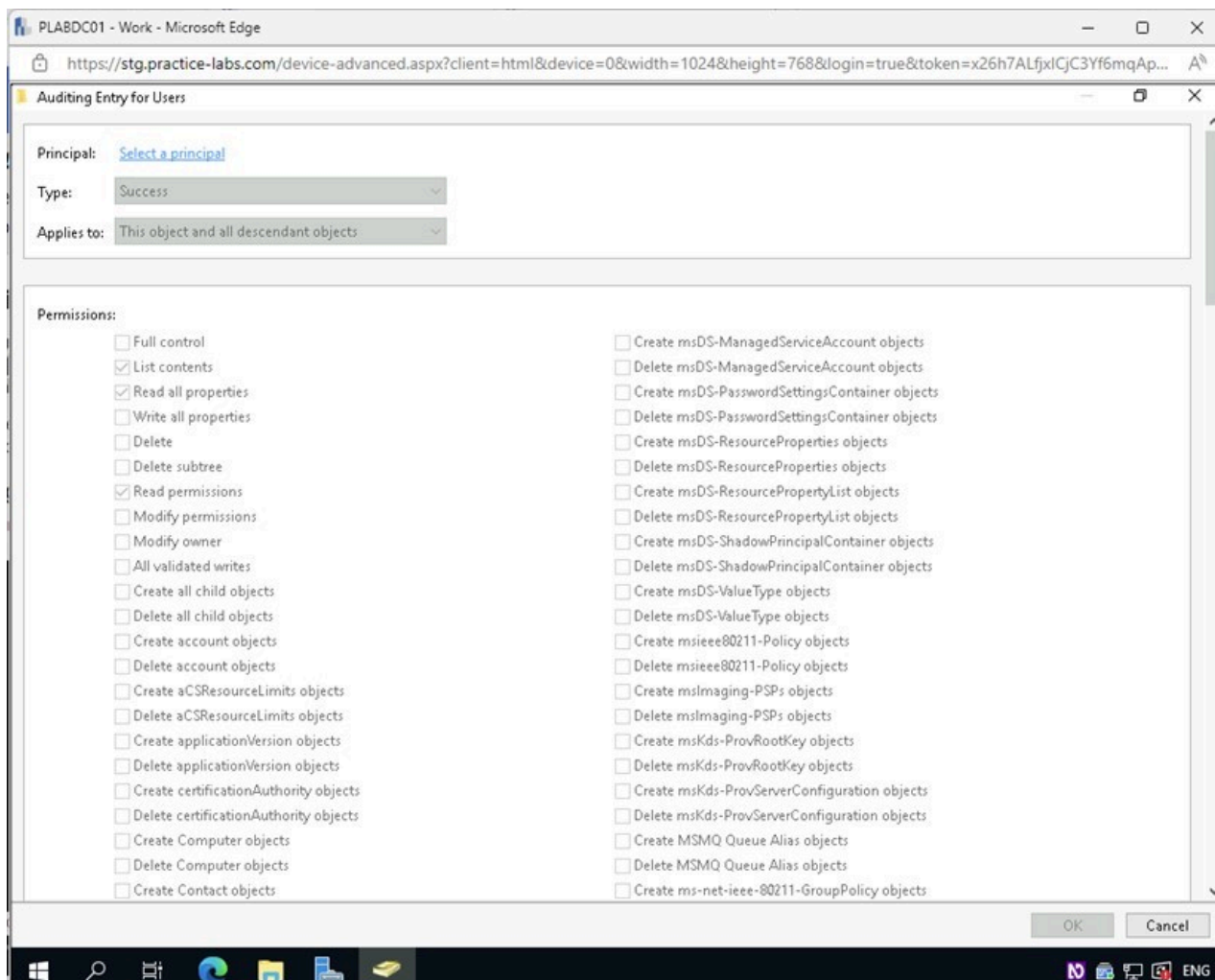


Figure 4.7 Screenshot of PLABDC01: Displaying the Auditing Entry for Users window, showing the Select a principal link selected.

Step 8

In the **Select Users, Computers, Service Account, or Group** window, type the following in the **Enter the object names to select** textbox:

test

Click **Check Names**.

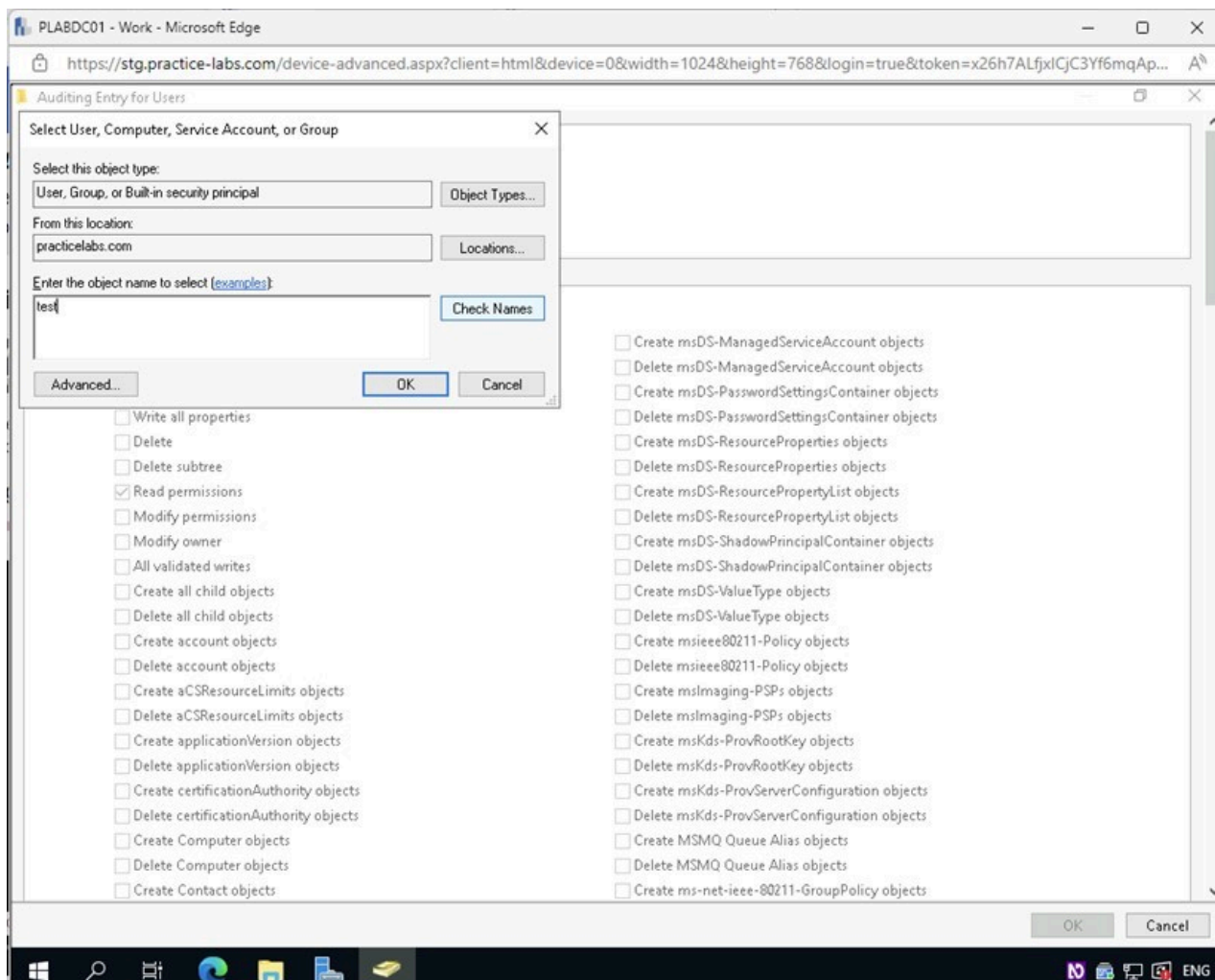


Figure 4.8 Screenshot of PLABDC01: Displaying the Select Users, Computers, Service Account or Group window, showing the required object name typed in and Check Names selected.

Step 9

Test User 1 now appears in the **Enter the object names to select** textbox.

Click **OK**.

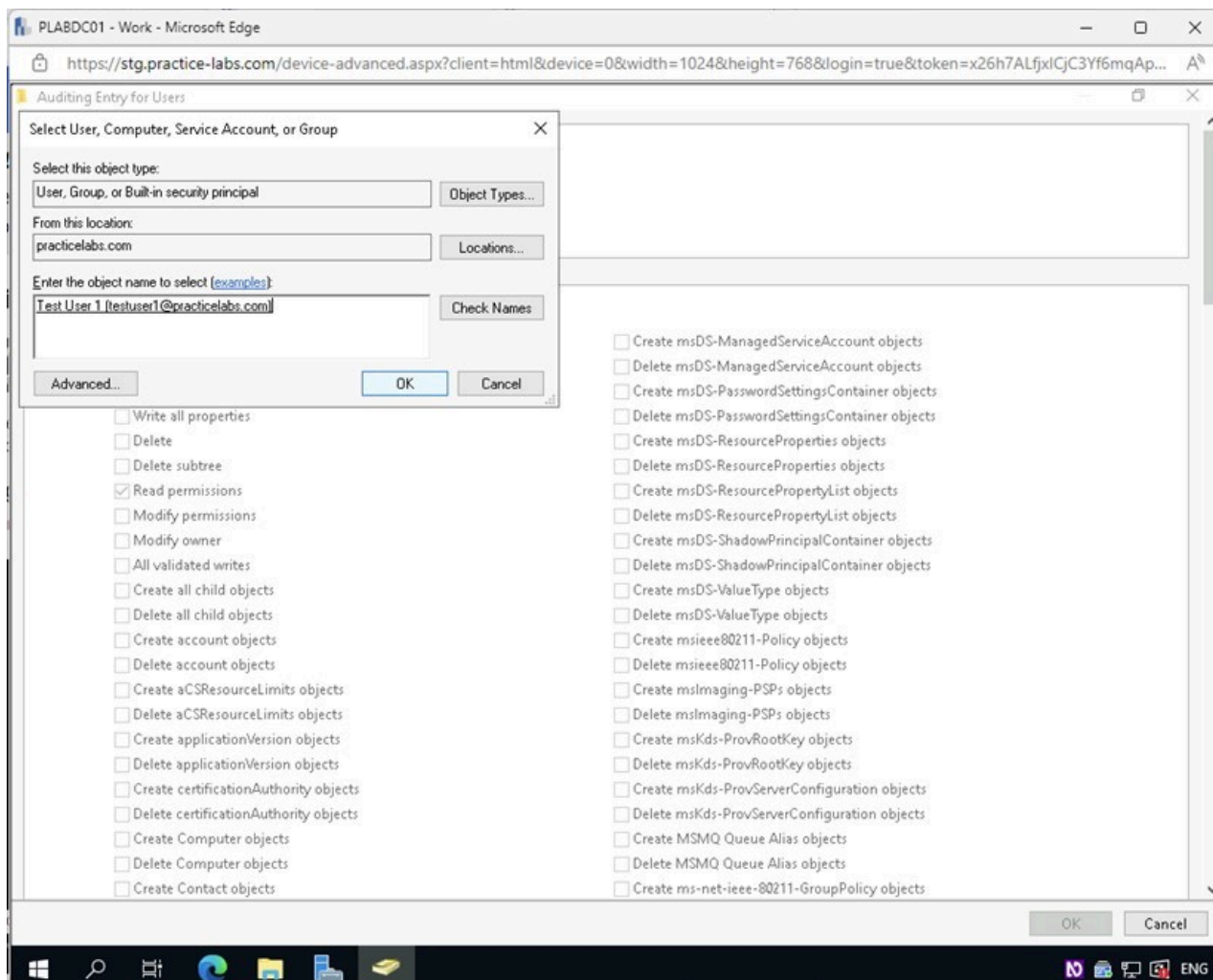


Figure 4.9 Screenshot of PLABDC01: Displaying selecting OK in the Select Users, Computers, Service Account or Group window.

Step 10

Back on the **Auditing Entry for Users** window, select **All** from the **Type** dropdown list.

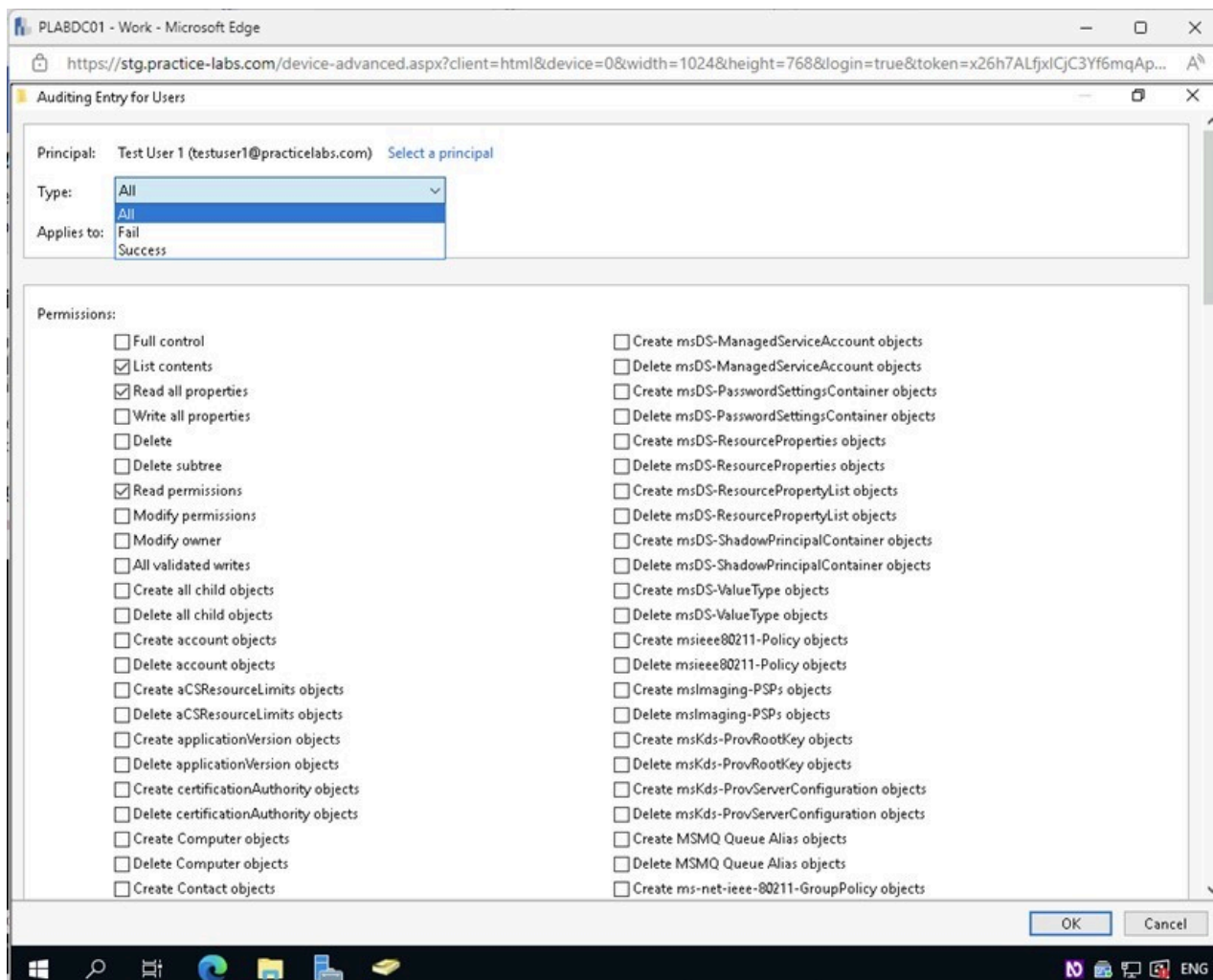


Figure 4.10 Screenshot of PLABDC01: Displaying the Auditing Entry for Users window. All is selected from the Type dropdown list.

Step 11

Under the **Permissions** field, tick the following checkboxes:

- **Delete**
- **Delete subtree**
- **Delete Group Objects**
- **Delete User objects**

Click **OK**.

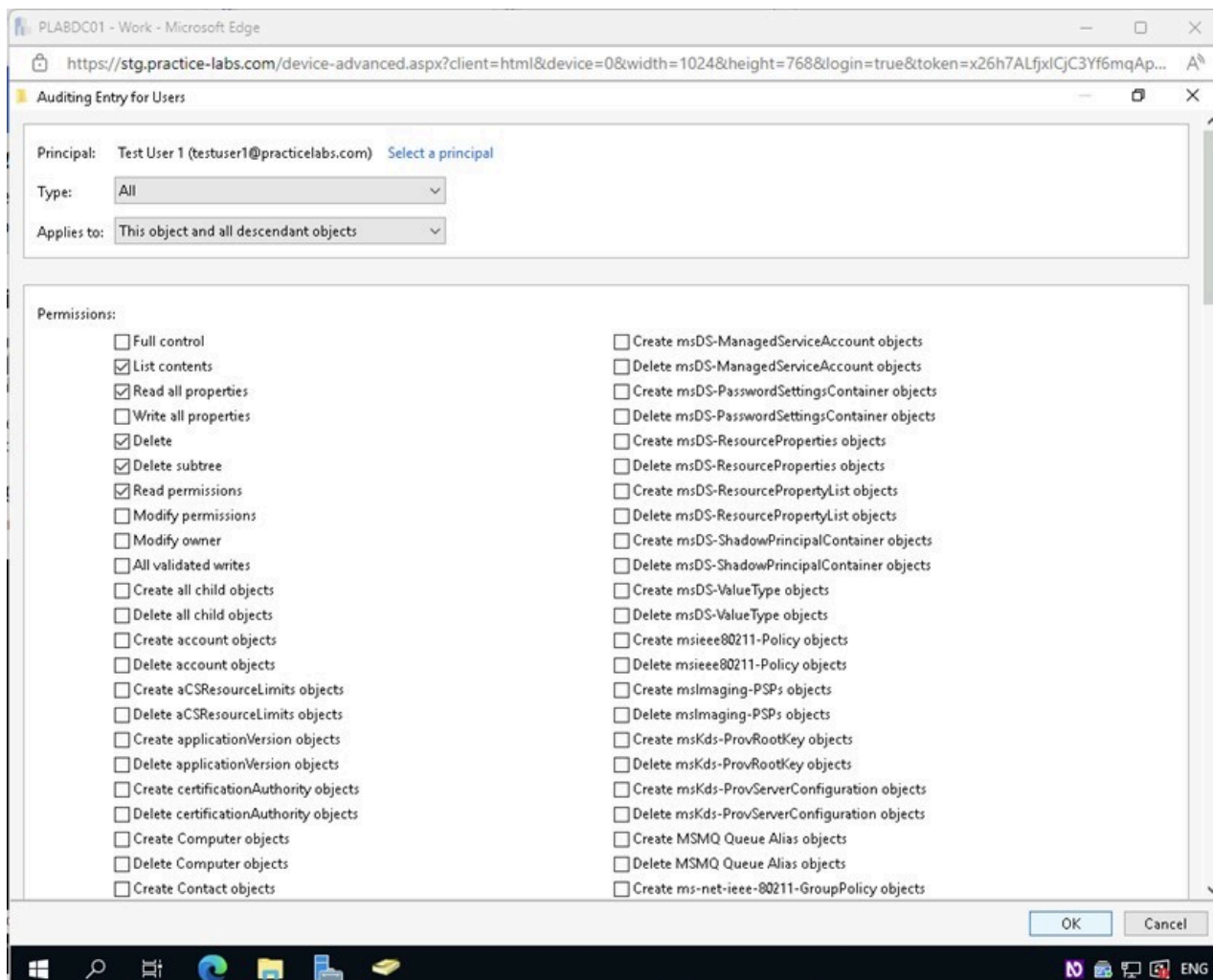


Figure 4.11 Screenshot of PLABDC01: Displaying the Auditing Entry for Users window, showing the required permissions selected and the OK button highlighted.

Note: Anything that is deleted within these categories will be logged. By setting this up, when users delete these types of files, logging will capture who did what action and when it all occurred, thus creating **non-repudiation**.

Step 12

Notice that **Test User 1** has now been added to the **Auditing** tab.

Click **OK**.

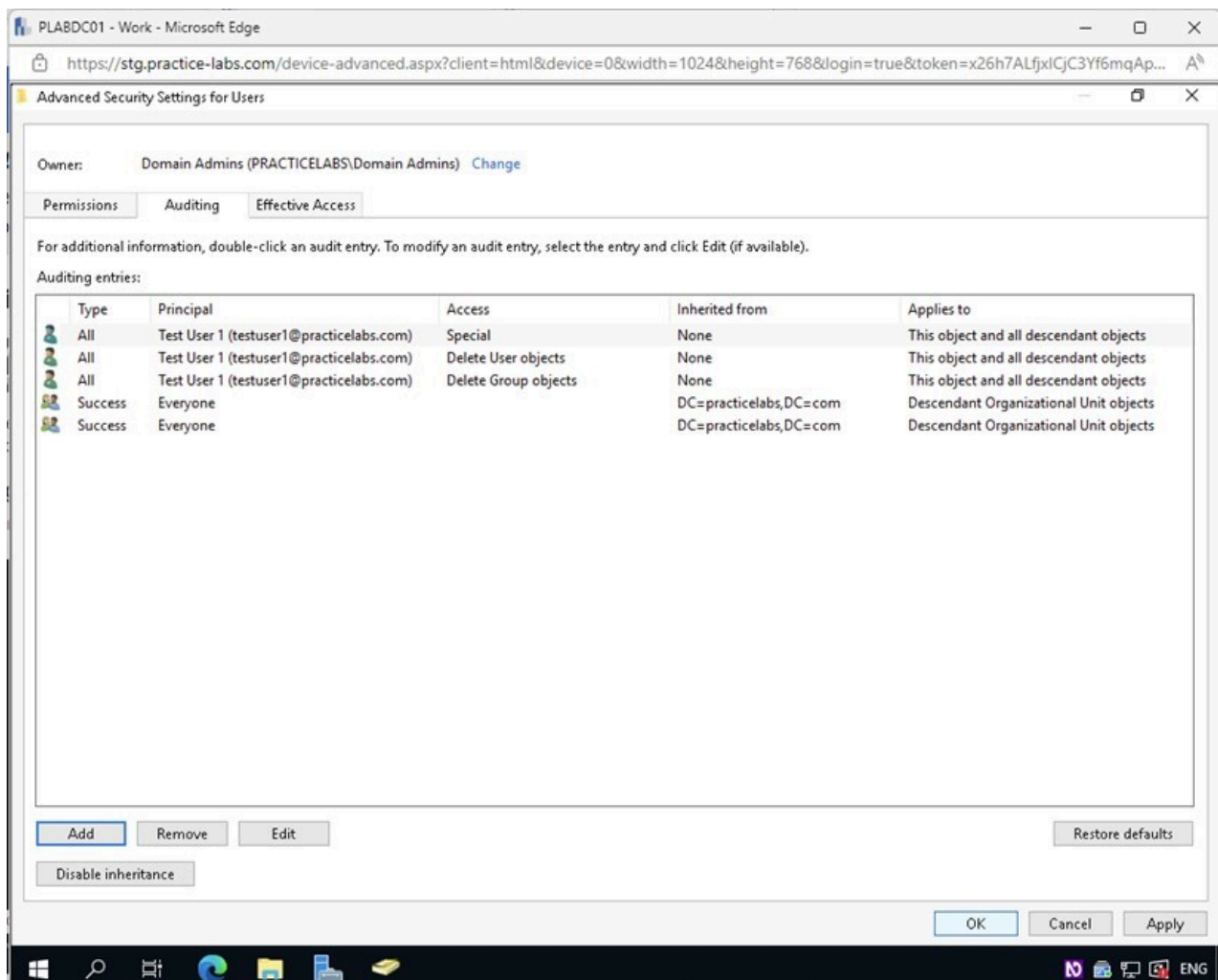


Figure 4.12 Screenshot of PLABDC01: Displaying the Advanced Security Settings for Users window, showing the OK button selected.

Step 13

Back on the **User Properties** window, click **OK**.

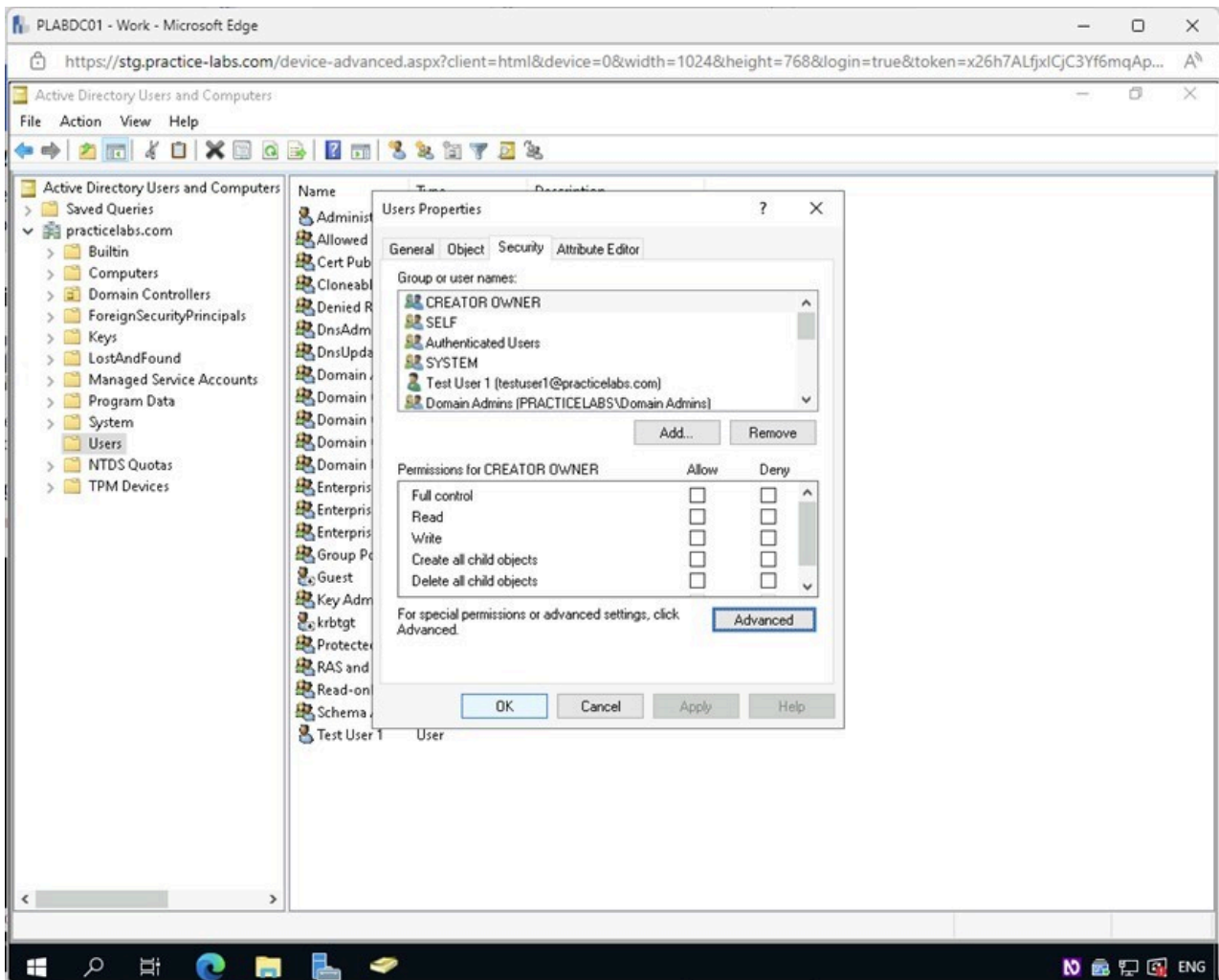


Figure 4.13 Screenshot of PLABDC01: Displaying selecting OK in the Users Properties window.

Step 14

Close the **Active Directory Users and Computers** window.

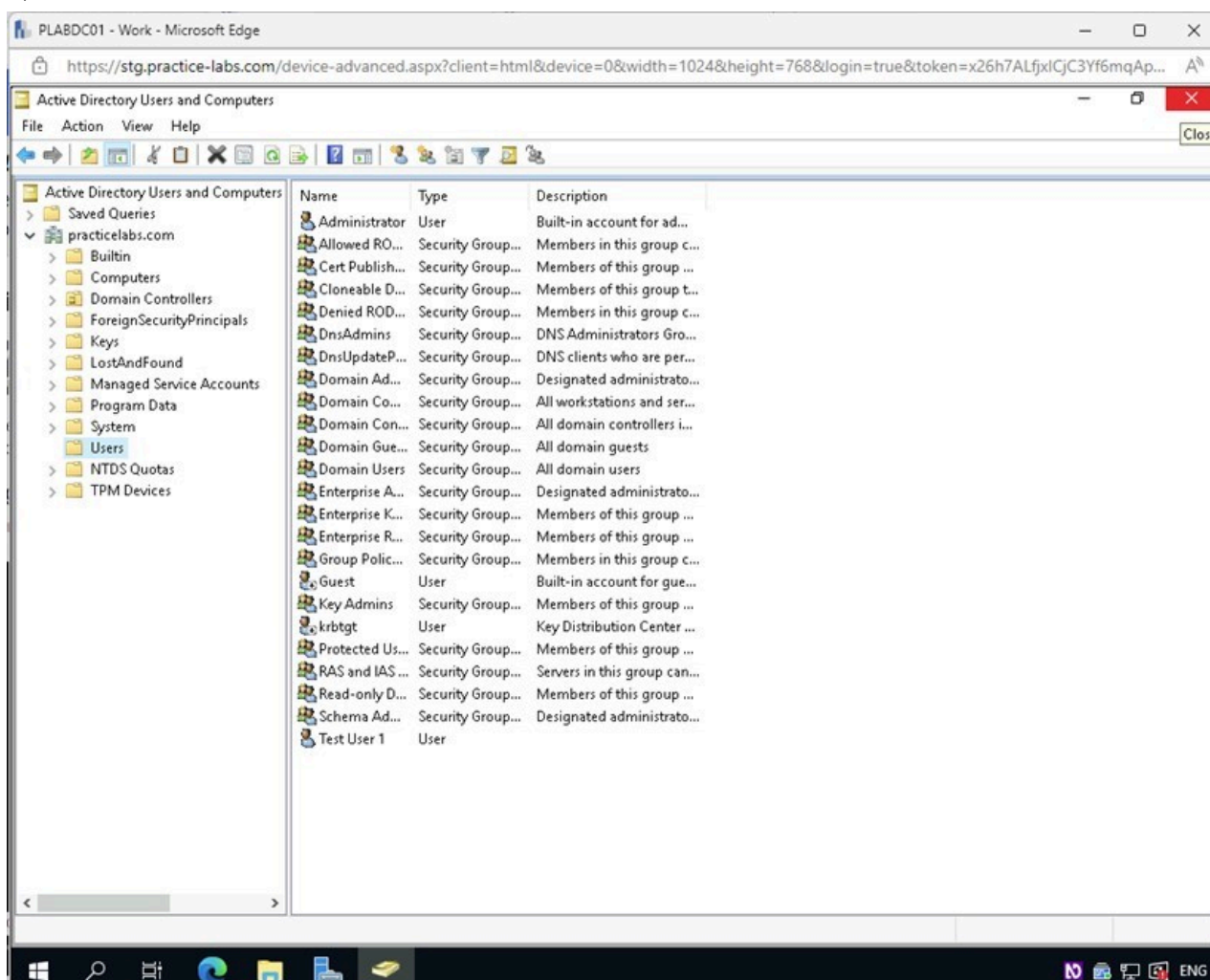


Figure 4.14 Screenshot of PLABDC01: Displaying closing the Active Directory Users and Computers window.

Keep the **Server Manager** window open.

Task 2 - Audit Logins

Keeping track of user logins through auditing can be quite helpful. Tracking successful logins is one way to show the type of accounts that have gained access to a system. Many failures to successfully log in can raise suspicions of an assault on the company, as does a consistent pattern of failure from a subset of individuals. When a single account displays many failed password attempts, there is a risk of Brute-force password cracking taking place. To prevent the account from being locked and to avoid being detected, scripts may be created that attempt a certain number of times with one user before switching to another.

In this task, you will audit user logins.

Step 1

Ensure you are connected to **PLABDC01**, where the **Server Manager** window is open.

Click **Tools** and then select **Group Policy Management**.

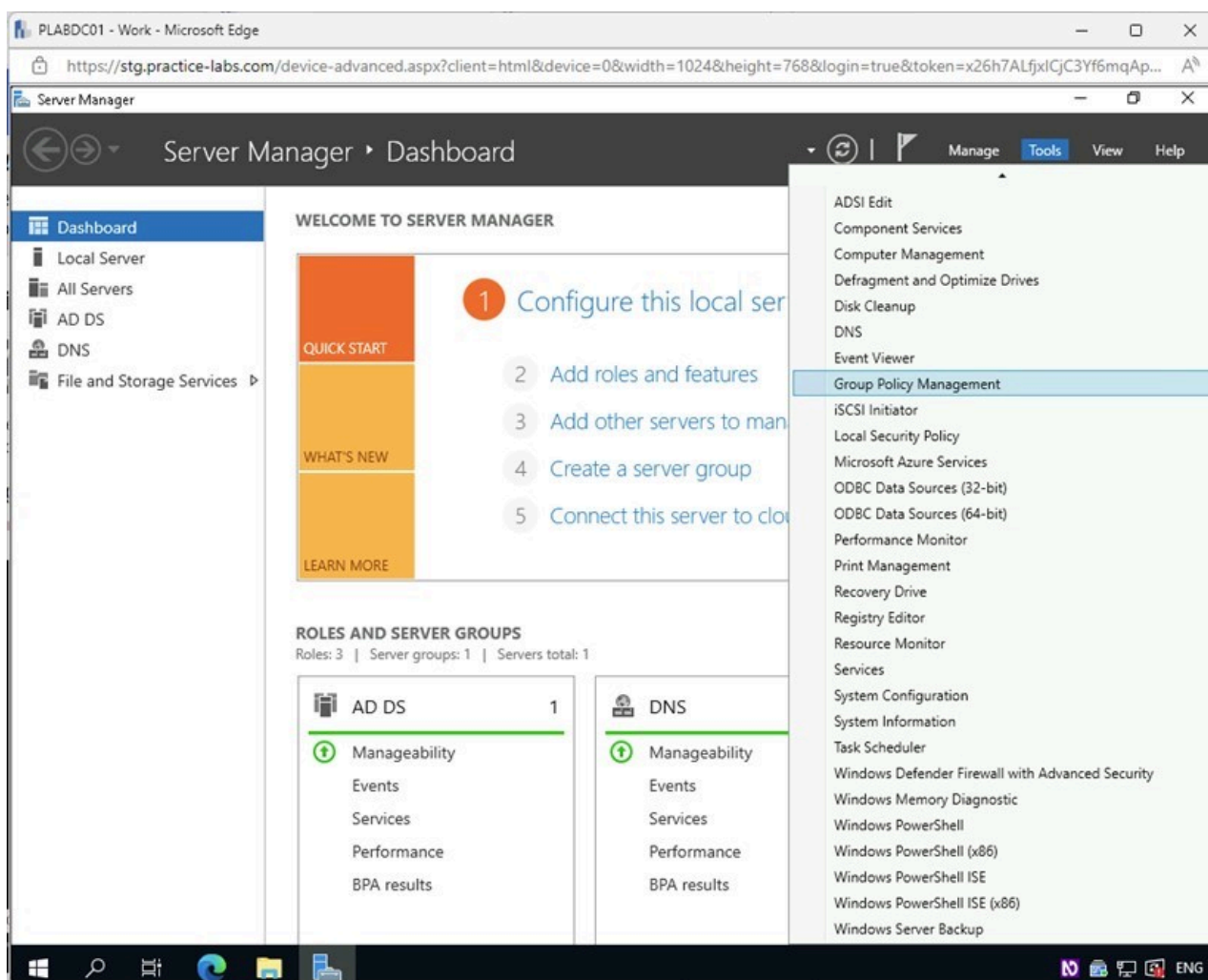


Figure 4.15 Screenshot of PLABDC01: Displaying the Server Manager window. The Tools menu is accessed, and Group Policy Management is selected.

Step 2

In the **Group Policy Management** window, expand **Forest: practicelabs.com > Domains**.

Right-click on **practicelabs.com** and select **Create a GPO in this domain, and Link it here**.

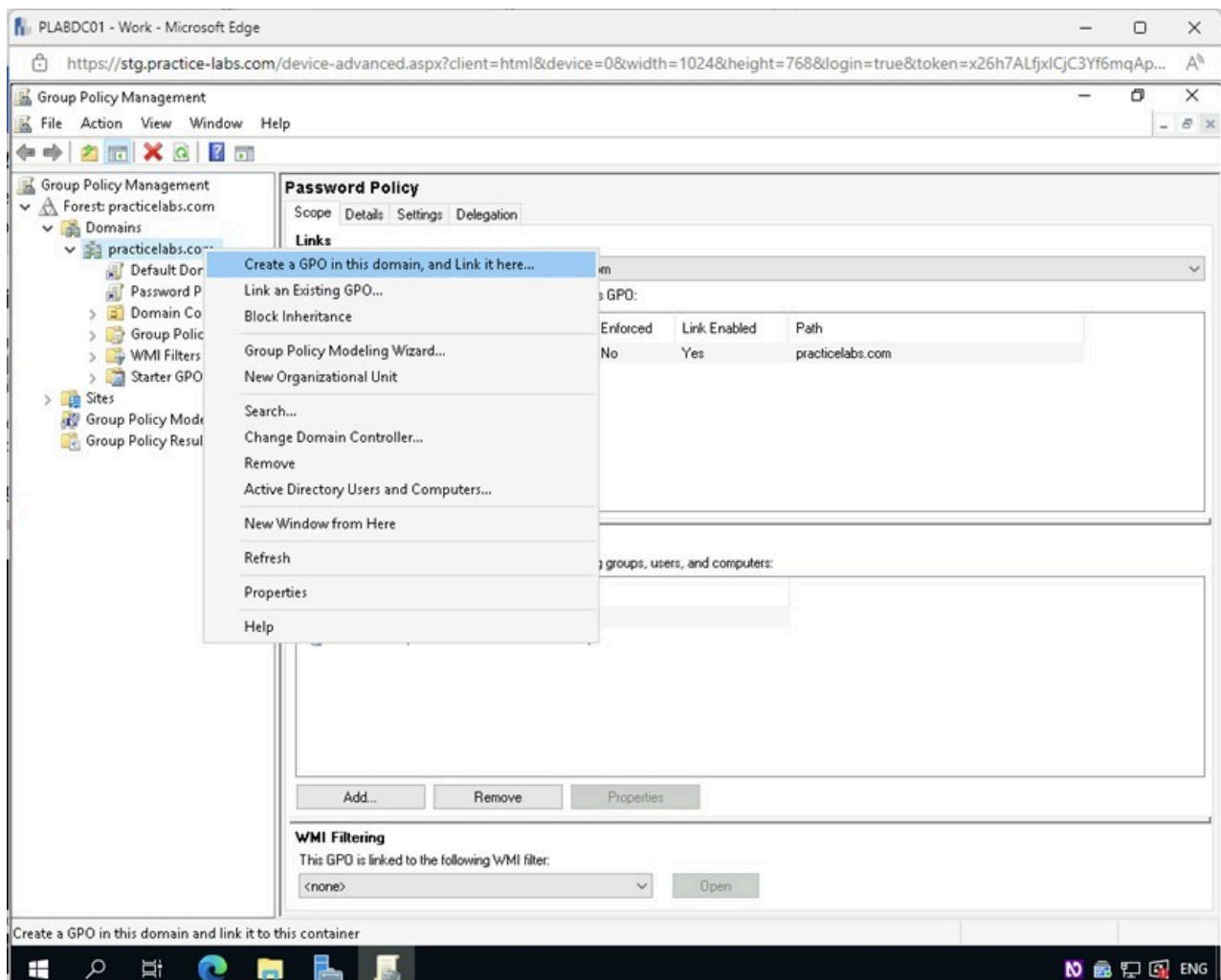


Figure 4.16 Screenshot of PLABDC01: Displaying the Group Policy Management window. practicelabs.com is right-clicked, and Create a GPO in this domain and Link it here is selected.

Step 3

In the **New GPO** window, type the following into the **Name** textbox and click **OK**:

Password Audit

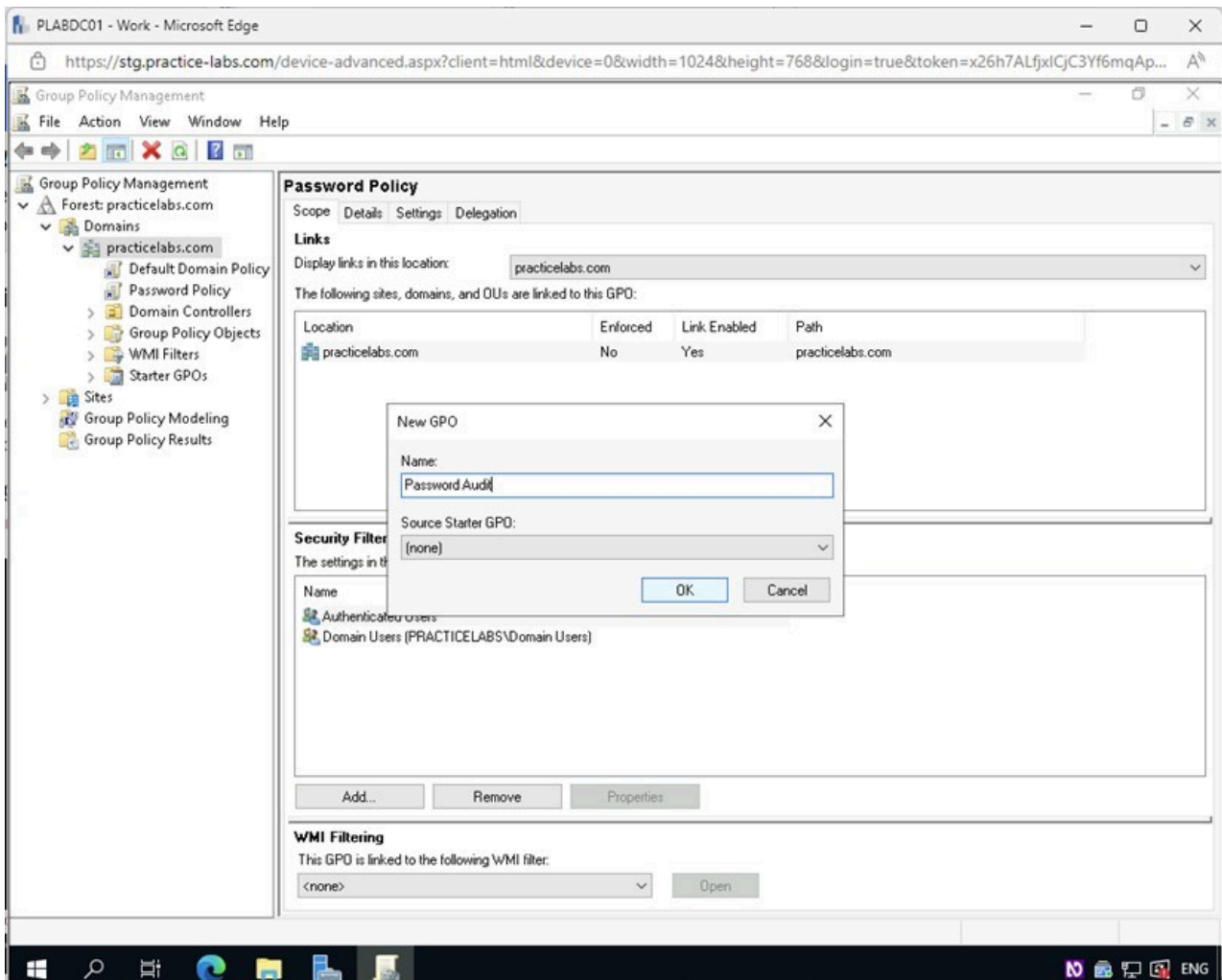


Figure 4.17 Screenshot of PLABDC01: Displaying the New GPO window, showing the required Name typed in and the OK button selected.

Step 4

Back on the **Group Policy Management** window, expand the **practicelabs.com** node.

Notice the **Password Audit** Policy has been created.

Right-click on **Password Audit** and select **Edit**.

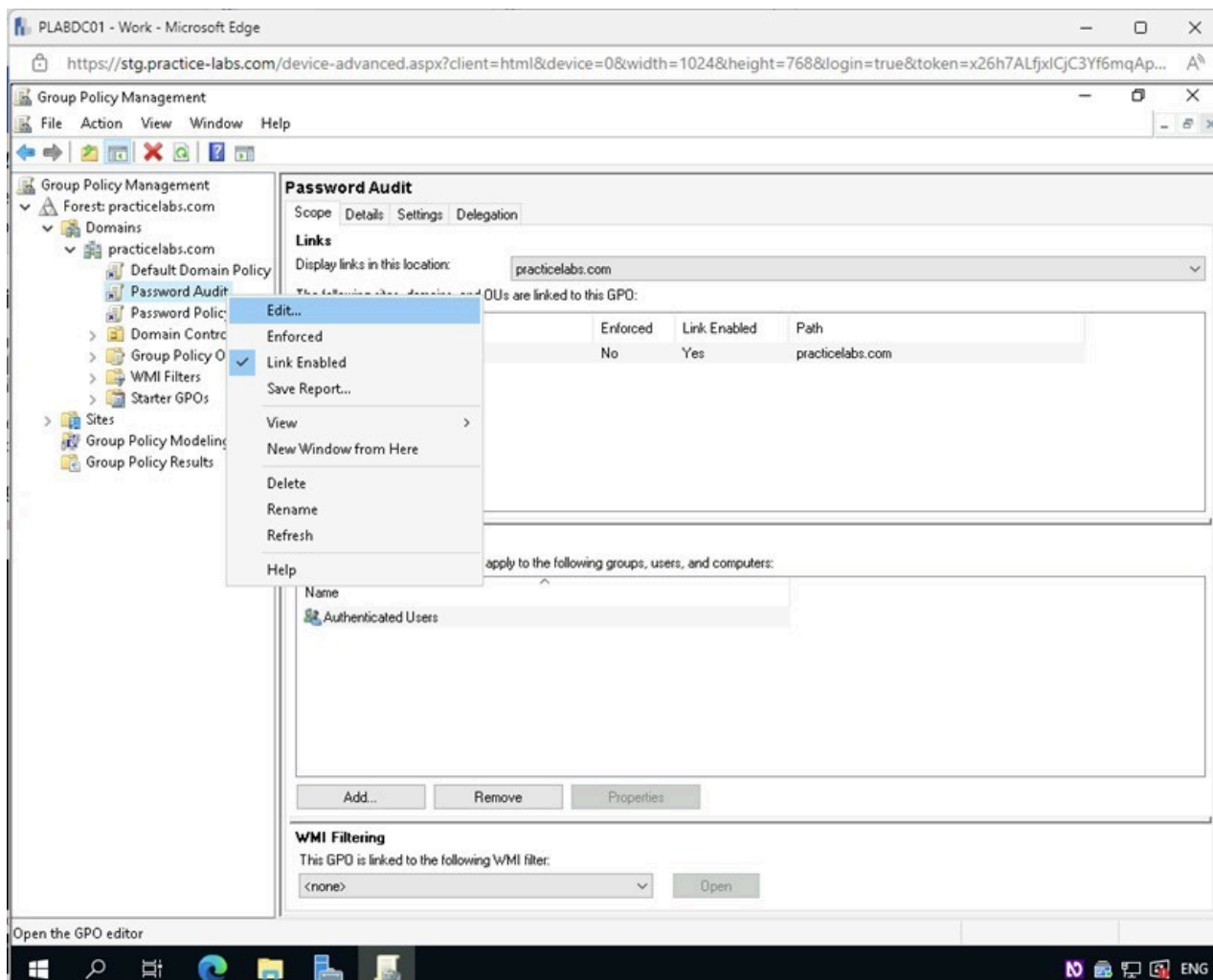


Figure 4.18 Screenshot of PLABDC01: Displaying the Group Policy Management window. Password Audit is right-clicked, and Edit selected.

Step 5

In the **Group Policy Management Editor** window, expand the following nodes:

Computer Configuration > Policies > Windows Settings > Security Settings > Advanced Audit Policy Configuration > Audit Policies.

Select **Account Logon**.

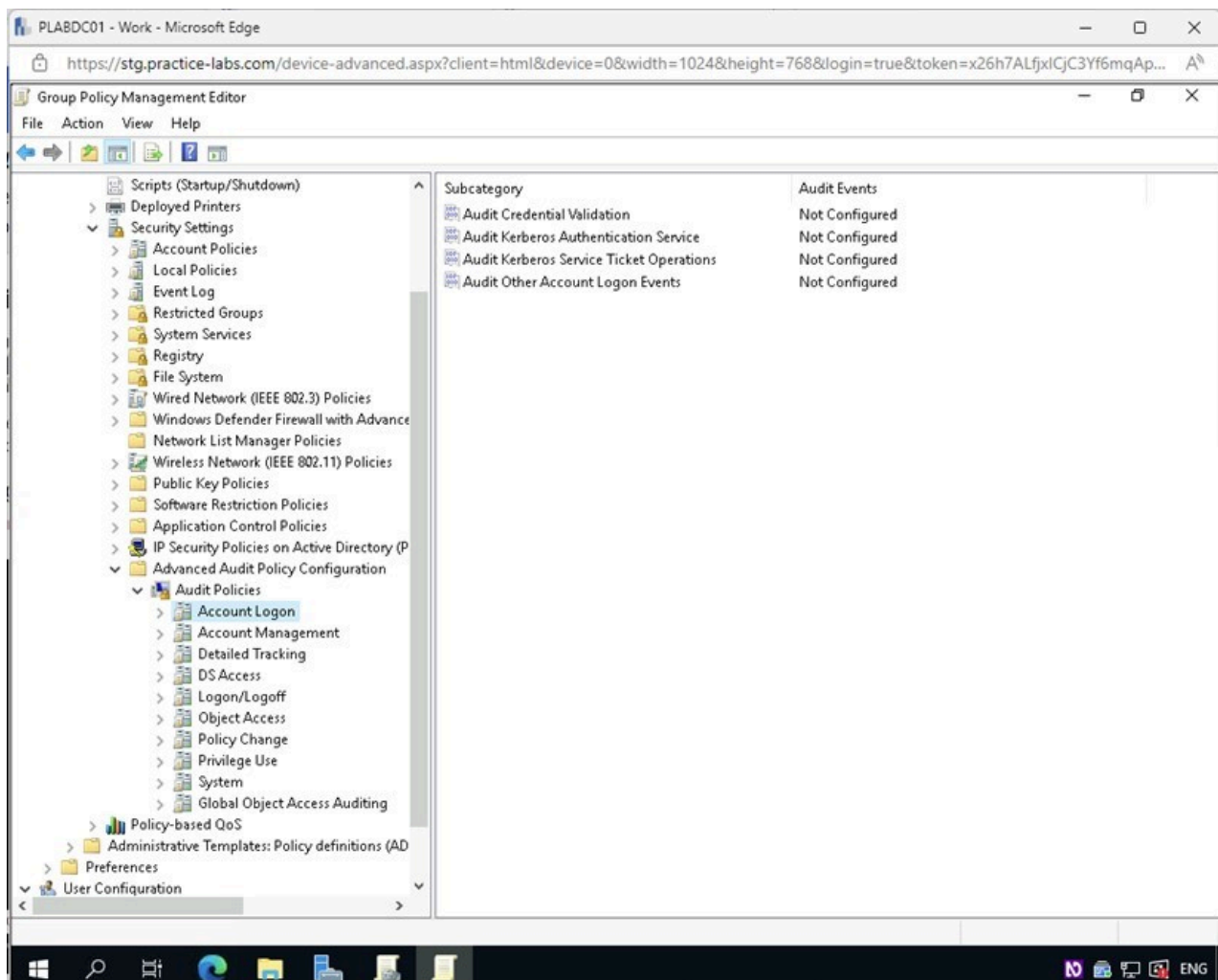


Figure 4.19 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Account Logon selected on the left pane.

Step 6

In the right details pane, right-click on **Audit Credential Validation** and select **Properties**.

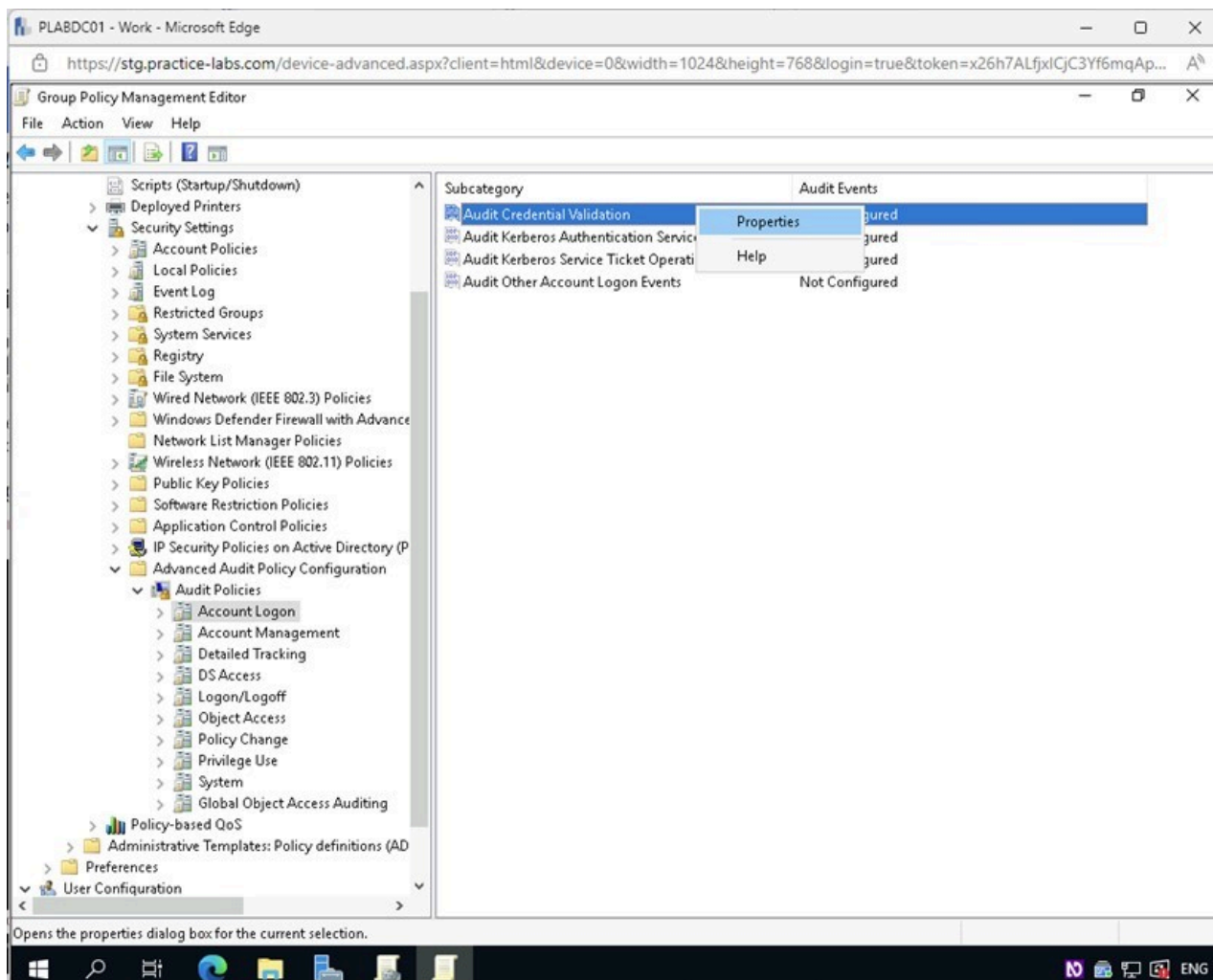


Figure 4.20 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Audit Credential Validation right-clicked and Properties selected.

Step 7

In the **Audit Credential Validation Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

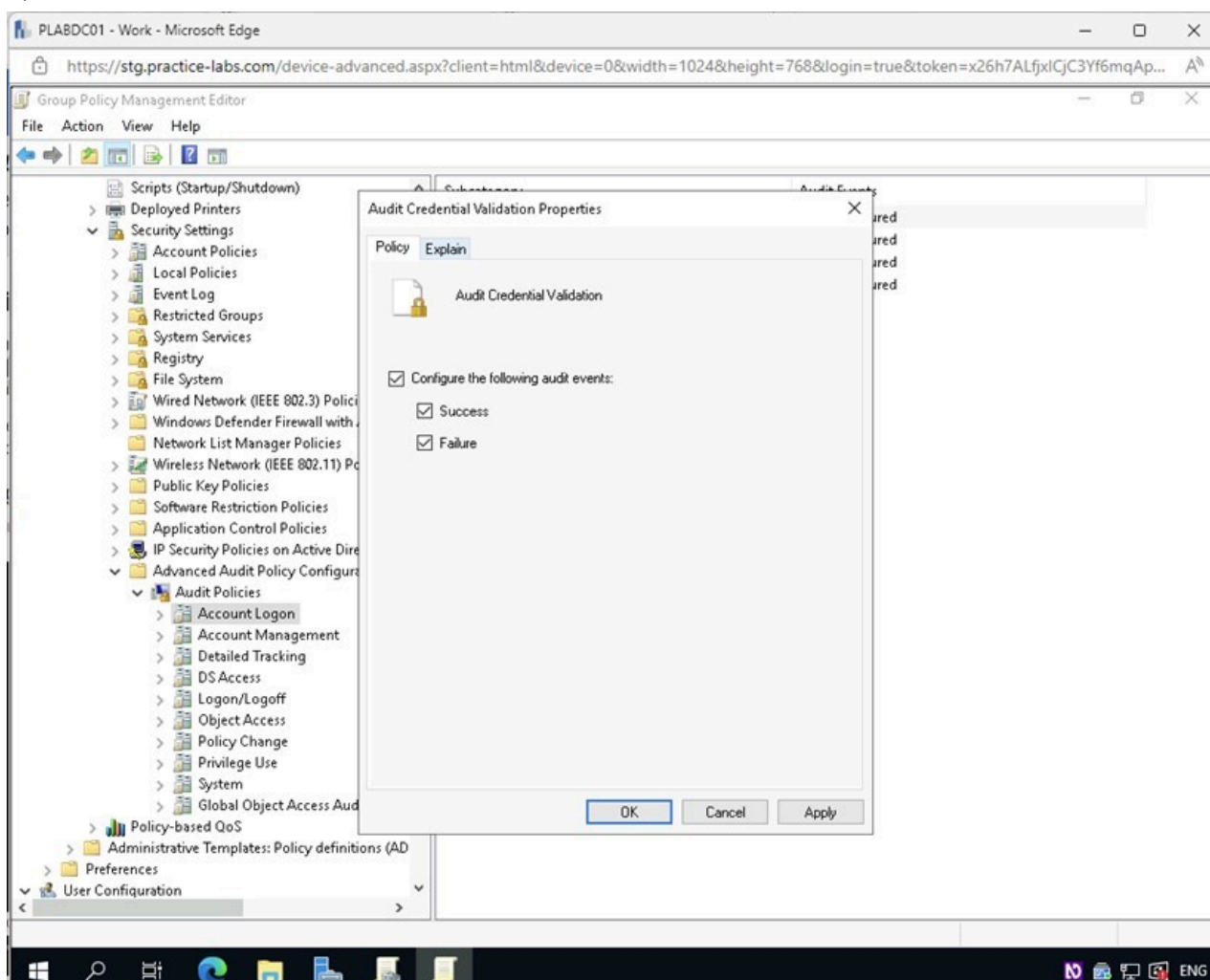


Figure 4.21 Screenshot of PLABDC01: Displaying the Audit Credential Validation Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 8

In the **Explain** tab, you can view detailed information on **Audit Credential Validation**.

Once you've read the information available, click **OK**.

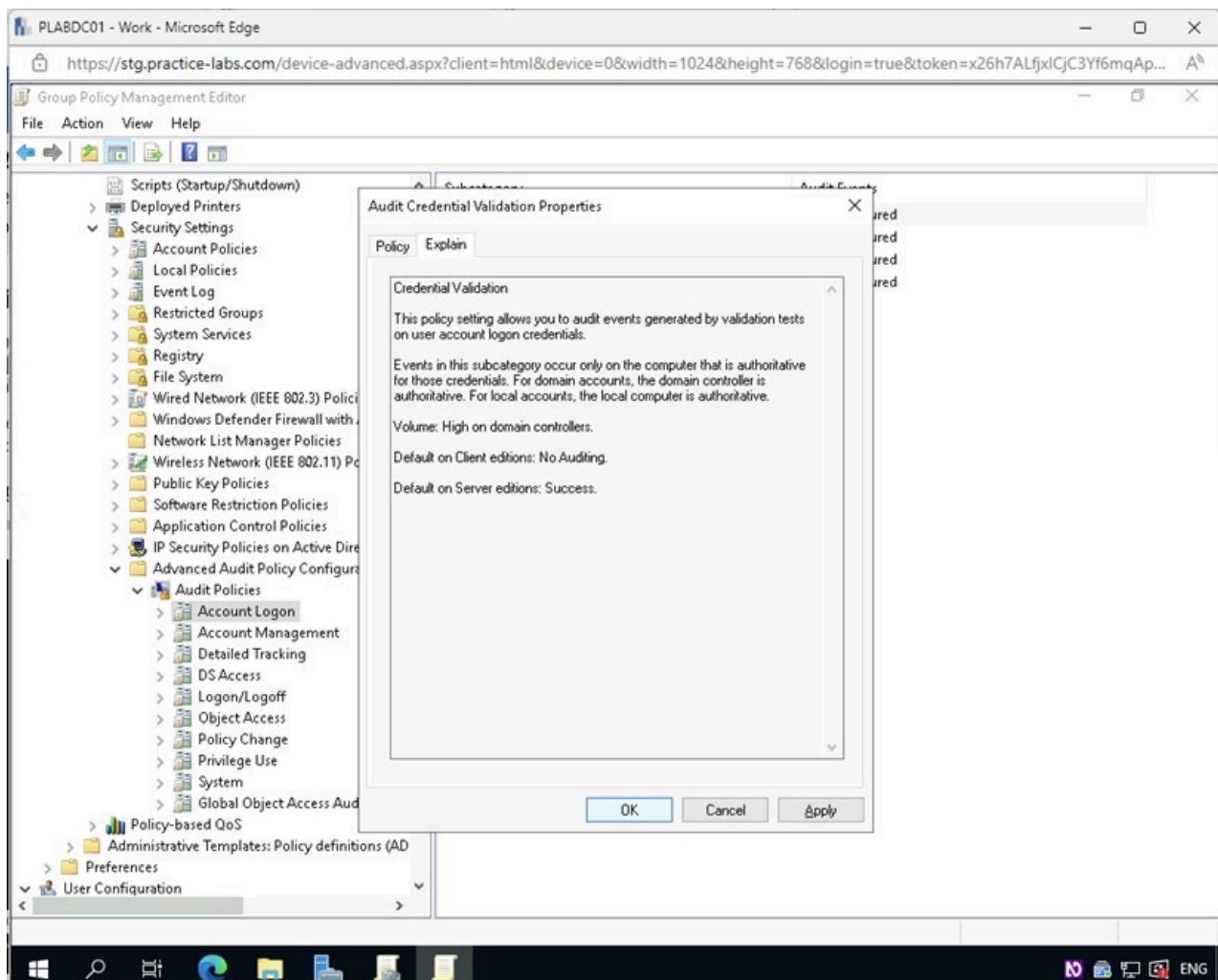


Figure 4.22 Screenshot of PLABDC01: Displaying the Audit Credential Validation Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 9

Back on the **Group Policy Management Editor** window, right-click on **Audit Kerberos Authentication Service** and select **Properties**.

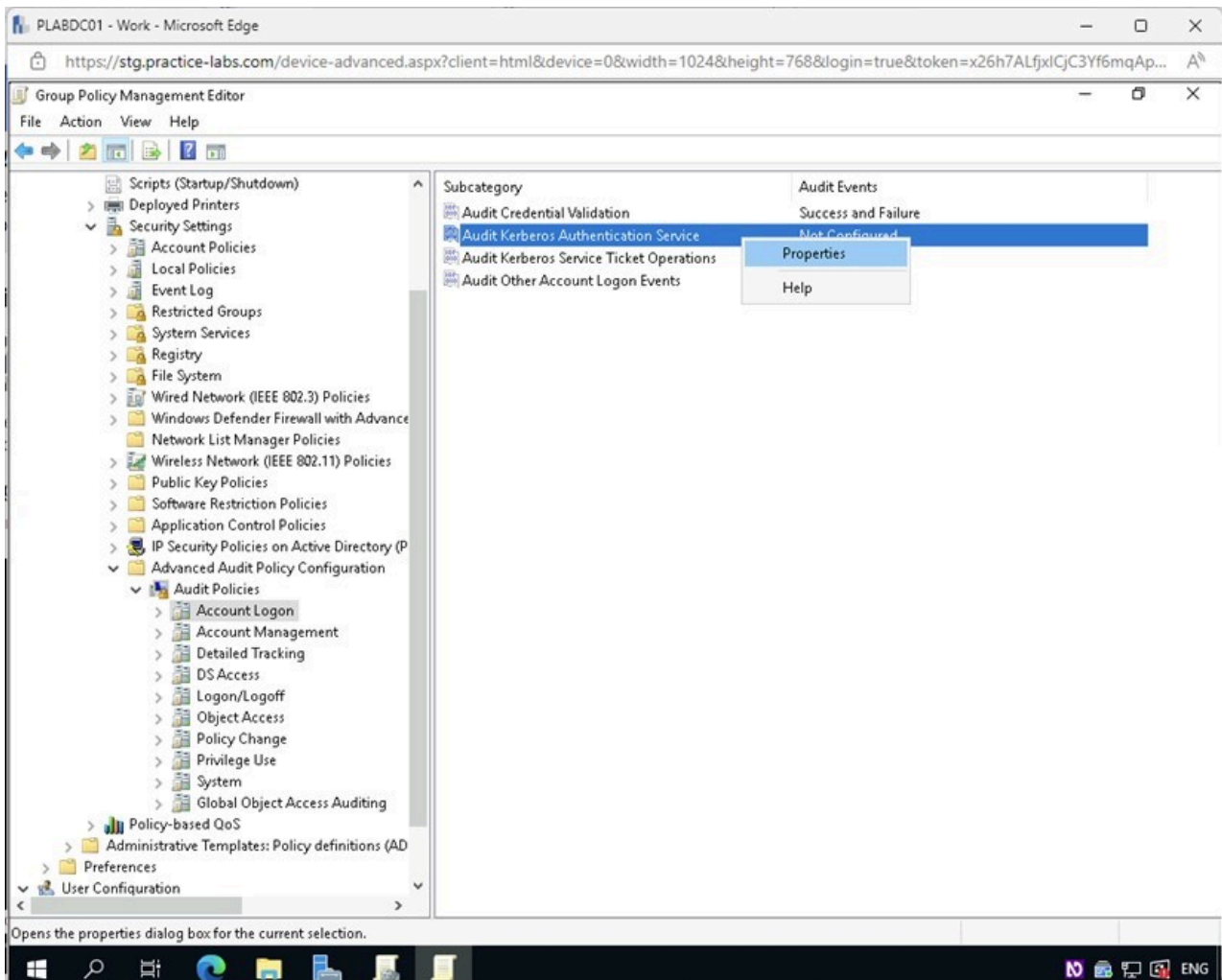


Figure 4.23 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Audit Kerberos Authentication Service right-clicked and Properties selected.

Step 10

In the **Audit Kerberos Authentication Service Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

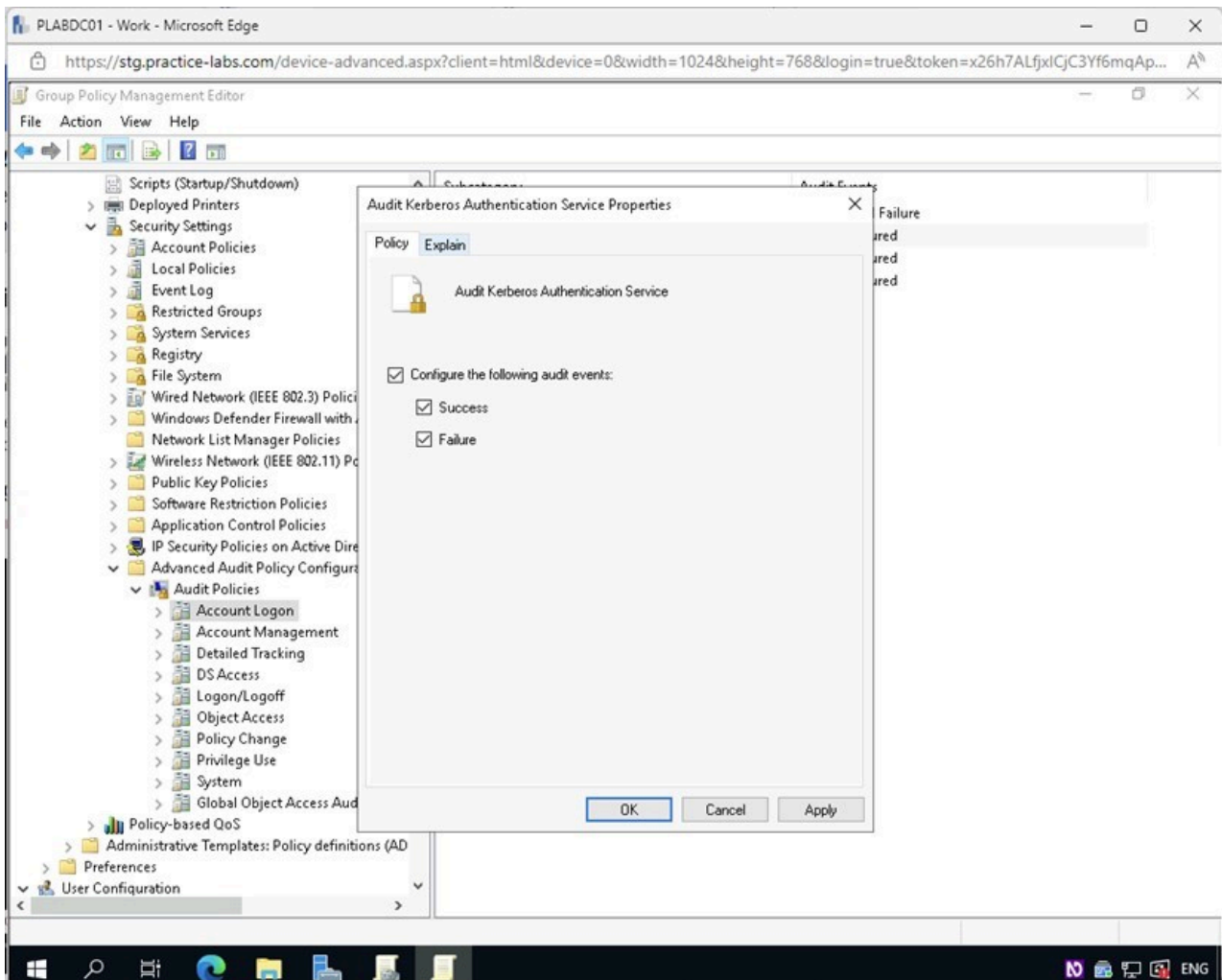


Figure 4.24 Screenshot of PLABDC01: Displaying the Audit Kerberos Authentication Service Properties window, showing the required settings performed and the Explain tab selected.

Step 11

In the **Explain** tab, you can view detailed information on **Audit Kerberos Authentication Service**.

Once you've read the information available, click **OK**.

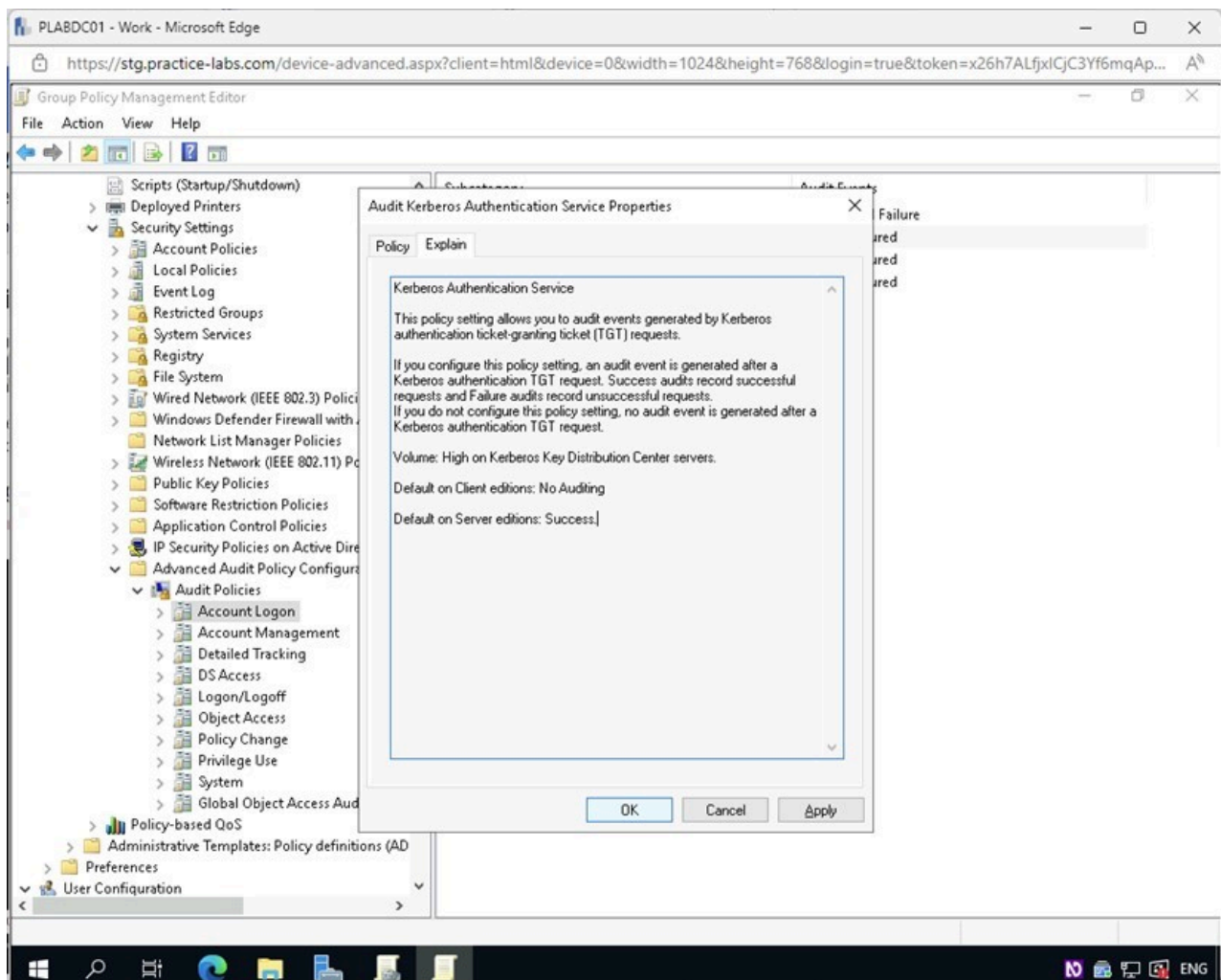


Figure 4.25 Screenshot of PLABDC01: Displaying the Audit Kerberos Authentication Service Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 12

Back on the **Group Policy Management Editor** window, right-click on **Audit Kerberos Service Ticket Operations** and select **Properties**.

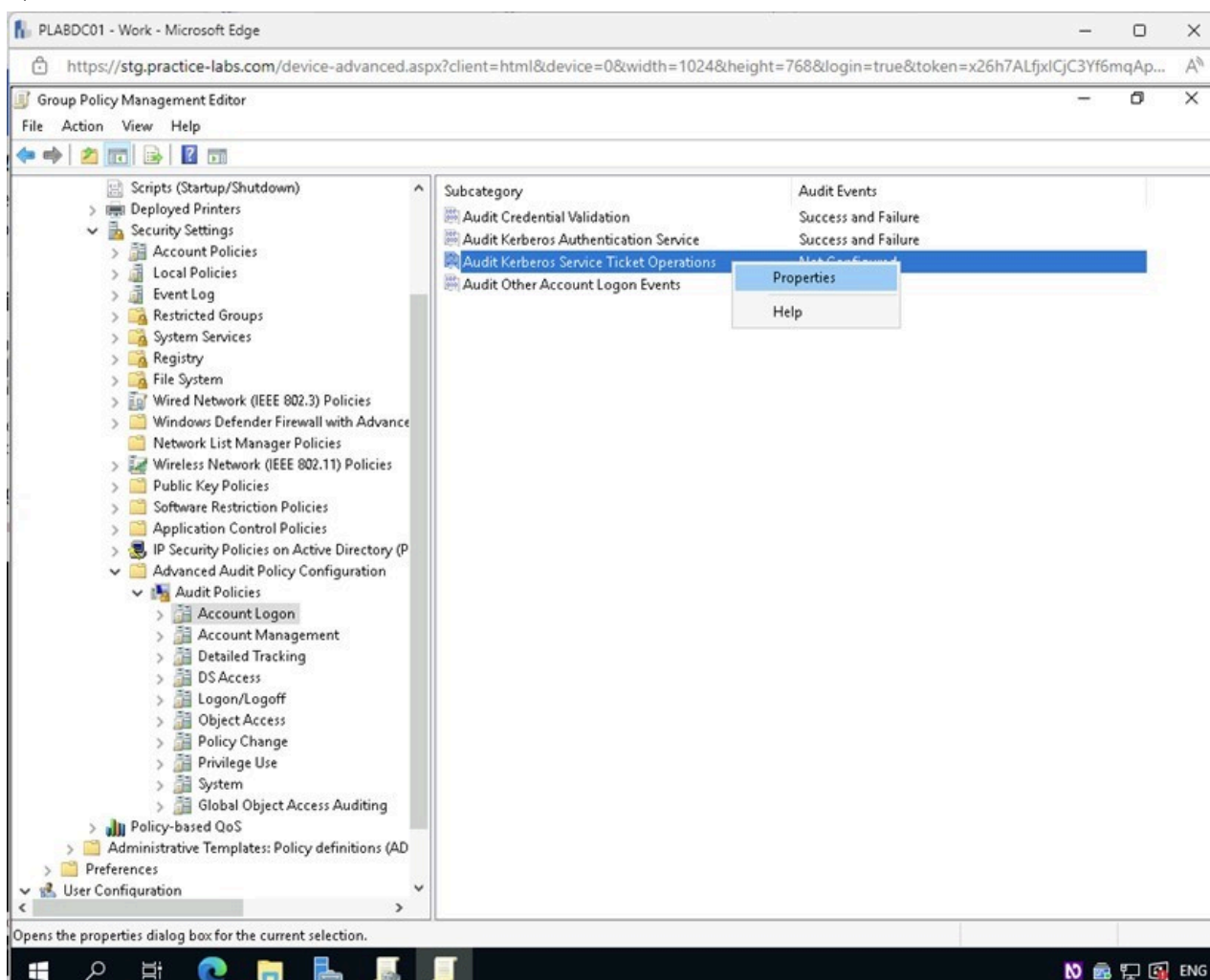


Figure 4.26 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Audit Kerberos Service Ticket Operations right-clicked and Properties selected.

Step 13

In the **Audit Kerberos Service Ticket Operations Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

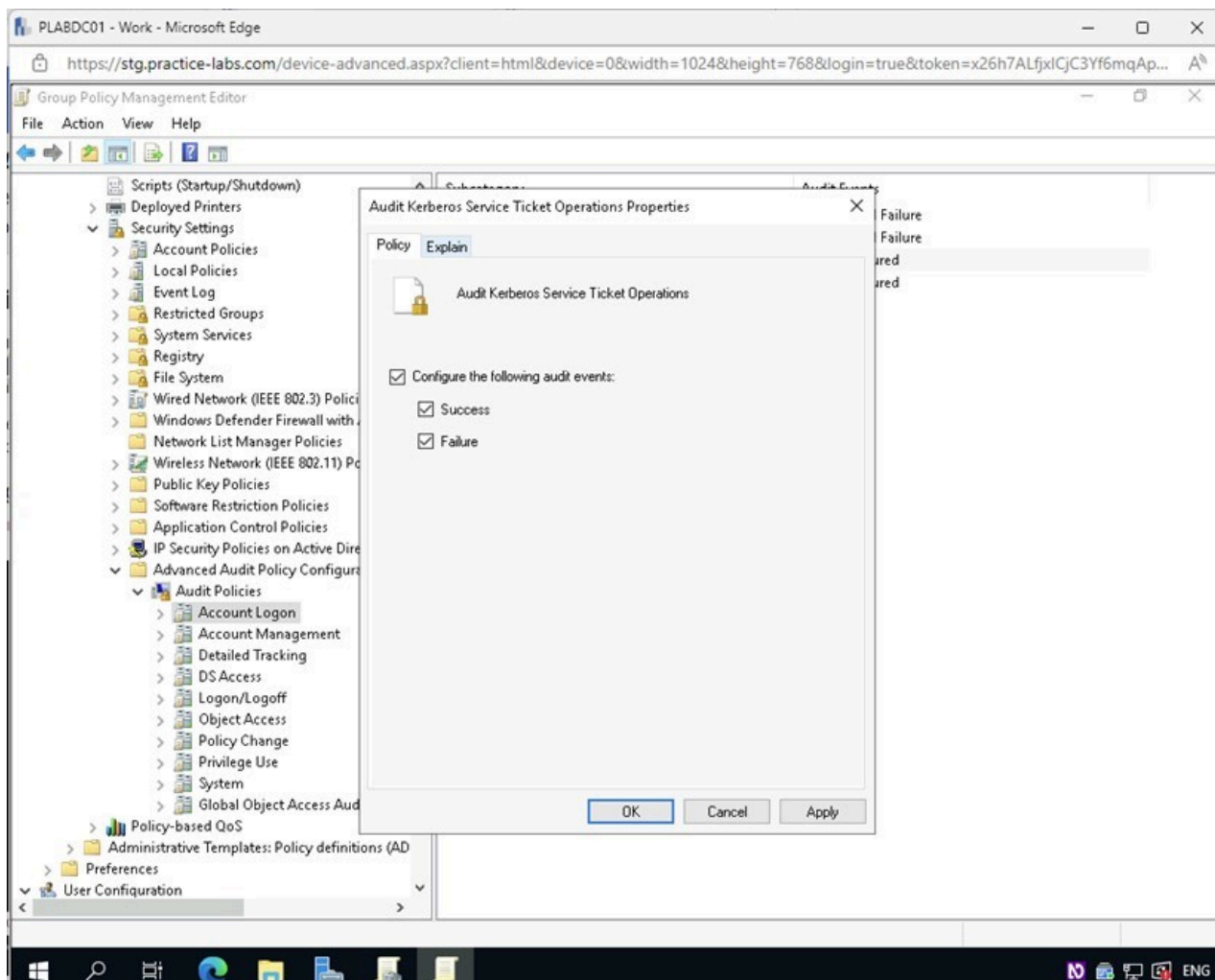


Figure 4.27 Screenshot of PLABDC01: Displaying the Audit Kerberos Service Ticket Operations Properties window, showing the required settings performed and the Explain tab selected.

Step 14

In the **Explain** tab, you can view detailed information on **Audit Kerberos Service Ticket Operations**.

Once you've read the information available, click **OK**.

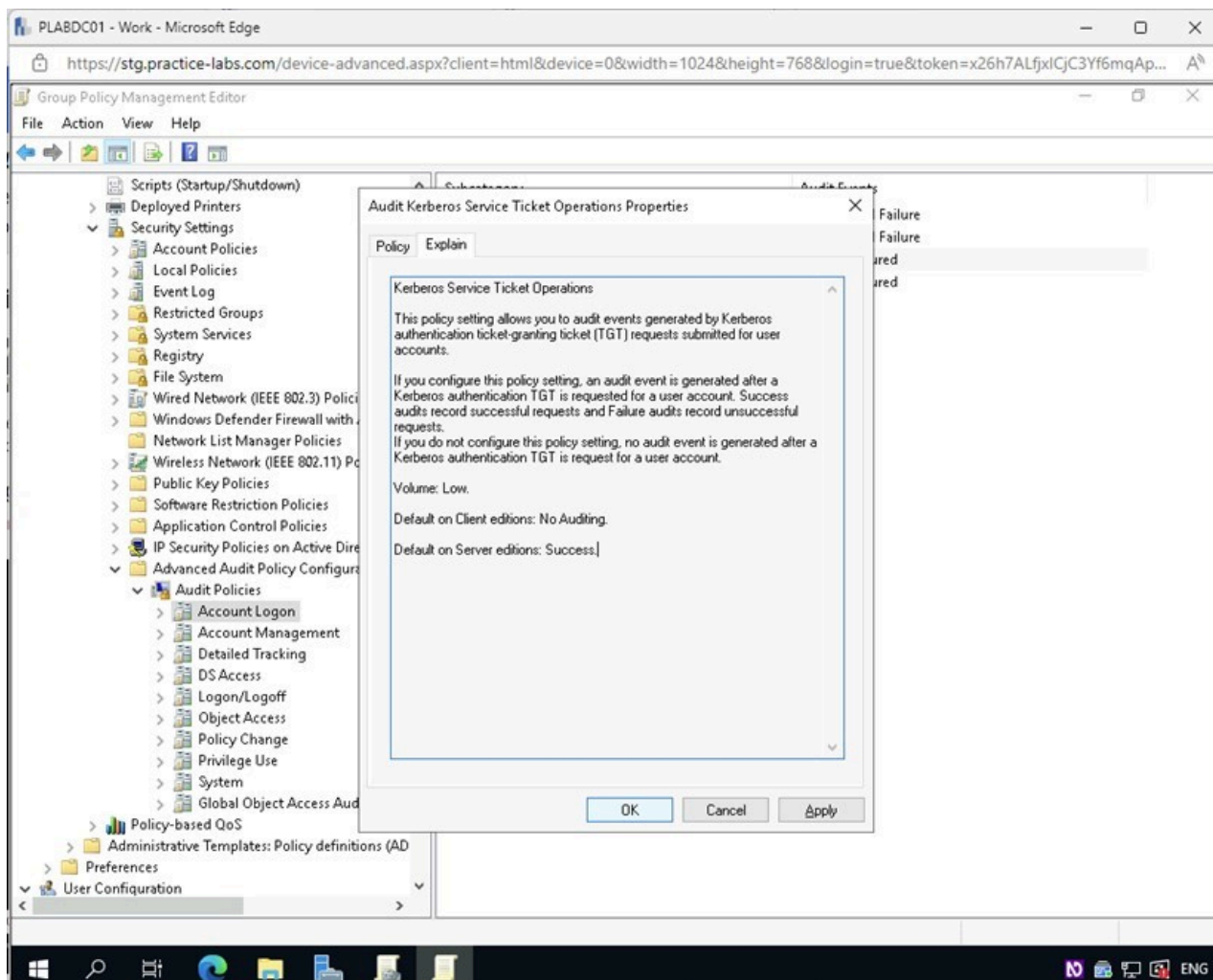


Figure 4.28 Screenshot of PLABDC01: Displaying the Audit Kerberos Service Ticket Operations Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 15

Back on the **Group Policy Management Editor** window, right-click on **Audit Other Account Logon Events** and select **Properties**.

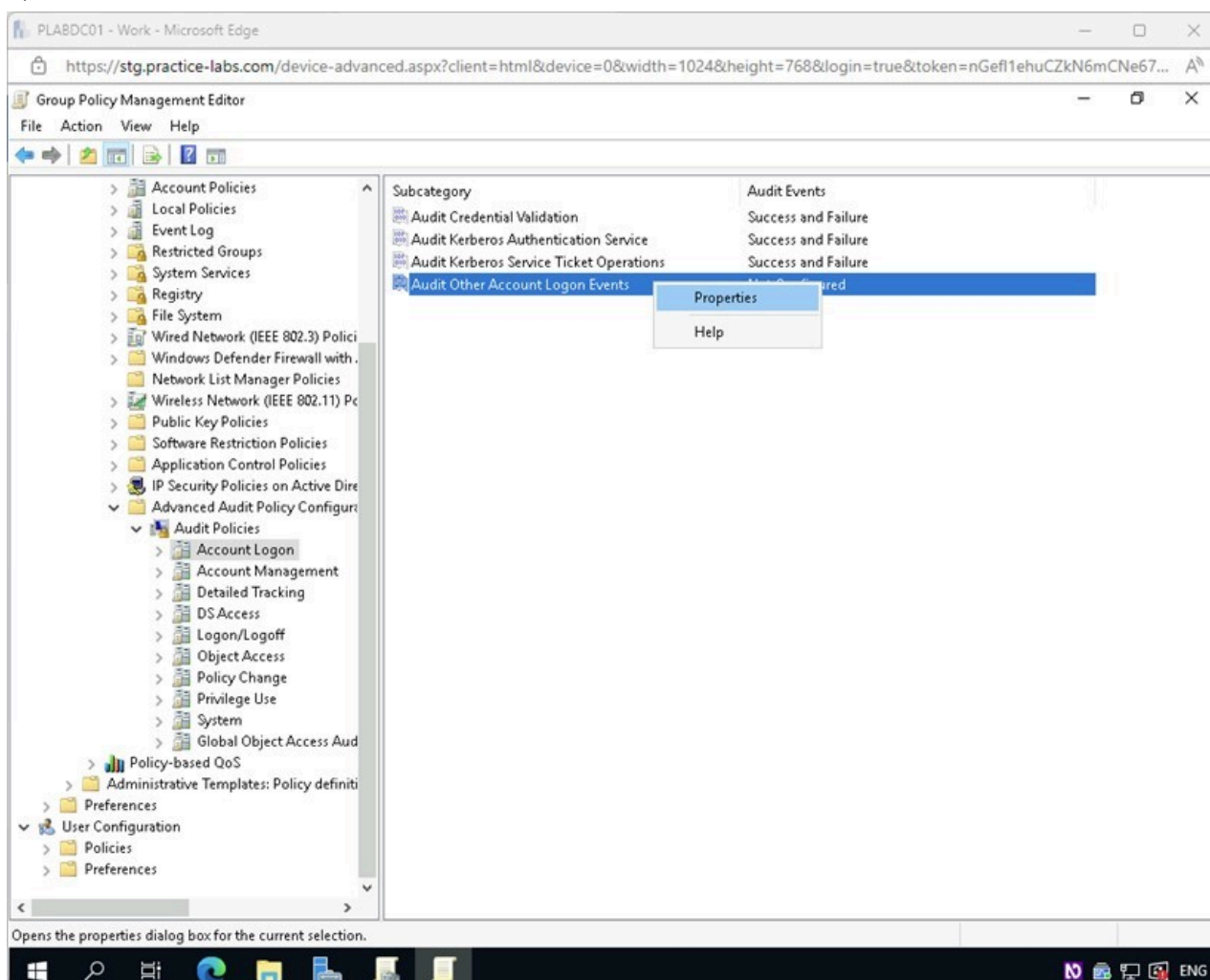


Figure 4.29 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Audit Other Account Logon Events right-clicked and Properties selected.

Step 16

In the **Audit Other Account Logon Events Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

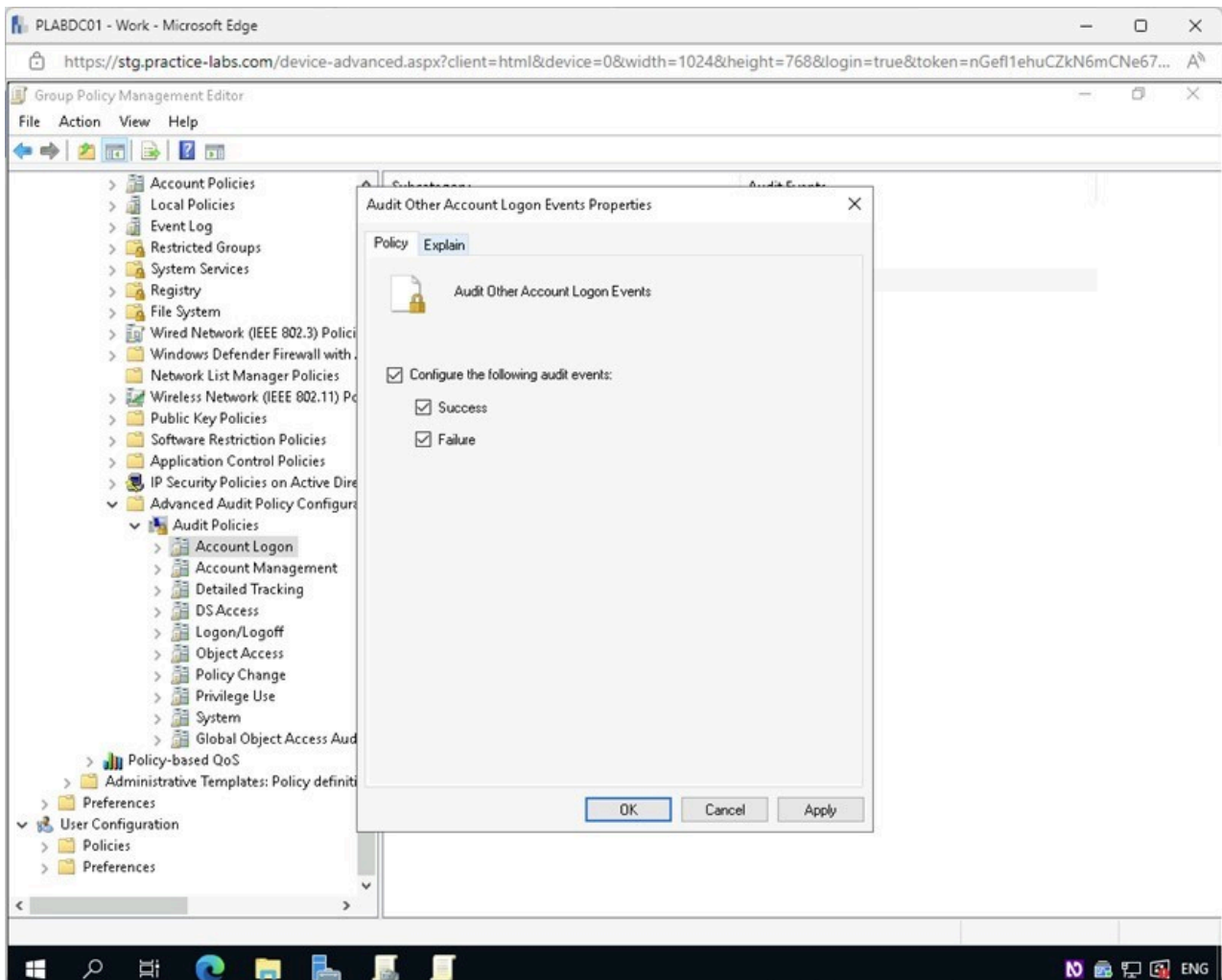


Figure 4.30 Screenshot of PLABDC01: Displaying the Audit Other Account Logon Events Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 17

In the **Explain** tab, you can view detailed information on **Audit Other Account Logon Events**.

Once you've read the information available, click **OK**.

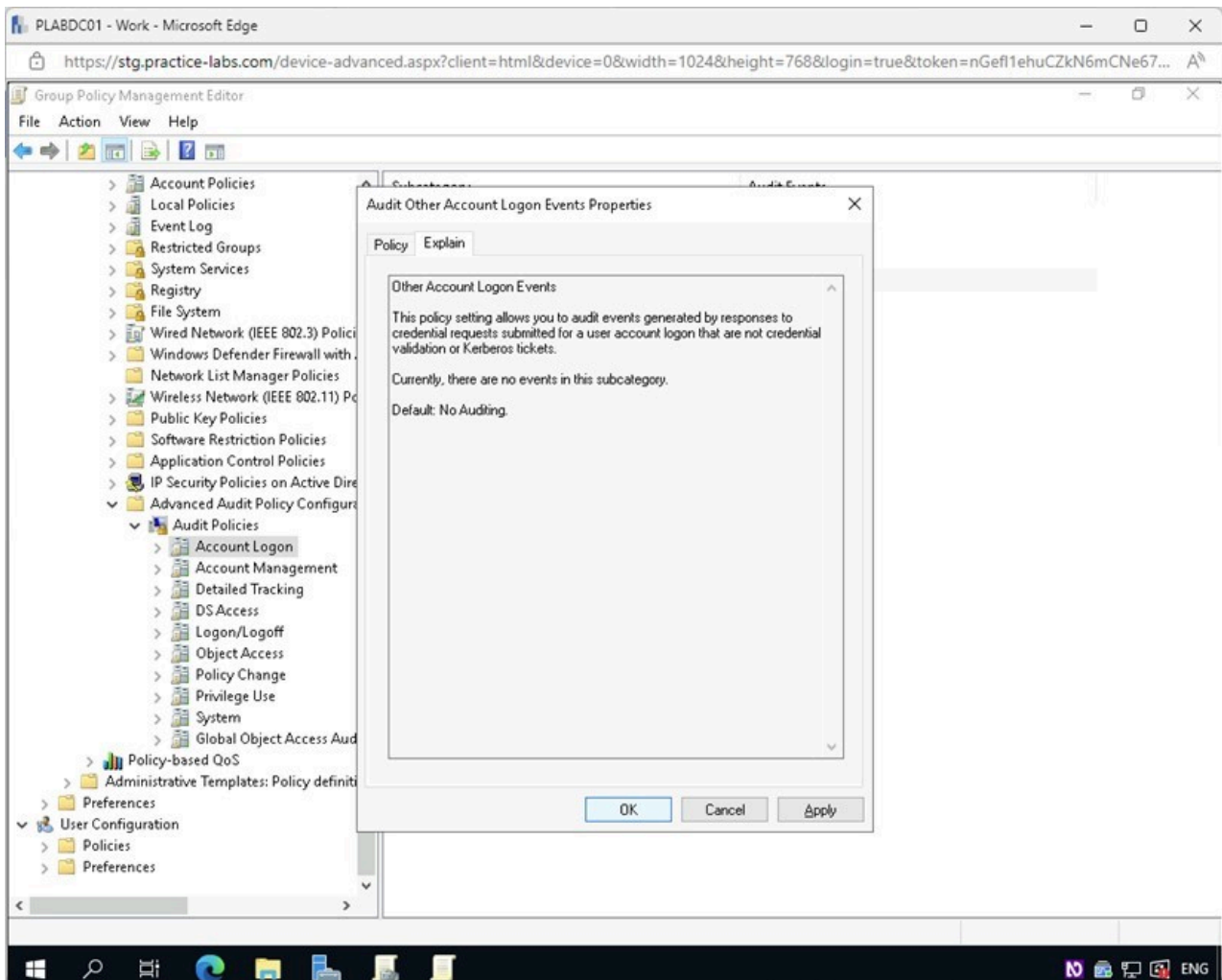


Figure 4.31 Screenshot of PLABDC01: Displaying the Audit Other Account Logon Events Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 18

Close the **Group Policy Management Editor** window.

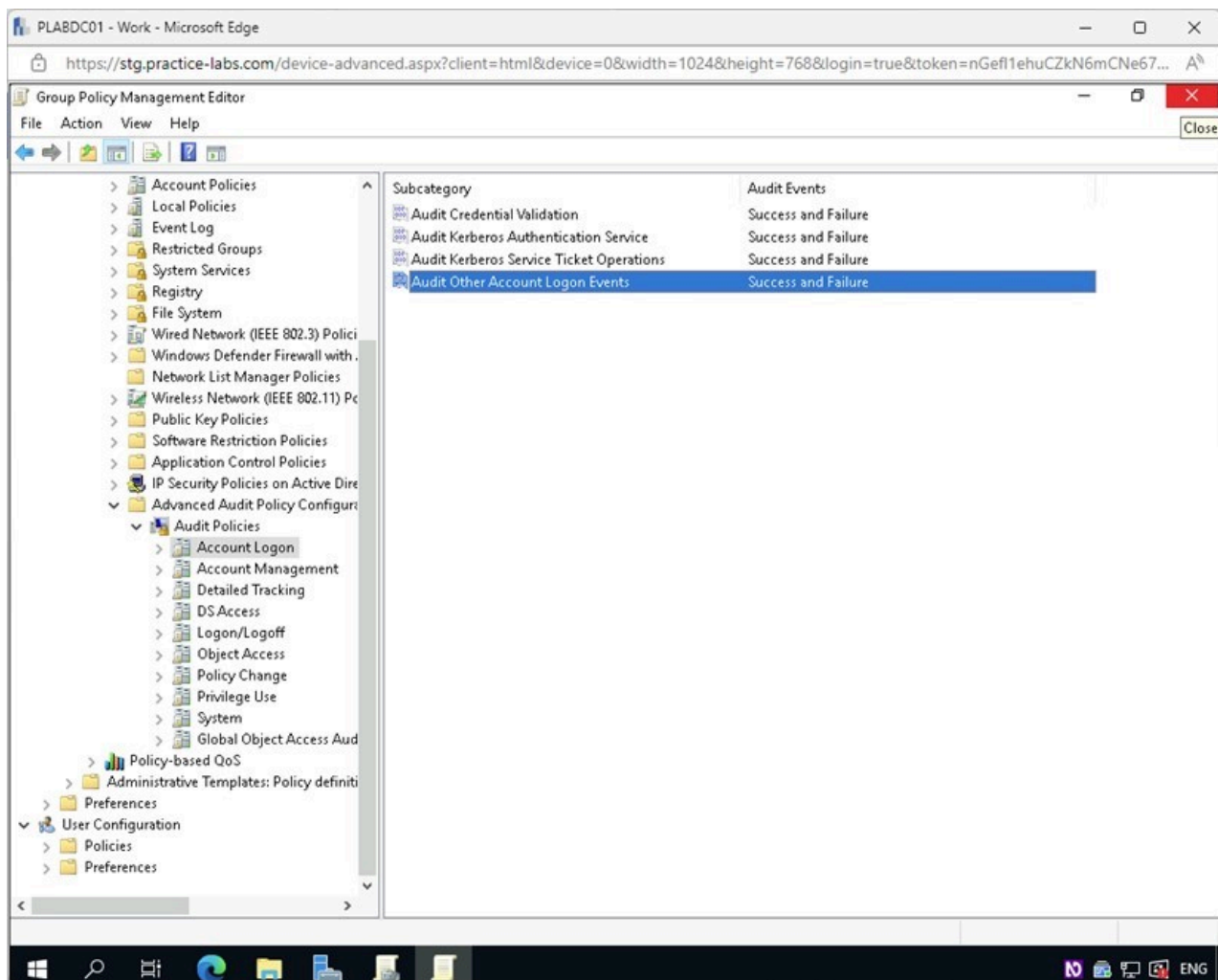


Figure 4.32 Screenshot of PLABDC01: Displaying closing the Group Policy Management Editor window.

Keep the **Group Policy Management** window open.

Task 3 - Audit Group Membership

By keeping track of who is a part of each group, administrators can ensure that no unauthorized individuals are allowed access to sensitive information or resources.

In this task, you will learn about auditing group membership.

Step 1

In **PLABDC01**, the **Group Policy Management** window is open.

Right-click on **practicelabs.com** and select **Create a GPO in this domain, and Link it here**.

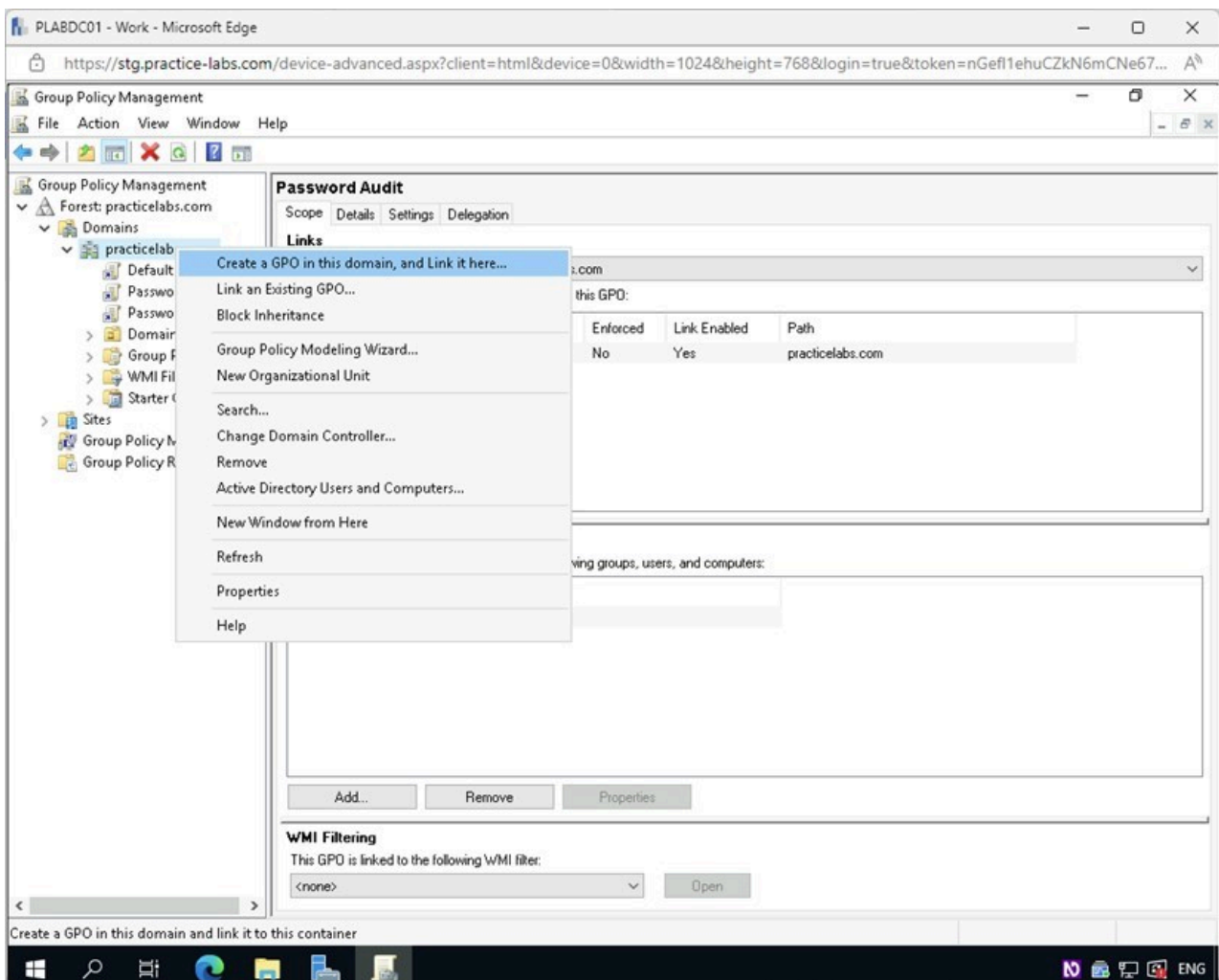


Figure 4.33 Screenshot of PLABDC01: Displaying the Group Policy Management window. practicelabs.com is right-clicked, and Create a GPO in this domain, and Link it here is selected.

Step 2

In the **New GPO** window, type the following into the **Name** textbox and click **OK**:

Group Audit

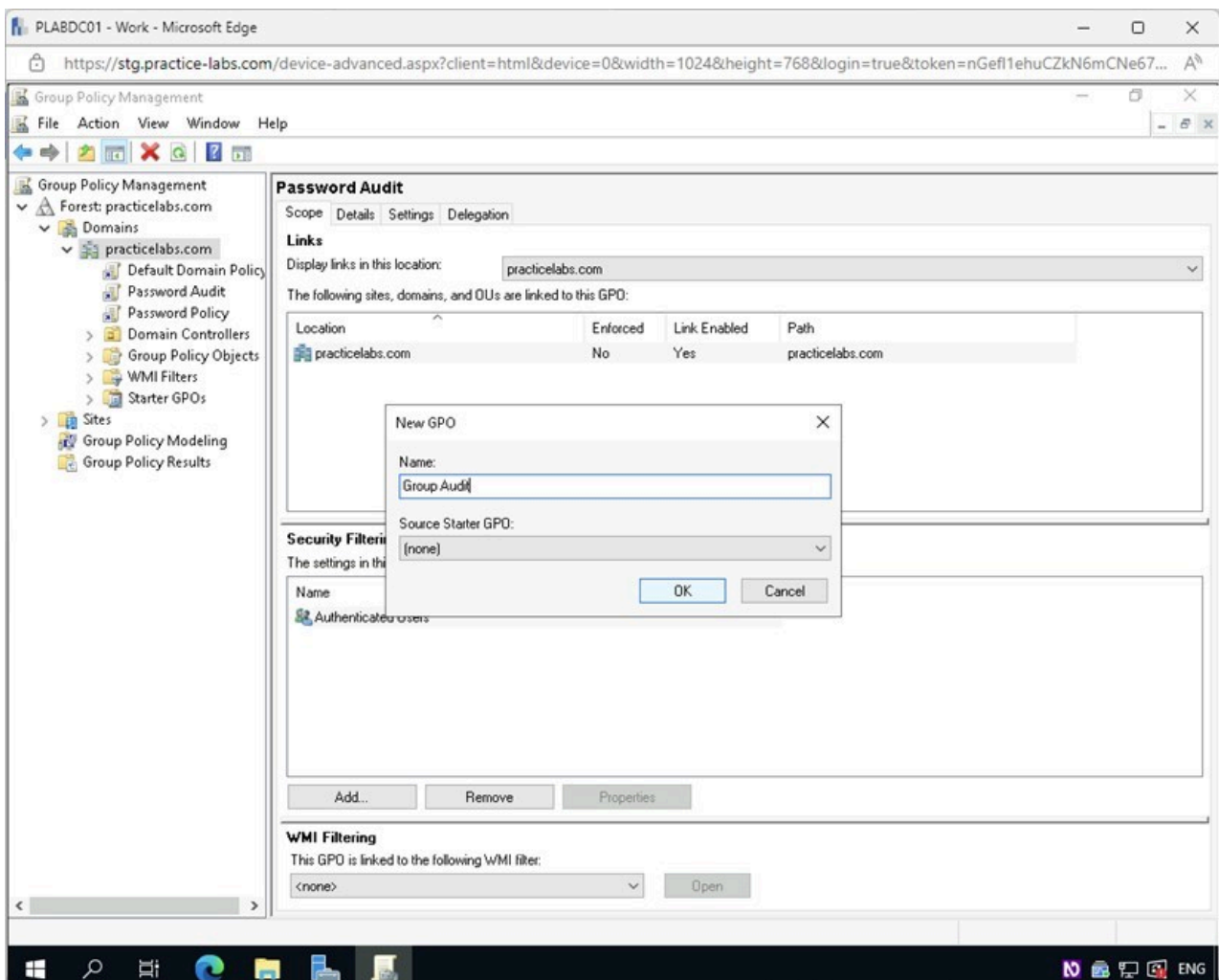


Figure 4.34 Screenshot of PLABDC01: Displaying the New GPO window, showing the required Name typed in and OK selected.

Step 3

Back on the **Group Policy Management** window, notice the **Group Audit** Policy has been created.

Right-click on **Group Audit** and select **Edit**.

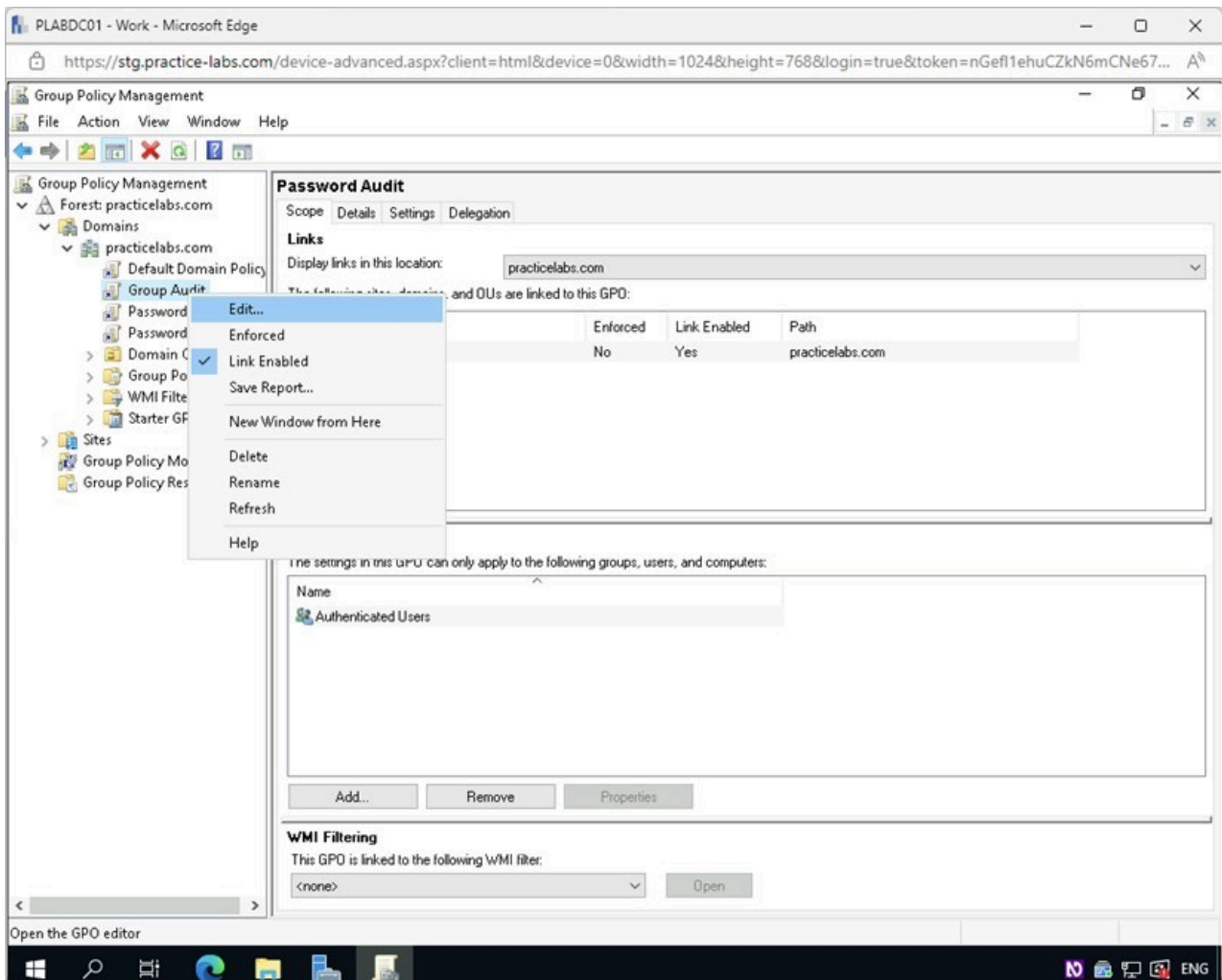


Figure 4.35 Screenshot of PLABDC01: Displaying the Group Policy Management window. Group Audit is right-clicked, and Edit selected.

Step 4

In the **Group Policy Management Editor** window, expand **Computer Configuration > Policies > Windows Settings > Security Settings > Advanced Audit Policy Configuration > Audit Policies**.

Select **Account Management**.

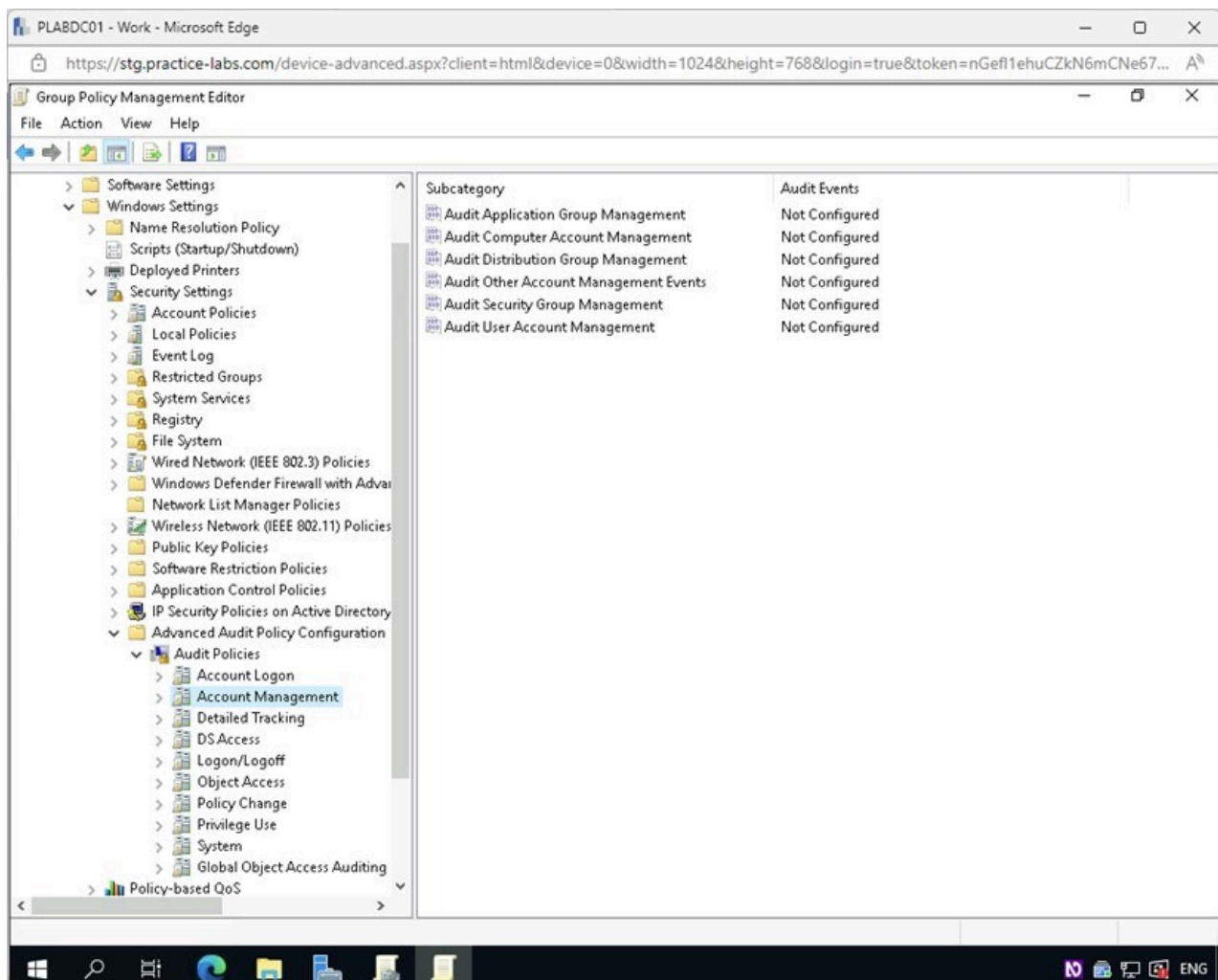


Figure 4.36 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window, showing Account Management selected.

Step 5

In the right details pane, right-click **Audit Application Group Management** and select **Properties**.

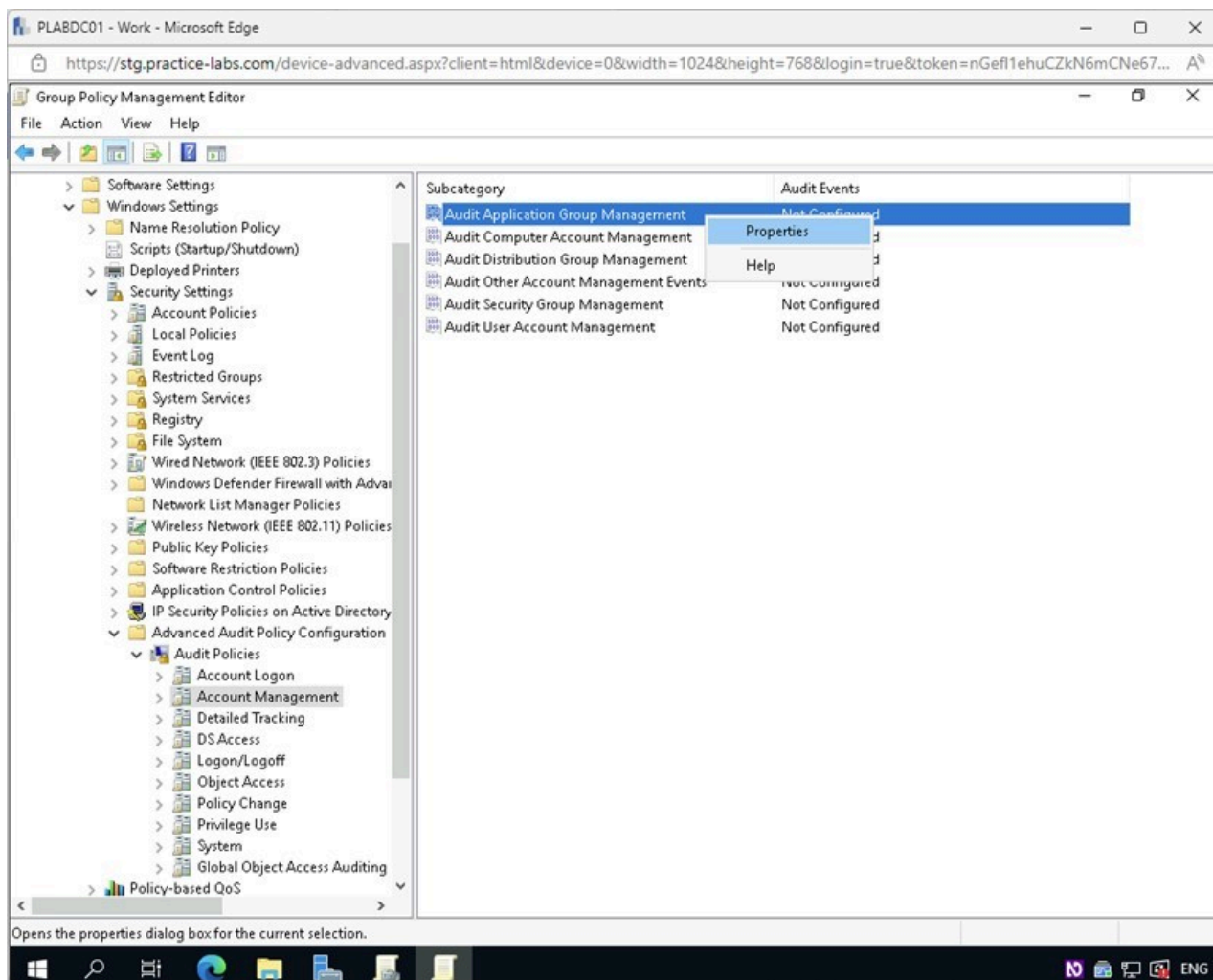


Figure 4.37 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Audit Application Group Management is right-clicked, and Properties selected.

Step 6

In the **Audit Application Group Management Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

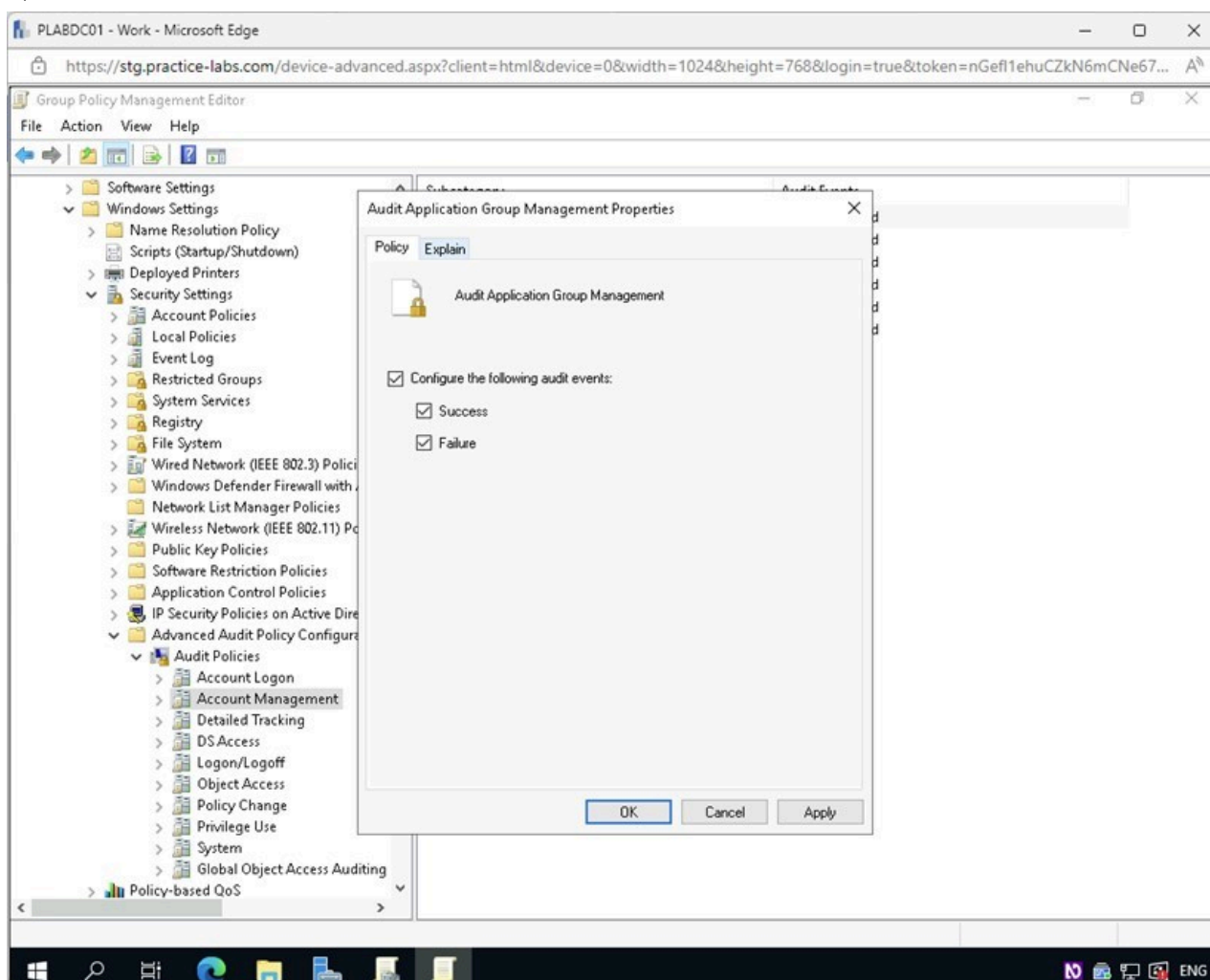


Figure 4.38 Screenshot of PLABDC01: Displaying the Audit Application Group Management Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 7

In the **Explain** tab, you can view detailed information on **Audit Application Group Management**.

Once you've read the information available, click **OK**.

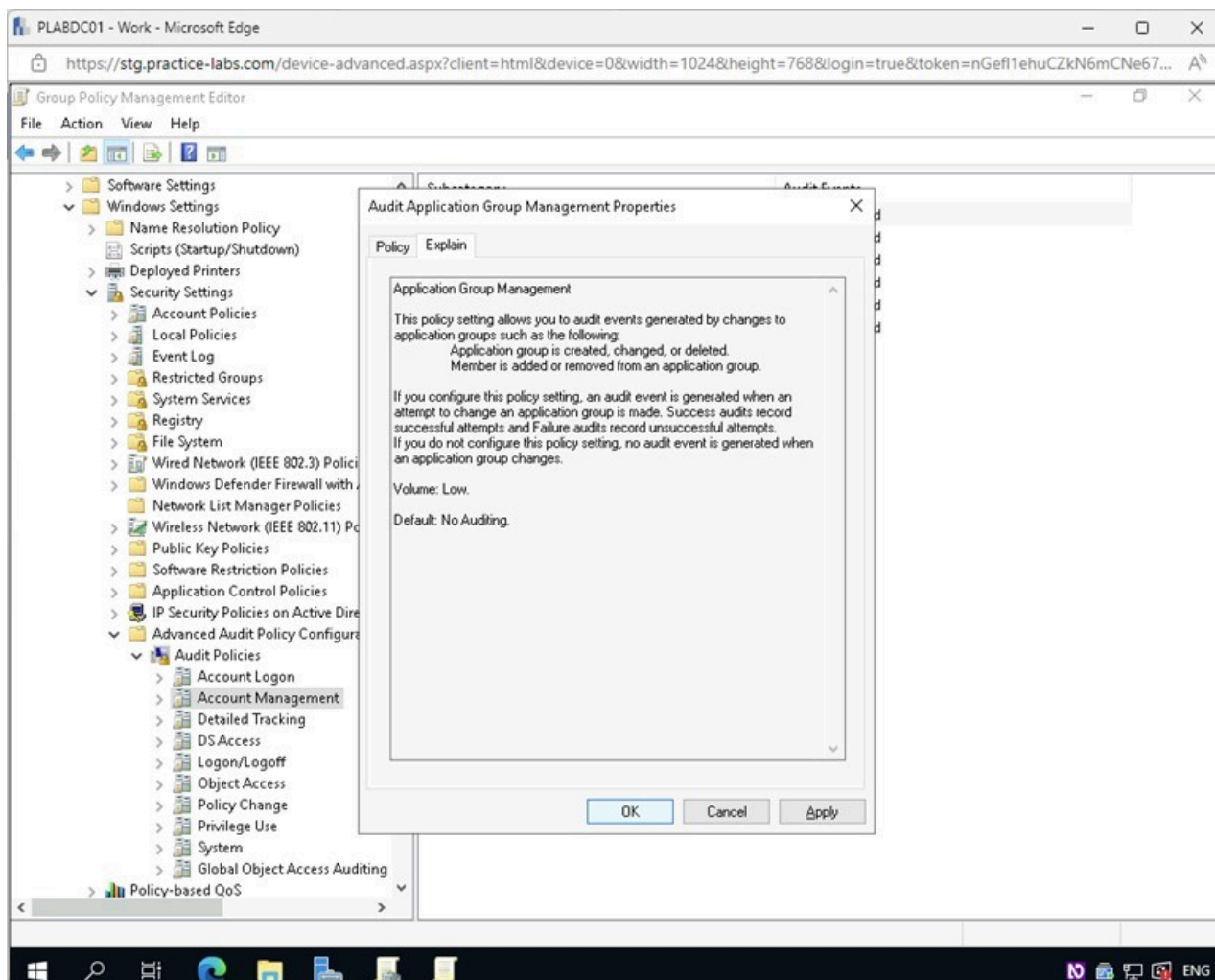


Figure 4.39 Screenshot of PLABDC01: Displaying the Audit Application Group Management Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 8

Back on the **Group Policy Management Editor** window, right-click on **Audit Computer Account Management** and select **Properties**

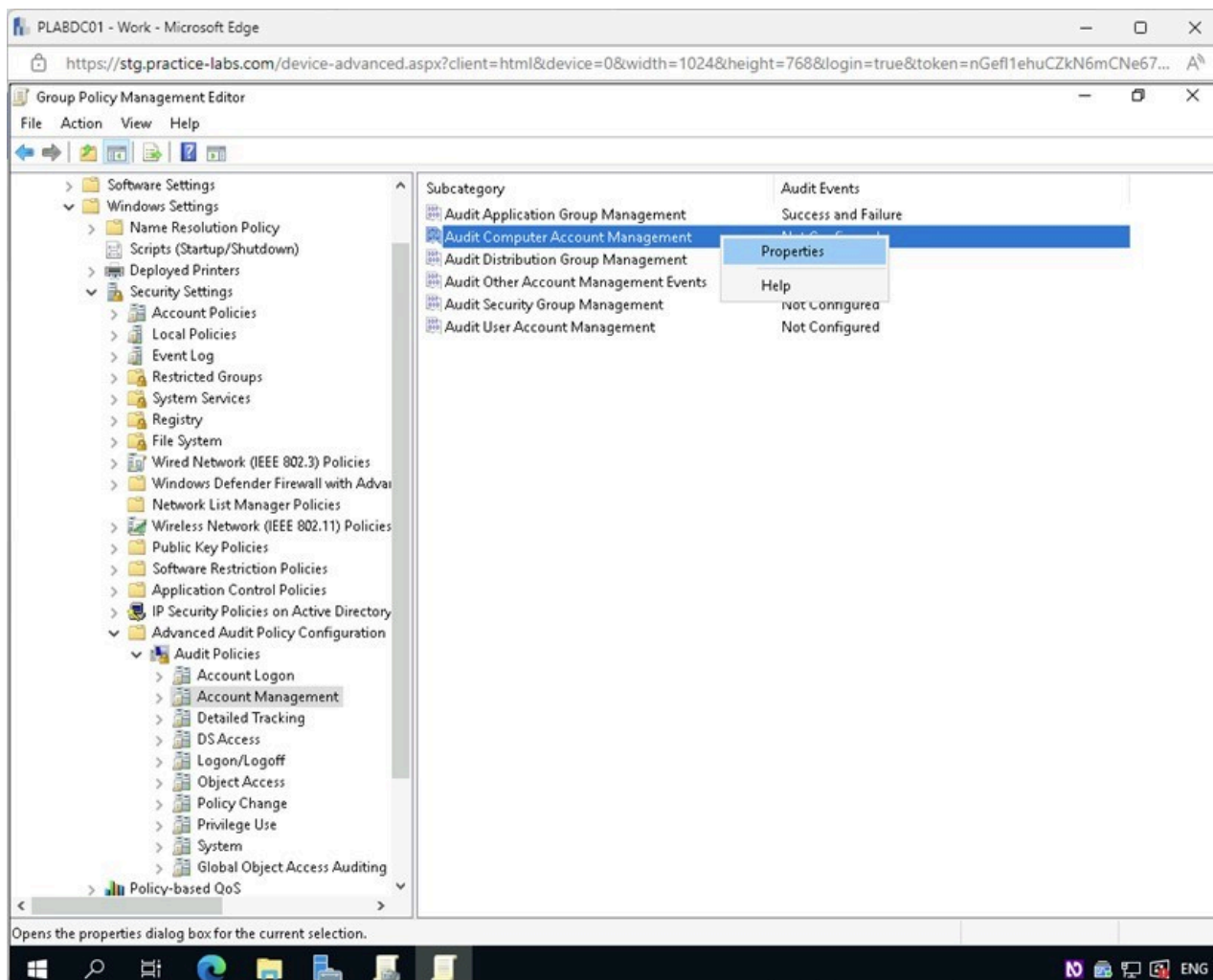


Figure 4.40 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Audit Computer Account Management is right-clicked, and Properties selected.

Step 9

In the **Audit Computer Account Management Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

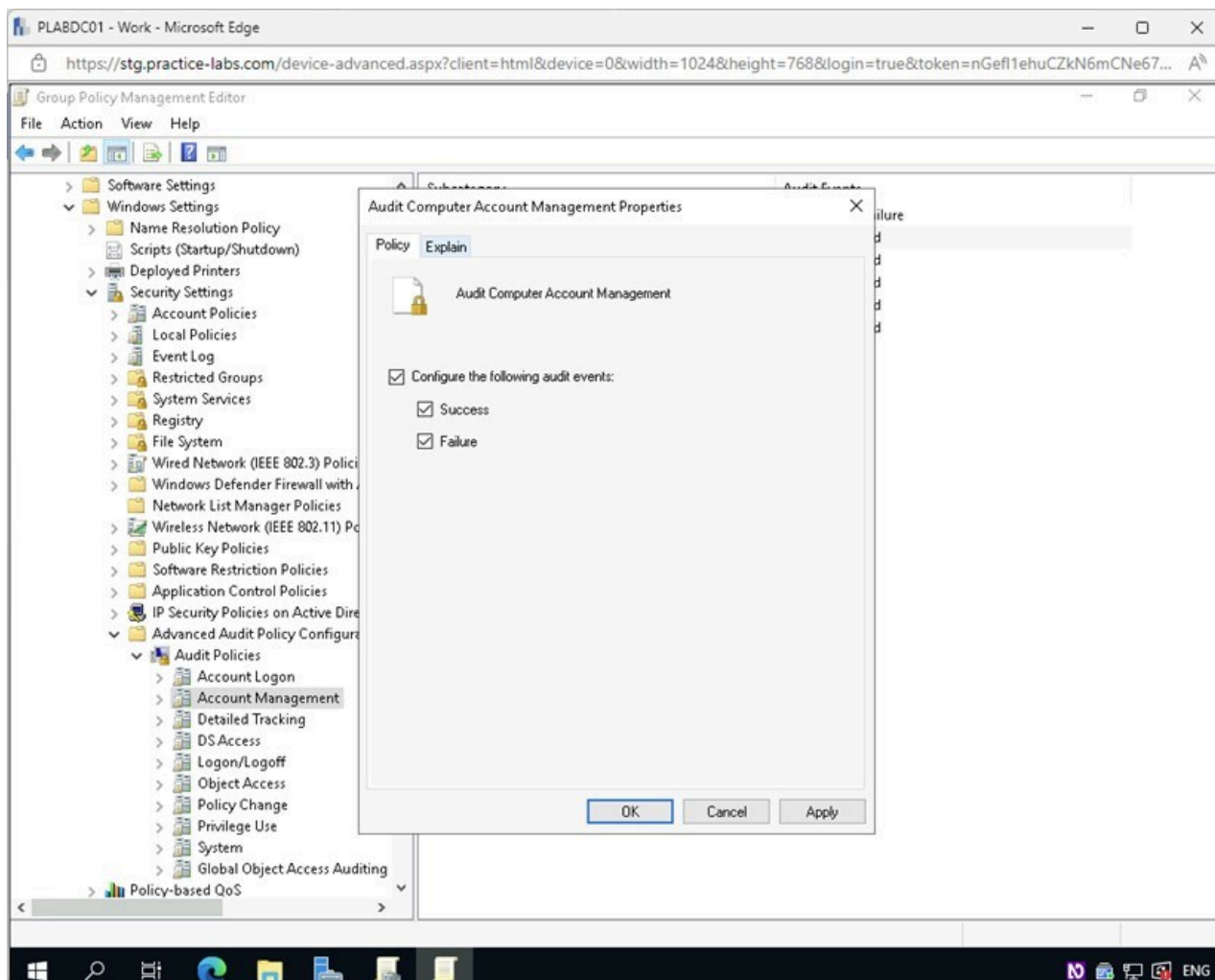


Figure 4.41 Screenshot of PLABDC01: Displaying the Audit Computer Account Management Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 10

In the **Explain** tab, you can view detailed information on **Audit Computer Account Management**.

Once you've read the information available, click **OK**.

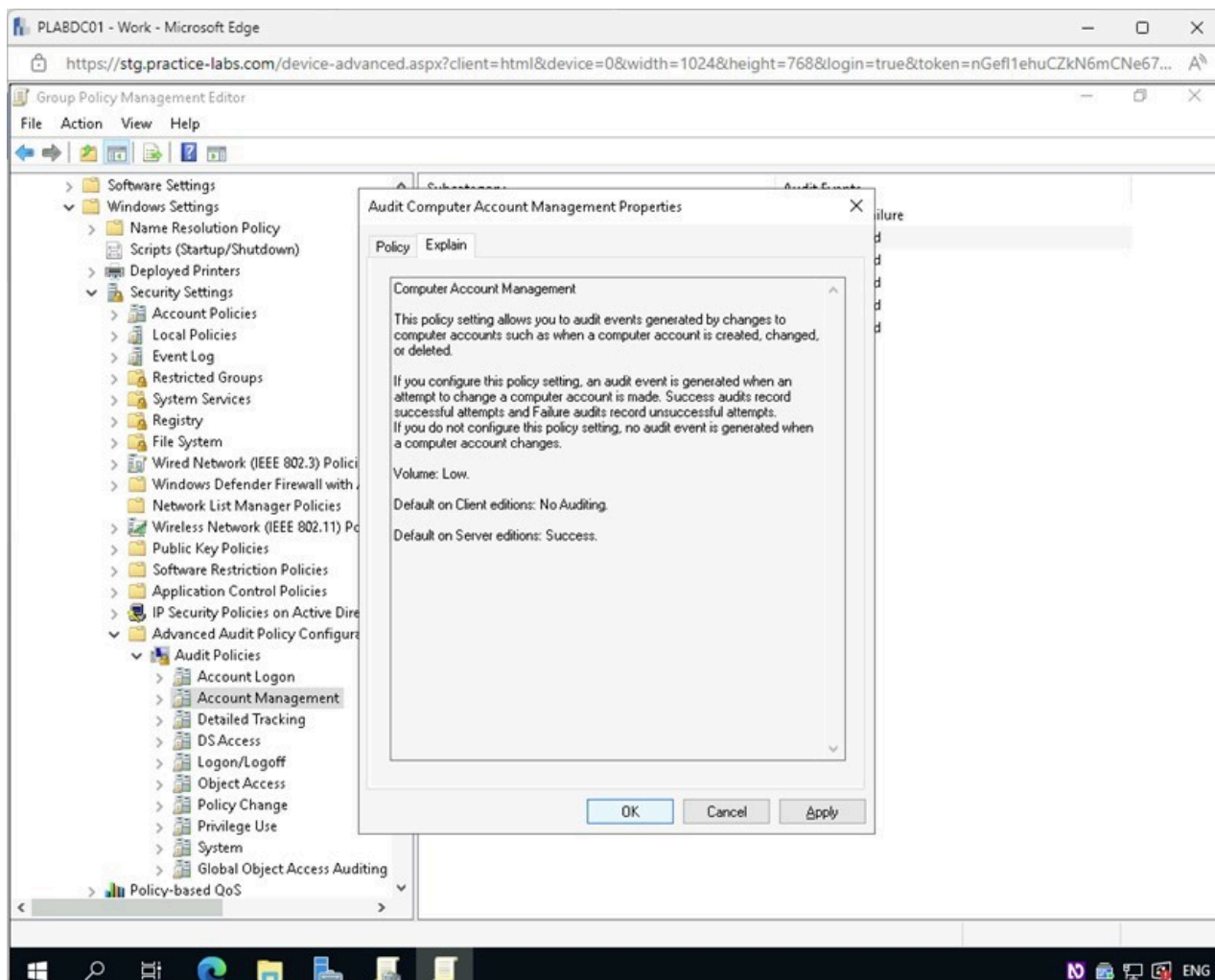


Figure 4.42 Screenshot of PLABDC01: Displaying the Audit Computer Account Management Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 11

Back on the **Group Policy Management Editor** window, right-click on **Audit Distribution Group Management** and select **Properties**.

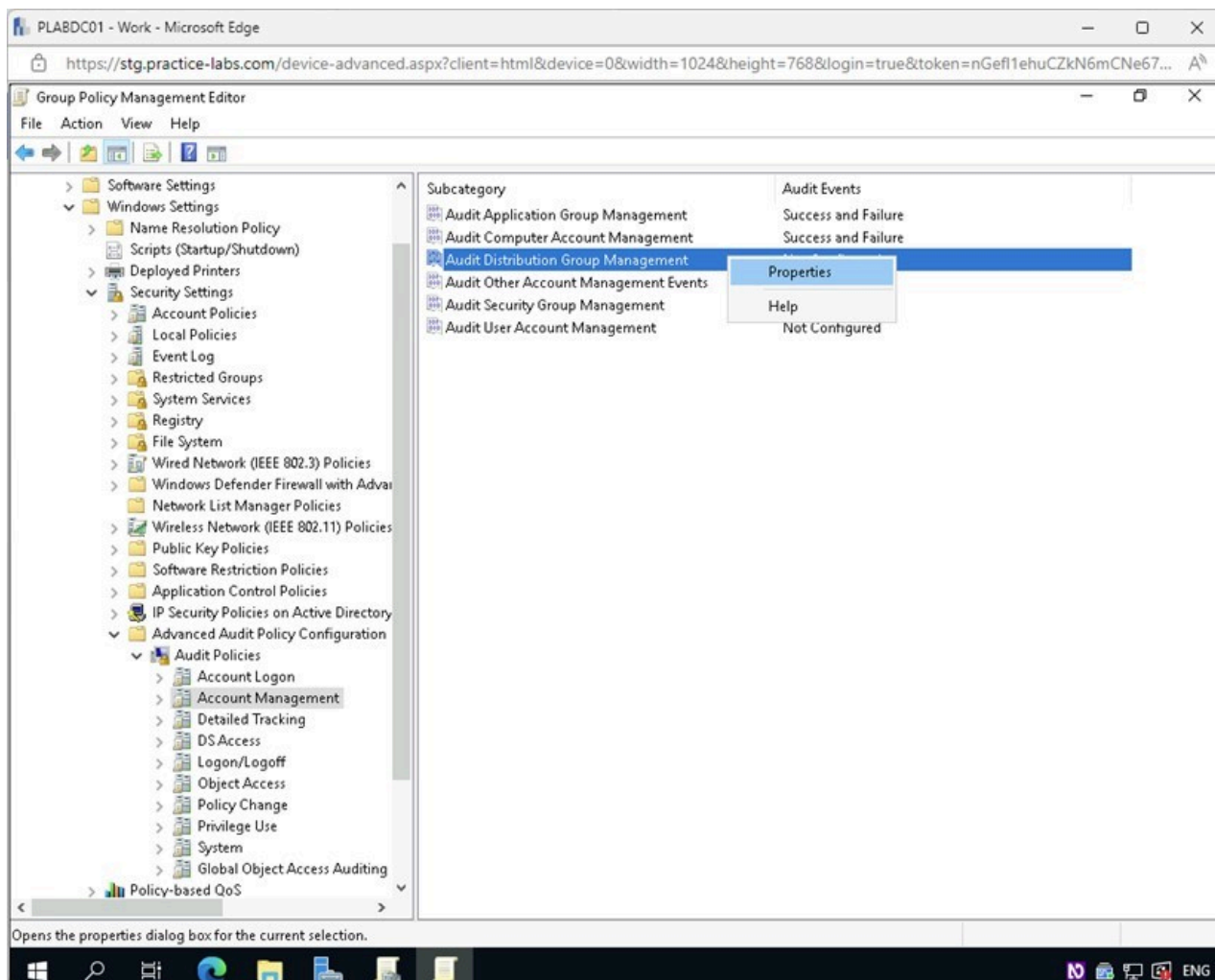


Figure 4.43 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Audit Distribution Group Management is right-clicked, and Properties selected.

Step 12

In the **Audit Distribution Group Management Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

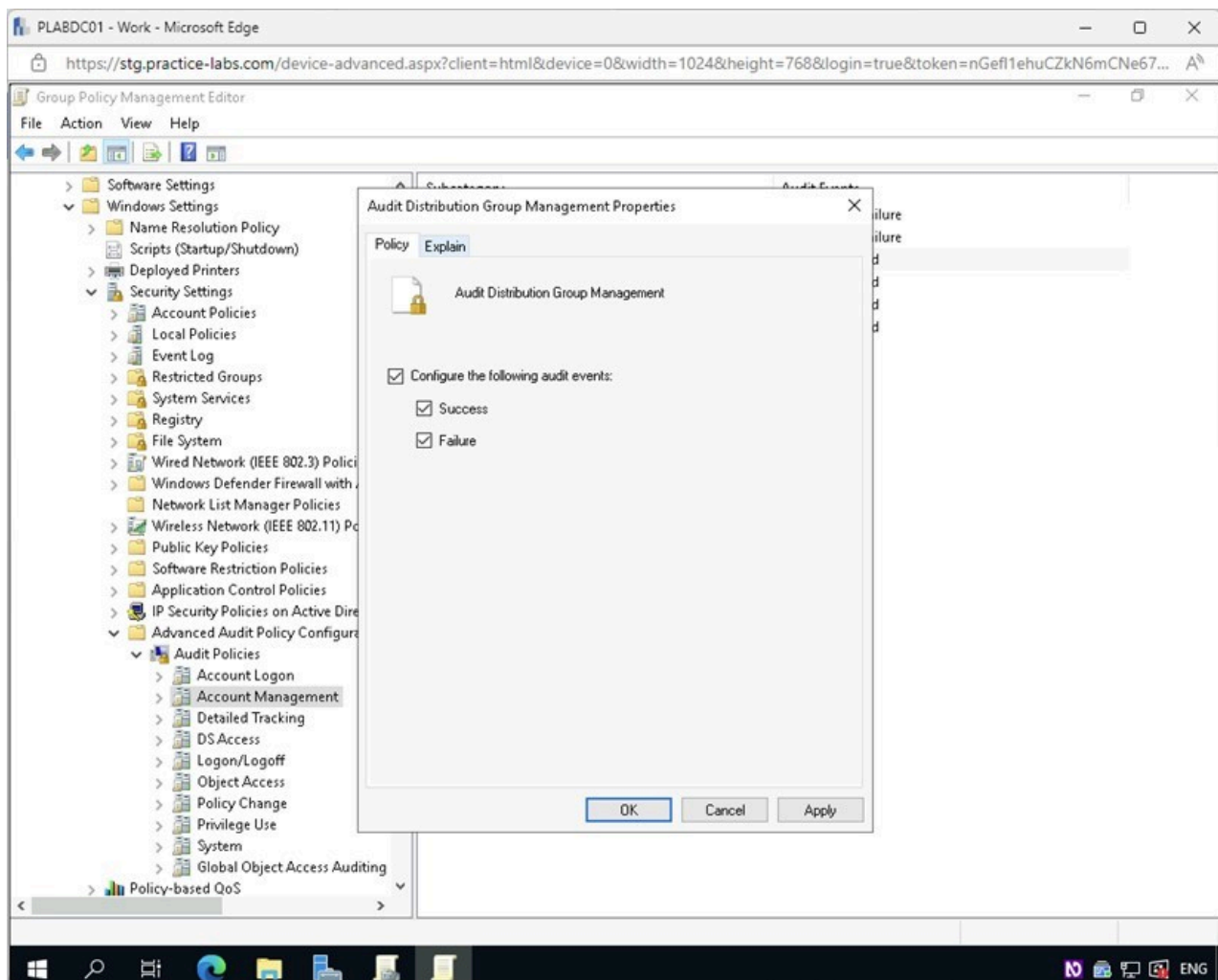


Figure 4.44 Screenshot of PLABDC01: Displaying the Audit Distribution Group Management Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 13

In the **Explain** tab, you can view detailed information on **Audit Distribution Group Management**.

Once you've read the information available, click **OK**.

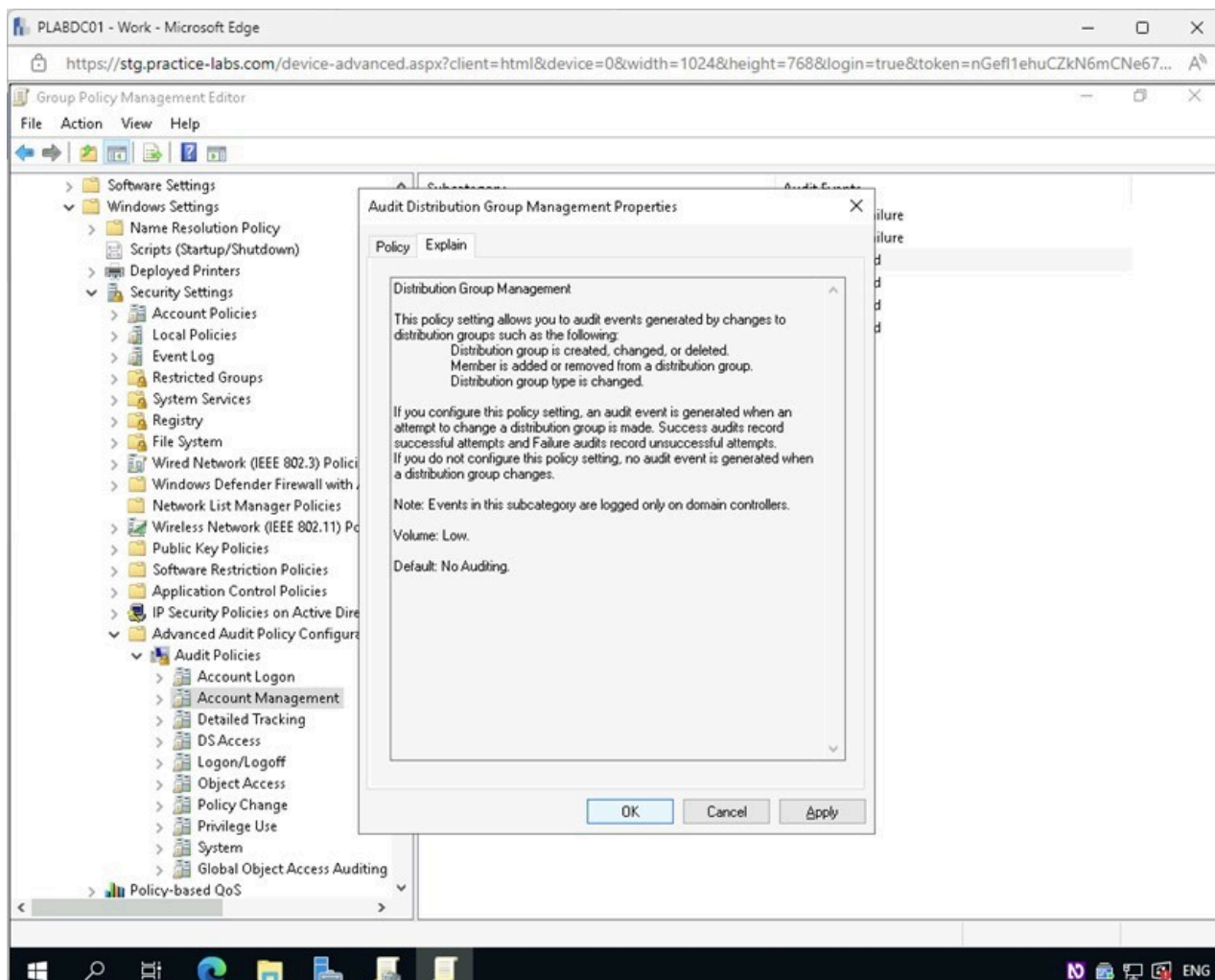


Figure 4.45 Screenshot of PLABDC01: Displaying the Audit Distribution Group Management Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 14

Back on the **Group Policy Management Editor** window, right-click on **Audit Other Account Management Events** and select **Properties**.

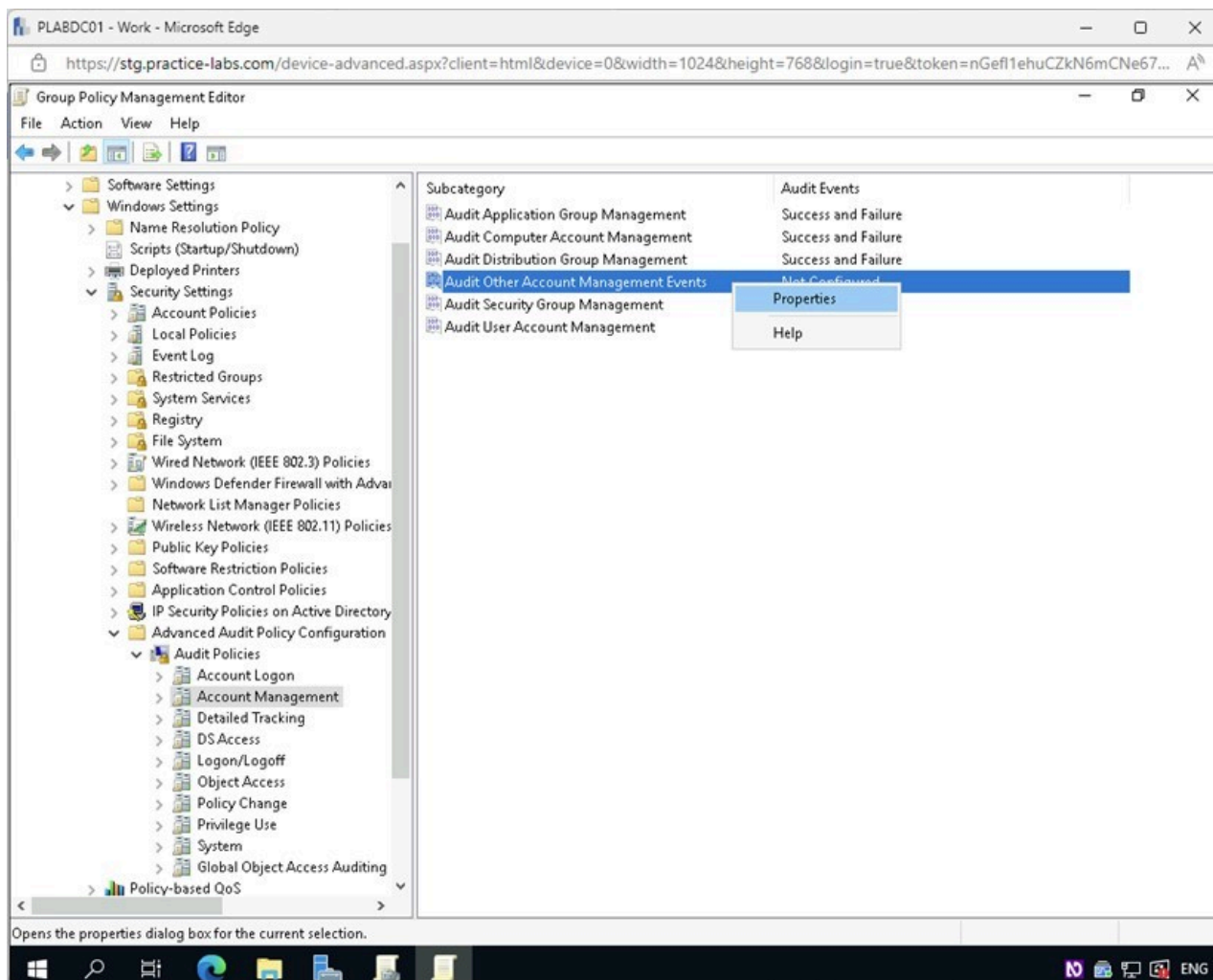


Figure 4.46 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Audit Other Account Management Events is right-clicked, and Properties selected.

Step 15

In the **Audit Other Account Management Events Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

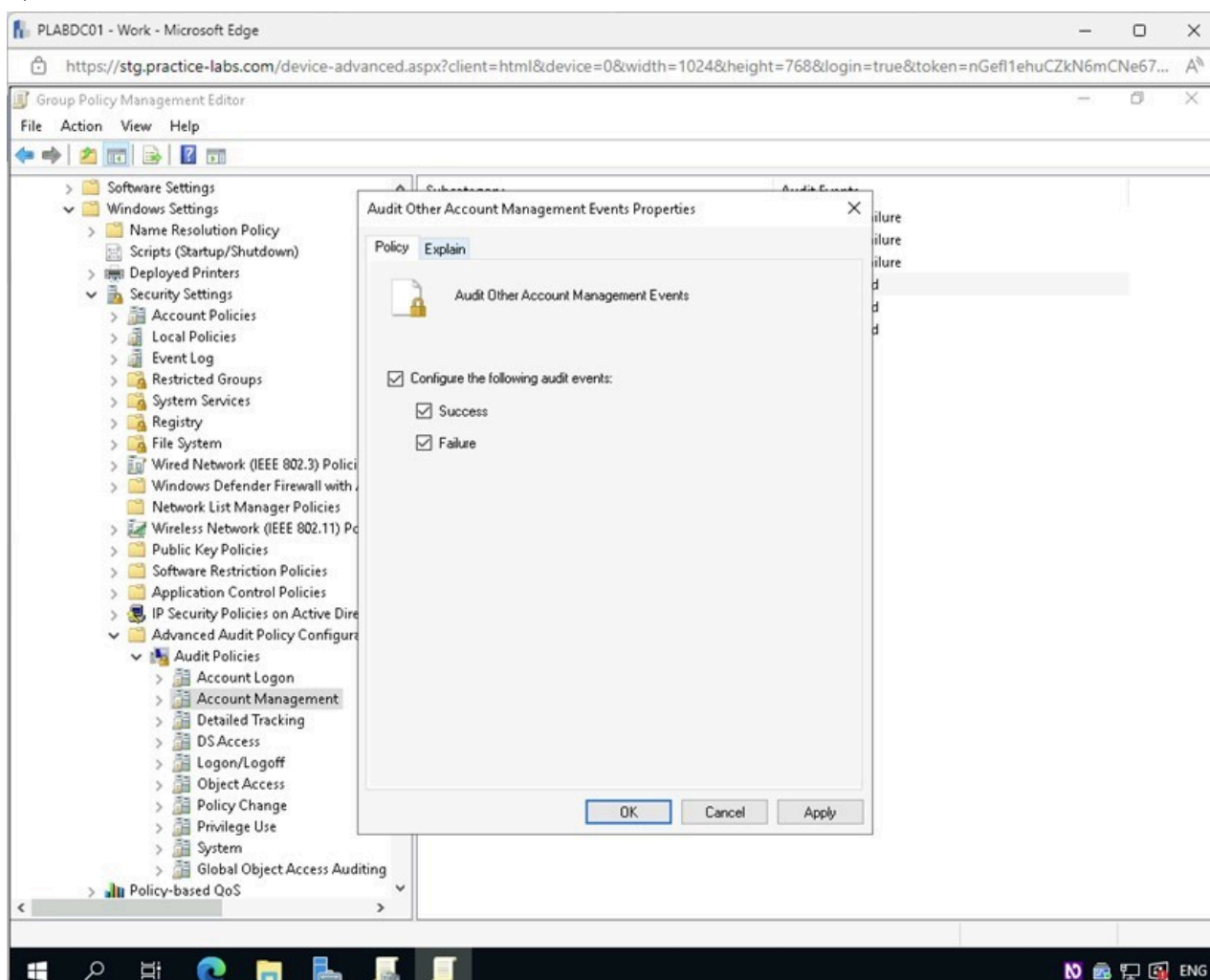


Figure 4.47 Screenshot of PLABDC01: Displaying the Audit Other Account Management Events Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 16

In the **Explain** tab, you can view detailed information on **Audit Other Account Management Events**.

Once you've read the information available, click **OK**.

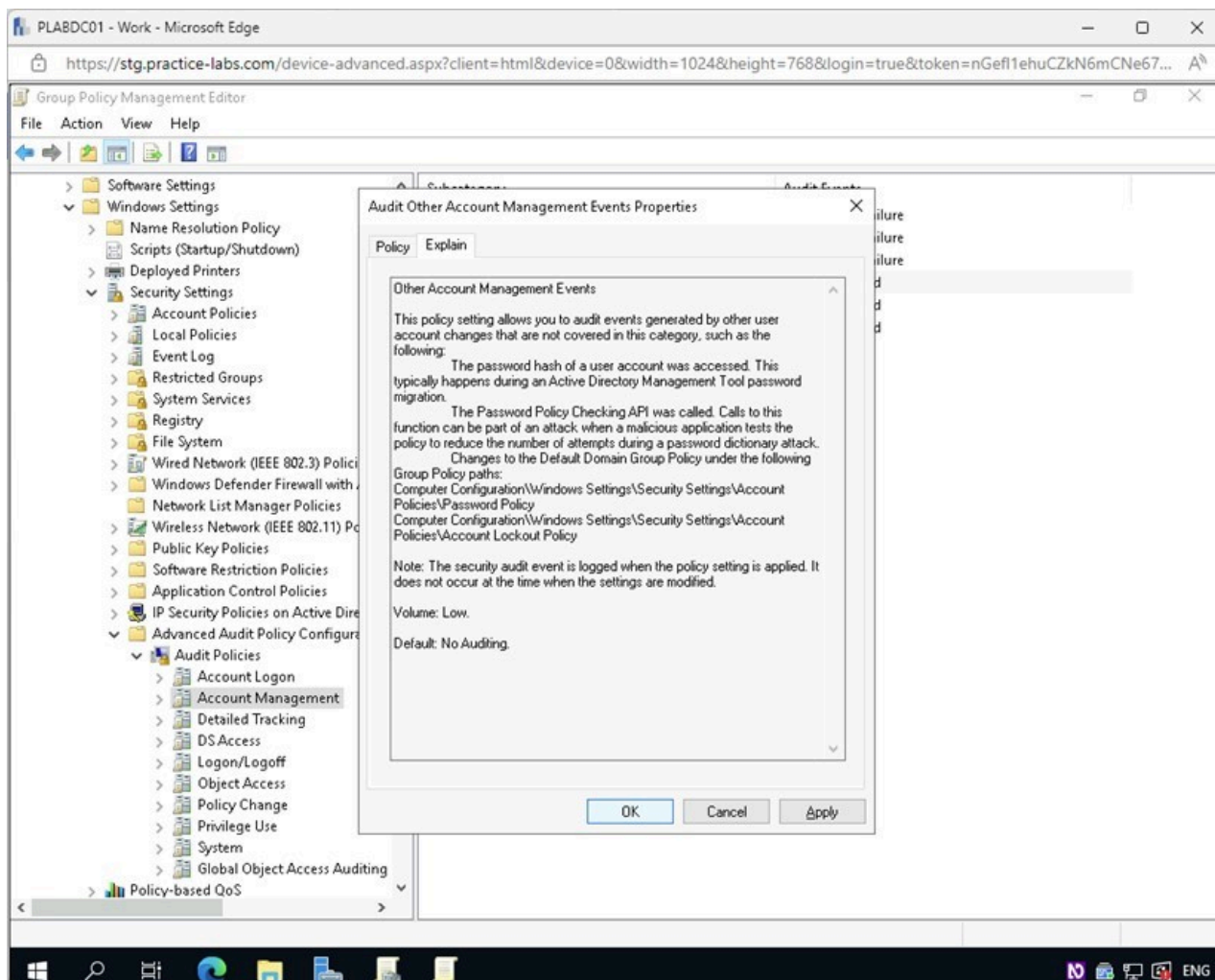


Figure 4.48 Screenshot of PLABDC01: Displaying the Audit Other Account Management Events Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 17

On the **Group Policy Management Editor** window, right-click on **Audit Security Group Management** and select **Properties**.

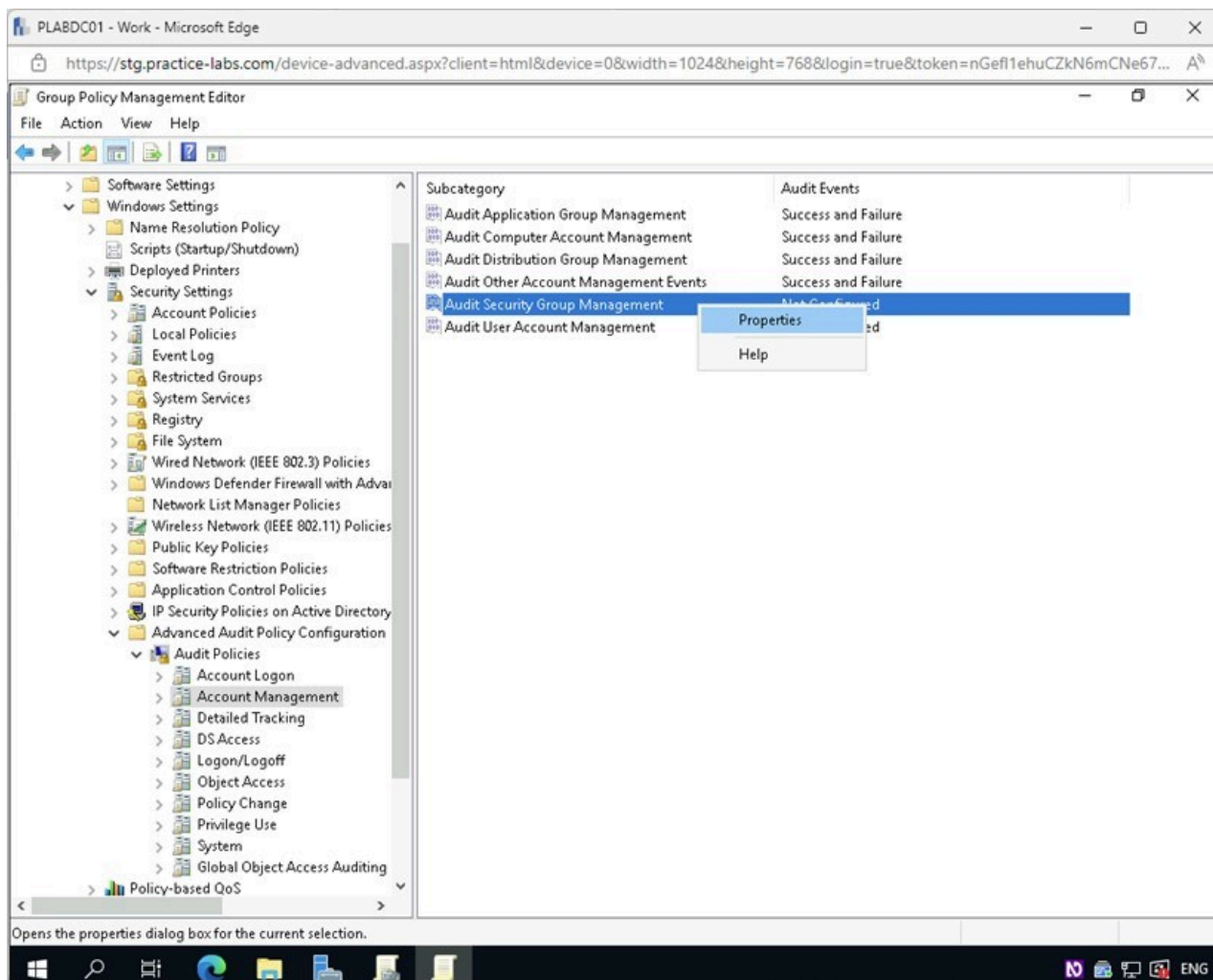


Figure 4.49 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Audit Security Group Management is right-clicked, and Properties selected.

Step 18

In the **Audit Security Group Management Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

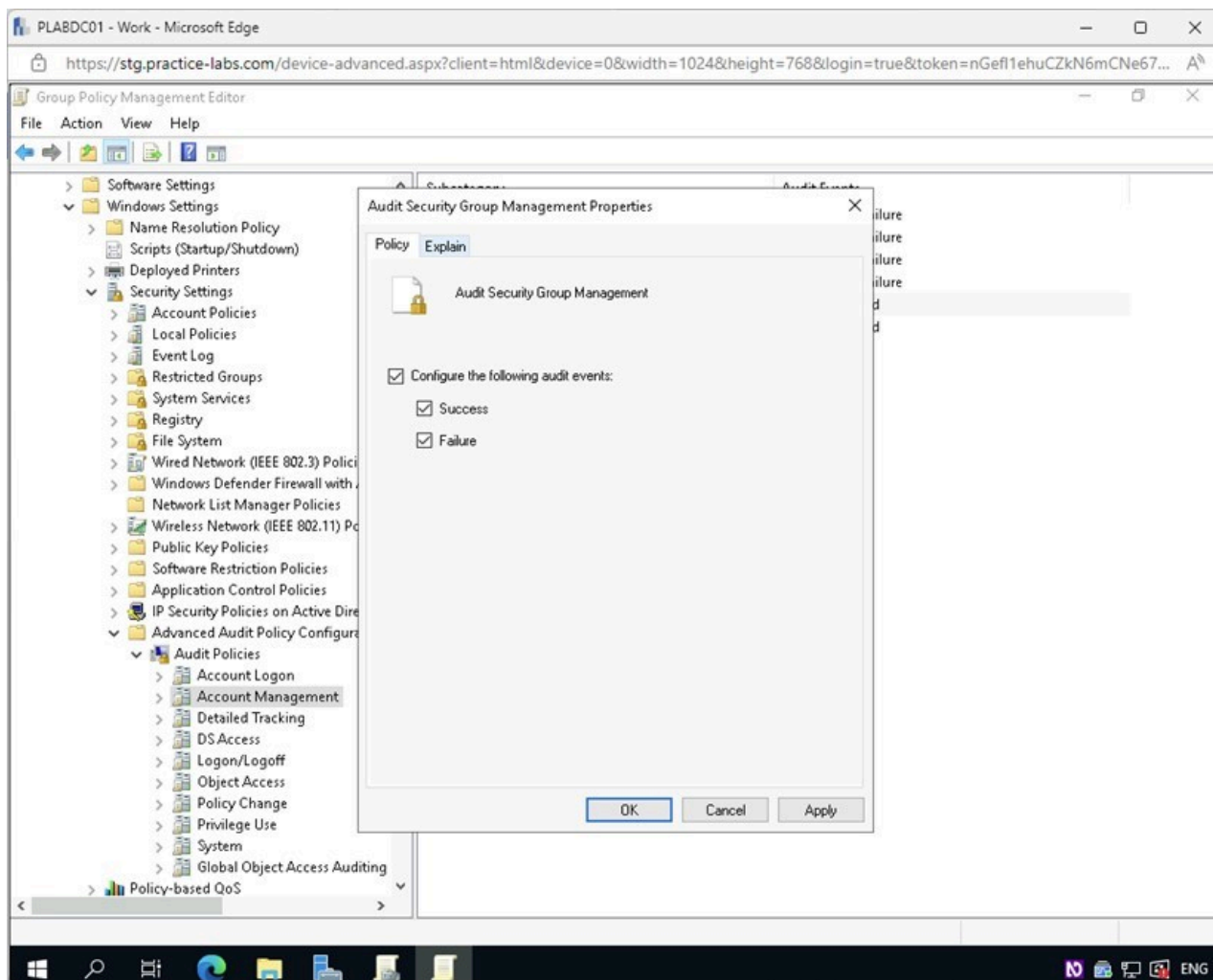


Figure 4.50 Screenshot of PLABDC01: Displaying the Audit Security Group Management Properties Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 19

In the **Explain** tab, you can view detailed information on **Audit Security Group Management**.

Once you've read the information available, click **OK**.

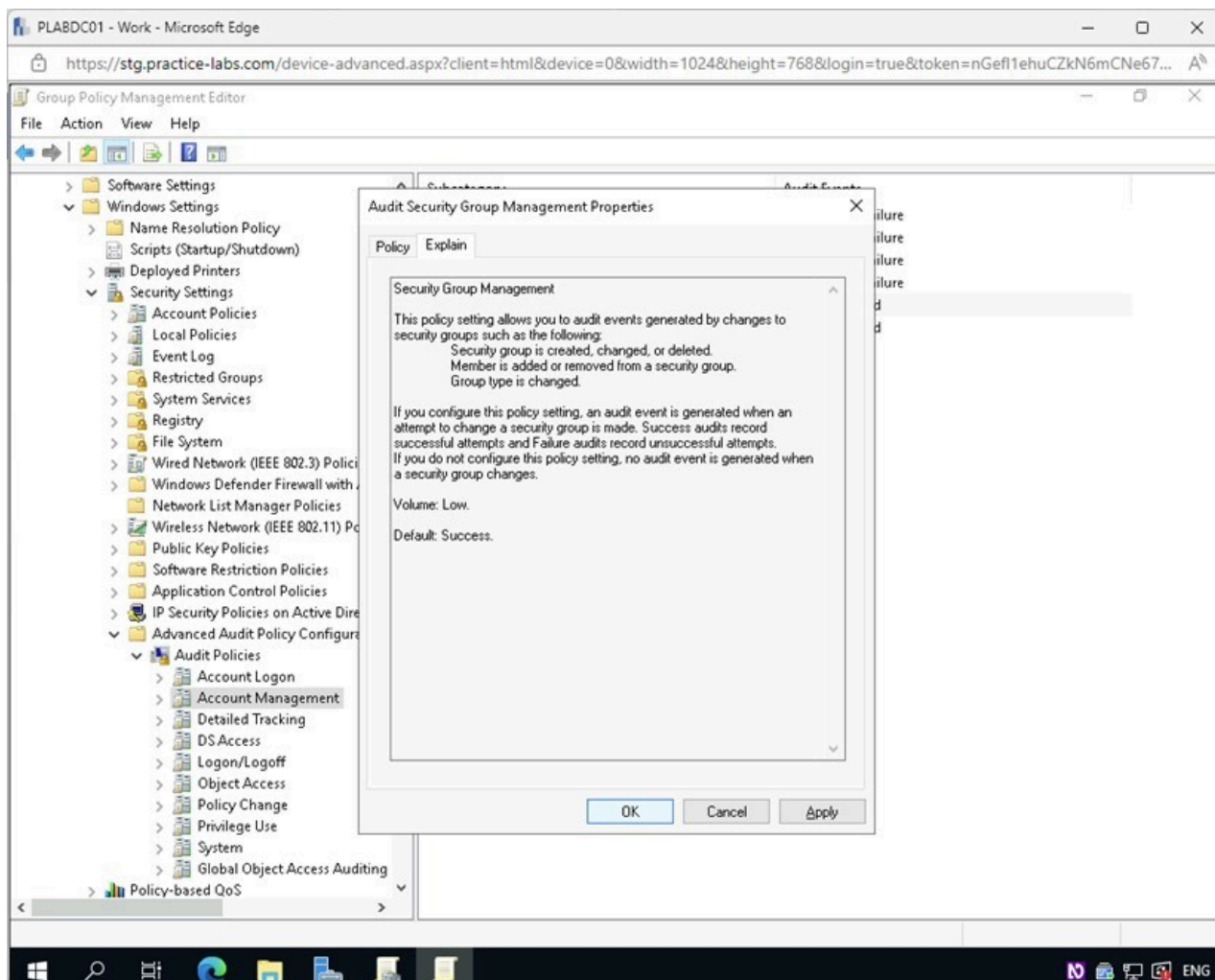


Figure 4.51 Screenshot of PLABDC01: Displaying the Audit Security Group Management Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 20

In the **Group Policy Management Editor** window, right-click on **Audit User Account Management** and select **Properties**.

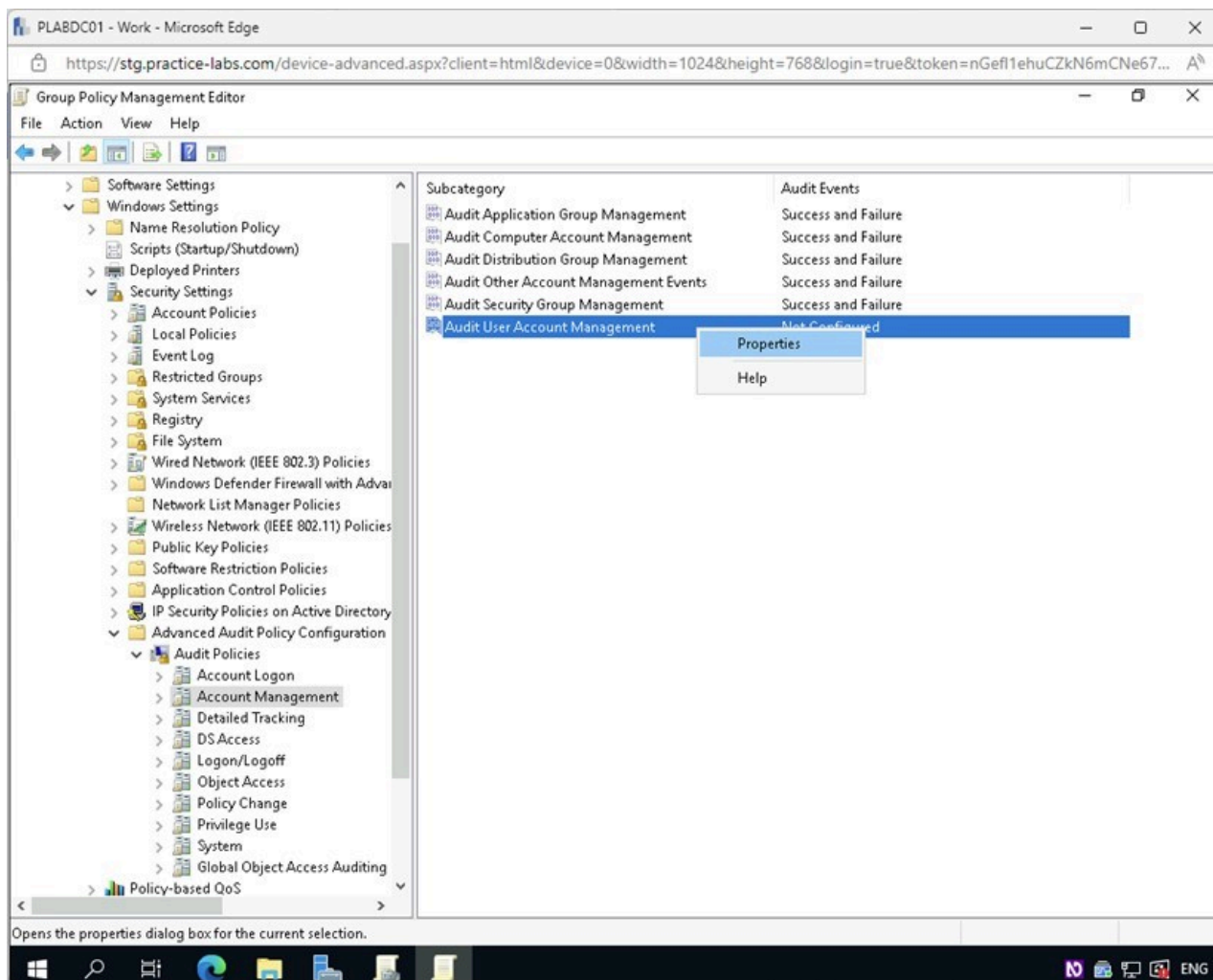


Figure 4.52 Screenshot of PLABDC01: Displaying the Group Policy Management Editor window. Audit User Account Management is right-clicked, and Properties selected.

Step 21

In the **Audit User Account Management Properties** dialog box, tick the checkbox next to **Configure the following audit events**.

Next, tick the following checkboxes:

- **Success**
- **Failure**

Click the **Explain** tab.

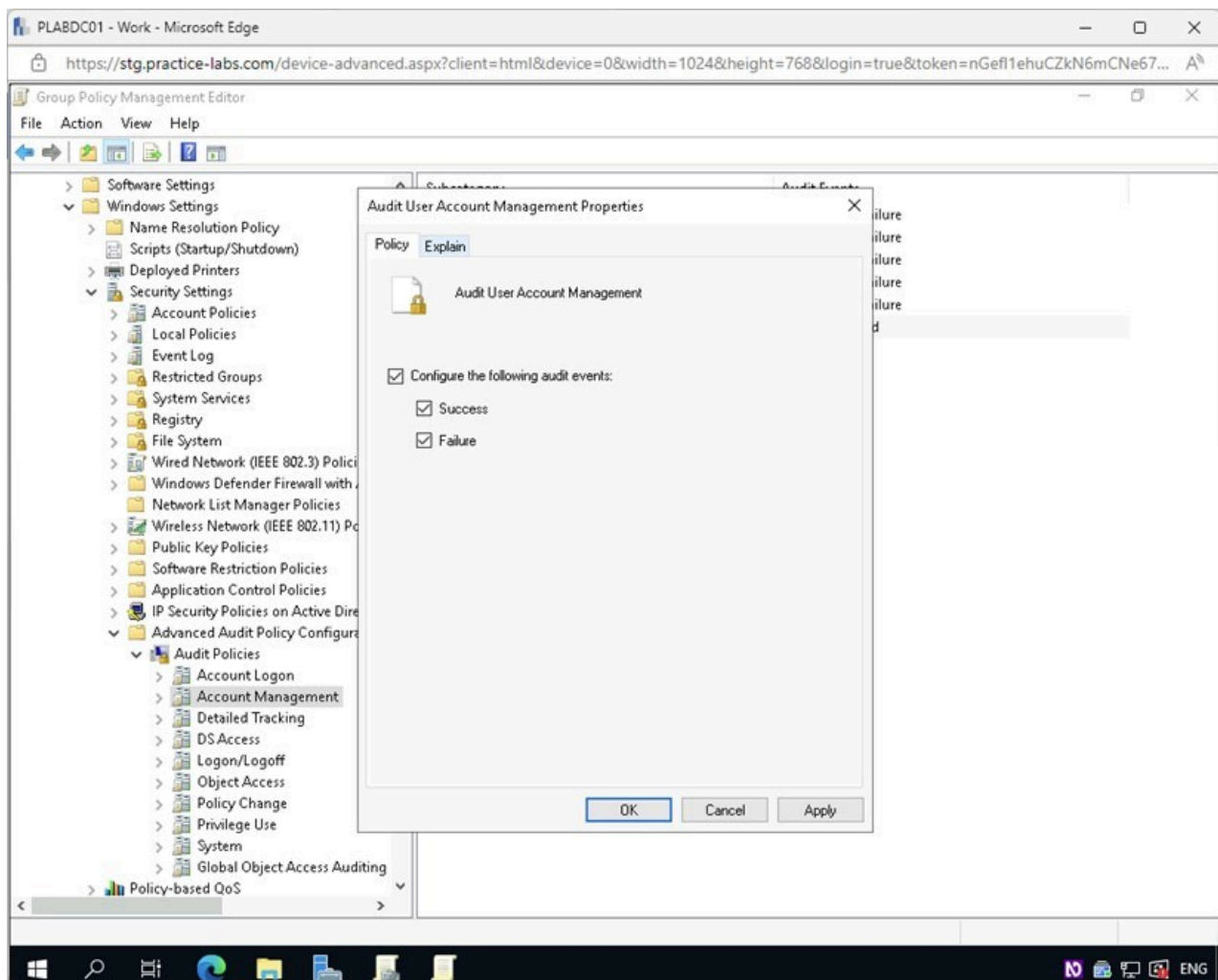


Figure 4.53 Screenshot of PLABDC01: Displaying the Audit User Account Management Properties dialog box, showing the required settings performed and the Explain tab selected.

Step 22

In the **Explain** tab, you can view detailed information on **Audit User Account Management**.

Once you've read the information available, click **OK**.

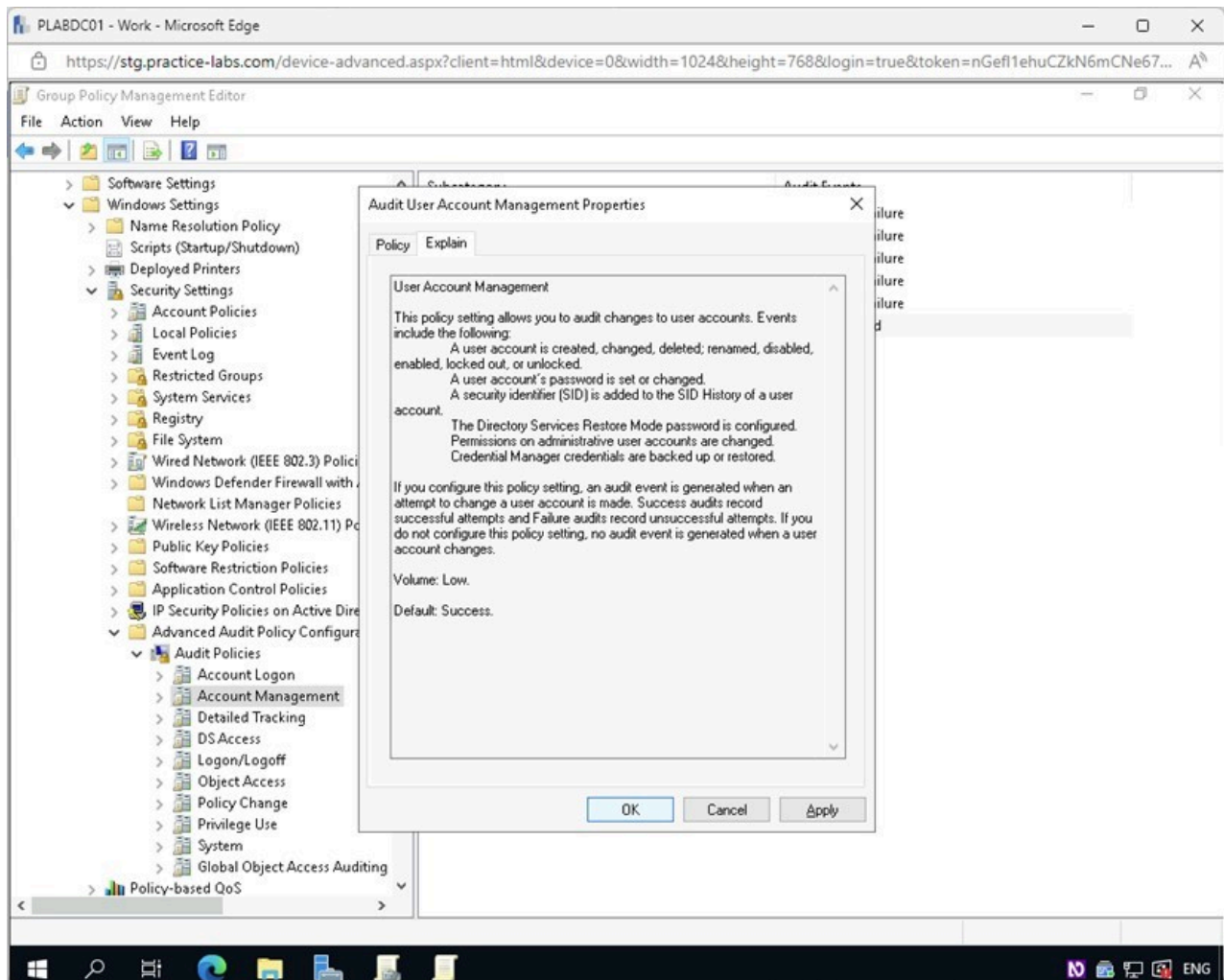


Figure 4.54 Screenshot of PLABDC01: Displaying the Audit User Account Management Properties dialog box, showing the Explain tab and the OK button highlighted.

Step 23

Close the **Group Policy Management Editor** and the **Group Policy Management** windows.

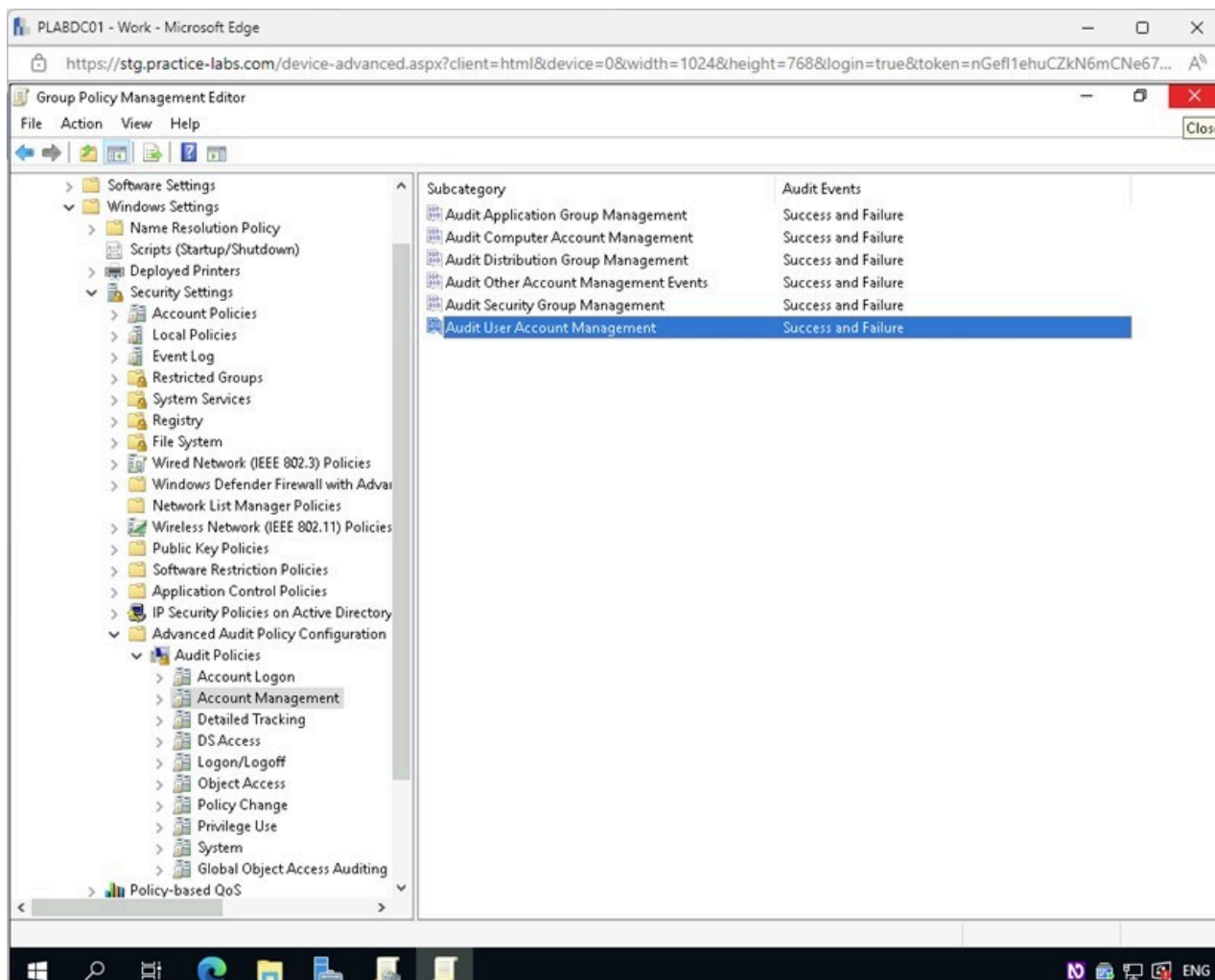


Figure 4.55 Screenshot of PLABDC01: Displaying closing the Group Policy Management Editor window.

Exercise 5 - Multifactor Authentication & Single Sign-on

Passwords are an essential feature of any secure system, but they can quickly become a source of frustration and inconvenience to manage for administrators and end users. Multifactor Authentication and single sign-on are two tools that can help alleviate these problems.

In this exercise, you will learn about Multifactor authentication & Single sign-on.

Learning Outcomes

After completing this exercise, you should have further knowledge of:

- Multifactor Authentication
- Single Sign-On

Your Devices

This exercise contains supporting materials for Server+ (SK0-005).

Multifactor Authentication

Multifactor authentication is a security technology that verifies a user's identity for a login or other transaction by using more than one method of authentication from various categories of credentials. Two or more independent pieces of information about the user, such as a password, a security token, or biometric methods, are used to verify who the user is. Several methods can be used, but they generally fall into a few categories:

- Something you know - Username, Password, Pin Code
- Something you have - Smartcard, Key Fob, Mobile Device
- Something you are - Fingerprint, Retina Scan, Facial Recognition
- Something you do - Push Authentication, Captcha

Most people now have their bank account information, contacts, messages, and more on their computers and mobile devices. Hence, there is more risk of people falling victim to cybercriminals. When it comes to keeping data safe, the traditional approaches aren't the most reliable, and they certainly won't hold up against today's sophisticated attacks. The number one priority should be the security of all data. Multi-factor authentication is an efficient and reliable security measure meant to secure beyond what a basic password allows.



Figure 5.1: Displaying a Two-Step Authentication method.

Single Sign-On

With Single Sign-On, a user must just remember one set of login credentials to access a wide variety of services, websites, and applications. Organizations of all sizes, from sole proprietors to multinational conglomerates, may benefit from SSO's ability to streamline the administration of user login credentials. The Open Authorization (OAuth) protocol is the framework that lets third-party services like LinkedIn, Twitter, Facebook etc., access a user's account data without having access to the user's password. By giving the service an access token that the user has approved, OAuth functions as a mediator between the user and the service. The service provider initiates communication with the identity provider when a user seeks to access an application. The supplier of the service then checks the user's credentials and grants access.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **Server Identity and Access Management** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Password and Account Lockout Policies
- Exercise 2 - User Accounts and Groups
- Exercise 3 - Permissions and Access Controls
- Exercise 4 - Auditing
- Exercise 5 - Multifactor Authentication & Single Sign-on

You should now be able to:

- Explore, Define and Configure Password Policies
- Explore, Define and Configure Account Lockout Policy
- Apply Password Policy
- Update Group Policy
- Explore, Define and Configure Active Directory User and Computers Accounts

- Explore, Define and Configure User Groups and Role Based Access Control
- Explore, Define, and Configure Delegation
- Audit User Activity
- Audit Logins
- Audit Group Membership

You should now have further knowledge of:

- Segregation of Duties
- Multifactor Authentication
- Single Sign-On

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.